

Psychology in Outline

on the Basis of Experience

Psykologi i Omrids paa Grundlag af Erfaring

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Chapter 1

The Object and Method of Psychology ^[1]

1. Psychology is the science of the soul — that is the shortest description we can give of the object of the inquiries we are here to undertake. But it is a purely provisional description, which yields no clear and distinct concept. By it we merely set up psychology as the science of that which thinks, feels and wills, in contrast to physics as the science of that which moves in space. Within these two domains lies everything that can be an object of human inquiry. Psychology is as little obliged to begin with an explanation of what the soul is as physics is obliged to begin with an explanation of what matter is. Nor does this general statement of the object of psychology contain any presupposition as to whether or not the soul exists as an independent being, separate from matter. Our endeavour is to keep psychology a pure science of experience, and to distinguish sharply between what is given and the hypotheses we use to order it and make it accessible to understanding.

But as soon as we assume that psychical phenomena have certain characteristic marks by which they can be distinguished from material ones, we presuppose a piece of knowledge which first appeared at a certain stage of intellectual development, and which even now cannot be said to have fully prevailed. It will throw light on the object of psychology to bring out some features characterizing the way in which the idea of the psychical has developed in the human race, and still develops in each individual.

2. It is with the mind's eye as with the bodily eye: from the first it is turned outward. ^[2] The eye perceives external objects, their colors and forms, and only by artificial detours does it see itself and what is within it. And as regards external objects themselves, the eye is naturally adjusted to the far point, the greatest distance of vision; while we feel a certain strain in accommodating the eye to nearer objects, it is with a feeling of relief and rest that we turn our gaze from the nearer to the more distant. In the same way we are

taken up with external objects long before we think of the very sensing and imagining through which alone external objects exist for us. Our immediate, natural life we lead in sense-perception and fantasy, not in subjective reflection. This is connected with the fact that man is a practician before he is a theorist. His weal and woe depend on his being able to forget himself in his surroundings. Observations of the doings of men and animals, of the appearance of plants and fruits, of the course of the heavenly bodies, etc., are more important in the struggle for existence than observations of oneself. Only at a higher stage of culture can the command be uttered: Know thyself! — and with it the way be opened for direct psychological inquiry.

3. Since language has developed under the influence of attention directed to the external world, we find that the expressions for psychical phenomena are originally taken from the material world. The inner world of the spirit is designated by symbols taken from the outer world of space. This observation had already been made by *Locke* and *Leibnitz*, and recent linguistics has confirmed it. "All roots, i.e. all the material elements of language, are the audible signs of sense-impressions, and of such only; and since all words, even the most abstract and the most exalted, are derived from roots, comparative philology fully confirms the conclusions which *Locke* reached."¹ Interesting examples [3] are found in the words that designate spirit and soul in the various languages, in the names of the mental activities, and in the meaning of the prepositions. "It may be taken as granted that all prepositions proper originally designate relations in space exclusively — not only because every tracing of their particular meanings back to the source leads there, but because all particular ideas are arranged for the imagination into connection in the schema of space, and because these spatial relations were the only ones that could be pointed out and that presented themselves in such a way that an understanding could form about words attached and applied to them. The task of language, i.e. the need and endeavour of speaking men, was therefore to make intuitable the non-spatial relation in which ideas were seen, through analogy with and subsumption under spatial relations."²

4. If, then, the first circle of ideas into which man lives his way draws its elements from external nature, how does it come about that we learn at all to distinguish between our own ego and the things outside us?

The beginning of conscious life must be placed before birth. The tender organism is, it is true, still separated from the great surrounding world; some of the most important

¹*Max Müller*: Vorlesungen über die Wissenschaft der Sprache. II. p. 320. — *Locke* gives as examples, among others, *to imagine, comprehend, conceive, spirit*. *Müller* adds, among others, *animus* from *anima*, breath; cf. Sanskrit *an*, to blow, Greek *aēnai*, and Sanskrit *anila*, Greek *anemos*, wind.

²*J. N. Madvig*: Sprogvidenskabelige Strøbemærkninger. 1871. (Universitetsprogram). p. 9. — *Leibnitz* had already stressed this in particular. *Nouveaux Essais* III, 1, 5. (Opera ed. Erdmann, p. 298).

sense-impressions (sight, hearing and smell) first become possible at birth; and the sensations that can be had (taste, touch and muscular sensations) no doubt stand out only faintly and dimly from the general feeling of vegetative well-being and ill-being. Yet the experiences that can be had before birth are perhaps already enough to form the basis for the consciousness of a surrounding world. The feeling of well-being or ill-being, together with the muscular sensations, or the sensations of energy expended, can already here come into a certain contrast with the sensations of resistance, touch and taste. It goes without saying that this first germ of world-consciousness is dim and dreamlike, and that from our waking, fully conscious standpoint we easily come to read too much into it. But these first stirrings must nevertheless be taken into account, especially since they give a hint of how difficult it is to assume any definite point as the [4] transition from unconscious to conscious existence. It is the merit of Cabanis to have shown the importance of this first period of the life of the soul.

The revolution that takes place at birth thus consists not in an ensoulment, but in a great change in the conditions of life, both inner and outer. The general sensation is altered by the fact that nourishment and breath no longer flow in directly from the mother's organism, with which the fetus was united in one living unity, but must be taken in from outside and assimilated by special organs (the alimentary canal and the lungs). Besides greater energy and independence in the internal functions, this also brings stronger fluctuations in the general sensation, since the inflow, which before was continuous, now becomes periodic and interrupted, so that the contrast between sensations of pleasure and of displeasure becomes more intense. At the same time the impressions of the larger surrounding world now storm in upon the delicate organism, which must be especially susceptible to impressions of cold. The cry of pain with which the newborn child begins its life finds its most probable explanation partly in the want of breath brought on by the separation from the mother's organism, partly in the sensations of cold.³

Although the general sensation still continues for the time being to play the largest part, such a multiplicity of impressions gradually streams into the infant consciousness that a contrast can arise between a subjective pole and an objective pole. The feeling of pleasure and displeasure, the general sensation, and the sensation of unimpeded movement do not lead consciousness beyond itself. But now, first of all, the sensations of sight gain a dominant influence. Light acts early on the newborn child, although in this as in several other respects individual differences at once make themselves felt. The

³*Adolf Kussmaul*: Untersuchungen über das Seelenleben des neugeborenen Menschen. p. 27 f.

child seems to feel pleasure in the impression of light and tries (already on the second day after birth) to turn toward it so as to keep hold of it. The ability to fixate definite objects develops from the third week on; it is of course near and conspicuous objects that [5] are chiefly held fast. Before that time sight is a mere staring without definite direction or expression. Bright, strongly colored and moving objects especially attract attention. Memory now also plays a prominent part: instead of continuing to cry as long as it feels hunger, the child is already quieted by noticing the preparations for satisfying its need (in the third week), and recognizes (from the third month) its mother as the source of this satisfaction — in which the sensations of sound also play their part, the head being turned toward the quarter from which the sound comes.⁴ — But neither in the sensations of light, of sound, nor of touch lies the full explanation of the consciousness of a surrounding world distinct from the individual itself. They show themselves, indeed, to be independent of the individual's feeling of pleasure and displeasure and of its active movement, but they do not come into outright conflict with these. That is first done by the sensation of resistance and limit. Such sensations of arrested and hindered movement are, as remarked, possible already in the fetal state, but now become more manifold and more vigorous. The great vegetative energy finds an outlet in movements of the limbs, and the child is thereby led to experiment with the things around it. The active experiment is for the adult, too, the surest means of finding his bearings. The child does not wait for the surrounding world to come to it; by its involuntary movements it reaches into the world of itself from the first, and thereby gains the best knowledge of where the boundary lies between the world and itself. At the points where movement meets resistance, especially when this resistance causes pain, the non-ego begins. — When the ideas of memory become frequent and connected, we get a third factor of importance, namely the contrast between the clearer and more vigorous impressions, which arise immediately, unexpectedly and often without connection, and the fainter images which are at the disposal of consciousness in all circumstances — that is, the contrast between observations and recollections.

The question now is how much is counted as belonging to the ego. If the sensations of resistance and the feeling of pain are what most strongly determine the distinction [6] between ego and non-ego, then the boundaries of the ego will for the time being coincide with the boundaries of one's own organism. Everything that takes part in the movement and meets resistance we count as ourselves. The limitation can only be learned by way of experience. The child discovers its own body successively. First the hands become

⁴Kusmaul. p. 26. 39. — Vierordt: Die Physiologie des Kinderalters. p. 154, 159.

a familiar member of its own organism; they are examined especially with the help of lips and tongue, the child sometimes putting its fingers in its mouth and sucking on them even on the first day. Later the child learns to fixate them with its eyes; a firm association of ideas then soon forms between the muscular sensation that accompanies the movement of the hands and the sight of this movement. Later the feet are discovered; this can only happen when the child can sit up and see them, or, lying on its back, can stick its legs in the air to look at them and grasp at them with its hands. The great interest with which the child regards its limbs and their movements may perhaps be derived from the strangeness of there being something here that can be seen and touched, and that offers resistance — and yet takes part in the active movement. It is an object that nevertheless belongs to the subject. The child here has the same experience as the dog that runs round after its own tail. When a child, even toward the end of its second year, offers its own foot a rusk, it no doubt still regards the foot as an independent being. The mutual touchings of the limbs, and their resistance to one another's movements, gradually clarify the idea of one's own body as both akin to and in a peculiar way different from other objects. This idea becomes clearest when the child causes itself pain by treating parts of its organism as a mere object.

One step more is possible here; but it is taken only at a later age, and not by all, or at all times. One's own body, which till now was distinguished from the non-ego, nevertheless has essential marks in common with it: it can be perceived by the senses and can offer resistance. It thereby comes into contrast with the feeling of pleasure and displeasure and with the inner stream of recollections and ideas. What we feel pleasure and displeasure at we can perhaps observe with the help of the senses, but not the feeling itself. What we remember and imagine can be an object of external observation, but not the recollection and the idea themselves. This contrast is so decisive that the idea [7] of the body passes over to the objective pole, to the non-ego, and what we then retain is the idea of the ego as the subject of thought, feeling and will. The contrast between the inner and the outer is now sharpened, or rather, we keep the expression "inner" as a figurative designation for the psychical domain in contrast to the bodily as the "outer." Inner experience, then, concerns thoughts, ideas and feelings as psychical states; outer experience concerns what can be intuited and can offer resistance to movement in space. Outer experience must necessarily be a part of inner experience, be enclosed by it. When at the primitive stage the ego is made one with the body, this really means: with the bodily sensations, which in reality, too, make up an essential part of the individual's history. Jeppe knew himself by his sore back and his hollow tooth, because these marked

outstanding points in the connection of recollections which made up his ego.⁵

5. Folk psychology teaches that the idea of the psychical has passed through stages in the history of the race similar to those in the single individual.

The current mythological theory ascribes to primitive man a tendency to apprehend and explain all natural phenomena as the intervention of personal beings. Everything acquires life and personality for him, because he puts his own personal life into everything. To explain means, after all, to refer the unknown to the known, and from where should [8] man at first take the elements for the explanation of the phenomena of nature, if not from himself? — So much, too, is clear: the mythological view of nature differs from the scientific by its personifying tendency. But might not this tendency itself need an explanation, and might one not ask whether there are necessary intermediate links through which the idea of personal powers is transferred to natural phenomena? The theory in question seems, besides, to ascribe to primitive man a creative fantasy that is possible only at a higher stage of development. Finally, if the theory were correct, one would have expected language to designate bodily things by originally spiritual expressions, whereas in reality it designates spiritual things by originally bodily expressions.

Recent inquiries into the mental life of savage peoples have therefore led to a modification, if not an outright rejection, of the theory of an original personifying faculty. *Tylor*, *Lubbock* and *Spencer* have shown that dream-images play an exceedingly large part in the formation of the primitive view of the world. In dreams man sees himself and others, and since he originally has no reason to suppose any difference between dream-images and observations, he regards the former as just as real as the observations of waking life. Just as a child sends greetings from those it has dreamt of, primitive man takes what he lives through in dreams for real experiences. He has thus been in distant regions, though it turns out that his body has not stirred from the spot, and others have visited him, though it is certain that they were far away or dead. Besides dream-images, mirror-images of the individual himself and of others also lend support to another form of existence than the one given with the body's presence at a definite place. A savage who was allowed to look in a mirror cried out: "I am looking into the world of spirits!" One of Darwin's children, at nine months old, turned toward the mirror when

⁵How much influence such bodily sensations have on self-consciousness is clearly seen in cases where one seems a stranger to oneself because of changes in the general sensations. The physiologist *Harless*, on waking from an anaesthesia with sulphuric ether, saw two persons about him, but missed with unease a third, who had been present before, without its occurring to him that this third was himself. Another time, on waking from the anaesthesia, he could not distinguish his personality from another person on whom his eye fell; only when a little later he looked in a mirror did he come fully to himself. (*Lotze's Medicinische Psychologie*, p. 502). In certain morbid states the individual seems to itself to be another; everything that moves in it is as if it took place in another being (*Revue philosophique* I, p. 289 ff.).

it was called by name. These experiences lead to the assumption of a double existence: as soul, man is a free, ethereal being; as body, bound to definite and limited places in space. This doubleness now forms a fixed point of support. The striking phenomena — change, coming into being and passing away, life and death — are now naturally explained by assuming a similar doubleness everywhere. The souls of the departed in particular furnish an important element of explanation; it is with them that fantasy is most occupied, and it is therefore inclined to find their intervention everywhere. It is from the world of dreams, then, that the whole of nature is peopled. [9]

Spencer, however, no doubt goes too far when he thinks he can do entirely without the presupposition of a personifying tendency. It is one thing that merely appealing to such a tendency is not enough, another whether it can be dispensed with altogether. Apart from the fact that without it one can hardly give a not overly artificial explanation of the worship of plants and animals,⁶ it is clear that man must after all recognize himself in the dream, and must put a life of feeling and will corresponding to his own into the images of other men that the dream shows him. If one denies the personifying tendency entirely, one cannot explain how man can assume the existence of personal beings outside himself at all. The dream theory marks an advance in so far as it makes it easy to understand that man puts his psychical life into his double; here he immediately recognizes himself under other conditions. But what happens here differs only in degree from what always happens when he acknowledges a psychical life in other beings that make movements resembling those he himself makes in certain moods. His instinct of self-preservation leads him early to interpret the movements of other beings, to read certain looks and features as signs of certain feelings. It is a natural inference by analogy through which even primitive man arrives at the assumption of psychical life outside himself. How far the analogy is extended depends on the standpoint; we shall see that it is not at all easy to fix measure and limit here. At the mythological standpoint man is very lavish in bestowing a psychical life of the same kind as his own on the things around him. The boundary between ego and non-ego is here, too, indefinite, just as for the child. Only advancing experience can draw sharper boundary lines.

At this stage the soul is thought of as an ethereal being in contrast to the body as a coarser and heavier being. This doubleness has still many changes to undergo before it becomes the contrast between an immaterial and a material being. Only slowly and successively are the physical characteristics in the concept of the soul worn away. In the case of Greek intellectual development, such a wearing away took place in the time [10]

⁶See Tylor's criticism of Spencer's theory in *Mind*, April 1877.

between Homer and Plato. For *Homer* the soul is only a fainter counter-image, a reflex of the body; in his childlike way of apprehending things, man's real self coincides with the body, which is why (in the 3rd and 4th verses of the first book of the *Iliad* in the Greek text) he makes the souls of the heroes go to the underworld, while they *themselves* become the prey of dogs and birds! — *Plato*, on the other hand, makes Socrates say to his friends, on the occasion of Crito's question how he wishes to be buried: "I cannot convince Crito that I am the Socrates who is now talking with you and has ordered the whole discourse; he thinks I am the one he will soon see as a corpse, and so he asks how he is to bury me." (*Phaedo*. Heise's transl. p. 119). — Here a purely spiritual idea of the soul is expressed, its essence being placed in the activity of thought. But this pure idea, which the ancient philosophers did not, moreover, carry through, was obscured again in the Middle Ages, whose faith had a decidedly materialistic stamp and, for example, imagined the souls burning in purgatory. As Plato stood to Homer, so in modern times does *Descartes*, who placed the essence of the soul in consciousness, stand to the medieval view. — Were we to explain this transition from Homer's to Plato's standpoint by way of psychological analysis, we should come back to a process like the one we have already described above within the individual consciousness (pp. 3–7).

6. That we know psychical life directly only in ourselves, and discover it outside us by way of analogy, we can finally convince ourselves by taking up a physiological standpoint. As long as the organic processes and movements were still explained by a special vital force or by the unconscious activity of the soul, one could of course extend [11] psychical life to all organic phenomena. But even then the question became one of the relation between the "soul" in the wider sense, as vital force, and the soul in the narrower sense, as consciousness. *Descartes* was the first to assert sharply the purely psychological criterion of psychical life, in contrast to the older, Aristotelian view, which also called the principle of nutritive life soul. He preferred to avoid the word "soul" (*anima*) because of its ambiguity, and put the word "consciousness" (*mens*) in its place when he wished to designate the object of psychology. The land of souls was thereby considerably reduced. *Descartes* himself found occasion to assume a consciousness only in man; the animals he regarded as mere machines. This was a paradox; but it points to a reform in the view of nature. Instead of appealing to mysteriously acting forces, a purely mechanical explanation of nature can now be introduced, once the "soul" has been separated out from the world of bodies. The phenomena of organic life, too, now come under the general mechanical principles. Where physiology has arrived at an explanation of an

organic phenomenon, it is by applying this principle. Physiology is organic physics.⁷ But if the soul can thus no longer be used to explain the bodily phenomena, if appealing to the soul as a cause is the same as giving up all explanation, how can one at this standpoint be led to assume a psychical life at all? Here we seem to have the complete opposite of the mythological standpoint: while that found the soul everywhere, at the physical-physiological standpoint the soul seems to vanish from existence altogether. The physiologist sees only movements of material parts, and it rests on an inference when at certain points he interprets these movements as the expression of an activity of consciousness.

The difficulties with which such an interpretation may be attended are seen from the following examples. — A frog whose head has been cut off does not thereby lose the ability to perform purposive movements. Some physiologists therefore think that one must ascribe a “spinal soul” to the headless frog; others deny that this is warranted. Precisely the purposiveness could, after all, point to a fixed mechanical connection; it [12] is in itself just as characteristic of conscious beings that they make mistakes as that they behave rationally or purposively. — In other cases physiological inquiry reaches a result different from what seems self-evident to the popular view. The death struggle is as a rule held to be very painful; but that rests on a false interpretation. “As a rule the pains diminish when death does not come suddenly but approaches slowly, and they cease altogether when the death struggle begins. This struggle, or this period of transition, may last a longer or shorter time, often even many long hours; and in most cases consciousness is lost only toward its end. Even after the circulation of the blood has stopped, the activity of the brain has ceased and life is therefore over, so that it may safely be assumed that the soul has left the body and the spiritual and bodily connection has ceased — even after that one can sometimes see convulsive movements of inspiration, with twitchings in all the inspiratory muscles about the nose and mouth, alternating with a faint, drawn-out, rattling sound of expiration (the so-called death rattle, which comes from mucus in the windpipe). These twitchings, which often give the face of the dying a suffering or even a frightful expression, are nevertheless not real manifestations of life, but a kind of galvanic reflex movements, which soon pass into a complete slackening of the muscles. Since they seem to be the last efforts of the body to hold on to life, it is no wonder that, with them in view, the transition to death has been regarded as a hard struggle and as a bodily suffering — which, as was said, it is not.”⁸ — In contrast to this, there are cases in which immediate apprehension would confidently assert that there is

⁷*Panum*: Indledning til Forelæsninger over Menneskets Physiologi. p. 13.

⁸E. Hornemann: Om Menneskets Tilstand kort før Døden. Kbhvn 1874. p. 18.

no suffering consciousness present, while a skilled examination nevertheless arrives at a quite different result. Some South American Indians poison their weapons with a poison called curare, which rapidly brings about a progressive paralysis of all the vital functions. Death comes without struggle or rattle, so that the onlooker must believe that it is the [13] mildest manner of death possible. But *Claude Bernard* has proved that curare attacks only the motor nerves, not the sensory nerves, so that consciousness and will are not abolished, and he thinks that this death, which seems so mild, must be precisely the most dreadful one can conceive. "Can one conceive a more fearful suffering than when the conscious spirit is thus witness to the successive sinking of all its organs, and in a manner finds itself shut up alive in a corpse?" — "The faculty of sensation," Bernard remarks, "can no more than any other of the faculties attached to the nervous system manifest itself by itself. These faculties necessarily need the contractile or muscular system to announce their presence or manifest themselves to the outer world. Consequently we can judge of the sensations of men and animals only by their movements. Yet we should have been in the most complete error if, in the animals poisoned by curare, we had inferred from the absence of movement the absence of the faculty of sensation (sensibility). This example proves that *only in our consciousness do we have an absolute criterion*, and that, when we set about interpreting external phenomena, we are everywhere surrounded by causes of error and illusion."⁹

If, then, we wish to come to know conscious life, we must first and foremost study it where it is *immediately* accessible to us, namely in our own consciousness. It is from this immediate experience alone that the physiologist, too, draws when he seeks to determine the significance of the various organs of the brain for psychical life. It is the fixed starting point for all our knowledge of the spiritual world.

7. But how much, now, lies in this starting point? The object of psychology is not something that can be presented to fantasy or to sense-observation. It is not something we can find directly outside ourselves, but something which, where we find occasion for it, we transfer consciously or unconsciously from ourselves to others. We have therefore no right to take as our basis more than what the inner observation of our own [14] consciousness teaches us. But here, too, is something whose reality no one can dispute. No one can deny that there are sense-sensations and ideas, feelings and resolutions; and when we say that psychology is the science of the soul, we provisionally understand by soul no more than the sum of all these inner experiences. In this sense the existence of the soul is no problem, but on the contrary a necessary presupposition. There is,

⁹*Claude Bernard*: La science experimentale, p. 299 f.

however, another sense in which a question may arise as to the position of science on the existence of the soul. Not content with the simple starting point in psychological experience, it has been thought necessary for psychology to take as its basis the idea of the soul as an absolutely independent individual being, subsisting by itself. It is an idea that historically points back to the mythological doubleness, as this was purified and given form partly by ethical and partly by theoretical motives. On the one hand, what was at work here was the feeling of the high worth of psychical life, the feeling that everything that interests us dwells in this inner world of thoughts and feelings, and that the outer world of matter itself has value for us only as an object of thought and feeling. This inner world was therefore raised high above the material world and sharply separated from it. On the other hand, the spiritualistic view also has its basis in an analysis of the characteristic marks of psychical states. It is peculiar to consciousness that it gathers what is scattered in time and space. In memory the differences of time are abolished, in comparing and combining thought the distances of space. Here is a thoroughgoing unity, an inner connection, to which we have no counterpart. Is that not warrant enough for ascribing to the soul subsistence in and for itself, and conceiving it as an immaterial substance?

However great the warrant may be, we cannot in any case acknowledge it right at the beginning of psychology. Here above all it is a matter of building on nothing but immediate observation. And that assumption does not even give us any help toward a better understanding of psychological experiences. One infers a soul-substance from the character of certain experiences, but knows only as much about it as lies in them — and in so far as one wants to know more, the warrant of the inference in any case ceases. Experience certainly teaches us that the peculiarity of consciousness, in contrast to mate- [15] rial phenomena, is the inner unity in all the multiplicity of the content of consciousness, a unity such as is not found in the world of space; but it does not teach us that this unity is absolute, unconditioned and independent. In so far as the concept of substance is taken in the strict sense, namely of that which subsists in and for itself, independent of and not resting on anything else, we have no right from experience to apply it to the soul. The most gifted defender of spiritualistic psychology at the present day, *Hermann Lotze*, has also clearly seen this. When he ascribes substantiality to the soul, he means merely to designate it as an independent element in the world, a center of acting and suffering, and not to assert anything about its absolute essence. He therefore also declares that from the essence of the soul as substance in this sense no further conclusions can be drawn — as the earlier spiritualistic psychology thought — in respect of the philosophy of religion;

no way is thereby opened to us to knowledge of the fates of the soul, its prospects for the future, or its origin. He even agrees with *Spinoza* that, if we wish to think the concept of substance in the strictest sense, there can be only one single substance, in whose essence everything else must be moments of content, whose reality rests on their significance within the nature of the primal substance. So even for metaphysical speculation the idea of the soul as substance is useless and unfruitful.

But even with the restriction Lotze has given the concept of substance in its application to the soul, misgivings remain for empirical psychology. What Lotze has chiefly in mind when he determines the soul as substance in the sense of an independent starting point and center is its relation to bodiliness. On his view, then, it is supposed to be clear from the very chief characteristics of the psychical nature that this is in itself something different from bodiliness, and that a relation of interaction must be assumed between them, however one may more precisely conceive this. Here again more has been put in than experience for the present warrants. From immediate experience of our inner states we can infer nothing whatever about their relation to other sides of existence. Psychological experience acquaints us only with the inner psychical phenomena themselves, not with the way in which they are connected with other phenomena. That is a question by itself, which psychology cannot settle one-sidedly and directly. Other circles of experience besides the psychological must be drawn in to settle this problem, and it is of great importance that, for each single circle of experience, no unwarranted ideas be put in that might anticipate the decision in one direction or the other. We cannot decide from the outset whether two different principles or substances underlie the psychical and the bodily. We see two domains of experience before us, each with its peculiarities, and study these, in order then, always under the guidance of experience, to attempt to determine their mutual relation.

This is also in the well-understood interest of metaphysical speculation itself. The human spirit will never let itself be kept from brooding over the ultimate principles of the existence of which it is a member. It will always seek to form certain highest, concluding ideas for its view of the world. But what it must learn, and what it already could and should have learned, is that speculation must not meddle in the daily housekeeping of empirical knowledge, must not anticipate the settlement of purely empirical problems. The point is not that speculation should wait until experience is exhausted; for that will never happen. But the truly superior metaphysician is he who lets his ideas move in the direction in which the leading features of empirical knowledge already point. He thus only utters the thoughts that more or less unconsciously underlie empirical inquiry

itself. He thereby does not come into conflict with it, although he goes beyond it. He seeks an ultimate, concluding hypothesis, but the foundation is common to him and the empiricist.

Psychology as we conceive it is a “psychology without a soul” in so far as it asserts nothing about the absolute essence of psychical life, or about whether there is any such absolute essence at all. Just as little as physics pronounces on the transcendent questions (those going beyond empirical knowledge) in the domain of outer nature, just so little does psychology do so in the domain of inner nature. This is not to say that psychology cannot make an essential contribution to the general view of the world. [17] On the contrary, nothing can be of greater significance for this than knowledge of the psychical phenomena, their mutual relations and the laws of their development. And precisely a view of these phenomena carried through empirically will be able to clarify the points of view and correct many prejudices.

When we assert the empirical character of psychology against metaphysical speculation, it is materialism just as much as spiritualism that we thereby exclude from psychology. In what precedes we have spoken especially of spiritualistic psychology, because it is of the greatest interest and has the most acute defenders. But it will easily be seen that materialism commits the same encroachment as spiritualism. It, too, infers a substance supposed to lie behind the phenomena of consciousness, but it finds this substance in matter, not in a spiritual principle. Spiritualism holds to the difference of psychical life from material phenomena, and infers from it a special soul-substance which in itself has nothing to do with matter. Materialism, on the other hand, infers from the connection of psychical life with the material that the “soul” must be a material being. “Let it suffice us,” says *Holbach* (*Système de la nature* I. p. 118), “to know that the soul is moved and modified by the material causes that act upon it. From this we have the right to conclude that it must be material.” *Broussais* defined the soul as “un cerveau agissant et rien de plus” (an acting brain, nothing more). Both in what it asserts and in what it denies, materialism, like spiritualism, goes beyond the standpoint of empirical psychology. It is another matter that psychology through its own inquiries can bring us to a point where we can pass judgment on these hypotheses.

Psychology is not in itself a part of philosophy, if by philosophy one means a striving for a general view of the world; like the empirical science of outer nature, it is a preparation for philosophy, a part of the foundation on which philosophy is to build, one of the witnesses philosophy must examine. Nor, if by philosophy one means a critical theory of the nature and limits of knowledge, is psychology philosophy: while psychology

has a purely natural-historical character, considering the psychical phenomena in their [18] development and mutual relations, the theory of knowledge seeks by critical analysis to set up the general principles of knowledge. The theory of knowledge, too, therefore presupposes psychology.

On the other hand, philosophical thought itself becomes one of the objects of psychology. Philosophy, as a form of mental activity, falls within the domain of psychological observation. And philosophical inquiry has in many ways played into the hands of psychological inquiry; consciously or unconsciously to itself, philosophical speculation always operates with psychological elements, and many profound psychological observations and ideas have been laid down in the philosophical systems.

It is the merit of *the English school* to have asserted the independence of psychology from metaphysical speculation. *Descartes* had, it is true, taken the decisive step toward freeing the concept of the soul from the mythological ambiguities, by stressing consciousness as the characteristic mark of psychical phenomena. Yet he did not hold fast to the empirical standpoint, but laid the very foundation of metaphysical-spiritualistic psychology by setting up the soul as “thinking substance” (*substantia cogitans*) instead of keeping to the phenomena of consciousness as the sure empirical foundation. Kant’s epoch-making philosophical reform became important for psychology through his criticism of metaphysical (“rational”) psychology, a criticism whose significance has been shaken only temporarily by romantic attempts at restoration.

8. We have attempted to set up psychological inquiry as an endeavour distinct from the view of outer nature and from metaphysical speculation. Immediate self-observation and self-consciousness is, for the physiologist and the metaphysician alike, the source from which they draw, but which they often overlook, because their essential interest is not this immediate observation itself, but what they think they can infer from it. Granted, then, that it is necessary to draw from this source, the question becomes whether independence as regards the *principle* also brings with it independence as regards the *method*. This leads us to a closer examination of the nature and limits of subjective observation.¹⁰

[19] a. The first difficulty that presents itself here springs from the fact that psychical states are not resting and fixed objects like those that are the object of physical observation. As space is the peculiar form of material phenomena, so time is the form of psychical

¹⁰On the method of psychology one may compare *Stuart Mill*: System of Logic. Book VI. Chap. 4. — *Bain*, Logic. II, p. 275–292. — *Ad. Horwicz*: Methodologie der Seelenlehre (Zeitschrift für Philosophie und philos. Kritik. 60ter Bd. 1872.) — *Delboeuf*: La psychologie comme science naturelle. Paris 1876. — *Wilh. Wundt*: Über psychologische Methoden. (Philosophische Studien, herausgeg. von Wundt. I, 1. Leipzig 1881.)

phenomena. What goes on within us, in our thought and feeling, is unstable and changing. When the botanist has laid a plant before him, or the chemist has got a substance into his retort, they can calmly observe how the objects look under certain quite definite conditions. But a state of consciousness cannot be isolated in this way; it has no boundaries in space; every moment brings, or can bring, new elements in. In the moment I wish to observe a state of consciousness, therefore, it is already gone, or has undergone fusion with other elements. Now it is true that several different series of ideas can move through our consciousness in one and the same moment; our ego is not a unity in the sense that different, even conflicting currents are excluded. One might then imagine that, while the main current consisted, for example, in the contemplation of a work of art and admiration of it, an undercurrent ran alongside it which in turn stood in a contemplative relation to the main current. In that way one would be able at once to admire and to study the psychology of admiration. Everyone no doubt knows from his own experience states of this kind, in which an inner spectator keeps talking the whole time, although one seems to be quite taken up with something altogether other than oneself. Such states can hardly be avoided entirely in men whose reflection has once been awakened; and a double current of this kind can be of importance especially in ethical respects, where it is a matter of judging, and gradually driving out, a reprehensible current in the life of the soul. Criticism then has its say as an undercurrent, as an opposing motive that seeks to overcome what for the time being prevails. But although such double currents can also be made fruitful for psychological observation, psychology [20] must nevertheless, in its own interest, warn against them. The capital of energy at the disposal of psychical life is necessarily divided in them, and each single current becomes weaker. If a mood is to be lived through fully and wholly, the undercurrent must be driven out, and one must not listen to one's inner stirrings, so as not to withdraw from them any of the energy at hand. Besides, the observing undercurrent is not indifferent, but always alters the main current more or less. Conscious attention necessarily affects the immediate state toward which it is directed, and can thereby in part destroy its own object. It puts art in the place of nature.

But what cannot happen in the very moment of experience can happen later. During the experience one should only haul in the net with everything that is in it, or, like the botanist, gather the plants that come one's way. What has been fully and clearly lived through will remain in memory and can be examined through it. It is the rhythmic alternation between self-forgetfulness and self-consciousness that makes psychological self-examination possible, and psychological talent rests on the ease and elasticity with

which one can pass from the one of these states to the other, so that one keeps what was immediately lived through clear and pure in the moments of memory and reflection, but, conversely, does not let reflection disturb the immediate stirrings. And yet the two states will not remain wholly untouched by each other. Unconsciously — and therefore without harm — practice in memory and reflection will work to make a stronger light, or a more vigorous accent, fall in the immediate experience itself on the elements that are of special psychological interest. It is the general law of practice that is at work here. In the same way the botanist, without thinking of it, passes by many plants that are of no interest for his plan of study. Involuntarily the observations pile up, and one fine day one finds oneself in possession of a rich psychological material.

Ludvig Feilberg's book Om størst Udbytte af Sjæleevner [On the Greatest Yield from the Powers of the Soul] provides a fine and illuminating example of the nature of psychological observation. The book warns against "Beflitning" [over-diligence], i.e. against the inclination not to let the events of the life of the soul develop freely and [21] vegetatively, but to intervene, hastening and regulating. "Such treatment is against the nature of thought, just as it is against the nature of the kitten to be mauled by the children. It must stand unheeded like the wild plant; then it develops normally and healthily. A sudden notion breaking in is a teacher to whom one must listen attentively and calmly, without noise of any kind. . . . Certain elves slip away when one looks for them." (p. 70 f.). But on the other hand the book contains precisely a multitude of psychical observations, of moods and states which — as the fruit of long spiritual botanizing — have fixed themselves in the author's memory and, without harm to their freshness and purity, are made the basis of reflections and inferences.

b. Even if one now succeeds in overcoming the difficulty just mentioned, a new one arises, namely that because of the individual differences between observers one has no guarantee that they really see the same thing. For here they do not have the object outside and between them; each has it within himself!

This holds more or less, however, of physical observation too. Observation is a subjective activity; what each observes exists for him just as he observes it, and only by comparison can he infer that others observe the same (cf., e.g., the perception of color). He who wants to show others something must lead them to see it themselves, must awaken their own activity of observation. What individual differences always make themselves felt here is seen, for example, from the fact that between two astronomical observers' determinations of the time of a star's movement a difference always appears, varying with the observing individuals and resting on the different speed with which

an impression is apprehended and noted. In so far as more perfect recording apparatus does not remedy this, observations are begun by determining “the personal equation” relative to other observers. It has then turned out that this mutual individual difference is not constant, but subject to fluctuations from day to day as well as in the course of years.¹¹ Such mutual control is — though of course in a far more imperfect form — [22] the only means of raising psychological observations above the merely individual, or rather of distinguishing between what is individual and what is of more general validity. The distinction between the typical and the non-typical must begin already within the individual: when the individual wants to know his own inner nature, there are many observations he must disregard, because they stand isolated and owe their origin only to single passing situations. Everyone makes such a sifting involuntarily, and just as involuntarily daily intercourse with other individuals leads to distinguishing between the common, universal and the merely individual in personal observations. Psychological inquiry in both respects only continues what was begun without conscious intent. But in its critical testing it must here seek supports that lie outside the purely subjective method, whose limitation here already comes plainly to light. Already in the thorough control of individual experiences the merely subjective starting point has been left. But before we go more closely into the objective psychological method, we must dwell on the way in which subjective observations are worked up.

c. The scattered observations form a chaos which is to be ordered into a world. The first working-up is a classification, by which definite groups or kinds of psychical phenomena are formed. According to their most prominent characteristic marks, the single states are referred to different classes. But such *classification* is not as easy to carry out as was for a time believed. Just as classification in the zoological and botanical domain has led to the idea of eternal and unchangeable species, so that it now costs a hard struggle to show that these species are the fruits of a natural history of development, so psychological inquiry, too, for a long time thought it had reached its goal when it referred the various inner phenomena to various “faculties” of the soul — a procedure that came into a strange conflict with the strictly spiritualistic view of the unity of the soul. At the same time these “faculties” were regarded as the causes of the phenomena in question, and the need for causal explanation was thereby satisfied in a way that [23] was certainly very easy, but purely illusory. What was chiefly overlooked was that in the classification only one predominant characteristic mark was taken into account — that it was thus not the real concrete *states* themselves that were classified, but the

¹¹ *Wilh. Wundt: Grundzüge der physiologischen Psychologie. 2te Auflage. Leipzig 1880. p. 269 f.*

elements of which on closer examination they turn out to consist. There is — as will later be shown in detail — hardly any single state of consciousness that is *merely* idea, *merely* feeling or *merely* will. The psychological rubrics can therefore be very useful for a provisional orientation; but for them to have scientific significance, they must be built on a thoroughgoing *analysis*, which seeks out the single elements and the laws of their combination and interaction. It is not enough to appeal directly to what one feels in oneself, or believes one feels; the psychological facts are not so simple. Just as little as immediate consciousness can teach us anything about whether it is the sun or the earth that moves, just so little does it contain in itself any psychological theory whatever. Our thoughts and feelings are highly composite products of the mind, fruits of a long, quiet and mostly unnoticed growth. The mere observation and description of them therefore has value only as a foundation.

Psychical states follow one another and call one another forth. Can laws and rules now be set up for this interrelation, and can it be shown which elements in the various states of consciousness lead from the one to the other? These are the questions with which psychological analysis is occupied. It therefore goes in two directions that stand in inner connection with each other: it seeks out the common features, what runs through the special cases, and by this route sets up general empirical laws (e.g. for the association of ideas, for the relation between idea and feeling, etc.) — and it seeks within each single state to discover the single elements of which it is composed. A thought, a feeling, a resolution is not an absolute unity; on closer examination it shows itself to be the fruit of a long course of development, in the course of which it has taken in nourishment from many sides. The feeling of love, conscience and — to take a purely intellectual example — [24] the idea of an external object stand as something whole and complete, and yet turn out to have their history and to rest on an interrelation between simpler psychical elements, which testing and appraising reflection brings to light. Analysis here goes from the composite to the simple, whereas there it went from the single cases to the general rule. One might call the one form *generalizing*, the other *elementary* analysis. But the laws of sequence and the nature of composition will turn out to belong closely together; for the most general laws must be those that hold for the most elementary activities, for the psychical functions that recur in all psychical phenomena.

d. It can never be known with complete certainty whether the explanation is exhaustive, and whether one really stands before elements that cannot be further reduced. This in itself is something that holds not only for psychology but for all our knowledge; the last thing we reach at any point is only something last for us. But we cannot even know

whether advancing inquiry will not lead further, so that the limits of human knowledge have not yet been reached. Where the single inquirer and the single age stop, later times with fuller experience and clearer principles can reach further. A remarkable example of a decisive change in the fundamental psychological view is the predominant importance that since Rousseau's time has more and more been ascribed to feeling in relation to the other sides of psychical life, whereas for a long time the elements of feeling were referred partly to the psychology of ideas, partly to that of the will. Apart from clarity in observation and analysis, the possibility cannot be excluded that human psychical life undergoes slow changes.

In the physical domain, however, it can in many single cases be proved whether in one's explanation of a phenomenon one has really taken account of all the co-operating elements. For there a quantitative determination of the elements is possible, and it thus becomes a simple arithmetical check whether one has got all the elements in. When the chemist has resolved a compound substance into its constituents, he knows that their weights together must equal the weight of the compound. In the transition from one physical action of force to another there is always one definite measure of the latter corresponding to one definite measure of the former. But such a quantitative [25] estimate seems to be excluded in the psychological domain. An exact check on the sum is therefore impossible, and further correction always possible. The only way in which a confirmation of the result can be obtained is to test the empirical laws found in as many typical and varied cases as possible.

This impossibility of an exact rational foundation for psychology is closely connected with the nature of the phenomena of consciousness. They do not arrange themselves, as external phenomena do, in the schema of space. One feeling does not lie to the right or left of another, one thought not above or below another. Different currents can, as already touched on, move simultaneously in our consciousness, but here we have no form of intuition that, like space in the domain of physical experience, can furnish the basis for a mathematical construction. We therefore get nothing here, either, corresponding to the mechanical laws of motion, which contain the most general principles of all exact explanation of nature as far as physical experience is concerned.

There is one property, however, by which psychical phenomena turn a side toward mathematics, namely their differences of strength. Psychical states can be present in infinitely various degrees, from the faintest glimmer to the most violent tension. Here it might seem that measurement was not excluded, and it is indeed at this point that attempts have been made to introduce mathematics into psychology. The difficulty in

these attempts always lies in finding a fixed starting point. *Herbart* wished to give the mechanics of the intensive states of the soul. He starts from the assumption that ideas (which for him are one with the simplest psychical elements) inhibit one another in a ratio determined by the degree of their opposition. But what, now, is the *absolute* intensity of an idea? Since this question cannot be answered, *Herbart* sets up as a hypothetical presupposition that the sum of the degrees in which the ideas present inhibit one another is at every moment as small as possible — which is the case when it is not all the ideas that inhibit one another mutually, but when all direct their inhibiting influence against a single one, namely the one least opposed to the others. This presupposition is, as [26] *Wundt* has shown (*Physiol. Psychol.* II. p. 315), improbable, since there is no reason why the tension between two ideas should cease because a third is added. *Fechner* has chosen a surer starting point. Although psychical magnitudes — so his reasoning runs — cannot be measured directly, they can still be measured by means of physical magnitudes with which they stand in lawful connection. It would be simplest to take as basis the physiological process in the nervous system (the psychophysical process), since it stands in the most immediate relation to consciousness; but we do not know its nature, nor can we measure its strength directly. We must therefore take the strength of the external stimulus as basis. The strength of light and sound, weight, etc., can easily be measured, and it then remains only to discover the ratio in which the psychical sensations grow or diminish when the strength of the external stimuli is changed. But in order now to get a real measure for psychical states, we must have a unit, a smallest unit, to take as basis. This psychical unit *Fechner* finds in the least noticeable degree of strength of a sensation, that point, then, in the process of development of a sensation where it just rises above the threshold of consciousness. The presupposition *Fechner* starts from, then, is that these *minima perceptibilia* are equal.

The question now is whether this presupposition is warranted. In the first place it holds only within one and the same modality or circle of quality of sensations. The least noticeable sensation of sight need not have the same strength as the least noticeable sensation of sound. This *Fechner* himself indicates (*Elemente der Psychophysik*, I. p. 63). In any case, then, we miss a universal psychical measure; each class of psychical elements must have its own special scale. In the second place, equal noticeability is not the same as equal strength. What can just be noticed may at different times and in different states have different psychical strength. Every change in sensation alters the state of the organism in greater or less degree, so that every new change in the strength of the impression meets new conditions, and the just noticeable sensations must therefore at

the various stages — of fatigue or of adaptation — be of different strength. Finally, it is a [27] question whether we do not immediately feel the difference between a weight of 29 lod resting on our skin being increased to 30 lod, and a weight of 29 pounds being increased to 30 pounds, although each of the two increases (according to *Weber's* experiments) is just noticeable. We probably have such an immediate sensation of difference when we make the two experiments so close after each other that the memory of the first has not weakened when we pass on to the second. If Fechner's presupposition seems to hold in some cases, it is only an illusion that is soon corrected, just as the poor boy soon discovers that it is a greater advance in value when 10 kroner rise to 100 kroner than when 1 øre rises to 10 øre.¹² Against such objections it is hardly of use to point out that it is not the sensation itself but our *estimate* of the sensation that is accessible to measurement, while direct examination can say nothing at all about the sensation independently of the comparison we make when we estimate it.¹³ For in the first place the difficulty pointed out returns in the case of the estimate: how do we know that what in the single cases we *estimate* as just noticeable always has the same strength? And in the second place the sensation — as will later be shown in a more special connection — is from its first emergence so interwoven in the connection of consciousness, which is always changing and is thus different for each new sensation, that the distinction between the sensation and its estimate is in fact impossible; the sensation does not exist at all except as a link in a chain, and always has other links as its relief. What already in the physical domain sets limits to the accuracy of measurement, namely the close [28] connection between the various phenomena, which in practice makes it impossible to isolate a single one entirely, holds in far higher degree in the psychical domain, where the phenomena are even more intimately interwoven. If a measurement is to be reached here, it can surely happen only by adopting an even more abstract point of view than Fechner adopted.¹⁴ What is significant in Fechner's investigations is that he rightly saw that one has a prospect of reaching a measurement of psychical phenomena only by using physiological experiment. The investigations of physiological time show how

¹²These or similar objections have been made chiefly by *Hering* (Ueber Fechner's psychophysisches Gesetz. — Sitzungsberichte der Akademie der Wissenschaften in Wien. 1875), *Delboeuf* (Revue philosophique 1877 and in separate writings) and *Funke* (Der Tastsinn und die Gemeingefühle. — Hermann's Handbuch der Physiologie. 3ter Bd. 2ter Th. Leipzig 1880).

¹³*Wundt*: *Physiol. Psychol.* I. p. 322, 332. — *Sigm. Exner*: *Physiologie der Groshirnrinde*, p. 215 ff. (Hermann's Handbuch der Physiologie. II, 2).

¹⁴*Delboeuf* has attempted a recasting of Fechner's theory in this direction. Fechner's error, he says, is that he takes no account of the change the organ undergoes as a consequence of the stimulation itself. But it might be that his theory had a quite abstract validity, namely if one starts from organs of perfect elasticity, able at every moment to function in a quite adequate way — that is, by making the conditions far simpler than they are in reality. (*Revue philos. Mars* 1877).

much of significance for psychology can be reached by this route, even if it should not prove possible to overcome the difficulty of determining a psychical unit.

e. The strictly psychological standpoint keeps to the phenomena of conscious life. We have stressed this so strongly in order to avoid ambiguities and confusions. Directly, we know only as much of psychical life as we know of the phenomena of consciousness. But we have already seen that consciousness is not a closed world: new phenomena constantly emerge which at that strictly psychological standpoint we cannot derive from anything earlier. Each new sense-sensation seems to arise out of nothing. We can perhaps follow its transformations and effects within consciousness, but can say nothing about how it first gets there.

Besides, we seek means of remedying the imperfection of the strictly psychological standpoint for other reasons as well. Comparison of our own observations with those of others is, as already stressed, necessary in order to avoid individual one-sidedness, and [29] further also in order to make sure whether all the elements that co-operate in the given psychological phenomenon have really been brought to light. Finally, it is not enough for concrete psychological knowledge to know the general laws of the connection of the phenomena of consciousness; in practice there appears such a multiplicity, such a welter of possibilities, that it becomes impossible to infer from the general psychological law which direction conscious life will take. For each single individual and in each single situation the result rests on the original temperament, on the circumstances of life and on special experiences. Whether an idea (b) will, by virtue of “the law of similarity,” call forth kindred ideas (b_2 b_3 b_4), or whether, by virtue of another law of the association of ideas, it will call forth ideas it has previously been together with (a c d...) — that can only be decided and understood by means of an all-round examination of the single case.

The strictly psychological standpoint must therefore be supplemented by physiological and historical (sociological) inquiry, or, as we may also say (using a term introduced by H. *Spencer*), subjective psychology must be supplemented by objective psychology. In accordance with what has been developed earlier, it must be held fast that objective psychology always rests in the last instance on an inference by analogy; only subjective psychology stands face to face with the phenomena themselves. What we as objective psychologists think we discover of psychical life outside our own consciousness we reproduce in ourselves by virtue of a sympathy closely bound up with the analogy. But these analogies can furnish indispensable correctives to our immediate observation.

The contrast between subjective and objective psychology does not coincide with the contrast between *abstract* and *concrete* psychology. The immediate subjective obser-

vations of inner states are concrete; it is only analysis that leads to abstract laws and rules. And on the other hand objective study (e.g. by physiological experiment) can lead to abstract psychological results, the conditions being made as simple as they never are in reality. In physical nature, too, the phenomena are so intricate that there is a great [30] distance between the general law and each single concrete case.

Objective psychology consists of *physiological* and *sociological* psychology. The former rests on the intimate connection of psychical life with organic life as a whole. Every piece of information *physiology* can give about the functions of organic life can from one side or another benefit psychological knowledge. What must especially be stressed here is the way in which conscious psychical life fades off into unconscious organic life. Physiology examines precisely the unconscious functions that precede psychical activities and form their constant basis. At the border points between the conscious and the unconscious, where the light of subjective observation falls only faintly, physiology by its physical method is able to establish definite relations. But throughout there is an intimate interrelation between the conscious and the unconscious: it is not only the infant that awakes to consciousness out of the night of unconsciousness; sleep is every day a relative renewal of this night; in instinct, impulse and habit we have forms in which the unconscious takes the conscious into its service, and in which conscious life in turn, by producing new habits and impulses, acts on the unconscious. The physiological study of these elementary psychical phenomena throws light on the more highly developed psychical life as well. The debate here comes to turn on the degree to which, and the modifications with which, what nerve physiology and the physiology of the senses reveal about elementary psychical life can be transferred to the higher — whereby it must never be forgotten that the physiological experimenter and observer himself, in his interpretation of elementary psychical manifestations, infers back from the fully developed consciousness to the elementary one. What interests the physiologist proper in the investigations of nerve physiology and sense physiology is not the states of consciousness as such, but the physical processes to which they are attached. For the physiologist psychological experiences are only criteria from which he infers physiological facts. He starts from the presupposition that to every psychological experience there corresponds a physiological process, which it is his task to demonstrate [31] and explain according to the general principles of natural science. For the present it is the more elementary phenomena of consciousness that prove most accessible to this explanation; but a principle is thereby given which physiological psychology with full right takes as its basis in the investigation of the higher psychical phenomena as well. Just

as a physiology of the senses has developed hand in hand with general nerve physiology, so a physiology of thought and feeling will develop.

When we consider that psychical life, as we know it, develops only under definite physical and chemical conditions and through a series of stages, of which the lower and the higher mutually illuminate one another, it is clear that — despite the independence we claim for psychology at the outset — we must nevertheless regard it as part of general biology. Biology must set up a concept of life that fits all its stages, from the organic process of nutrition in its simplest forms to the most ideal process of feeling or thought. It seems that the biology of our time is approaching such a comprehensive biological view when it regards life in all its forms as an adaptation of the inner to the outer. Conscious life marks the highest point of the development of life, shows us the highest forms in which beings fight the great struggle with the conditions of the world and in this struggle unfold their nature. A purely subjective treatment of psychology would overlook the great truth that everything that moves in the soul is conditioned by its position in the great connection of the world.

Besides nerve physiology and sense physiology, the study of mental diseases is also an important auxiliary source of psychological insight, both by what it shows about the connection between psychical and bodily disturbances and by its investigations of the forms and course of development of morbid psychical life. In the latter respect it forms a transition from physiological psychology to *sociological psychology*, which considers psychical life as it reveals itself in movement and action, in word and image. The material of sociological or comparative psychology is animal life, child life, the [32] savage tribes of men, the whole of human history, works of poetry and biographies. Sociological psychology can therefore be divided into a great number of branches of study (child psychology, animal psychology, folk psychology, the psychology of language, the psychology of literature, etc.), all of which lead into the great historical connection within which the individual consciousness develops, just as physiological psychology led over into the physical connection by virtue of which psychical life takes part in the life of the universe. The way in which thoughts and feelings take shape in the individual at a given time is conditioned not only by his inherited original organization but also by the atmosphere of cultural history in which he develops. The physical and the historical connection stand in inner relation. The study of the influence of heredity forms an intermediate link between physiological and sociological psychology, especially when we widen our view to the great horizon that the hypothesis of evolution opens for us. Organic nature, indeed the whole connection of nature and its laws, thereby acquires

a historical character, just as, conversely, the organic forces and laws are seen in their influence on the historical development of human nature and of the conditions of human life.

f. One needs only a quick glance at all these various sources to know that psychology cannot be a sharply bounded science. It can be pursued in many ways and along many roads; here we have wished only to impress the relation in principle between the various roads. There is, then, not one psychology but *many psychologies*. But in accordance with the position in principle that subjective psychology occupies, it will — despite the growing importance of objective psychological study — always be a natural and warranted endeavour to take it as the basis and to gather the contributions of the other sources of knowledge about it as *center*. This is in fact the road psychology has taken since *Aristotle* founded empirical psychology. It is only temporarily that psychology, led by a misunderstood spiritualistic interest, has attempted to separate its cause from physiology and the other objective sciences, so that it could look like a new discovery when the connection was asserted again. Subjective psychology has had to wait for objective psychology; once it had become properly clear about its principle, it soon reached a [33] certain completeness in its broad features, before physiological and sociological study had developed far enough to play seriously into psychology's hands. In this respect a turning point may be said to have come in our time, which has broken down so many barriers and united so much that did not seem to belong together.

He who studies psychology from a philosophical interest will have a special reason to hold fast everywhere to the relation in principle between the various psychological sources of knowledge, so that the contribution psychology has to make to the general view of the world may not be distorted. Here, as everywhere, the task is to spread out a content of experience for consideration in such a way that the given and the hypothetical are sharply distinguished, while on the other hand the general and pervasive features, the common laws, stand out clearly and distinctly; for it is on them, and not on the gaps that always remain in human knowledge, that further philosophical inquiry must rely.

9. Psychology thus stands at a point where natural science and the sciences of the spirit intersect, where the one passes over into the other. In its principle it has the center about which the currents from both sides circle, since all knowledge — as knowledge on the basis of human nature and organization — also becomes, directly or indirectly, knowledge of man.

Psychology forms the foundation on which the ideal sciences of the spirit, logic and ethics, build. What is true and good can be determined only from the human standpoint

and cannot be understood without knowledge of real human nature. Logic and ethics set up models for human striving in thought and action. But if these models are to hold in reality, they must also have their root in it, and must therefore be formed on the basis of a knowledge of the deepest and most pervasive elements and forces of human nature. *Logic*, as the theory of method, studies the special methods of inquiry and seeks to trace them back to a fundamental method that follows immediately from the nature of human [34] consciousness; as the theory of knowledge, it seeks to set up the general principles of human knowledge and the limits connected with them; but this endeavour cannot succeed without psychological insight into the development of the life of ideas, although it does not coincide with it. *Ethics* seeks to find the practical principle that makes possible a harmonious development of the powers of personal life, in the individual as in the race; but in this endeavour it must always start from the really given human nature and its possibilities as determined by the psychological laws. It is among these possibilities that ethics makes its choice, after having appraised them.

On the other hand it leads to a distortion of psychology when logical and ethical ideals are confused with psychological reality. Psychology has to do only with what is, not with what *ought to be*. Of course the state of consciousness in which it dawns on us that there is something we ought to do itself falls under psychology; it is examined here like any other. But psychology makes no appraisal; it asks only about the factual connection and about the way in which, and the laws by which, it has developed, and leaves it to ethics to pass its judgment on the various states. Psychology regards psychical phenomena as natural phenomena and examines them all with the same calm and impartiality. This independence of psychology from ethics *Spinoza* already asserted energetically (see my work *Spinozas Liv og Lære*, p. 120 ff.), but it is far from sufficiently acknowledged even now. There is still a constant inclination to regard certain forms of psychical life as raised above explanation and analysis because of their exalted and valuable character. But precisely the psychological phenomena most valuable in ethical and aesthetic respects are not single and uncompounded, since they stand as the culmination of a long and rich process of development. From their very value, then, what follows is precisely the opposite of an exemption from the general psychological conditions. It is altogether a false presupposition that the determination of value and causal explanation necessarily [35] exclude or work against each other. Theoretical inquiry can certainly expose illusions; it calls in any case for a renewed testing of the correctness of the appraisal; but in itself it can very well be united with the determination of value. It is only mythological superstition or blasé satiety that thinks a phenomenon loses its value because it is understood. Yet

it must be granted that the harmony between appraisal and causal explanation is still only coming into being; the life of feeling has not yet adapted itself in this way, and knowledge on its side has not yet come so far that H. C. Ørsted's words fully hold: "The full truth always brings its comfort with it."

Chapter 2

Soul and Body

[36]

1. The foregoing inquiry has established that the knowledge of the psychical and the knowledge of the bodily spring from two different sources. The question that now arises concerns the relation between these two different domains of experience. This question, as it is here put, does not lead us into metaphysical inquiries. We take the word soul merely in the sense of consciousness, the expression for the unity of all our inner experiences (sensations, thoughts, feelings and resolutions), and ask what indication experience itself gives us concerning the connection of these experiences with the material phenomena that present themselves to sense-intuition. *Our* standpoint is thus for the present purely *empirical* or *phenomenological*, not metaphysical or ontological. On the view given in the foregoing, the work of metaphysics begins only when the domain of experience has been traversed and the tendencies and possibilities appearing there have been discussed.

What here, as everywhere, marks the difference between the popular and the scientific view is that the former is a mixture of experience and metaphysics. The popular view often seizes with instinctive sureness upon certain salient experiences; but it interprets and paraphrases them under the unconscious influence of metaphysical ideas, partly handed down, partly growing wild. The scientific view, on the other hand, seeks first and foremost to become conscious of the source of its assumptions, and to distinguish sharply between experience and interpretation. Difficulties and problems thereby arise for it which the popular view does not feel.

As long as neither the phenomena of consciousness nor the material phenomena are [37] sharply apprehended, each with their own peculiar character and inner coherence, the problem of the relation between soul and body does not really exist. When by soul one understands, in all indefiniteness, a moving principle, an inner force in things, one has no

reason to see any difficulty; for such principles and forces may be posited with equal right in every possible domain. Only when the concept of soul is definitely restricted to the life of consciousness and its facts, and when on the other hand the material phenomena close themselves off as a world that has its own principles and laws — only then does the difficulty of connecting these two domains arise. The problem thus does not arise from a dogmatic and absolute separation of two substances or beings; whether the difference is a difference of essence or not, we do not as yet know; we know only that here there is a difference, and we ask what lies in and follows from this phenomenal difference given in experience. To answer this question we bring together the most decisive features that characterize the two domains of experience.

2. The first chief characteristic mark of *material* phenomena is that they appear in the form of space, that all of them can directly or indirectly be traced back to a motion in space. By this they are distinguished from the states of consciousness, which can only symbolically be represented as something spatial. But in this characteristic mark there lies in itself as yet nothing that sharply delimits and closes off the material as a world by itself. For those spatial motions themselves might, after all, be conceived as brought about by something that was not spatial. The material world would then lie open to intervention from without.

Natural-scientific research cuts off such a possibility more and more. It does not regard it as an explanation of a material phenomenon that one appeals to a non-material principle. It finds a thoroughgoing explanation only when the given experiences have, by continued analysis, been traced back to the laws of mechanical motion. What cannot yet be reduced in this way remains, so far, obscure and not understood.

[38] All research rests on the principle of causality. The difference lies in, and the disputes turn on, where one seeks the causes, which causes one recognizes as real. Now natural science carries out the general principle of causality in such a way that for every material change it seeks a material cause. Its principle is therefore that nothing arises of itself, and nothing can absolutely perish. This principle, which was already pronounced as an axiom by the founders of modern natural science, has in the last century received a wider, empirical grounding.

Since Lavoisier's reform of chemistry it has been shown that the sum of the particles of matter always remains the same, however much the compounds vary. The elements neither lose nor gain by entering into chemical compounds: as soon as the compounds are dissolved, one gets exactly the same number of elements (atoms) back again. When, therefore, a new substance apparently arises, there takes place only a conversion into a

new form of what was present beforehand, in the same quantity, under another form. Oxygen and hydrogen combine under certain circumstances into water, and when the water is decomposed again, one gets the same number of particles of oxygen and hydrogen back.

In chemical combination and decomposition other forces come into play besides chemical attraction. In chemical combination heat arises. But just as much heat as arises in combination, just so much disappears again in decomposition. Whence, now, does this heat come, and whither does it go? It arises as the equivalent of the elastic force that held the parts apart before the combination, and it again gets its equivalent in the tension with which the parts are held separated from one another after the decomposition.

For it is not only with respect to the *fundamental parts of matter*, it is also with respect to the *fundamental forces of matter*, that the material principle of causality holds. The force with which a stone falls to the earth depends on the height from which it falls; but the height depends in turn on the force with which it was lifted up into the air. When the stone is stopped by the earth, force seems to be lost; for the stone is not able to move the earth; but here too the disappearance of the force will only mean its [39] conversion into another form — into heat. The same holds where the motion does not stop entirely, but is checked by frictional resistance. The force that a body loses through frictional resistance is not lost, but becomes heat. The wheel of the water-mill produces heat by striking against the water. Conversely, heat can produce mechanical motion, as when the steam, expanded by the heat, drives the piston which in turn sets the wheels of the locomotive in motion. And repeated, ever-confirmed experiments have shown that the quantity of force or energy¹ that disappears under the one form gets its definite equivalent under the other form, so that the same quantity of the same form of energy can be restored again.

However much the various forms of energy may each vary, their sum thus always remains the same. But by energy we must then understand not only the actual performance of work (living force, actual energy), but also the possible performance of work (tensional force, potential energy), the stored-up work that under certain conditions can be released and applied. When Sisyphus has got his stone rolled up the mountain, he has really accomplished something; his labor is not wasted; for the stone represents a greater energy when it lies up on the mountain than when it lies at its foot. In both cases it is at rest; but the potential energy is greater in the first case than in the second, which shows itself as soon as it is set in motion. Sisyphus's misfortune consists merely in this, that he

¹Since the expression force is ambiguous, one as a rule uses the word energy, by which nothing else is meant than the capacity to perform work or to overcome a resistance.

cannot apply the greater potential energy to anything useful to himself or others; he must always begin over again. — It is thus the sum of actual and potential energy that — in so far as we regard the universe as a closed whole — always continues to be the same.

But it must be firmly held (though even physicists do not always hold it firmly) that the law of the *conservation of energy* does not hold for limited beings or relative wholes. Every single system can, as experience amply shows, lose energy that it never gets back.

[40] When Sisyphus was condemned for all eternity to roll the stone up the mountain, only to see it roll down again every time, he can to that extent console himself: there will come an end to his sufferings; he and the stone and the mountain will all be worn out in the end. They gradually give up their energy to other systems. As soon as there is anything outside a being or a system, this will receive influences from it that raise or lower the sum of its energy. Neither our earth, nor our solar system, nor our stellar system have we any right to regard as a finished, closed system. They have limits, with the limits the possibility of influence from without, and with that in turn the possibility of loss or increase of energy. Only for the *whole* universe — if we were able, as in reality we are not, to think it as an absolute whole — would the law of the conservation of energy hold completely. This law thus has the same validity as so many other laws and principles (e.g. the law of inertia) which cannot appear purely in our experience, but which are nonetheless proved by showing their validity the more, the more we are able to simplify and isolate the phenomena.

3. Yet there is one class of beings which not only the popular view, but for long periods science too, has been inclined to regard as exceptions to this general law of the world. The organisms, with their peculiar development of form and their capacity for self-preservation over against the surrounding world, seem to be small worlds that can draw their life out of inner sources in themselves. For a long time the explanation of the phenomena of organic life by means of the general forces of nature was regarded as materialism. But since, on the other hand, it was clear that the conscious soul is not directly the cause of the organic processes, which so often take a direction wholly opposed to the conscious will of the individual, a so-called vital force was inserted between the conscious soul and the body to explain those phenomena. Here, then, something new was supposed to have come in which was found in no sense in the rest of nature. This view, so-called vitalism, could of course not overlook that living beings stand in manifold and constant interaction with the surrounding world, at every moment receive and give off influences, and that their process of development and of dissolution

[41] is conditioned throughout by this relation. The “vital force” had thus to run through a

similar course. But what dazzled vitalism was the peculiar way in which the organism answers all influences from without. That a ball rolls when we push it seems to us quite natural; but that the plant turns toward the light, that food is converted into flesh and blood, that the fingers bend together at a gentle touch of the inner palm of the hand — that seems to us very wonderful, since the answer here does not seem to stand in proportion to the address. Here vitalism commits the error of making the organism an absolute unity, whereas in reality it is an extraordinarily composite whole. An influence received is propagated from part to part within this whole, and thereby gradually changes its appearance entirely. Partly it is converted into other forms of energy, partly into potential energy; partly, finally, it serves to release potential energy or tensional force in the organism itself. As research seeks step by step to follow these conversions, it begins to understand that something quite different can come out than went in. It understands that the organism, on account of the store of potential energy that is accumulated in its tissues and circulates through it in its fluids — which are as it were a surrounding world within the organism (a *milieu intérieur*, to use Claude Bernard's apt expression) — must stand more independently over against the surrounding world than the inorganic forms of existence. The organic answer to the address from the surrounding world (irritability) must become the richer and more peculiar, the more bound forces there are to release. The doctrine of the "vital force" is really only a mythological expression for the wonder that the peculiarity of the organic phenomena called forth. Modern physiology has advanced beyond this standpoint by the analysis of the individual links in the life-process. The idea of the one, indivisible vital force has thereby given place to the picture of an extraordinarily intricate interaction, in which the individual manifestations of force can be traced back to the general forces of nature, and the individual particles of matter to the common elements. That is the principle with which physiology now works, and to which we must therefore keep here as well, even if no one dares to assert that it is, or perhaps ever can be, carried through everywhere. The main thing is that all the advances of physiology point toward it. However many riddles still remain, especially in [42] the morphological domain, no explanation is any longer accepted that does not rest on the stated principle. The burden of proof lies in any case on those who appeal to the intervention of immaterial causes.²

²Cf. *Panum*: Indledning til Fysiologien [Introduction to Physiology]. København 1865. — *Charles Robin*: Anatomie et physiologie cellulaires. Paris 1873. Introduction. — *Claude Bernard*: Leçons sur les phénomènes de la vie. Paris 1878. (Note here especially the following definite statement: »Quel que soit le sujet qu'il étudie, le physiologiste ne trouve jamais devant lui que des agents mécaniques, physiques ou chimiques«, p. 52). — *Exner*: Physiologie der Grosshirnrinde. 1879. (Hermann, s Handbuch der Physiologie. II, 2). p. 189–191.

Organic life is thus subject to the law of the conservation of energy. For everything that arises and disappears in the organism, physical and chemical equivalents must be sought, either in the organism or outside it. Organic life is thereby drawn into the great circulation of nature. It is especially in the green plant cells that the transformation of inorganic substances into complicated organic substances takes place.³ The organic [43] matter thus accumulated is consumed by the functions of plant and animal. Metabolism rests on the conservation of energy, and on metabolism in turn rests the activity of organic beings. The form under which, and the manner in which, the accumulated tensional force is applied depend on the structure of the organism. Here the relations of form become of great importance. Every organic cell contains a capital of energy; but it depends on the structure of the organs what application this capital receives.

The law of the conservation of energy thus closes in on us ever more nearly. Vegetative life, the functions of nutrition, one would perhaps even hand over to it with a light heart. But neither can the nervous and muscular system be exempt from coming under it. The energy that every nervous and muscular activity consumes has been stored up during the activity of nutrition. The nervous and muscular systems are themselves only peculiarly shaped (differentiated) apparatuses for the exercise of functions that also go on, in extraordinarily simple form, in the homogeneous, structureless protoplasm. Already here an impression at one place on the surface of the organism can propagate itself through the mass and release movement from quite other places or from the whole mass. It is the progressive division of labor that makes different systems necessary in the higher organisms. But this richer elaboration (differentiation) cannot emancipate from the general elementary laws.⁴ These laws still hold, only under extraordinarily

³“Only the green parts of plants are able to produce a new organic substance from carbonic acid and water, by means of other substances already found in their cells... all animal substances are descended from those which the green plants have formed. The substances that compose our blood, our muscles and nerves, our brain, are only slightly modified plant substances, which are produced in the chlorophyll-bearing cells... But the means that enables the green plant to convert inorganic substances into organic ones is light; only under its influence are the chlorophyll-bearing cells able to produce carbon-containing compounds from the carbonic acid of the air, compounds which, after having undergone new changes, are just as fit to form new parts of plants as to nourish our body.” J. Sachs (Tidsskrift for popul. Fremstillinger af Naturvidenskaben [Journal for Popular Expositions of Natural Science], 5te Række, 3die Bind, p. 111). — According to Pringsheim’s theory (see the journal cited, 1879, p. 316 f.) chlorophyll is not directly active in the process of assimilation, but serves only as a protection against the light and as a regulator of respiration. — Sachs expresses himself too absolutely when he denies to the animal organism the capacity to form organic compounds from inorganic elements. This synthetic process is not entirely lacking in the animal organism, although it finds especially favorable conditions in the plant organism. Cf. Pflüger: Ueber die physiol. Verbrennung. (Archiv für Physiologie. XI. p. 345).

⁴“The nerves are to be regarded as places in the tissues which are more easily traversed by the train of entering influences, without our needing at once to seek in them more mysterious forces than in other parts of the body.” Lotze: Allgemeine Physiologie des körperlichen Lebens. Leipzig 1850. p. 386. — A proof of this has been sought in the fact that the effect of narcotic substances on nerve tissue differs only

composite and often inscrutable conditions. Both nerves and muscles behave differently in chemical and physical respects during and after their function than before it. The blood that flows through active muscles contains several per cent less oxygen, and on [44] the other hand more carbonic acid, than the blood in resting muscles. The nerve tissue, both the fibers and the central organs, cannot function without an ample supply of blood containing the material that is necessary for the metabolism heightened by the function. The brain is affected by every change in the circulation of the blood; both anemia and hyperemia bring disorder into its activity. Brain-work draws on the organic capital just as the activity of every other organ does.

What it really is that goes on in the nervous system during its activity has not yet been cleared up. Only so much is clear, that it cannot be a substance that is propagated from one end to the other (as was earlier assumed of the so-called “animal spirits”). The nerve process probably consists in a change advancing through the nerve fiber, a release of tensional force, which is called forth by an impression from without (the irritant) and is continued successively from part to part, so that the one nerve element serves as irritant in relation to the next. The tensional forces thus successively released seem to be of a chemical nature; yet a purely chemical theory of the nerves founders on the difficulty of explaining why not the whole of the tensional force, but only a part of it standing in a definite proportion to the irritation, is released.⁵ But everything indicates that we nowhere here come upon any exception to the great law by which physical nature now appears to us with a magnificent unity and coherence — a unity which had indeed often been divined and pronounced in the form of principle and postulate, but which only the research of our own time has supported by the control of experiment.

4. a. The plant is wholly absorbed in the life of nutrition. It takes up and excretes matter, grows and propagates itself. What it needs in order to lead this life it finds in its immediate vicinity, and it can live only when it finds it there. Air, water, light, etc., must bathe the surface of the plant if it is to be able to subsist. The plant is like a fetus that still rests in the womb of nature and has not emerged from it into independent, individual life. The fetus, after all, receives its nourishment immediately from the mother organism. [45] The properly animal life is conditioned by the fact that everything is not thus laid ready. The animal must seek, work and fight to get its needs satisfied; it must therefore act as a whole over against the surrounding world, must be able to gather its energy and

in degree and speed from their effect on other organic tissues. Cf. *Laycock*: Further Researches into the Functions of the Brain. (The British and Foreign Medico-Chirurgical Review. July 1855). p. 185. *Claude Bernard*: Leçons sur les phénomènes de la vie. p. 289. *Herbert Spencer*: Principles of psychology. I, p. 631 ff.

⁵L. *Hermann*: Allgemeine Nervenphysiologie. 1879 (Hermann's Handbuch der Physiologie II, 1). p. 186–193.

direct it upon definite points. It must also be able to take account of relations and facts that do not immediately touch it at the moment. These demands are satisfied by the *nervous system*: through it the various parts and regions of the organism are brought into close mutual connection, so that the organism becomes a whole in a more intimate sense than the plant could be said to be; and through it there arises the possibility that the conditions in the surrounding world can determine the movements of the organism not only directly but also indirectly.

It is true that nerves have not yet been demonstrated in the lowest animal organisms, and on the other hand many plants perform actions similar to those which in higher animals are performed by means of the nervous system; but on the whole plant and animal may nevertheless be characterized as two types of life, of which the one stands merely in immediate interaction with its surroundings, the other in mediate and indirect interaction as well. The higher we go within animal life, the greater the part played by the nervous system, because the interaction with the surrounding world spreads in ever wider and wider circles, and thus becomes less and less immediate and momentary.

b. The simplest form of nervous activity is the so-called *reflex movement*, in which an influence is conducted along an ingoing (centripetal) nerve fiber to an inner center (a ganglion), and from there in turn releases an impulse which, through an outgoing (centrifugal) nerve fiber, sets a muscle or another organ (a gland, for example) in motion. We have here the simple schema that seems to repeat itself at all stages of the development of the nervous system, only in extraordinarily many coordinate and subordinate layers. For as a rule the ganglion sends out not only outgoing fibers to the organs that are to be roused to movement; there also go out from it ingoing or upgoing fibers to higher centers, which thus receive impulses from several sides, impulses that can partly reinforce and [46] partly inhibit one another. The ganglion itself exerts a checking influence on the impulse, since experiments show that the nerve process goes on more slowly during its course through brain and spinal cord than through the peripheral *nerves*. This checking seems to make it possible for the impulse, before it is propagated further, to be altered by the influence of other impulses. It is this central working-up of the peripheral impressions that brings it about that the movement they release is not determined by merely local and momentary effects, but to a certain degree by influences from the whole organism. The nervous central organs are thus organs of modification and combination.

A very simple example of this relation is found in the suckers⁶ of the cuttlefish. Each sucker on this animal's arm has its own ganglion, and can therefore be brought

⁶I cite it here after *Carpenter*: Mental physiology. p. 50.

to contract and fasten itself by suction when an object is brought into contact with it alone. This can also happen when the arm is severed from the rest of the animal. Here we have a nervous function in its simplest form: propagation of the impression to a simple central organ, and release in this organ of an impulse to contraction. But now the ganglia of all the suckers are connected both with one another and with the animal's highest centers (the pharyngeal ring), so that the animal, when it clasps an object with its whole arm, can set all the suckers in activity at once. The single elementary nervous function then appears as a link in a whole system of functions.

In higher animals too this relation between subordinate and superordinate centers can be demonstrated, although the more intimate connection and mutual dependence of the organs here makes it the more difficult to see through the relations, the higher one goes up in the series of development. In cold-blooded animals (such as the frog, which for just that reason must especially serve in physiological experiments) the independence of the subordinate centers is greater than in warm-blooded ones, and within the series of the latter this independence is greater in birds and rabbits than in the ape, and especially than in man. The complete removal of both cerebral hemispheres can be survived only by animals whose cerebrum has not reached a higher degree of development. Higher mammals, on the other hand, quickly perish when they are deprived of the whole mass [47] of the hemispheres.

c. The vegetative *organs* are connected by ingoing and outgoing fibers with the spinal cord and the brain, and are regulated from these central organs. Yet some of them seem to have special nerve centers, to a certain degree independent, in or near them. An excised frog's heart continues to beat for several hours, and thereby proves its relative independence of higher centers. Experiments (on dogs and rabbits) have shown that when the heart was freed, by severing the *nervus vagus*, from its connection with the medulla oblongata, the pulse beat faster. In violent fright the rabbit's heart stops, only to beat more rapidly than before; but after the *nervus vagus* had been severed no influence on the heartbeat was to be traced. With the intestines the case is similar. The peristaltic movements can continue after the connection with higher centers has been cut.⁷

The *spinal cord* is an important seat of reflex movements. In the headless frog one can, by sufficiently strong stimulation of one or another part of the skin, release reflex movements in all directions. The remarkable thing about these movements is their coordination and purposiveness. In so far as it has been possible, by severing the spinal cord and applying an irritation below the cut, to call forth reflex movements in mammals,

⁷M. Foster: Text-book of Physiology. London 1877. p. 81. — Eckhard: Physiol. des Rückenmarks (Hermann II, 2), p. 71. Exner: Physiol. der Grosshirnrinde (Hermann II, 2), p. 289.

these proved indeed to be coordinated to a certain degree, but little purposive. In the higher animals the spinal cord seems more and more to be absorbed in the activity of forming the connecting link between the brain and the extremities of the organism.⁸

In the *medulla oblongata* a number of centers important for the maintenance of life are localized, which can function independently of the higher parts of the brain, and which set very intricate mechanisms in motion: thus the respiratory center, the center for the regulatory cardiac nervous system, and those for the secretion of saliva, the act [48] of swallowing and the secretion of urine. They seem to be set in activity partly reflexly, partly automatically (i.e. by the constitution of the blood directly affecting the central nerve cells).

A frog that has been deprived of the cerebrum, but still has the *mid-brain* (the cerebral ganglia lying in front of the medulla oblongata), proves still to command the motor apparatus that is necessary for the performance of independent movements; but it moves, as it seems, only when a definite sense-impression rouses it. The capacity to take the initiative is lacking. It has this advantage over the mere spinal frog, that it can be determined by sense-impressions and is therefore not wholly passive. Whereas the spinal frog is of course unaffected by light, and sinks to the bottom when it is thrown into the water, the mid-brain frog avoids a very dark shadow, and when it is thrown into the water, the impression of the movement of the particles of water makes it swim. But it always needs an impulse from without in order to move. — Similar features are found in birds and mammals when these survive the removal of the cerebral hemispheres. There is a lack of initiative and of the capacity to help themselves in cases that are at all difficult, whereas single elementary sense-impressions can call forth movements that are in part very composite.

d. The part that can be reserved for the most important portion of the brain, *the cerebrum*,⁹ can be no other than that of working up and combining the elementary impressions received in the medulla oblongata and the cerebral ganglia (perhaps also in the cerebellum), and of employing the motor apparatus lying ready in these lower parts of the brain in accordance with the result of this working-up. The cerebrum stands as the keystone of the ingenious structure of the nervous system. The nearer we come to it, the more intricate the relations become, the more numerous the nerve cells and the connecting fibers. Here paths are laid ready which make the most composite interaction

⁸M. Foster, p. 420.

⁹The cerebellum is not discussed here, since the results of the investigations concerning it are very unclear. Its significance in the various vertebrates is probably different, and moreover no sharp boundary can be drawn between it and the adjoining parts of the brain. Eckhard, p. 102.

between the various impulses possible. When we remember that every impression acts [49] by releasing tensional force in the organic cells, and that the result of this release in the single cell can be combined in the cerebrum with the results from millions of other cells,¹⁰ the mind reels at the combinations that here become possible.

The question whether the cerebrum functions as a unity, or whether its various functions are each bound to a region of their own, has received changing answers and is still an object of debate among physiologists. *Gall*, the founder of phrenology, taught a very far-reaching localization of all the higher and lower faculties of the soul, but brought the idea of localization into discredit by his uncritical procedure and his fantastic cranioscopy. The reaction against his doctrine is represented by *Flourens*, who concluded from his experiments that any parts whatever of the cerebral hemispheres could be injured or removed without the functions of the brain suffering thereby.

This theory prevailed for nearly half a century, during which it was contradicted only by *Broca's* demonstration (1861) that the most important central organs for language and speech are found in the third frontal convolution of the left hemisphere. A new period in the physiology of the brain begins with the investigations of *Fritsch* and *Hitzig* (1870). These researchers believed they could show that irritation of definite points on the surface of the cerebrum calls forth definite movements of definite parts of the body. Later *Hermann Munk* in particular has sought to demonstrate definite organs in the cerebrum for the apprehension and recognition of the elementary sense-influences (a visual sphere, an auditory sphere, etc.). There would thus be reason once more to assume a localization, a division of labor within the cerebrum — yet with the great difference that it is only the elementary psychical activities that are localized, not thinking itself, [50] or “intelligence.”¹¹ But even with this restriction the new theory of localization does not stand unchallenged. In the first place it has been shown that the local motor centers do not act in the first 8–10 days of the newborn dog's life (not before the animal has begun to see). There might thus be reason to assume that they are first formed under the influence of experience and practice.¹² But in the second place *Goltz* has come forward with an objection to the new theory of localization. *Goltz* takes up an intermediate position

¹⁰*Meynert* and *Bain*, independently of each other, estimated the number of nerve cells in the cortex of the human cerebrum at a thousand millions. (*Meynert*: *Zur Mechanik des Gehirns*. Wien 1874. p. 7). — “A portion of the gray substance on the *surface* of a cerebral convolution, only as large as a very small pin's head, contains parts of several thousand nerve fibers.” *Beale*, quoted by *Maudsley*: *Physiologie de l'esprit*. *Trad. de l'angl.* p. 110.

¹¹»Die Intelligenz hat überall in der Groshirnrinde ihren Sitz und nirgend im besonderen; denn sie ist der Inbegriff und die Resultirende aller aus den Sinneswahrnehmungen stammenden Vorstellungen«. *H. Munk*: *Ueber die Functionen der Groshirnrinde*. Berlin 1881. p. 73.

¹²*Soltmann's* experiments. See *Vierordt*: *Die Physiologie des Kindesalters*. p. 134 ff.

between Flourens's doctrine and the new theory. He does not deny the possibility of a localization of the various functions of the cerebrum, and he disputes the correctness of Flourens's doctrine that any part of the cerebrum was able to act vicariously for any other. On the removal of large parts of both hemispheres a lasting weakening of the intelligence sets in. But the weakenings of definite sensory and motor functions that follow the removal of certain areas of the cerebral surface Goltz explains as phenomena of inhibition brought about by the operation. If the injury is not too extensive, the animal finds its bearings again — which the theory of localization explains only in an unnatural way, by assuming that new special centers develop in regions of the brain which had hitherto not possessed any such at all. The whole of what Munk calls the visual sphere can, according to Goltz, be removed without the animal becoming blind.¹³

Whereas, then, there is dispute as to how far there is a localization of the individual sensory functions in the cerebrum, both of the contending parties agree that the higher psychical manifestations of life (understanding, feeling and will) are not bound to definite regions of the cerebrum. In holding that "the most important functions of the cerebrum, the actions from which we infer intelligence, feeling, passion and instinct, cannot depend [51] on limited sections of the cerebrum,"¹⁴ both Goltz and Munk agree with Flourens. The cerebrum thus stands before us as an extraordinarily composite organ which — however it may be with the distribution of labor in its interior — takes up the more elementary influences from the lower central organs, works them up and combines them, and which thereby makes possible a more many-sided and more concentrated reaction upon the surrounding world.

e. A couple of examples will show us the physiological relation of the cerebrum to the other nervous organs. — The study of the disturbances of speech seems to have led to the result that in the regions of the brain lying beneath the surface of the cerebrum are found only the arrangements for the mechanical execution and connection of the movements of sound, whereas the properly linguistic formation of syllables and words goes on in the surface of the cerebrum. The primitive sounds of the infant have perhaps even the conditions for the mechanism that orders them in the medulla oblongata alone, whereas the developed sounds of the national language, which are joined into syllables and words and are bound up with the development of the intelligence, presuppose the intervention of higher centers.¹⁵ — The movements that are initiated from the mid-brain, the medulla oblongata and the spinal cord have the character of simple reflex movement.

¹³ Goltz in Pflüger's Archiv für Physiologie. 20de og 26de Bd.

¹⁴ Goltz in Pflügers Archiv. XXVI. p. 35.

¹⁵ Ad. Kussmaul: Die Störungen der Sprache. Leipzig 1877. p. 71, 80.

Actions, on the other hand, that are guided by the will, since they presuppose more or less distinct ideas of movement, come about only with the cooperation of the cerebrum.¹⁶ — While elementary sense-impressions, as stated above, also affect an animal that has been deprived of the cerebrum, the proper apprehension and understanding of the impression takes place only when the cerebrum is intact. After extensive injury to both hemispheres of the cerebrum a dog no longer understands the meaning of what it sees and hears. It does not care if it is threatened with the whip, pays no heed to its food when this is not set in the usual place, is not frightened by noise, does not obey when it is called, does not mind tobacco smoke, and devours a dog's carcass without any sign of disgust. On the [52] other hand, such a dog goes around obstacles in its way and shuns glaring light. *Goltz* and *Munk* describe the condition of such an animal in much the same way, but differ in the explanation. According to *Munk* this "psychical blindness," as he calls it, is bound up with an injury to a certain limited region of the surface of the brain; according to *Goltz* every extensive injury to the cerebrum brings such a condition with it. *Munk* explains the condition psychologically by the loss of the ideas of memory by means of which the new impressions can be recognized and understood, while *Goltz* explains it by a general intellectual dullness, perhaps especially a want of attention. On both explanations the significance of the cerebrum in relation to the subordinate centers comes clearly to light.¹⁷

But the cerebrum stands not only in a positive but also in a negative relation to the subordinate nervous organs, since it can inhibit their activity. The vegetative functions therefore go on more briskly during sleep, when the cerebrum has laid down its scepter. Even in lower animals, where its position is not so dominant as in the higher ones, even there this inhibiting influence is noticeable. When the headless frog has recovered from the operation, its mobility becomes even greater than before. The subordinate centers yield more easily to impressions than the higher ones. This is in itself a direct consequence of the fact that in the higher centers the impressions have a long process to go through, have, so to speak, to be confronted with so many other parties, that the single impression does not so easily and so completely get its will as in the less composite organs. Thus every new impression modifies — either reinforcing or inhibiting — the movement that preceding impressions have called forth.

The increased briskness with which subordinate nervous processes go on after the removal of higher centers is explained by some from the fact that the quantity of nervous activity which has been roused in the lower center through the ingoing nerve now

¹⁶*Munk* p. 51 f. Cf. *Goltz* in *Pflügers Archiv*. XXVI. p. 36.

¹⁷*Goltz* in *Pflügers Archiv*. XXVI. p. 42 ff. — *Munk* p. 29.

[53] spreads over a smaller area, and must therefore produce quicker and more powerful effects. But all the phenomena cannot be explained in this way. Strong brain-activity, as in sudden and powerful sense-impressions, emotion, or the labor of thought, seems directly to affect the subordinate centers, so that the direct impressions upon these do not have the effect that they would otherwise have.¹⁸

Through such inhibitory activity, what goes on in the higher centers acquires significance for the lower. It is well known, to take a simple example, how a sudden sense-impression can prevent sneezing. Violent emotion or pain inhibits the movement of the heart (by an influence from the brain through the medulla oblongata and the *nervus vagus*), and thereby causes fainting. Strong fear can prevent the secretion of saliva, which lies at the bottom of that “judgment of God” by which the accused was held guilty when he had rice in his mouth without its being moistened. A sudden and gripping impression can stop the pains of labor, and an influence on the *nervus splanchnicus* the peristaltic movements of the intestinal canal. It is not only single sense-impressions that release inhibitory effects in this way; more intricate functions of the brain also exert this influence, and we shall see in a later section that an important part of the dominion of the will is connected with it. The only thing still worth noting here is that the phenomena of inhibition appear the more strongly, the more vigorous the organism is, whereas they are weakened by fatigue. The condition of the central organ is of decisive influence here: if it is exhausted, badly nourished, affected by cold or by strychnine and certain other poisons, then the reflex movement gains in rapidity, strength and extent. In “nervously weak” persons, whose morbid condition is bound up with disturbances of nutrition in muscles and nerves, there is a strong tendency to reflex movements and spasms.¹⁹ — In so far as one wishes to speak of reflex movements (in the sense of immediate motor discharges) within the brain itself, the brain is indeed a whole little world, which in its myriads of cells and fibers has means enough for inner reinforcement or inhibition, for inner debate and struggle for dominion among all the impulses that can arise in it.

[54] 5. From the purely *physical* point of view, which physiology too acknowledges as its own, everything that goes on in the nervous system, even in its highest centers, is a conversion of force, in which an impression, from the organism’s surrounding world or from its own interior, releases the tensional force stored up in the nerve tissue. The *physiological* expression for this is that the impression (the irritant) calls forth a reaction, which consists either in a muscular movement, or in a glandular secretion, or else in a

¹⁸Eckhard: *Physiol. des Rückenmarks*. (Hermann. II, 2). p. 37.

¹⁹Wundt: *Physiol. Psychologie*. I. p. 260–263. — Ranke: *Physiologie des Menschen*. 3 Aufl. Leipzig 1875. p. 871 f.

more comprehensive process within the centers of the nervous system itself. But now it turns out that in the case of some of the phenomena that fall under these points of view a third point of view must also be applied, namely a *psychological* one, since certain states of consciousness are bound up with the physical-physiological processes. The question then becomes one of the relation between these different points of view. Is the one subordinate to the others, so that, just as the physiological point of view falls within the physical when this is taken in the widest sense, the psychological point of view in turn falls within the physiological as a special form of it? Is what shows itself to us from the one point of view perhaps the cause or the effect of what shows itself to us from the others?

We still lack something here for the discussion of this question. For while we have given an outline of what physiology can teach us that is of significance for our problem, we have not yet given a fuller characterization of the phenomena of consciousness. The psychological point of view has thus not yet been distinctly indicated. Now it is really the task of all the following inquiries to give a characterization of the psychical life, and to that extent it will be impossible to present one already here. But since I have nevertheless chosen to treat the general problem of the relation between the psychical and the bodily ahead of the more special psychological inquiries, which in many ways presuppose a definite view of that problem, there is nothing for it but to give here a provisional characterization of the psychological phenomena, reserving for the following sections the task of furnishing fuller proofs of its validity. —

It is with consciousness in general as it is with special forms or elements of consciousness (e.g. colors and tones) in particular: a description or a definition of them is [55] impossible, since they are fundamental facts that cannot be traced back to anything still simpler and clearer. Now this does not exclude that one can nevertheless bring out the most important characteristic marks. For one can direct one's attention to limiting cases, where consciousness is about to glide over into unconscious states; and one can attend to the transitions from weaker and dimmer to stronger and clearer consciousness, and examine what it is that conditions the higher degrees of consciousness.

Peculiar to all life is a constant unrest, a struggle with changes, a want of equilibrium. With complete equilibrium death sets in. This general mark of life we find in the life of consciousness too. When we become conscious of something, a movement, a change must have taken place. Something must in one way or another have made itself felt which was not there before. A completely uniform existence abolishes consciousness. Without a resistance to our accustomed and uniform movement the spark of consciousness is not

struck.

Thomas Hobbes, the founder of English psychology, already taught: it comes to the same thing to feel one and the same thing constantly and not to feel at all. An experimental proof of the correctness of this proposition is found in the fact that the sustained direction of attention upon one single sensation or idea produces a sleep-like state (hypnotism). As the condition for the onset of such artificial sleep the discoverer of hypnotism, James *Braid*, therefore gives “monoideism,” i.e. absorption in a single idea. (Cf. *Preyer*: *Die Entdeckung des Hypnotismus*. Berlin 1881. p. 41 f. 81). — The history of mysticism also teaches, in many examples, how one approaches the limit of consciousness the more change, movement and alternating ideas are kept away.

Strong impressions are needed to wake one from sleep or from absent-mindedness. When consciousness is awake, it is sharpened and heightened by contrasts and changes. We feel the cold more strongly when we come from a warm room; light appears with a special brilliance when we come from deep darkness. Of stillness and rest we become properly conscious only when we come from the noisy town or from hard work.

[56]

But change and contrast are nevertheless not enough in themselves. They call forth a sudden shock, a surprise; but if their effect were not preserved, it would be only like a flash of light that vanished at once. One could imagine a living being so organized that from time to time wholly isolated sensations arose in it. Such separate flashes would not correspond to what we notice as sensations in our consciousness; in us the single elements of consciousness are not isolated, but stand in firmer or looser connection from first to last. Such a connection is necessary if the single impressions themselves are each to be able to make themselves felt. For a change to be able to have its effect, it is necessarily presupposed that the preceding state has not entirely vanished, for in that case the background, against which the new was to stand out, would also be lacking.

For consciousness there is therefore required, secondly, the preservation of the impressions and *interaction* between the elements that make up the content of consciousness. — This is confirmed by the fact that incoherence and want of inner mutual interaction between the elements of consciousness is a sign of the beginning dissolution of the life of consciousness. In advancing mental illness “fixed ideas” form, which make the free, natural movement and debate among ideas impossible. Later on not even such fixed ideas can be held fast and carried through. No comparisons and combinations can be made. Finally there sets in a complete absence of images and thoughts. The sense-impressions are no longer worked up, memory is almost extinguished, and language for the most

part lost.²⁰

This second mark can of course be present in countless degrees. Seen from an ideal standpoint we are really never fully conscious. We always have a multitude of elements in our consciousness which are not brought into distinct mutual connection, are not measured against one another, and are therefore not organized as their nature would require. But there nevertheless always asserts itself an involuntary tendency to compare and order all the content of consciousness according to definite principles. For every newly emerging element of consciousness calls forth — and this is the third characteristic [57] mark of the life of consciousness — according to certain laws, remnants in memory of elements given earlier. To become conscious of something is to *recognize* it, to refer it to a group of similar elements, to distinguish it from elements of a different kind. It is never the single elements (sensations, feelings, ideas, etc.) that determine the state of consciousness; what matters is the relation between the new impressions and those preserved in memory. The single impression is known, exists in consciousness at all, only through its relation to the other elements of consciousness. In unconscious beings the one moment falls outside the other; the past is gone and exists only in its effects. The conscious being, on the other hand, can reproduce earlier states and thereby alter the present state. In the moment of recognition the difference of time is abolished.

There are thus three chief properties that characterize consciousness: *change or contrast* as the condition for the arising of the single elements of consciousness (sensations, feelings, etc.), — *preservation and reproduction* of elements that have arisen earlier, — and *connection* between these and the new elements.

When we think back on our earlier states of consciousness, they stand before us as a series of sensations, ideas and feelings, as a stream whose higher or lower waves succeed one another. It can often look to memory as if this series were composed of independent, separate links that were connected with one another only in an external way. Some psychologists (especially *Hume*) have therefore depicted consciousness as a mere succession of ideas without inner bond and coherence, or more definitely as “the series of our possible and actual sensations” (*Stuart Mill*). What speaks decisively against the correctness of this description is the impossibility of the single elements occurring in *absolute* separation within what we know as consciousness. Each single element belongs to consciousness only through its connection with other elements. The emphasis must thus be laid on the connection, the coherence, and not on the links in their singleness. The peculiarity of the phenomena of consciousness, in contrast to what is [58]

²⁰Cf. *Griesinger*: Die Pathologie und Therapie der psychischen Krankheiten. Stuttgart. 1861. p. 323–351.

the object of the outer apprehension of nature, the material phenomena, is precisely the inner connection between the single elements by which they stand as states of one and the same subject — a connection that has its typical expression in *memory*, which may therefore be called the fundamental phenomenon in the spiritual domain. What has fallen out of memory can still exert a great influence on our life of consciousness indirectly, through its unconscious after-effects; but it is no longer itself a part of that life. In the physical world one can speak of memory only figuratively. Wherever a development takes place, what comes later is determined by what went before; and by virtue of the law of the conservation of energy nothing happens, be it ever so insignificant, that does not have its effect on what happens afterward. But only on the presupposition of a consciousness is what is past itself lived over again, itself made a link in a later spiritual connection, so that the differences of time are abolished. — Of the various stages and forms in which this fundamental psychological phenomenon appears, a following section will treat.

What must still be added here is that this inwardness and unity can appear in many degrees, and that in our psychological experience we never meet with it in the highest conceivable degree. Beneath the clarity and coherence of our consciousness there is always a more or less obscure chaos, whether it appears as a barrier to clearly conscious spiritual activity or as a nourishing, refreshing source for it. The primordial elements of our consciousness must always arise in it as something given to it as material for its activity. There is thus a passive and an active side in the nature of consciousness, the first corresponding most nearly to the multiplicity of the elements, the second to the unity and coherence in all the content of consciousness. The energy of consciousness shows itself in the way in which the single elements are connected and brought into mutual interaction. Kant has therefore rightly characterized consciousness as a synthesis. But the synthesis presupposes, at whatever stage in the development of the life of consciousness we look for it, a given multiplicity. It unites the single sensations into observations and ideas, the ideas into concepts, and so on. Such work goes on right from the threshold [59] of consciousness: it lies already in the fact that the more consciousness approaches the point where only one single element holds sway, the more it approaches unconsciousness. To that extent consciousness, as soon as it awakens to clarity, finds its own work already in full swing. The synthesis is, to use the words of the thinker whose genius in research has especially thrown light on these obscure paths, “the mere effect of a blind, though indispensable, psychical activity, without which we should have no knowledge at all,

but of which we are only seldom conscious”²¹

It is above all this fundamental property of the life of consciousness that makes its first arising so great a problem. For even if one were to presuppose elementary sensations bound up with forms of development standing lower down, what is quite decisive is, after all, precisely the connection, the unity of these sensations. The first sensation cannot, after all, attach itself to any other spiritual element; but how, then, can there be synthesis and consciousness here? — Such questions arise everywhere when we go back to the beginning. Possibly, however, the difficulty here is connected with the fact that we speak quite unwarrantably of single sensations. As will be discussed in a later section, there are circumstances which might indicate that what we regard as psychologically single is not so in itself.

The history of psychology shows that the various schools have laid different weight on the two sides of the nature of consciousness. *The German school* (Leibnitz, Kant, Hegel) lays predominant emphasis on the synthesis, the activity, the unity. *The English school* (and in Germany Herbart) has especially brought forward the passive or mechanical side, the multiplicity of the elements and their mutual interaction. Each school has its strength in the treatment of different problems. The English school has more eye for the elementary, real side of the life of consciousness, for the way in which the spiritual edifice is raised by the joining together of the fundamental elements; the German school, on the other hand, more for the coherence and unity that is the characteristic mark of consciousness from first to last. The *more recent* English school seems to meet the German in the recognition that the single sensation or idea exists only by being a link in [60] a connection of consciousness, for which reason consciousness can never be conceived as a mere sum or a mere product.²²

German psychology has often shown a tendency to approach metaphysics; English psychology, on the other hand, has approached mechanical natural science, and has carried analogies from it over into the conception of the psychical phenomena.

In the characterization of consciousness now given we have kept chiefly to its morphological side. Only in the more special psychological discussion can we go more closely into the more real side. In the present connection we must content ourselves with pointing out that the unity in the psychical life has its expression not only in memory and synthesis, but also in a dominant fundamental feeling, characterized by

²¹*Kant*: Kritik der reinen Vernunft. 1st ed. p. 78. (Kehrbach's edition p. 95). — Kant calls this faculty “imagination” [*Einbildungskraft*].

²²See on this my book *Den engelske Filosofi i vor Tid* [English Philosophy in Our Time]. København. 1874.

the contrast between pleasure and displeasure, and in an impulse to movement and activity proceeding from this fundamental feeling. At the simplest stage of consciousness that we can conceive, the psychical life consists in an oscillation between pleasure and displeasure, such that each of these states, while it holds sway, all but displaces the other, for which reason memory plays a very subordinate part. At this stage the life of consciousness plainly shows its original practical character, which at higher stages is more hidden and indirect. Like all life, the life of consciousness too is intelligible only through the living being's relation to, and position in, the world. Life is everywhere a struggling harmony between the inner and the outer. The rhythm of pleasure and displeasure must therefore be connected with a rhythm in the organism's conditions of life. This practical character is never entirely lost. Even those of our thoughts and feelings that do not concern our own personal existence nevertheless spring from our relation to other beings and our position in the universe. The thinking and feeling ego is always determined by its relation to the non-ego, whether this non-ego is a world of sense or a world of ideas.

[61] **6.** With these last remarks we have already overstepped the strictly psychological characterization and hinted at a parallelization of the life of consciousness and the nervous system. This hint we shall now carry further. Here one parallel will present itself after another, so that the craving which on the naive standpoint so often makes itself felt, to get an intuitable picture of the soul, is properly speaking satisfied by nature itself in the form and mode of functioning of the nervous system. The next task then becomes that of interpreting the picture, of finding out the relation between the symbol and the thing itself.

a. The great significance of the nervous system was, as we saw in an earlier connection, to serve as a connecting central organ for the various parts of the organism, to guide their activities in mutual harmony, and to make possible a unified action over against the surrounding world. But consciousness solves precisely the same task in its own way. In it what is scattered in time and space is united; the surge of the conditions of life expresses itself as the rhythm of pleasure and displeasure; and in memory and the activity of thought the most intimate concentration that the whole circle of our experiences can show is revealed.

b. That one becomes conscious of something presupposes a change, a transition, a contrast. The content and the energy of consciousness must be brought out of their equilibrium; attention must be aroused. An arousal, an influence (an irritant) is likewise the condition for the functioning of the nervous system. The impression acts by releasing

bound force, by abolishing the equilibrium in nerve fibers and nerve centers.

c. But the impression does not act merely on a single center; owing to the manifoldly branching connection between the various nerve centers, it releases a series of processes which mutually promote or inhibit one another, so that the total effect depends on the result of this physiological debate. Psychologically seen, what corresponds to this is that the single sensation calls forth the ideas with which it has most kinship. The single sensation therefore has not a single but a very composite effect. The psychological relation between sensation and idea has its physiological parallel in the relation between the arrival of an impression at the nervous central organs and their mutual interaction. Not only from the purely physical, but also from the physiological and the psychological [62] point of view, the effect of the spark on the gunpowder is the most apt image of what goes on.

d. The formation and arising of sensations and ideas take a certain time. None of our movements take place so quickly as the unconscious ones. The greater the deliberation, the slower the action. The more intricate the spiritual operations that are carried out, the longer the time required. The nerve process, too, takes a certain time, which physiology has begun to measure. What interests us in this connection is that the movement in the nerve fibers goes on more quickly than the movement through the nerve centers (the gray matter), and especially that the central nervous functions to which the activity of consciousness seems to be bound (the psychophysical functions) take longer than the merely physiological ones.²³ It agrees with this that actions which at first are performed with consciousness come, through frequent repetition and practice, to be performed unconsciously — and more quickly. The child who is learning to read looks at each letter carefully before it recognizes it, and applies special attention and effort to pronounce it correctly. But gradually it learns to read aloud without thinking of the form of the letters and the character of the sound. It is the same with dressing and undressing, walking, dancing, swimming, and many of our daily occupations. The shorter the time between the impression and the movement released by it (the time of reaction, the physiological time), the more unconsciously the action takes place.²⁴

e. To the physiological hierarchy of superordinate and subordinate nerve centers, and the relative independence of the latter, there corresponds the fact that activities can go on in our organism which under normal conditions are not bound up with consciousness,

²³See *Wundt: Physiologische Psychologie*. II. p. 227. Cf. *Spencer: Principles of Psychology*. I. p. 107.

²⁴Sometimes, to be sure, it is precisely a straining of the attention, a concentration of the will, that can lead to greater speed in thinking and acting. But the presupposition of this is previous practice, or else an originally adapted organization which needs only an impulse to be set in function.

but become so when they depart from the normal conditions. The functions of nutrition, [63] for example, as a rule go on without our sensing anything of them. Only when they are especially favored or hindered do sensations arise in us, bound up with pleasure or displeasure. When food is lacking, the flow of blood to the stomach, which now no longer has anything to work on, also ceases, and in the opinion of some the feeling of hunger arises on account of the deficient nourishment of the nerves thereby brought about in turn. The *nervus vagus* forms an intermediate link between the organs of feeling and the brain; in a high degree of hunger the sensory nerves of the small and large intestine also seem to play a part.²⁵ The sensation of hunger is precisely a good example of a transition from unconsciousness to consciousness, since it runs through a whole scale of degrees, from the first indefinite feeling of discomfort to the most violent torment. The opposite feeling does not stand out with so sharp a character.

From another side the psychological parallel to the relation between higher and lower nerve centers appears in the phenomena of movement. We have just seen how new reflex movements can arise through the constant and repeated functioning of the higher centers. But the higher centers also, as was brought out earlier, act to inhibit the involuntary movements that can be released by lower centers. Just as the training of new reflex movements corresponds to the positive work of the will, so the inhibition of original and involuntary movements corresponds to the negative work of the will. Both our education and our self-education consist partly in accustoming ourselves to something, partly in breaking ourselves of something. This will be treated more fully in the section on the psychology of the will; here it was only to be hinted that even the struggle between “the spirit and the flesh” has its physiological counterpart.

f. Finally there appears a parallel between the various sides of the life of consciousness and the various organs of the nervous system. In knowledge and feeling the life of consciousness turns, as it were, an open side out toward the non-ego; in this openly [64] exposed region it behaves predominantly as receiving and appropriating. In the will (at its various stages as impulse, desire and resolution), on the other hand, we have the response of the life of consciousness, the psychical reaction. The nervous system shows precisely the same two-sidedness in the contrast between the sensory and the motor organs. It would be to renew the misunderstandings of phrenology if one were to imagine knowledge, feeling and will each localized in its own place in the brain; none of them is, psychologically seen, a single and simple process that could be conceived

²⁵Ranke: *Physiologie des Menschen*. p. 219 f. — It has, however, been found that the sensation of hunger persists after the *nervus vagus* has been severed. — According to E. H. *Weber* the contraction of the muscles of the empty stomach also contributes to the sensation of hunger.

as going on in a definite organ; they interpenetrate one another, as will be shown later, to such a degree that it is only in abstract usage that we can speak of them as different processes. We therefore keep here merely to the general schema of the nervous system: an ingoing movement, a central working-up of it, an outgoing movement. The same schema fits the life of consciousness.

7. These parallels must have a real significance; there must be an inner connection between the life of consciousness and the brain.

The matter could be settled most simply of all if we could feel immediately that the life of consciousness was bound to the brain, so that simultaneously with a certain state of consciousness we had a sensation of a certain state in the brain. In strenuous spiritual activity we do, to be sure, believe that we sense something in the brain; but it is not then the function of the brain itself that we sense. "When," says *Griesinger*,²⁶ "in strong alterations of the life of ideas one sometimes seems to sense something going on in the head (as though something opened or closed, or the like), there can be no question of changes in the processes of the brain being sensed; they seem rather to be processes connected with the membranes of the brain and their supply of blood."

It turns out, too, that it took a long time before men became fully convinced that consciousness was bound to the brain. In antiquity the seat of the soul was thought to be in the blood, in the diaphragm, or in the heart. Among the older Greeks only *Alcmaeon* and *Plato* taught that we think with the head. It was first the great Alexandrian anatomist *Herophilus* (c. 300 B.C.) who placed the soul in the brain, relying on definite facts, namely [65] the observation that the nerves, especially the sensory nerves, gathered in the brain as their final center.²⁷ Yet this anatomical proof has not been sufficient to establish the conviction of the real connection between consciousness and brain. What has been of decisive importance here, on the other hand, is a series of comparative observations and experiments.

In the lowest animals no nervous system has yet been found. In mollusks and arthropods there is found only slight centralization of the nervous system; the central nervous system consists at most of a ring of nerve ganglia. The lowest vertebrate, *Amphioxus*, has only a spinal cord, no brain, and in the lower classes of vertebrates the brain is developed in a far smaller degree than the spinal cord.

The greater the weight of the brain in relation to the rest of the body, and the more prominent the cerebrum in relation to the other organs of the brain, the more highly developed the life of consciousness proves to be. The higher centers, again, occupy

²⁶Die Pathologie und Therapie der psychischen Krankheiten. 2 Aufl. p. 26.

²⁷*Exner*: Physiologie der Grosshirnrinde. (Hermann's Handbuch. II, 2). p. 193.

a far larger place in the human cerebrum than in that of the animals, in which the immediate centers for sense-sensations and muscular movements seem to preponderate. The greater or lesser richness of the brain in convolutions also proves to be connected with the higher or lower stage of development of the life of consciousness. The brain of the more intelligent breeds of dogs has more convolutions than that of the less intelligent; man in this respect far surpasses the apes, which otherwise stand so near him in structure; eminent men have very large cerebral hemispheres, rich in convolutions.

The constitution of the brain of the fetus and of the newborn child also agrees with this. At the earlier stages of fetal development the cerebrum in man, as in all vertebrates, lies in front of the other parts of the brain without covering them. In the course of growth it comes, in man (and in part in the apes), to cover the mid-brain and at last the cerebellum. The cerebrum of newborn children is little developed as regards both [66] structure and capacity to function, whereas subordinate cerebral apparatuses can be taken into use at once.

Finally it has been shown by experiment that sensations arise only when the impression on the outer surface of the organism is propagated to the brain, and that voluntary movement is possible only when the motor centers in the brain are uninjured. If the cerebrum is removed, the life of consciousness in those animals that can survive this operation is weakened; deliberation and initiative fall away. Conversely, a dull and undeveloped life of consciousness (as in idiots) is bound up with deficient development and nourishment of the brain; and the progressive dissolution of the life of consciousness in mental illness is accompanied by a progressive dissolution of the brain, first and foremost of the cerebrum. If the vessels that supply the brain with arterial blood are tied off, an unconscious state sets in which leads to death.²⁸

8. Whither, now, are we led by this formal agreement and real connection between the life of consciousness and the life of the brain?

Only that hypothesis can be justified which lets all the factors brought out in the foregoing come into their own. Now, from the nature of the case, four possibilities can be conceived here. a) Either consciousness and brain, soul and body, act on each other as two different beings or substances; b) or else the soul is a form or a product of the body; c) or conversely the body is a *form* or a *product* of the soul; d) or finally soul and body, consciousness and brain, develop side by side with each other without our being able to derive the one from the other. These various possibilities we now test, relying on what has been developed and set forth in the foregoing.

²⁸Exner. p. 193–206. — Wundt: *Physiol. Psychol.* I, p. 150. 204 f. — Griesinger. p. 418–444. — Tardieu: *Etude médicolégale sur la folie*. 2 éd. p. 85–89. 119. — Carpenter: *Mental physiology*. Chap. 2.

a. The common view is that the soul acts on the body and the body on the soul. One perhaps even believes that one can sense this immediately, although the very fact that not everyone agrees that there exists an independent soul, separate from the body, and that it was in any case only by many detours that knowledge was reached of which part [67] of the body the soul is especially bound to, seems to tell against it. — “But are there not indisputable facts from which that assumption can be inferred? An impression on a sense-organ is propagated to the brain and there passes over into sensation; conversely, our will is able to set our body in motion!” But the question concerns precisely the relation between what goes on in the brain and the states of consciousness; and if what has been adduced were facts, we should have no reason to ask; we should then know enough for the present. When the state of the brain to which the sensation or the resolution is bound does not itself become an object of consciousness, it is impossible to discover whether a relation of cause and of interaction really exists between brain and consciousness.

This view rests, moreover, on the presupposition that a causal relation can take place between the spiritual and the material. But such a presupposition is in the highest degree improbable. For it would conflict with the *law of the conservation of energy*. At the point where the material nerve process was converted into psychical activity, a certain sum of physical energy would disappear without being replaced by any corresponding sum of physical energy. It has been answered to this that it is, to be sure, incomprehensible how a material activity can pass over into spiritual activity, but that every transition, every causal relation and every conversion of force is at bottom incomprehensible to us — and that the law of the conservation of energy, moreover, requires only that a certain corresponding sum of energy come into action in place of the one that has disappeared, no matter whether this equivalent is of a physical or a psychical nature. But this would be a bold and unwarranted extension of the law of the conservation of energy, which in the form in which it stands is a purely physical law. Such an extension would presuppose that it was possible to obtain a *common measure* for the spiritual and the material. Now what common denominator is there for a thought and a material motion, what common form contains them both? Until such a common form has been demonstrated, all talk of interaction between the spiritual and the bodily is, scientifically seen, meaningless. So long as we move within the material, we move safely; and so long as we move within the spiritual, we move safely; but as soon as we are required to imagine a transition from [68] physical laws to psychological laws or conversely, we stand before the incomprehensible. In the concept of cause, as in all the metaphysical concepts that condition our view of

reality, there is contained an epistemological problem; as everywhere when we approach the boundary-posts of our knowledge, difficulties crop up; but it becomes a difficulty raised to the second power when the concept of cause is to be used to connect two terms that have no common measure. It is not too strong a word that *Griesinger* has used about this: that even if an angel came from heaven and explained it to us, we should not be able to understand it.

Now in reality the usual view also involuntarily brings about a homogeneity, namely by now conceiving spirit as something material, when it lets it be affected by material motion, now making matter into something spiritual, when it lets it be accessible to spiritual influence (as when in Plato reason “persuades” matter).

It will easily be seen that it is of no use to say that the soul cannot, indeed, increase the *sum* of physical energy in the world, but that it can alter the direction in which energy is applied. A physical motion does not alter its direction except under the influence of a physical force of a certain strength. This way out, too, therefore necessarily makes the soul into a physical energy.

The application of the concept of cause presupposes not only a common measure, but also a *difference of time*. The question that has been raised, whether cause and effect are simultaneous or not, rests in part on a misunderstanding, and in any case does not concern us here. But no one can deny that we should never have formed the concept of cause if no changes took place in the phenomena. If everything were uniform and unchangeable, we should have nothing whose cause we could ask about. The causal relation presupposes that an event has taken place. If the relation between soul and body, or consciousness and brain, is a causal relation, there must therefore be a difference of time between the process in the brain and the act of consciousness. But this presupposition conflicts with the pointers that physiology gives. As we have [69] seen, modern physiology aims at conceiving all processes in the organism as physical or chemical. It does not pretend to have explained the origin of organic life; it asserts only that every time it has arrived at an understanding within the domain of organic life, this has happened by tracing the organic phenomena back to physical and chemical laws. If, then, there is a transition from physiological function to psychological activity, from body to spirit, physiology as such cannot in any case discover it. And yet such an assumption presupposes that at certain points the physiological process is broken off by a psychological one (when the impression becomes a sensation in the soul), only to begin again under altered conditions (when the soul has recovered from the material influence and answers it with an act of will). This assumption thus forces such a break upon

physiology. But physiology will hardly ever agree to concede this break. Apart from the physical difficulty of the violation of the law of the conservation of energy — which physical difficulty is at the same time a physiological one — the nerve process must, from a physiological standpoint, be conceived as a coherent circuit. A very great deal here is still unexplained. The relation between nerve fiber and nerve cell is very obscure;²⁹ the physical properties of the ganglion cells themselves, and with them the physical origin of the simplest reflex movement, are not yet understood; it is not even quite certain that it is the ganglion cells that form the intermediate link between the ingoing and the outgoing nerve fiber. Nor has it been possible to demonstrate the anatomical connection between the centers of the centripetal and the centrifugal nerves in the spinal cord.³⁰ But in spite of all this physiology cannot let its boundary be staked out for it from [70] without. Its fundamental thought is the thoroughgoing coherence in organic life; the most intricate processes it seeks to explain by tracing them back to the simpler; it learns from the lower phenomena how the higher are to be understood, since the principles of structure and activity are the same throughout. Thus it is, for example, the doctrine of reflex movements that for physiology throws light on the way in which the highest functions of the brain are to be explained,³¹ without its therefore being right, without further ado, to say that all activity of the brain is reflex activity. In so far as one can speak of available and settled results in the physiology of the brain, it presents to us the picture of the brain as a republic of nerve centers, each with its function and in mutual interaction; but there is no trace of the physiological *process* anywhere being able to cease in order to pass over into a *process* of a wholly different kind.

The usual doctrine of interaction (the doctrine of *influxus physicus*, as it was called in older times) appears in a stricter, more metaphysical form and in a vaguer and more indefinite one. In its metaphysical form it appeared in Descartes, who conceived soul and body as two wholly heterogeneous and yet mutually influencing substances. Here, then, that hasty fixing of the soul as an independent substance which was mentioned earlier

²⁹Cf. *Ditlefsen: Menneskets Histologie* [Human Histology]. København 1879. p. 582: "The more recent discoveries concerning the structure of the nerve fibers have not taken us further toward knowledge of the relation between them and the cells; for the present, therefore, we must presumably hold fast to the view that the nerve cells are centers for the nerve fibers, and continue to direct histological investigations toward furnishing a satisfactory morphological understanding of this physiological fact." — See also *Wundt: Physiologische Psychologie*. I, p. 35.

³⁰C. *Lange: Rygmargens Patologi* [Pathology of the Spinal Cord]. p. 24. — *Eckhard: Physiologie des Rückenmarks und des Gehirns*. (Hermann. II, 2). p. 7–19. 61 ff.

³¹After Marshall *Hall* had set up the theory of reflex movement in 1833, *Laycock* showed in 1840 that its principles must also find application in the physiology of the brain. (On the reflex action of the brain. — *The British and Foreign Medical Review* 1845). Apparently independently of him, *Griesinger* expressed the same thought in his treatise »Ueber psychische Reflexactionen« (*Archiv für physiologische Heilkunde*. 1843).

has been carried out, whereby the standpoint of the psychology of experience has been exchanged for metaphysics. Yet precisely the sharp and clear form in which Descartes presented the current doctrine has contributed extraordinarily to laying bare its weak sides. It falls as soon as one tries to present it in its full consequence. Descartes therefore has the merit of having set up the problem of soul and body. For to the current view in [71] its vaguer form there exists no difficulty in this relation. With justified thoughtlessness practical usage disregards the theoretical difficulties. Just as little as the current usage of language respects Copernicus's *doubt* whether the sun really moves around the earth, just so little does it take account of the fact that physiology and psychology shrink from letting brain and consciousness influence one another. In addition, usage has been formed under the influence of a metaphysics partly spiritualistic, partly materialistic.

b. An end is made of this inconsistency and vagueness when one simply strikes out one of the terms that are to be connected. Now since the intuition of the outer, material world plays an overwhelming part in the common circle of ideas, while inner self-consciousness is not easily worked out to the same degree of clarity and distinctness, it lies perhaps nearest to hand to make the material one with the real, and to conceive the spiritual as a form or an effect of it. Historically, materialism is no doubt also older than the current doctrine of interaction. Homer and the oldest Greek philosophers (before Socrates and Plato) are materialists; even among the Christian Fathers before Augustine materialistic ideas prevailed. In modern times materialism has its firm standing ground in the law of the conservation of energy and in the doctrine of physiological coherence. It is fully justified against every idealistic line of thought that leads to setting external limits to the series of physical and physiological causes. As a natural-scientific method materialism is unassailable. It is another matter when the method is rewritten into a system. It is fully entitled to regard every change in and function of the organism, and especially of the brain, as material; but as a metaphysical theory it goes further and maintains that the phenomena of consciousness are only changes in or functions of the brain, and in this consists its encroachment.

Carl Vogt in his day aroused great offense by declaring (in his "Physiologische Briefe") that "in the same way as contraction is the function of the muscle, and as the kidneys secrete urine, in the same way the brain produces thoughts, movements and feelings." [72] He could thereby seem to give a choice between two ideas: either to conceive thought as a substance or as a motion. The first way of thinking, which lies nearest to immediate apprehension, and to which ancient materialism therefore especially kept, has, however, for any closer reflection something so baroque about it that it needs no further discussion.

Nor, indeed, was it probably Vogt's meaning; the chief weight must be laid on the *activity of secretion*, not on the product. But in principle nothing is thereby altered. Even among sober-minded physiologists, some of them philosophically trained, one does sometimes meet the proposition that the activity of consciousness is a function of the brain. It seems, however, as if precisely the strictly physiological use of the expression function must conflict with such a proposition. That, for example, contraction is the function of the muscle means only that it is a certain form and state of the muscle in motion. As *Goethe* has said: »Function ist Dasein in Thätigkeit gedacht«. The muscle in function is just as material as the muscle at rest, and what does not have the properties of matter cannot be the form of activity of anything that is material. The concept of function (in the physiological sense)³² points, just as well as the concept of substance or product, to something that appears to us as an object of intuition in the form of space. But thought and feeling cannot be pictured as objects or motions in space. We come to know them, not by outer intuition, but through self-sensation and self-consciousness — a source to which the physiologist too, without always being clearly conscious of it, has recourse when he wishes to investigate the relation of the life of consciousness to organic life. It is by many detours that it is discovered that certain definite phenomena of consciousness are bound to the function of certain definite parts of the brain. There is no doubt that even the very highest activities of consciousness have their corresponding functions of the brain — just as even the most beautiful melodies are not too exalted to be expressed in notes. But the activity of consciousness and the function of the brain we always come to know by using different sources of experience. The encroachment of materialism [73] consists in simply drawing a line through this essential difference. By simply ascribing to the brain a capacity for consciousness, or perhaps even making the brain the subject of the manifestations of consciousness,³³ it goes back at bottom to a mythological, fantastic standpoint.

We have regard here especially to *empirical* or *phenomenological* materialism, i.e. the view that regards it as a direct result of the science of experience that the phenomena of consciousness are a form or an effect of the material phenomena, so that all reality

³²In the mathematical sense, on the other hand, it can with full right be said that consciousness is a function of the brain, since experience shows us a certain proportionality between the degrees of development of consciousness and of the brain. The activity of consciousness would then, strictly, have to be said to be a mathematical function of the physiological function of the brain.

³³Ch. *Robin*, for example, defines »sensibilité« thus (*Anat. et physiol. cellul.* p. 540): »Ce mode de la névrité est caractérisé par ce fait, que *les éléments* nerveux qui en jouissent, après avoir reçu une impression du dehors, la transmettent de ce point à un autre où ils [sic] la *perçoivent*.« A similar mode of expression occurs in *Broussais*. But what idea can really be connected with nerve elements observing or perceiving something?

can be traced back to motion in space. We thereby move not only in the domain in which we ourselves would most like to move, but also in that in which materialism has always believed itself to move. It has not noticed that even if it were right in all its assertions, there would still be something that it always overlooked, and by which a new problem, and for it a terrible one, arose: namely the circumstance that motion in space is known to us only as an object of our consciousness. *Epistemologically*, then, concepts such as consciousness, idea and intuition lie deeper than concepts such as matter and motion. An absolute and resolute materialism was therefore possible only in antiquity, before the awakening of more penetrating philosophical reflection. *Democritus* is the only consistent materialist. None of the modern materialistic writers can speak with a calm and sureness like that with which *Lucretius* in his majestic verses sets forth the doctrine of Democritus. The moderns for the most part concede that even if we can trace everything back to matter, we still do not know what matter is in itself. So La Mettrie, [74] Holbach, Cabanis, not to speak of the wildly fluttering inconsistencies in the writers of the most recent time (*Büchner, Moleschott*).³⁴

What we have here urged against materialism is not, however, the *metaphysical* and *epistemological* inconsistency that it commits in demanding that consciousness should acknowledge as the absolutely original and alone real something that is given only as an object for consciousness, and that can be imagined and known only by consciousness's own activity. Our task here, on the contrary, is always merely that of finding which view the given itself drives us toward; and the result of our critique of materialism is that it already sins against the very concepts of experience.

c. When, as the third possible hypothesis, we discuss spiritualism, we must constantly hold fast the difference between a *phenomenological* and a *metaphysical* mode of consideration. A very great many obscurities regarding our problem stem from this difference being overlooked. Like materialism, spiritualism has almost always mixed together metaphysical and *empirical* results. Spiritualism is the view according to which the spiritual is the only reality; everything material, all motion in space, is only an outer and, in the last instance, illusory form of a spiritual life. It relies on the impossibility of explaining the spiritual from the material, and on the fact, always partly overlooked and partly underrated by materialism, that our concept of matter is a spiritual product, and that we do not know what matter is apart from our concept of it. The spiritual is thus a *prius*, a presupposition on which all thinking rests; a reasonable hypothesis results only when we trace the less known back to the better known. The spiritual is really the only

³⁴Cf. my work *Filosofien i Tyskland efter Hegel* [Philosophy in Germany after Hegel]. Kbhvn. 1872. p. 94–117.

thing that is fully intelligible to us, where we have not merely a knowledge of external circumstances and relations (*cognitio circa rem*), but also a knowledge of the thing itself (*cognitio rei*).

Even if spiritualism is right in all this, it nevertheless does not belong here. For granted that everything is at bottom spiritual, and that nothing exists but thoughts and ideas: there still always remains a difference between *ideas of material motion* and *ideas [75] of phenomena of consciousness*, and the problem then arises again of how these different ideas, which have arisen in experience, are to be combined. In other words, the empirical problem is independent of the metaphysical one. We do not discuss here whether spirit or matter is what lies at the ground, the truly real; we investigate how the spiritual and the material phenomena are connected in the experience that every metaphysics, consciously or unconsciously, presupposes.

The foremost representative of spiritualism in the philosophy of our time, *Hermann Lotze*, has with great clarity impressed upon us the difference between the phenomenological (or, as he called it in his earlier writings, the occasionalistic) inquiry and the metaphysical one.³⁵ But he has not held fast to this difference, but on the contrary has pushed it aside in an arbitrary way.

Lotze is one of the writers who in recent times have most energetically maintained the mechanical conception of organic phenomena, and have shown the concept of mechanism to be the one called to govern our whole conception of nature. All the more remarkable is it that he refuses to concede the full consequence of the law of the conservation of energy. "The physicists," he says in one of his earlier writings (*Allg. Physiol.* p. 461), "seem convinced that every motion of masses in space and time can be annihilated only by another equally great and opposite one, and from this the task presented itself of following every motion, once it has arisen, through all its changes of form, right up to its final passage from the organism to the surrounding world, or to its cancellation by opposite motions. I am not fully convinced of the unconditional correctness of this principle; on the other hand, from my view of the philosophy of nature I cannot meet with any conclusive counter-argument the possibility that spatial motions may be absorbed by passing over into intensive states." At the point where this transition from the spatial to the non-spatial world takes place we must, according to a later statement of his (*Mikrokosmos. I. 2 Aufl.* p. 326), seek the seat of the soul. In his most recent work (*Drei Bücher der Metaphysik*), the last that it was granted to [76] this brilliant researcher to give us, he expresses himself more precisely on his position

³⁵Cf. *Filosofien i Tyskland efter Hegel*. p. 293–295; 299–302.

regarding the law of the conservation of energy. He lays the weight on the point that this law in itself states only an equivalence between the manifestations of force, promises us only a certain replacement for the energy that ceases, but says nothing about what kind this energy is. He therefore does not think it impossible to apply it to a case where a spiritual energy takes the place of a physical one, or conversely. Phenomenologically, then, he takes his stand on the standpoint of the natural doctrine of interaction, and to that extent his theory falls before the objections directed against that doctrine (see above, p. 67–68). But he does not remain standing on the phenomenological standpoint. The distinction between spiritual and material phenomena is for him only the point of departure; the spiritual is — as he endeavors to show by metaphysical inquiries — the only reality, the material phenomena only the manifestation of the interaction of spiritual elements; and a transition from the spiritual that underlies the material phenomena to that which appears in the phenomena of consciousness entails no theoretical difficulties: the heterogeneity, the want of a common measure, has fallen away through metaphysical reflection.

Lotze really always evades the question whether a transition actually takes place. He builds his metaphysics on the assumption of such a transition, as though experience immediately offered one. The form in which spiritualism appears in him is thus bound up with — not to say encumbered with — an assumption which a consideration based on the analogy of our experience must reject. The idealistic view of life is in principle independent of the form that Lotze has given it. It can very well be combined with other phenomenological assumptions than those its eminent philosophical champion has taken as his basis.

d. There then remains only the fourth possibility. If a transition from the one domain to the other conflicts with the law of the conservation of physical energy, and if the two domains nevertheless stand in our experience as different, they must unfold, each according to its own laws, simultaneously with each other, so that to every phenomenon [77] within the world of consciousness there is a corresponding one within the world of matter, and conversely (so far as there is reason to assume a life of consciousness connected with the material phenomena). The parallels we have drawn earlier point directly to such a relation; it would be a strange coincidence if the marks should thus repeat themselves were there no inner connection lying at the ground. Like the *parallelism*, the *proportionality* between the activity of consciousness and the activity of the brain also points to an underlying *identity*. The difference that nevertheless remains in spite of the agreements compels us to assume that one and the same principle has found its

expression in a twofold form. We have no right to assume soul and body to be two different substances in mutual interaction. But we are led to conceive the material interaction between the elements of which the brain and the nervous system consist as an outer form of the inner, ideal unity in consciousness. What we become conscious of in our inner experience as thought, feeling and resolution thus has, within the material world, its representation in certain material processes in the brain, which as such come under the law of the conservation of energy, whereas this law can find no application to the relation itself between the processes of the brain and the processes of consciousness. It is as if one and the same content were expressed in two languages.

Only experience can decide whether the two forms really extend equally far. We have already touched on the difficulty of fixing the limits of the life of consciousness downward; the following section will give us occasion to take up this question again. On the other hand there are still those who think that the noblest spiritual manifestations of life are not bound to material processes. That sense-perception and so-called physical pleasure and pain are bound to certain nerve processes no one will deny; it is only in the case of the higher phenomena of consciousness that it is considered necessary to assume a wholly new principle. But already the general characterization of the life of consciousness given above leads us to see the impossibility of drawing any boundary line between a lower and a higher within consciousness. The same type prevails from the simplest to the highest forms. However high the ideal world of thoughts and feelings seems raised above the series of single, momentary sensations, it is nevertheless the [78] same principle that prevails in both; it is only the degree of development that is different, not the plan of construction or the building material. The more special psychological inquiries that follow will show this more closely.

Just as little as a sharp boundary can be drawn between a lower ("sensuous") and a higher ("spiritual") content, each with its own conditions of existence, just so little is it justified to regard the material or content of consciousness as bound to physical processes, while the forming and elaborating activity of the spirit was to have no physical parallel. Already in the simplest observation, right at the threshold of consciousness, we find the result of an activity of the spirit, a connection of elements into a unity, a synthesis. An absolutely single state would not be conscious. At no point, therefore, can matter and form be separated. The physiological coherence and interaction between the extraordinarily numerous centers in the brain furnishes, moreover, a sufficient basis for the assumption that not only the spiritual elements but also their connections have their physical expression.

Both within the spiritual and within the material world we hold fast to the law of continuity. The parallel theory, or *identity theory*, sees the two worlds as two expressions, given in experience, of one and the same being. They are two languages in which the same thought is expressed, but whose roots we are unable to trace back to any common root. Moreover — so long as we keep strictly to experience — the one domain stands before us as a fragment, while the other extends into the infinite and rounds itself off into a closed whole. The law of the conservation of energy makes the material world a totality which we can never, to be sure, measure out, but within which we can in principle follow the fate of the single forms and elements. The spiritual world has no corresponding law to show. The spiritual elements arise and disappear in experience without any equivalent being demonstrable that is consumed in the first case and takes their place in the second. The mere circumstance that spiritual states cannot be measured as physical energies and chemical substances can is enough to destroy the hope of finding a spiritual parallel [79] to the law of the conservation of force. But in addition to this, the very fundamental concept of spiritual existence brings difficulties with it here. Material existences can pass over into one another, in such a way that all the energy that is lost with the one is preserved in the others. The law of the conservation of energy shows us the unity and eternity of nature beneath the arising and perishing of all material beings. But spiritual existence has, as we have seen, memory, synthesis, for its fundamental form; — and synthesis presupposes *individuality*. The material world shows us no real individualities; these are first known to the psychological point of view, from which inner centers of memory, of acting and of suffering are discovered. Now if we were to imagine that the single spiritual elements (sensations, thoughts, feelings, etc.) were converted into other combinations, like the chemical atoms, we should be conceding that they could have an existence outside an individual consciousness, which according to the foregoing is meaningless. Sensations, thoughts and feelings are spiritual activities, which cannot subsist when the individual connection in which they occur has ceased. They correspond to the organic functions, not to the chemical elements. When the organism is dissolved into its elements, the organic function is impossible. The spiritual individuality has its physical expression in the sum of energy that the organism commands in its germ and in the course of its development, and in the organic (especially neurophysiological) form under which this energy finds application. Even the most spiritualistic view must concede that here something is given which can indeed be altered, but not wholly created, by the later development.

The theory to which we have here been led is no complete solution of the problem

of the relation between soul and body. It is only an *empirical formula*, a designation of how the relation provisionally stands when, following the pointers of experience, we attend at once to the close connection between the spiritual and the material, and to the impossibility of tracing the one back to the other, and of assuming a transition from the one to the other. Of the inner relation itself between spirit and matter we learn nothing; we know only that one being works in both. But which is this being? Why has it a twofold form of manifestation, why is one not enough? These are questions that lie [80] outside the domain of our knowledge. Spirit and matter stand before us as an irreducible duality, like subject and object. But the spiritual does not on that account become a superfluous addition. On the contrary, from a psychological standpoint — which is, after all, the one from which we finally judge existence — we can find meaning and value in the world only in this, that the same thing which works, spreads and forms itself in the outer world of matter opens itself in its inner being to itself as thinking, feeling and willing. All determination of value is bound to the spiritual, and has the material for its object only indirectly. Even when we admire the play of masses striking one another in space, what we really admire is the reflected image of a connection of thought that the lawfulness of physics shows us. When we ask about the inner connection between the physical and the spiritual world, we stand at the limit of our knowledge, and no view whatever has yet answered the question of the necessity of spiritual life in existence except by a teleological postulate, whose justification it is not the business of psychology to discuss. Only so much is certain, that if we are right in assuming that duality in the world of phenomena, it must also have its ground. It is not the only place in our experience where we are compelled to assume a connection that actually subsists without being able to demonstrate its necessity.

The empirical formula with which we here conclude does not exclude a further-reaching metaphysical hypothesis. The fundamental thought of idealism, of the spiritual as the true essence of existence, can thus very well be reconciled with the empirical assumption of the parallel theory. For this theory, as an empirical formula, says nothing whatever about the two forms of existence being absolute or possessing validity when we disregard the human point of view. It was a hasty metaphysics when *Spinoza* taught that spirit and matter were two equally eternal and infinite attributes of the absolute substance. We do not know the absolute substance, and so cannot know whether spirit and matter are equally essential to it. On the contrary, epistemology leads (as was indicated under c) to regarding the phenomena of consciousness as the most fundamental facts in our experience, since the subjective point of view always lies deeper than the objective.

[81] It can therefore easily lead to misunderstandings when the identity or parallel hypothesis is designated as neo-Spinozism. It is, to be sure, bound up with Spinoza's name; it is he who has the honor of having first set up this theory and thereby led beyond the contending materialistic and spiritualistic theories. He was guided in this by two different motives. On the one hand he wished to remove all imperfect ideas from the idea of the infinite being; there was to be nothing outside it, nothing that it did not pervade. But besides this religious-philosophical or metaphysical motive, the conviction of the unbroken coherence of the physical series of causes also made itself felt. If the spiritual activities cannot intervene in this series of causes, there is nothing for it but to assume that the spiritual and the material activity do not succeed each other, but go on simultaneously (*simul naturā*. Eth. III, 2. Schol.), especially since they cannot be traced back to any common measure. Spinoza anticipated in a brilliant way the triumphal progress of the mechanical conception of nature; he saw that principles had been set up in Galileo and Descartes under which the whole of material nature must be brought. It is therefore no wonder that the attention of many researchers, now that the law of the conservation of energy passes one empirical test after another, has also turned to this profound thinker. But from the nature of the case it is the experiential rather than the speculative side of his doctrine that is here the most important for us.³⁶

Spinoza's theory has really never disappeared from philosophy since it was once set up. Even before the renaissance that Spinozism celebrated at the end of the last century, Kant had come by his own path to the identity hypothesis. In certain ways Kant and Spinoza are antipodes; what was for Spinoza the clearest of all, the absolute being or the substance in all things, is according to Kant precisely what our knowledge can never reach, "the thing in itself." This contradiction, however, points to a unity lying deeper. One need only put X in Spinoza wherever he speaks of substance, and one has Kant's

[82] doctrine. Kant teaches, quite like Spinoza, that we have no right to regard spirit and matter as two different substances. The duality is only empirical or phenomenal and does not exclude a unity in essence.³⁷ Kant therefore excludes, just as Spinoza does, both spiritualism and materialism. "The ego, represented through the inner sense in time, and

³⁶Cf. moreover my book *Spinozas Liv og Lære* [Spinoza's Life and Teaching]. Kbhvn. 1877.

³⁷Kant hardly knew Spinoza's theory at first hand. At least he does not mention it at all. He reproaches the earlier theories with having regarded matter as a "thing in itself" instead of regarding it as phenomenon. This criticism does not strike Spinoza, for whom matter was not substance (fundamental being) but attribute (fundamental form). — For the rest, Kant himself declares, in spite of the subjectivistic turn he gives the matter, that "in all problems that can occur in the domain of experience we treat those phenomena [soul and body] as objects in and for themselves, without troubling ourselves about the first ground of their possibility" (*Kritik der reinen Vernunft*, Kehrbach's edition. p. 330). This is precisely the point of view I have held fast with regard to the relation between psychology and physiology on the one side, and theory of knowledge and metaphysics on the other. See above, 74 ff.

objects in space, outside me, are indeed specifically quite different phenomena, but they are not thereby thought of as different things. The transcendental object which underlies the outer phenomena, and likewise that which underlies the inner intuition, is neither matter nor a thinking being in itself, but a ground, unknown to us, of the phenomena that give us the empirical concept of the first as well as of the second kind.” “Thus the very thing that in one respect is called bodily would in another respect be at the same time a thinking being, whose thoughts we cannot indeed intuit in the phenomenon, but whose thoughts’ expression we can intuit. The expression that only souls (as particular kinds of substances) think would thereby fall away; it would instead be said, as in everyday speech, that men think, i.e. that the very thing which as outer phenomenon is extended is in its inner being (in itself) a subject, which is not composite, but is simple and thinks.”³⁸

For Kant the epistemological point of view stood in the foreground; therefore he does not lay the weight on the principle of physical continuity, although he acknowl- [83] edges it. In his successors, too, we meet the identity hypothesis. But in them the empirical-phenomenological way of looking at things was more and more displaced by the speculative; the task then was no longer to formulate the ultimate principles of empirical knowledge, but to deduce everything from an absolute principle. In *Schelling* and *Hegel* the identity hypothesis passes over into a spiritualism that breaks off the connection with the science of experience. Only in researchers who worked in Kant’s spirit and with sober regard for the given, such as *Fries*, and among us *Treschow* and *Sibbern*, do we find the hypothesis clearly set forth. It obtained a firmer foundation when the law of the conservation of energy received one empirical confirmation after another. So far as I know, it is *Fechner* who in our time first saw the consequences of that law for this problem.³⁹ Numerous psychologists and physiologists have now come over to the identity hypothesis, by different paths and with different degrees of clarity. It surely

³⁸Kritik der reinen Vernunft. Kehrbach’s edition p. 320 and 306. — It is in the first edition that Kant speaks most fully and most clearly about this question, in the section “on the Paralogisms of Pure Reason” (the critique of rational psychology).

³⁹In his psychophysical law for the relation between physical impression and sense-sensation, according to which an increase in the strength of the impression acts in proportion to the impression already present ($dy = d\beta/\beta$), Fechner also saw a law for the relation between soul and body, since he thought that a simple proportionality probably obtains between nerve process and physical impression. In the course of the discussion of Fechner’s psychophysical law it seems, however, to have become clear that *in so far as* Fechner’s law has validity (see on this below, V A, 4), it holds for the relation of the impression to the nerve process (Fechner’s “psychophysical motion”), whereas the relation itself between nerve process and consciousness (body and soul) must be assumed to be one of direct proportionality. As *Hering*, Fechner’s most important critic, has remarked (Ueber Fechners psychophysisches Gesetz. — Wiener Sitzungsberichte. 1875. — § 4–5), this also agrees best of all with Fechner’s own fundamental view. — Cf. further on this question: *J. Ward*: An Attempt to interpret Fechner’s Law. (Mind. 1876). *G. E. Müller*: Zur Grundlegung der Psychophysik. Berlin 1878. p. 224–374.

also gives the most natural settlement of the boundary dispute between physiology and [84] psychology.⁴⁰ Their object is one and the same, only they approach it from different sides. They therefore play into each other's hands and can make use of each other's results. The contradictions between them must rest on misunderstandings and errors. Every phenomenon of consciousness gives occasion both to a psychological and to a physiological inquiry. The psychological and the physiological point of view belong together just as closely, and are at the same time just as different, as the convex and the concave side of an arc of a circle. The identity hypothesis thus becomes fruitful for special psychological research, since it enforces the necessity of illuminating and correcting the psychological results by the facts of physiology.

⁴⁰Dr. Kromann has, in his work *Den exakte Videnskabs Lære om Sjælens Existens* [The Doctrine of Exact Science concerning the Existence of the Soul] (Kbhvn 1877), given a critique of the identity hypothesis (or, as he calls it, "duplicism"). But his objections do not hit the mark, chiefly for two reasons. In the first place he confuses the empirical and the metaphysical point of view, and in the second place he constantly imputes to the adherents of the identity hypothesis the view that the material is what is really real and the spiritual a powerless appendage to it. (See my review of this work in *Nordisk Tidsskrift* 1878).

Chapter 3

The Conscious and the Unconscious [85]

1. In the characterization of the life of consciousness given earlier, weight was laid on two main features: the occurrence of a change, whereby new elements of consciousness arose, and the connection among all the elements of consciousness. If this characterization is correct, consciousness can cease for two reasons: either because the individual elements do not possess strength enough to assert themselves, or because the connection among them ceases.

So long as we hold firmly to the fact that we know the soul only through the manifestations of consciousness, the domain of psychical life is of but small extent. It is not all nerve processes to which we have reason to think of consciousness as attached, and even the nerve processes with which this is the case can also take place without consciousness when their intensity is not great enough. A physical impression can thus act upon the nervous system without any sense-sensation arising; the sensation arises only at a certain definite strength of the impression. The nerve process, on the other hand, must already be going on at lower degrees of strength, so that when the sensation crosses the threshold of consciousness, the nerve process has already reached a certain strength. Let x , for example, denote the degree of strength of a nerve process which is just strong enough that a just noticeable sensation, which we will call y , corresponds to it. We then face the peculiar relation that while the degrees of strength on the physical side decrease continuously downward from x , the psychical side goes blank, stops suddenly at y . This is how the relation presents itself, whatever fundamental assumption regarding the relation between the psychical and the bodily we may start from. It is the same with [86] composition as with degree of strength; for there is only a difference of degree between the structure and mode of operation of the lower and of the higher nerve organs. Is it now reasonable that at a certain step of the scale something should arise which did not

exist at all on the lower steps? When the series is continuous within the one sphere, should it not also be so within the other? We have, after all, no right anywhere in nature to assume leaps or gaps; at least the progress of knowledge consists above all in the filling up and bridging of intervals and chasms.

The question we here face is whether the unconscious can become anything other than a purely negative concept. In everyday speech (and more than is proper in scientific usage as well) we use expressions such as unconscious sensations, unconscious ideas, unconscious feelings. But since sensations, ideas and feelings are states of consciousness, the expression is really meaningless. If by an unconscious idea is meant an idea that I have, then the predicate "unconscious" says only that I do not think of or attend to the fact that I have it. This use of the word unconsciousness is connected with a double use of the word consciousness. For this word is used not only to designate the inner appearance of our sensations, ideas and feelings, but also self-consciousness, the attention expressly directed toward our sensations, ideas or feelings. We naturally *have* many sensations and ideas of which we are not *conscious* that we *have* them; many feelings and impulses *stir* in us without our clearly *apprehending their nature and direction*. In this sense one can, e.g., speak of unconscious love; he who has this feeling does not know what it is that stirs within him; others perhaps see it, or he discovers it himself by degrees; but he has the feeling, his life of consciousness is determined in a peculiar way.

When, on the other hand, we here wish to investigate the relation between the conscious and the unconscious, we mean by unconsciousness a state which lies below the threshold of our consciousness altogether (not merely below that of our self-[87] consciousness). We wish to investigate whether the unconscious might not prove to be akin to the conscious, so that the difference, on the psychical as well as on the physical side, is merely a difference of degree. For the time being, however, we hold fast to its being a purely negative concept. It is not our intention to follow *Eduard v. Hartmann* along the mystical paths he believes he has cleared in his *Philosophie der Unbewussten*. Hartmann not only makes the unconscious without further ado into a positive concept, but uses it as a principle of explanation wherever he thinks the physical and psychological causal series gives out. — Psychology proceeds safely only when it keeps to the clear and certain phenomena and laws of consciousness. But from this standpoint it now discovers precisely the unconscious, and sees to its astonishment that psychological laws hold sway beyond the domain of the life of consciousness as well. In what follows we bring forward some examples to illustrate this.¹

¹In *Carpenter*: Mental physiology. p. 515 ff. a long series of examples is to be found. Some of those adduced in what follows are taken from his collection.

2. Repeatedly, memory has been pointed to as the typical phenomenon of consciousness. But memory presupposes that the elements of consciousness change. What, then, becomes of the vanished ideas until they are called up again? In everyday speech memory figures as a storehouse or treasury where the ideas are kept until they are to be used again. It is only in a figurative way that we can thus ascribe a real existence to the ideas that have vanished from consciousness. But the remarkable thing is that it looks as if they nevertheless play a part in the actual activity of consciousness. When we want to call something up in our memory and cannot hit upon it, it is a well-known remedy to leave off searching for it and think of quite other things; the idea sought may then suddenly emerge. Here we give up the conscious search and let an altogether unconscious process take its place. A similar case is when we have collected material for a piece of work, e.g. for the treatment of a scientific question. We are then often so overwhelmed by the particulars that we are unable to order and combine the material. Here too, then, occupation with quite other things can be a good remedy. In the midst of [88] the new pursuit the right way of treating the question may then suddenly, as if of itself, present itself to consciousness. The unconscious working has accomplished what the conscious, direct and strained labor perhaps would never have managed. Naturally one never comes to such results in one's sleep; the strenuous labor is the presupposition, the unconscious working puts the crown on the work.

Some psychologists explain such phenomena simply by the greater freshness of brain and mind when one has "slept on" the matter. But this explanation can fit only cases where one straightforwardly takes the matter up anew after the interruption, not those where the interrupted subject suddenly presents itself in full clarity to thought in the midst of occupation with other things. Here the subject itself has undergone further treatment since last time, but this treatment has taken place below the threshold of consciousness; it has taken place in us, not by us. And yet this unconscious labor bears the stamp of the same principles and laws that govern conscious labor.

3. What thus holds for many of our thoughts holds — as the physiology of the senses has shown — also for our apparently simple and immediate sensations and perceptions. We form a coherent visual image, although there is a point on the retina (the blind spot) which receives no impression. Many of our sensations of color, which we take to be immediate, are determined by the effect of contrast with other *colors*. The apprehension of space, which unfolds before us at a single stroke, has arisen through the combination of various impressions. Our two eyes do not have the same field of vision, and yet we believe we immediately survey the binocular field of vision with both. The direction of

the line of sight we judge (according to Helmholtz) by the effort of will with which we try to change the position of the eyes. But all these combinations are not conscious to us, and when it has been said that our sense-perceptions are the results of “unconscious inferences,” it must be well noted that this expression is admissible only because we have no other designations for these processes, known to us only from their results, than [89] such as are taken from the analogy with the higher life of consciousness.² It is a kind of inverted anthropomorphism that we here employ.

4. Not only can conscious results be formed from unconscious preliminary work, but in the midst of conscious work unconscious intermediate links may be found. When the idea *a* is associated with the idea *b*, and *b* in turn with *c*, *a* will at last be able to call up *c* directly without taking the way round by *b*. A proposition we have learned to see by way of proof goes on living in our consciousness after we have forgotten the proof. All education rests on intermediate links being able to sink below the threshold in this way. The authority of the educator is indispensable at first, but by degrees steps back. Often the intermediate links are so numerous or hidden that they are not found, or are found only with great difficulty. What leads us from the one thought to the other we do not always become conscious of. Many psychological paradoxes and sudden inspirations can find their explanation in this unconscious motivation of conscious ideas.

5. Everything we call instinct, tact, self-acquired or inherited disposition, works in this way. Habits and tendencies which we have ourselves acquired or fallen into, or which we have received as an inheritance from earlier generations, live on long after their causes have fallen away. The ideas, feelings and actions to which these tendencies lead do not find their full explanation in the life of consciousness itself. There are always intermediate links that are skipped, and which only physiological and sociological inquiry discovers. The conscious motives have fallen away, but their effects remain. Instinct has therefore been defined as a working toward ends of which one is not conscious. Conscious intervention is determined in part by unconscious motives and likewise leaves unconscious effects behind it. Both in the individual and in nations sudden upheavals avail but little; subterranean tendencies persist, which are subdued only late. That is [90] why the Israelites had to wander forty years in the desert. *Herodotus* relates (IV, 3–4) that while the Scythians were away on distant campaigns, the slaves had married their women and seized dominion; when the masters came back, they could not subdue by

²*Helmholtz*: *Physiologische Optik*. p. 811. — Later Helmholtz has avoided using this expression on account of the abuse Schopenhauer and Hartmann have made of it. He maintains, however, that we have here to do with an elementary process that lies at the basis of all thinking properly so called. »Die Thatsachen in der Wahrnehmung«. Berlin 1879. p. 27 f.

force of arms the young generation sprung from these marriages, but brought them to submission as soon as they cracked the whips with which the slaves used to be punished. This story can at any rate serve as a poetic expression of the power of inherited habits. — In the lives of eminent and pioneering men we often see how they must struggle to overcome what childhood impressions and practice have implanted.

Every spiritual upheaval at first shakes only what moves in clear consciousness; the unconscious undercurrents can often continue their course for a long time without the movement propagating itself from the surface down to them. The reaction after a revolution often reveals how little deep the movement has gone. What has been consciously won by struggle first becomes rooted when it works unconsciously, or, as one says, passes into flesh and blood.³ Conscious labor breaks the path; but after that the task is to get the unconscious machinery going. Conversely, one can practice something purely mechanically which then by degrees gains power over the life of consciousness. Forced conversion can lead to living faith, if not in the same generation then in later generations. Compulsion works against what moves in clear consciousness; forced conversion therefore succeeds only where there is no clear and developed life of consciousness present.⁴ But mechanical practice can by degrees weaken consciousness. We are, in *Pascal's* expression, automaton as well as spirit, for which reason Pascal [91] advises beginning by taking holy water and going through the customs, and the rest will surely come. That is his notorious principle: *Il faut s'abêtir!*

6. Simultaneously with a conscious activity an unconscious one can go on. The spinner turns the wheel and draws out the thread while all her thoughts are far away. He who reads aloud can be wholly absorbed in the content, or perhaps even in other thoughts, while he sees the letters and pronounces the corresponding words. In these examples the subordinate action at any rate approaches the unconscious, and there is hardly any question that the boundary can be crossed. And yet what has thus taken place unconsciously can later make itself felt in consciousness. *Fechner* relates (*Elemente der Psychophysik*. II. p. 432) that one morning in bed he was surprised that a white image of the stovepipe stood before him when he closed his eyes. While he lay speculating

³In H. Bröchner's work »Om Udviklingsgangen i Filosofiens Historie« [On the Course of Development in the History of Philosophy] (Kbhvn 1869) one will find many interesting demonstrations of how ideas can unconsciously influence the successors, and even the opponents, of the age that fostered them. — See also his essay in »Nyt dansk Maanedsskrift« [New Danish Monthly] (1871) on the relation between the conscious and the unconscious.

⁴Children's impulse to imitate is so strong that they take pleasure in joining in movements and ceremonies they do not understand. *Egger* (*Observations et réflexions sur le développement de l'intelligence et du langage chez les enfants*. Paris 1879) relates: »Un enfant élevé dans une maison religieuse donnait un jour cette définition de la *méditation*: On va à la chapelle, on se tient comme cela, sans bouger, et on ne pense à rien«.

with open eyes, he had, without being conscious of it, looked at a black stovepipe with a white wall as background, and it was the negative after-image of this that now showed itself. The physical impression had thus occurred in such a way that the visual sensation *could* arise; it had remained unconscious so long as attention was turned away, but could nevertheless still come forward into consciousness without further ado. Likewise, when we listen absent-mindedly to someone speaking to us, we may become conscious of what he has said only long afterward. It is then only the express turning of the mind that raises the unconsciously received impressions above the threshold. That we can remember something is therefore no decisive proof that we really apprehended it with consciousness at the time. By its connection with what was consciously apprehended, an unconscious impression too can be called up again in memory.⁵ To this belongs also [92] the well-known fact that warriors in the heat of battle do not notice even very serious wounds, but only later are reminded by the pain of how things have gone with them. This was already known in antiquity. "It is related," says Lucretius (*De rerum natura* III, 641 ff.), "that scythe-bearing chariots cut off limbs so swiftly that one sees the severed part writhe on the ground before the consciousness, or the man himself, has time to feel the pain."

7. In the development of the feelings unconscious impressions play an especially large part. This development is determined not only by clear and definite sensations and ideas, but also by imperceptible influences whose sum first makes itself felt in consciousness. Hence the mysterious and unexplained in the nature of so many feelings; in their first stirrings especially they are incomprehensible to the individual himself, since he does not know the definite causes. The vital feeling is due to the action of the organic functions on the brain; but the individual impressions do not stand out clearly here, they merely unite in producing a dim and shifting fundamental mood of well-being or ill-being. The feeling of love, especially in its first dawning, has a mysterious character through the awakening of uncomprehended organic instincts and their influence on the vital feeling and on fantasy. But in other feelings too there is something of the same, inasmuch as we never become fully conscious of the influence of our experiences and circumstances of life on our life of feeling until the feeling takes on a pronounced character, or perhaps even breaks out in action. It is with such influences as with the air

⁵Koch: Vom Bewusstsein in Zuständen sogenannter Bewusstlosigkeit. Stuttgart 1877. p. 19: "Memory can be present after preceding real unconsciousness when, e.g., of a series of processes that have not come to consciousness, the last link reaches into the conscious state, and now, starting from this, the earlier links that ran their course unconsciously are also raised into consciousness by reproduction; or when at some later time, during the conscious state, a brain state sets in which has a certain kinship with the one that released processes in the unconscious state, so that now, starting from it, the same processes can be reproduced as conscious, as memory."

we breathe in without thinking of it. A quiet growth thereby goes on in us, often the most important and most decisive thing in our spiritual life. We are here referred back to the general presupposition of the life of consciousness and of the nerve process, that only a more or less sudden change rouses them to activity. A slowly rising change in heat or electricity can cause the death of a frog without its stirring from the spot.

The inner coherence in the history of the single human being and of the whole race [93] is preserved by these unconscious increments, whose results determine so large a part of the content and energy of the life of consciousness. It is only when we keep to the pronounced states of consciousness that there seem to be sharp boundaries and sudden revolutions; in the depths the infinitely ramified transitions are discovered. Thus the coral animals always build beneath the surface of the sea, and their building is first discovered when it rises above the level of the sea.

It is Leibnitz who first, in a brilliant way, drew attention to the significance of infinitely small elements in psychology (just as in mathematics and in physics). He has moreover brought this doctrine into connection with the law of continuity, which he so energetically maintained. By means of the unconscious impressions (what he called “*petites perceptions*”) he explains the connection between past, present and future: “*En conséquence de ces petites perceptions le présent est plein de l’avenir et chargé du passé.*”⁶

8. In the state of dreaming we have an intermediate stage between the purely unconscious and the conscious state. The analogy that proves to hold between dream consciousness and waking consciousness can to that extent throw a light on the relation between the conscious and the unconscious. Here too we see the same laws prevail, only farther down the scale.

We are fond of setting sleep and waking over against each other as sharp opposites. But just as within the waking state there are countless degrees of energy, clearness and coherence in the life of consciousness, so there are also many nuances of transition from waking to sleep, and conversely, and within sleep, too, there are various degrees. The following descending steps on the scale of psychical energy have thus been distinguished: 1) reverie with half-slumber, 2) the light morning sleep with lively dreams, 3) deep sleep with dim, indefinite dream images, 4) the deepest sleep without dreams [?] and 5) in illness a still lower degree, the soporous state or fever stupor.⁷ The deepest sleep we have [94] just after having fallen asleep. Whether we always dream is a question that has been

⁶*Opera philosophica*. Ed. Erdmann. p. 197.

⁷P. *Hedenius*: *Om drömmen* [On Dreams]. (Vetenskap för alla [Science for All]. Bd. III. Stockholm 1880). p. 609–611.

answered differently. Those who assume it appeal partly to metaphysical grounds, namely that the soul by its nature cannot refrain from acting, partly to physiological grounds, namely that movements must continually be going on in the brain and impressions continually be received. In any case it is clear that the dream stands a step nearer to waking life than dreamless sleep, if there is such a thing.

The connection of dream consciousness with waking consciousness shows itself, in the first place, in the fact that what is the object of waking interest often occupies dream consciousness; difficulties and vexations that were insurmountable in the waking state one overcomes, or imagines one overcomes, in dreams. The elements out of which the dream world is built up are, next, for the most part taken from the experiences of waking life, although they are placed in new, often fantastic combinations. Yet impressions are constantly received in sleep, not only from the interior of the organism (from the respiratory and digestive organs) but also from outside (impressions of touch, sound, light, etc.). The connection with the surrounding world is thus not wholly broken off. These impressions are now woven together with the after-effects of waking life into a new world of ideas. But the formation of this world takes place under other conditions than in the waking state. There is lacking the firm concentration of attention and the many-sided control which waking life calls forth or enforces. The individual impressions, especially the vital sensations, thereby gain a power that pushes unity and continuity aside. A free and bold interpretation of each individual impression is the consequence. The dream takes on what has aptly been called a mythological character. If the breathing is particularly light and free, one believes one is flying; if it is labored, one is ridden by the nightmare. If one comes to be cold through being uncovered, one is on a North Pole expedition or walking naked in the street. A man who had a hot-water bottle under his feet dreamed that he was walking on the crater of Etna.

In the waking state too we interpret the individual impressions according to their relation to our other experiences. Dream consciousness follows the same method, but [95] cannot get the better of the individual impressions; each sets its own stream of thought in motion, which then for a moment dominates the whole of consciousness, until it is replaced by the one roused by the next impression. Here there is not power of resistance enough against the individual elements. Hence the shifting and arbitrary character of the dream, by which it comes to resemble insanity, which is indeed also a state of dissolution. The shifting of the dream images is, as has rightly been remarked⁸, not “transformation”; it is to be understood as the emerging of one image after another.

⁸*Delboeuf* in *Revue philosophique*. Octobre 1879. p. 341.

The state of dreaming, then, shows us the psychological laws at work, but below the threshold of consciousness proper. It is a station on the way from the conscious to the unconscious life.

9. The act of waking up sometimes presents circumstances that can throw a light on the relation between the conscious and the unconscious. When one is woken, it is not always the physical strength of the impression that tips the scale, but its relation to the weal and woe of the individual, his interests in the waking state, *what* the physiologist *Burdach*⁹ has called the psychical relation of the impression. An indifferent word does not wake one when it is spoken softly; on the other hand the mother wakes at the child's softest movement. A very miserly man was woken by having a coin put in his hand. A naval officer who slept despite loud noise was woken on hearing the word "Signal!" whispered. From such cases we are not entitled to conclude with *Burdach* that "the soul in sleep distinguishes among the sense-perceptions"; they point, rather, to the fact that an individual impression comes to consciousness only through its connection with other experiences. It releases a whole series of effects in the brain, and with that consciousness is given. The act of waking, which is a transition from an at least relatively unconscious state to a conscious one, takes place through the individual impression's acquiring, by combination with others, the background it needs in order to become conscious. It agrees with this that the life of consciousness proves to be bound to very composite nerve organs, where many currents can meet.

This circumstance could perhaps throw some light on a psychological paradox mentioned earlier, namely that an altogether single and isolated sensation cannot be conscious, and on the consequent difficulty of imagining the beginning of consciousness. One could think of this beginning as conditioned by a single impression's calling up several elements of consciousness at once. Just as it seems that one can remember what nonetheless happened unconsciously, so an impression would be able to rouse consciousness by releasing foreground and background at one stroke. It is not thereby to be said, however, that the origin of consciousness can be thought of as a single, instantaneous act. [96]

10. Despite the intimate connection and interaction between the conscious and the unconscious, the unconscious nonetheless remains for us a negative concept. The unconscious processes are brain processes just as the conscious ones are; but whether they, like these, are something more, we do not know. Instead of unconscious thinking or unconscious feeling it would — if one wishes to avoid all hypotheses — be safest to speak, with *Carpenter* and *Stuart Mill*, of unconscious brain activity (unconscious

⁹Physiologie III. p. 460. — *Carpenter* also adduces a great many examples (§ 479–480).

cerebration), were this expression not unfortunate in that, in the first place, it could give rise to the misunderstanding that we ever become conscious of the brain activity itself, and, in the second place, could seem to settle that unconscious activity has nothing at all akin to what we know from ourselves as conscious states. Precisely the impossibility of drawing a sharp boundary line between the conscious and the unconscious, as well as the thoroughgoing analogy in mode of operation and results, might perhaps make it a justified hypothesis to maintain the law of continuity within the ideal world, just as everything in the material world seems to obey it. To all the various material phenomena of motion there would then correspond various degrees and forms of the same thing that appears in us as thought and feeling. Just as the organic world is built up of elements and by means of activities which also appear in inorganic nature, but more scattered and without unity and harmony, so we would have in the sensations, feelings and thoughts of conscious beings higher forms of development of something which, in a [97] lower degree and form, occurred on the lower steps of nature. We would avoid the paradox that the life of consciousness should begin wholly without preparation. Leibnitz drew precisely this conclusion from the law of continuity: *Rien ne saurait naître tout d'un coup, la pensée non plus que le mouvement*.¹⁰ He set up an analogy between the relation of living force to tension (potential energy) and the relation of the conscious to the unconscious. Just as tension (potential energy) is living force that has come into equilibrium, so unconsciousness could be resting or neutralized consciousness. It would agree well with this that change, or the disturbance of equilibrium, is so essential a condition of consciousness. Just as there is no absolute rest in the external universe, so — one could then say — there is no absolute unconsciousness either. Unconsciousness would then be not a negation of consciousness but a lower degree of it, lying in the continuation of the descending series of degrees of consciousness.¹¹

Empirically, the life of consciousness appears bound to certain forms of the functioning of the nervous system. But the nervous system itself has arisen by differentiation out

¹⁰Op phil. ed. Erdmann. p. 226. — Cf. already *Spinoza* Eth. II, 13. Schol. — Among more recent psychologists *Beneke* must especially be singled out, who in the first volume of his »Psychologische Skizzen« (Göttingen 1827) has investigated the relation between the conscious and the unconscious in a clear and penetrating way.

¹¹*Zöllner*, in his brilliant book »Ueber die Natur der Kometen« (1872), has taken up the thought of *Spinoza* and *Leibnitz*. One will also find it in *Häckel*, *Nägeli* and other adherents of modern "monism." — *Virchow*, who polemicizes against this hypothesis in his address »Die Freiheit der Wissenschaft« (Berlin 1877), nonetheless makes the highly significant concession that *if* one is determined to bring the spiritual processes into connection with what goes on in the rest of the world, one necessarily comes to extend the domain of psychical phenomena far down below the domain of human consciousness — to the lowest animals, to plants, cells and atoms (p. 27). — Monistic speculation cannot be said to be unjustified if it really is the only logically possible way in which connection can be established between spiritual phenomena and the rest of the world.

of homogeneous protoplasm; the properties of the nervous system must then likewise be higher degrees of something already attached to organic matter in general. (Cf. above p. 43–44). There is nothing here that justifies wholly new points of view at a certain [98] given step of material development. The nervous system is only, so to speak, the highest flower of material existence; what distinguishes it from other forms of material existence is only its higher degree of development. For this reason recent physiologists (thus *Claude Bernard*) trace sensibility back to the irritability of organic matter (the capacity to receive and respond to impressions). Conscious reaction is thus only a higher form of unconscious reaction. Bernard draws a proof of this view from the action of numbing substances (*anaesthetica*). When such substances (opium, chloroform, etc.) abolish consciousness, it is because they act first and foremost on the part of the organism most susceptible to impressions, and this is the nervous system, the most differentiated organic matter. But with stronger doses or more prolonged action the rest of the vital activity is by degrees attacked as well. What is thus affected in the same way, only in different degrees, by one and the same thing can also differ only in degree. We thus ascend gradually from the lowest manifestations of life to the highest vital activity, to which consciousness is attached.¹²

11. One more important consideration must be made here. In the foregoing, weight was laid on the law of the conservation of physical energy as the law which expresses a magnificent connection in the world of matter, but prevents the assumption of a causal relation between matter and consciousness. But the law of the conservation of energy is only the special, precise form the general law of causality takes in the physical domain. The law of causality is satisfied when physical causes have physical effects. Now consciousness nevertheless appears as a surplus, which at certain points is added to these physical effects, as an extra which cannot be explained from the physical causes. The famous physiologist *Dubois Reymond* has drawn from this, in his essay »Ueber die [99] Grenzen des Naturerkennens« (1872), the conclusion that spiritual phenomena stand outside the law of causality and signify a breach of the principle of sufficient reason. After all that has gone before we must grant him this, so long as one takes up a purely *deductive* standpoint and takes the premises of the deduction from the fundamental laws of physical mechanics. But we showed in the first chapter that the independence in principle of psychology in relation to natural science (in the narrower sense) is marked by its building

¹²*Bernard*: *Leçons sur les phénomènes de la vie*. p. 280–290. — See also his essay »La sensibilité dans le règne animal et dans le règne végétal« (in the collection of Bernard's essays published under the title »La science expérimentale«) as well as his »Rapport sur les progrès et la marche de la physiologie générale en France.« Paris 1867. p. 180.

not merely on consequences of physics and physiology, but drawing on a new source of knowledge which is not directly used by natural science, namely inner subjective observation. If, then, experience is not exhausted with physical experience, we see the necessity that one — after having drawn the last consequences of the fundamental laws of physical experience and seen that they do not lead us to consciousness — undertake a new *inductive investigation*, take up a *new empirical starting point*. Two tasks here present themselves: the one, to find a lawfulness within the world of psychological experience itself; the other, to find the relation in which psychological experiences must be placed to physical ones. The first task we shall occupy ourselves with in the following chapters; the second task we have treated in the preceding chapter. It proved impossible there to insert the life of consciousness into intervals or gaps in external nature, and we were led to conceive it as another form of manifestation of the same principle that works in the material world. Herewith Dubois Reymond's paradox falls away. For now there is the possibility of a hypothetical assumption of an equally thoroughgoing connection in the one domain as in the other. It is then only a semblance when consciousness seems to arise from nothing, just as it is a semblance that anything in external nature arises from nothing. The apparent arising of consciousness is then only a transition from one ideal form to another, just as every new material motion comes into being by conversion from another form of motion.

Such a hypothesis must be taken for what it is. The unconscious is a limiting concept in science; and where we stand at such a limit it may have its significance to try, by way [100] of hypothesis, to survey the possibilities that present themselves as consequences of our knowledge; but a real extension of our factual knowledge is here impossible. We proceed here like the philologist who by conjectural criticism supplements a fragment of an ancient author. The spiritual world stands — compared with the physical world — before us as a fragment, and only by way of hypothesis is there any possibility of supplementing it. Such a supplementation we have here indicated from a purely psychological point of view. Ethical motives *might perhaps* be able to lead to further-reaching speculations, which would show us the *law* of the conservation of energy as the means and *bearer* of the conservation and unfolding of *spiritual* values; but this would take us beyond the domain we have here staked out for ourselves.

12. This fragmentary character in the position of psychological phenomena within our empirical world-view entails that psychology will never be able to become an exact science, as physics has become and physiology is in the course of becoming. This difference between physics and psychology Lotze has clearly and aptly developed in

the following statement (Mikrokosmos. I. 2 Ausg. p. 218 f): “The difficulties of inner observation are increased by the fact that we cannot lean on a general doctrine, certain in and of itself, of the necessary interactions in which the states of every being must stand to one another. Most of the fundamental features we notice in the relations of spiritual life we can regard only as factual arrangements, whose value for the higher development we frequently comprehend perfectly, without, however, being able to show that precisely these forms of operation had to be the necessary consequences in every spiritual being susceptible to an indeterminate plurality of external impressions. One easily sees how unfavorable this order of things is to what belongs to an explanation. When we are referred to a collection of empirical facts, we may not go beyond what experience itself tells us; could we, on the other hand, follow the same facts in their necessary issuing from the nature of the soul, we would easily be able to give them a stricter and deeper expression, which would open to us access to a multitude of inferences now inadmissible. These difficulties one is inclined to rate lower than they deserve; spoiled by the results of natural science, one is often accustomed to regard propositions [101] that possess undisputed validity in the explanation of physical processes as general and necessary truths, and forgets on that account that all unprejudiced observation of the inner life presents us with quite other forms of becoming and acting, forms that can scarcely be compared with natural phenomena. With regard to the motion of matter we possess a sum of scientifically exact laws, with regard to the manifestations of the soul a number of empirical views; but there is still lacking the third and higher thing we should need: a general doctrine that could show the laws by which the states of beings in general are governed, and from which the science of the spiritual life could issue as two different applications.”

The inner imperfection of psychology is connected, as is here rightly said, with our inability to specify a principle to which both our psychological and our physical knowledge could be traced back. Philosophical speculation has sought such a principle, from which both the world of spirit and that of matter should be derivable. Such a principle *Spinoza* found in “Substance,” the infinite ground-being, *Schelling* in “the absolute identity,” *Hegel* in “the absolute Idea.” But it has always turned out that every such principle was only apparently of a positive nature, and that the deduction rested on unjustified presuppositions. (Cf. above p. 81). Starting from experience, we are indeed led to formulate such a goal, but see at the same time that it cannot be reached.

But although we cannot reach an exact deductive psychology, let alone a higher knowledge from which psychology as well as physics could be derived, we can neverthe-

less by way of induction demonstrate significant laws which in fact hold both for the life of consciousness and for external nature. It is *Herbert Spencer's* merit to have been the first to carry through this point of view.¹³ He has shown that all the phenomena we know, in the domain of inner as well as of outer experience, are subject to development, and has attempted to set up common laws for all development. In all domains [102] development consists in the transition from an incoherent, indefinite and homogeneous state to coherence, definiteness and heterogeneity.¹⁴ By these common laws and forms psychical life shows itself, from a new side, closely bound up with the general life of the world; its paths are interwoven with it. Here we will merely point out that progressive individualization can be named as the common feature of development in all its forms. Everywhere in nature smaller wholes form themselves within the great, infinite whole, each with its own peculiar interplay with the surrounding world.¹⁵ An individual is a being so separated from and independent of its surroundings that it can react upon them with a certain peculiarity of its own. But fully pronounced, as already touched on earlier, we find individuality only in the domain of the life of consciousness, where inner centers of suffering and acting are given. This world-law thus receives its most striking expression in the spiritual domain — as a kind of compensation for the fact that the more elementary law of the conservation of energy cannot be demonstrated there. If these two laws could be brought into inner harmony or reduced to a more deeply lying principle, all riddles would be solved.¹⁶

¹³First Principles. (First edition appeared 1861). Principles of Biology (1864). Principles of Psychology. (First edition 1855).

¹⁴See »Den engelske Filosofi i vor Tid« [English Philosophy in Our Time]. p. 137–139. — Among us, *Sibbern* has laid particular weight on the fact that all development takes place sporadically from various starting points and only by degrees leads to coherence, and he has demonstrated this in a very fine way as regards psychical life in an essay in »Filosofisk Arkiv« [Philosophical Archive] (1830). p. 263.

¹⁵See my work »Om Grundlaget for den humane Etik« [On the Foundation of Humane Ethics]. (Kbhvn. 1876). p. 110–112.

¹⁶On the difficulty that lies in the idea of an analogue to the conservation of force in the spiritual domain, see above p. 78–79.

Chapter 4

The Classification of Psychological Phenomena

[103]

1. The preceding sections have been occupied with the relation of psychological phenomena to other domains. Before we enter on an investigation of special psychological relations, it will be useful to cast a glance at the life of consciousness in its whole extent and at the main differences that make themselves felt there. We thereby expose ourselves less to losing sight of the whole for the particulars. Here, to be sure, as with the psychological developments and results used in the preceding sections, we must take much for granted and conceded which only the special investigation can establish. But we gain the advantage that the exposition can proceed from outside inward and from the main features to the particulars. Add to this that the special psychological investigations presuppose an abstraction from the great, living connection in which every psychical fact arises and subsists; all the more necessary is it, then, first to establish the totality out of which psychological abstraction singles out particular elements.

It has not always been clearly recognized that psychological distinctions and concepts have an abstract character. Reflection early discovered various elements in the conscious states, but was inclined to fix them as independent divisions or faculties of the soul, lying outside one another. Thus already in *Plato* (in the fourth book of the *Republic*) we find a distinction among various “parts” of the soul, built on an acute analysis of those cases where inwardly struggling oppositions show themselves in the mind (1) reason, 2) feelings of courage and indignation, 3) sensual impulse). In modern times there [104] has been talk, in an equally external way, of various “faculties,” which could act in dependence on one another and in opposition to one another. Besides the split thus introduced into psychical life, between various parts or faculties — a split refuted by the

thoroughgoing unity of the life of consciousness, without which precisely the strong oppositions could not be felt and apprehended — people were also caught in the illusion that by tracing things back to various principles an explanation had been reached — that, e.g., knowledge and feeling each became easier to understand when one assumed a special faculty of knowledge and a special faculty of feeling. It was an illusion similar to deriving life from a vital force, and one which Molière mocks *at*, when in the epilogue to *le malade imaginaire* he has the doctoral candidate explain the soporific power of opium by ascribing to it a *virtus dormitiva*. What was overlooked — as it is still popularly overlooked — was the purely empirical character of such distinctions. They state only that certain differences obtain among certain states of consciousness. But from this there follows no warrant for assigning them each to its own rubric. It must first be investigated whether the same elements are not found in all actually given states of consciousness, so that the differences are grounded in the preponderance of certain elements and the subordinate importance of others. It is thus not the phenomena of consciousness or the states of consciousness themselves that are grouped and classified; it is the elements we find in them on closer consideration.¹ When we therefore set knowledge and feeling in opposition to each other, this can mean only states with predominant elements of idea over against states with predominant elements of feeling. It will appear that this view is [105] the only tenable one, since no state can be pointed to that is altogether pure ideating or feeling or willing. The question of classification, then, properly concerns whether we are right to assume different kinds of psychological elements.

2. The now usual psychological classification is the tripartition into knowledge, feeling and will. After the bipartition into knowledge — will had been followed since *Aristotle*, the German psychologists *Sulzer* and *Tetens* in the last century took up feeling as an intermediate member. It was especially Rousseau's insistent assertion of the right and significance of the life of feeling that here gained influence on psychological classification. Kant's use of this tripartition procured it general acceptance. The attempts made since Kant's time to trace all manifestations of consciousness back to thought or idea (*Hegel*, *Herbart*) have not succeeded, and besides, they really recognize the threefoldness as given, but seek to trace it back to a unity in the principle. The older psychologists conceived feeling either as obscure thought or as impulse and will. It was natural that

¹Among older psychologists this has been impressed with great clarity by *Fries*: *Psychische Anthropologie*. Leipzig 1820–21. II p. XXVIII: »Die Begriffe von Geistesvermögen klassificiren *allgemeine Beschaffenheiten* von Lebensäusserrungen aber nicht *die einzelnen wirklichen Lebensäusserrungen selbst*.« — When we use the expression "elements" instead of "sides" or "properties," this must not be misunderstood as if the states of consciousness could really be resolved into parts. It is only by abstraction that we distinguish such elements.

attention should first be directed to knowledge and will and should overlook the more deeply lying elements. This is analogous to the fact that attention is directed outward toward the surrounding world before it is directed toward the inner. For knowledge and will designate the outward-turned sides of the life of consciousness. In knowledge (in which in psychology we include sensations, ideas and *thoughts*) an image is formed of the surrounding world and of the individual himself as a part of the world. In will (to which impulse, desire and resolution are reckoned) the individual in turn reacts upon the surrounding world. The elements of feeling, the inner rhythm of pleasure and displeasure, are always so intimately bound up with certain images and thoughts or with certain actions that they are easily confused with them. The element of feeling is, moreover, so difficult to characterize in and of itself. No real element of consciousness can be defined, since it is precisely the expression for the last thing to which investigation leads us. But a simple sense-sensation, e.g. a color or a sound, can nevertheless be pointed out more directly; the feelings of pleasure and pain lie deeper and cannot be elements in immediate sense-perception. One could perhaps define feeling as *that in our* inner states which can never become an element in a perception or an image. It is an inner illumination that falls on the stream of sensations and ideas. On the other hand, not all [106] pleasure and displeasure issues in action. We have given examples in the first section of how difficult it is to infer inner feelings from outward behavior. Just as we express spiritual relations by designations taken from the material world, so we express our feelings by means of the thoughts and actions to which they are attached.

The independence of the elements of feeling over against the other elements of consciousness shows itself in this: even if there should be no state that could be said to be mere feeling without knowledge and will, feeling is nevertheless not necessarily bound to any definite theoretical or practical relation. In different individuals, and in the same individual at different times, pleasure and displeasure are found connected with different objects. Something that at first arouses displeasure can later arouse pleasure, and conversely. There appears, moreover, a clear opposition between states where thought or action lays so strong a claim on consciousness that the wave of feeling is hardly noticed, and states where violent emotional commotion pushes back clear and coherent thinking and deliberate acting. Physiologically the last-named states are clearly marked by the propagation of the movement from the central nervous system to internal organs and by reactions from these on the brain.

3. But although it will thus be justified to take the tripartition as the basis of psychological investigations, it does not follow that it must be regarded as original. In our

characterization of the life of consciousness we take it as it shows itself at a higher stage of development, where it has attained a certain pronounced form. We have no right to suppose that the threefoldness of the elements should stand out equally clearly at lower stages of development as well. On the contrary, it is indeed one of the general laws of development that the indefinite and homogeneous precedes the definitely and many-sidedly pronounced. The first germ of the organism is a homogeneous mass in which no definite structure can yet be distinguished. If the life of consciousness follows the general laws of life and of development, we must expect that the three different kinds of elements do not stand out as distinctly at the lowest stages as later.

[107] We are thereby brought to adopt another point of view than before. Instead of a classification in breadth, according to the dissimilar but simultaneously active elements, we now get a classification in height, according to stages, one of which develops out of the other. Such a point of view, too, was adopted early in psychology, and it is again *Aristotle*, the founder of empirical psychology, who must be named first. *Plato* did distinguish between higher and lower forms of psychical life; but with him it was chiefly ethical motives that lay at the basis, and he denied that the higher forms of soul developed out of the lower, since these arose only through the spiritual being's being implanted in a mortal body. *Aristotle*, on the contrary, seeks, with acute use of the material at his disposal, to show how the one form of psychical manifestations of life forms the basis for the other. This view has ever since been laid at the basis of psychology and has received a new foundation in our day through the hypothesis of evolution and the challenge it poses to find the connection between the stages of development of psychical life not only in the single individual, but in the race and in the succession of generations.

4. Great weight has often been laid on learning of what kind the first manifestation of consciousness is, since it was thought that one could infer from this which side of the life of consciousness was the most deeply lying and the most important. Here it was assumed, in the first place, that the rank of psychical manifestations of life (like that of noble families) is determined by age, whereas experience shows that something can have the highest ideal value and yet be the fruit of a long course of development from an insignificant beginning. In the second place it was assumed that the first psychical manifestation of life must be assignable either to knowledge, feeling or will. But this is, as already mentioned, improbable in view of the general laws of development, and it is contradicted by experience.

The life of consciousness no doubt begins already in the fetal state (see I, 4). By all that can be inferred, various more or less definite sense-sensations (thus elements of

knowledge) must already make themselves felt there. There are likewise surely fluctuations of pleasure and displeasure, according to the course of the vegetative functions. In the often very lively movements the fetus makes, one can see an expression of a [108] primitive will, whether one wishes (with Bain) to explain them by an unconscious urge to discharge the accumulated tension, or already here to speak of a real impulse to movement. For even if these movements at first take place purely unconsciously, they will soon bring consciousness with them through the sensations they give rise to. We have here, then, elements of all three kinds; but we have them in close connection. The transition from impression to reaction is here immediate; even if the movements of the fetus are more than mere reflex movements, they still do not rise much above the level of instinct. Instinct differs from mere reflex movement by containing an obscure urge of feeling, thus a kind of consciousness, though not consciousness of the actual aim of the action; it differs from the involuntary discharge of energy assumed by *Bain* by being directed toward a definite aim useful to the individual or the species. Instinct need not unfold at a single stroke; it does not exclude the necessity of certain elementary experiences; but these are made easily and naturally by means of the original organization. The movement in which well-being or ill-being issues must naturally go by the paths that, in accordance with the organization of the individual, are most passable. This original organization is a given starting point where the unconscious and the conscious, the inheritance from the race and the individual's own experience and activity, hang together. — The newborn individual, then, is not merely in possession of sensory and motor apparatus, but has already begun to practice them in the mother's womb.² But a definite distinction between the various elements cannot yet be made at this stage. The sense-sensations fuse immediately with the feelings of pleasure and displeasure, which just as immediately break out in movement.

The experiences from earliest infancy correspond to what can be inferred about the life of consciousness of the fetus. It is the predominant importance of vegetative life that prevents differentiation from advancing rapidly. The immediate, instantaneous [109] transition from impression to movement is characteristic of the initial stage of the life of consciousness; only by degrees is an interval formed in which inner differences and oppositions can make themselves felt. "The absolute strength of the metabolic exchanges in the brain substance is considerably greater in childhood than in the adult. The greater water content and decidedly softer constitution of the brain substance likewise favor

²*Kussmaul*: Untersuchungen über das Seelenleben des, neugeborenen Menschen. p. 35. — Cf. also *Cabanis*: Rapport du physique et du moral. éd. Peisse. Paris 1843. p. 114 ff. — *Burdach*: Physiologie als Erfahrungswissenschaft. 2ter Bd. Leipzig 1828. p. 693 f.

metabolism, whose strong preponderance may in part explain the greater irritability of the child's nervous system."³ Where adults merely tremble, small children get convulsions.⁴ The wholly involuntary and instantaneous character of the transition from impression to movement shows itself generally in the slight importance the cerebrum has in the first period of life. Removal of it or disease in it does not have the same consequence for newborn children as for older individuals.⁵ When one remembers the connection between the functions of the cerebrum and the higher activities of consciousness, one will see the psychological significance of these facts.

Just as animal life here slowly develops out of vegetative life, so it also dissolves back into it in the enfeeblement of old age, at the approach of death and in advanced mental illness. The developed, diversified life of consciousness gives way again to instinctive and reflex movements, the mediate transition from impression to movement gives way again to the immediate. Language, e.g., frequently dissolves again in such states into mere exclamations of feeling or involuntary repetition of words heard by chance. Animal life dies before vegetative life, the brain before the heart, which is *ultimum moriens*, just as it is *primum vivens*. Instinct stands on the boundary between the two domains. It is characteristic that taste is one of the sense-sensations that lasts longest, just as it is one of those that are first to appear.⁶

[110] The life of consciousness thus describes an arc from the fetal state to death. Both end points of this arc are relatively simple, little differentiated and articulated states. It is only in the middle and at the summit that ideas, feelings and manifestations of will make themselves felt in their specific peculiarity.

5. What thus holds for the development of the single individual holds also for the race. A definite difference between knowledge, feeling and will presupposes a stage of culture where constant and instantaneous reaction upon the surrounding world is not demanded. Directly or indirectly the whole life of consciousness is determined by the individual's position in the universe and its need to orient itself and to adapt its surroundings to itself, or itself to its surroundings. Even in the thoughts and feelings apparently most independent of practical considerations such practical motives can be traced. If it is not the physical world that motivates the life of thought and feeling, then it is the social world; if it is not the given physical or social world of experience, then it is an ideal world built up of elements from it, of which man in his inner states

³Vierordt: Physiologie des Kindesalters. p. 133.

⁴Darwin: The Expression of the Emotions. London 1872. P, 67.

⁵Vierordt p. 137.

⁶Cf. F. Papillon: Physiologie de la mort. (In his book »La Nature et la Vie«. Paris 1874.) p. 379.

feels himself to be a member. But every independent development of the life of thought and feeling presupposes that the elementary practical demands of life are satisfied. Art and science, the forms of the life of thought and feeling emancipated from immediate practical motives, do not unfold in a war of all against all. Where life is an immediate struggle for existence, thought is not isolated from feeling, nor feeling from will. It is the dangers that threaten, or the goods that are hoped for, that fill consciousness and immediately set the will in motion. The content of thought is what the impulse (Schopenhauer's "will to live") demands, and feeling is one with the desire to attain it. The more rapidly and instantaneously the transition from influence to reaction takes place, the nearer we approach to instinct and reflex action.

6. The formation of an interval between influence and reaction presupposes sufficient energy, sufficient organization and sufficient time. — A certain resistance must be offered to the influence; its immediate effect must be checked in order that further-reaching inner activities can be called forth and unfold. And these inner processes themselves [111] lay claim to an energy that could otherwise have been used immediately to react in response to the impression. If we start from the assumption that a conscious being at every stage of development disposes of a certain sum of energy, whose limit is also the limit of the individual (as regards intensity), then it is clear that this sum must be greater when it is to be distributed over various, complicated functions than when it is merely used for the exercise of a single simple function.⁷ — That these inner activities presuppose a richer organization needs no special demonstration. — It is likewise clear that when several different activities unfold before the reaction takes place, a longer time is required. A certain independence of the demands of the moment is therefore, as already remarked, the condition of a higher spiritual development. In simple and clear form this has been shown by the investigations of physiological time, i.e. the time taken up by the apprehension of an impression and the response to it. The stronger and the more familiar an impression is, and the more natural or practiced the movement with which it is to be answered, the shorter is the physiological time (or, as it is also called, the reaction time), and the more nearly we approach the certainty and unconsciousness of reflex movement and of instinct. The more the individual is prepared for the kind and strength of the impression, and for the movement with which it is to be answered, the more quickly the reaction can take place. Already when he does not know which of two different possible impressions is coming, the physiological time increases; there is then inserted a "discrimination time," taken up with deciding the nature of the impression.

⁷The distribution over the various functions will not always, perhaps even seldom, be even. What the consequence of this is in psychological respects we shall investigate in later sections.

When, in addition, the different possible impressions are each to be answered by its corresponding movement, so that it has to be decided which movement is to be made, a special “will time” is required.⁸ Even quite simple movements last considerably longer [112] than twitches that are produced by direct stimulation of a motor nerve. When the index finger of the right hand is set in motion by stimulation from an induction current, the movement is performed on average in 0.166”, whereas a voluntary movement requires 0.296”.⁹

7. It must not be forgotten that differentiation merely means the preponderance of different elements in different states, not complete separation.

a. Despite all independence of practical demands and of the claims of the moment, thought is nevertheless always connected with a certain mood. Elements of feeling are present, which are overlooked so easily only because they do not push themselves into the foreground, but subordinate themselves to the play of thoughts and are determined by it. A wholly feelingless thinking (a “pure thinking,” as speculative philosophers have so often demanded) does not exist. Lotze has aptly shown that even the most abstract thinking is accompanied by emotional movements. “We do not apprehend the concept of unity without at the same time enjoying the happiness of satisfaction that its content encloses, nor the concept of opposition without at the same time feeling the displeasure of hostility.” (*Mikrokosmos*. I. 2. *Ausg.* p. 273). It is by virtue of these emotional movements, connected with all ideas and thoughts, that knowledge becomes a power in the soul. When one speaks of the struggle of reason with the passions, what is really meant is a struggle between the feelings attached to rational considerations and the more vehement feelings, united with fewer elements of thought, which are designated by the expression passion. A feeling can be very strong and intense without being vehement, but is then more easily overlooked. The feelings attached to ideal ends and relations are far less able to bring about instantaneous affect and outburst than the primitive feelings attached to the physical functions of life. There is in the passions connected with self-preservation and the propagation of the species an animal ardor which often cannot be subdued by any other influence. Ideal enthusiasm is distributed over longer periods of time and works more hiddenly. And yet step by step it is able to take possession of the center of [113] the soul and to press into its service the collected energy, originally directed by those primitive impulses.

b. Just as little is knowledge ever completely emancipated from will. In all memory

⁸Donders’s experiments. See *Exner*: *Physiologie der Grosshirnrinde*. (Hermann. II, 2). p. 278–281. A detailed account by *Wundt* (*Physiol. Psychologie*. II. p. 206–279) of his own and earlier experiments.

⁹*Wundt*. I. p. 259. (Bast’s experiments).

and synthesis an activity manifests itself which we become especially conscious of, and call attention, when for inner or outer reasons a strong claim is laid upon it, but which in reality plays a part even in the simplest sense-perception. We must *will* to see in order to see rightly. All mental labor is connected with a certain feeling of strain. Already *Leibnitz* has shown that in the transition from one idea to another a striving or desire makes itself felt (*Monadologie* § 15). *Hume* and, in our day, *Helmholtz* have traced the principle of causality back to an instinct or impulse according to which, when something happens, we involuntarily look about for its cause. But it is with this striving just as with emotional movement: when it does not grow to a higher degree of strength through resistance or in some other way, it is as a rule overlooked. This moment of activity in all knowledge has been especially brought forward in recent psychology, at first chiefly as a reaction against the attempt of *Condillac* and of the one-sided “association psychology” to trace all knowledge back to a mechanical interaction between originally purely passive sense-sensations and their after-effects.

c. While for a long time there was an inclination to overrate knowledge at the expense of feeling — a tendency first overcome through *Rousseau*’s influence on modern intellectual development — attempts have been made, especially in more recent times, to conceive feeling as the primitive form of consciousness, so that the life of consciousness at the lowest stage would be a pure life of feeling and the other elements would only by degrees have developed out of it. Such attempts are explained partly by the intimacy with which the life of feeling is bound up with the very conditions of our existence, while the elements of knowledge seem rather to move on the periphery of our being, and partly, and above all, by the fact that the elements of feeling stand out more distinctly and strongly and play a larger part in relation to the elements of knowledge the more we descend from the higher forms of the life of consciousness to the more primitive. But this assumption becomes untenable already from the fact that as soon as a state of [114] pleasure and displeasure endures — and it must endure in order to be properly felt¹⁰ — memory already plays a certain part; and as soon as a pulsation takes place — and states of feeling always fluctuate in intensity — an involuntary comparison will already

¹⁰Patients anesthetized with chloroform often utter cries of complaint during the operation, without, however, being able after waking to remember having felt pain. »A vrai dire«, says *Richet* of this case, »cette douleur si rapide qu’on n’en conserve pas de souvenir, n’est rien, et c’est un moment presque mathématique dont il n’y a guère à tenir compte. Ce qui fait la cruauté de la douleur, c’est moins la douleur elle-même, si intense qu’elle soit, que le retentissement pénible qu’elle laisse après elle«. *Recherches sur la sensibilité*. p. 256. — *Maudsley* thinks that those cries arise purely by reflex. He cites a case where a lady, one of whose breasts was amputated, remembered after waking that she had heard herself scream, although she declared that she had not felt any pain. (*Physiologie de l’esprit*. Trad. de l’angl. par A. Herzen, p. 195).

make itself felt. Here, then, there are already elements of knowledge that can serve as a starting point for further development, so that knowledge does not arise through a kind of *generatio æquivoca* from wholly formless and blind states of feeling.

Such a view has recently been advanced by Ad. *Horwicz* (*Psychologische Analysen auf physiologischer Grundlage*. I. Halle 1872. p. 350 ff.). According to this investigator's very interesting account it is the movement called forth by feeling that paves the way for knowledge. Pleasure and displeasure lead to certain movements, which are tried until the most satisfying is found; this is then practiced and thereby acquires a special distinguishing mark that makes it an object of consciousness. But something new is thus added after all, besides the element of feeling itself: namely the sensation of movement, and one has no right to suppose that it should always be derived; on the contrary, involuntary movement occurs as early as consciousness can be assumed to occur at all. At the basis of this theory, then, lies an abstraction or, if one will, an anticipation: feeling is conceived as independent and resting in itself, which it first becomes — to a certain degree — when at a higher stage of development it has emancipated itself from will.

[115] It is impulse rather than feeling that is the right expression for the primitive stage of consciousness. Pleasure and displeasure occur originally only as elements in desire and striving. Even in the lowest animal beings, the monera, movements of extension and contraction have been demonstrated; the former lead to the taking in of nourishment, the latter to protection against attack. Already here, it seems, elements of consciousness other than pure feelings of pleasure and displeasure must be presupposed, such as sensations of touch and of movement, perhaps also sensations from chemical influence, an analogue of the sensations of taste. "In the protozoa's search for nourishment we observe distinctly that they are capable of a certain discrimination, without which capacity the feeling-about would have no purpose, could not be called a feeling-about at all. — With locomotion in search of nourishment there necessarily goes hand in hand the development of a discrimination between different directions, i.e. a distinguishing of the pleasant direction (leading to the object of nourishment) from the relatively less pleasant direction; once this distinction is present, the direction from which danger comes is also easily distinguished from the opposite one."¹¹ Even if one should think that in these little-formed beings, in which no nervous system has yet been found, the sensations could have no distinctness and definiteness, the facts adduced show nonetheless that the animal must apprehend a difference between the impressions. Pleasure and pain would indeed help the animal very little if they merely drove it to the performance of movements, without the kind

¹¹G. H. *Schneider*: Zur Entwicklung der Willensäußerungen im Thierreich. (Vierteljahrsschrift für wissenschaftliche Philosophie. 3. Jahrg.) p. 183. 301.

and direction of the movement being able to be more closely determined by the nature of the impression.

What holds for the lowest conscious beings no doubt holds also for the higher beings at the first stage of development. (See on this I, 4 and IV, 4).

In higher and fully developed beings too we have an approximation to a pure state of feeling, namely in the vital feeling or common feeling, the fundamental mood that arises from the whole state of the organism, from the normal or abnormal course of the vital movements, especially of the vegetative functions. Only rarely and imperfectly are we [116] able to localize the impressions that arouse this feeling. They do not appear separated, nor with such qualitative peculiarity as the impressions received through the external senses. Even differences of degree cannot be apprehended here so definitely as with the senses proper. The vital feeling consists in an unclear mood, of whose cause we do not, at least, become conscious at once. Diseases of the heart and of the mind can produce unrest and melancholy without the sufferer discovering the cause of these moods. We have seen what part the vital feeling plays in the fetal state and in the newborn child. In the years of puberty there is awakened, simultaneously with the maturity of the sexual organs, an indefinite presentiment and longing, an obscure impulse which, incomprehensible to the individual, leads him out beyond himself. The temperaments *manifest themselves* especially in the fundamental moods that dominate the vital feeling. — However few elements of knowledge can be demonstrated in all these states, they nevertheless each appear with such peculiarity that the transition from one to another, and with it the difference between them, must come more or less distinctly to consciousness, and none of them can be as simple as a pure state of feeling would have to be.

Self-observation shows us at most only an approximation to a state in which all elements of knowledge have vanished. Such an approximation we get the more the strength of the element of feeling increases, and this strength in turn depends on the strength of the impression. Knowledge and feeling here come to stand in inverse relation to each other: the more strongly the one manifests itself, the less strength the other commands. Exceedingly strong joy or sorrow can drive away almost all ideation, all reflection; but such an ecstasy also stands on the border of consciousness.

That the higher feelings contain elements of knowledge within them needs no further demonstration. Feeling acquires a content or an object only by being attached to memories and thoughts.

d. The intimate relation between feeling and will is given with the fact that only a strong and living feeling is a motive for the will. Elements of knowledge in and of

[117] themselves do not lead to movement of will. It is from this point of view that *Sibbern*, who otherwise follows the tripartition in his psychology, points out that it can be transformed into a duality, since feeling and will have this in common, that in both there shows itself “a personal being-gripped and being-moved by what is known, whereby we attach ourselves to it and are brought to act and strive for it.”¹² The close connection with movement both have in common: feeling finds outlet in movement and likewise initiates movement. The expressive movements are, to be sure, in part such as are withdrawn from the direct influence of the will, and arise through the strong movement in the brain propagating itself further to larger or smaller regions of the organism. Heart and lungs, intestinal canal, vascular system and other internal organs thus feel the effect of the emotional movement. But organs and muscles too that otherwise stand under the dominion of the will can be set in motion by strong feelings, and it is difficult, if not impossible, to distinguish between emotional movement and voluntary movement. What we call instinct and impulse lies in this intermediate region. Feeling naturally finds vent precisely in such movements as have frequently (in the same individual and in earlier generations) been employed in the service of the will. Already the monera perform contraction and recoil in order to protect themselves against danger. When beings at a higher stage, too, now start together and back in sudden fright, it is an old instinct that still stirs dimly in the *expression* of the emotion. The Greek word φόβος, fear, originally means (thus often in Homer) flight. In a similar way anger finds vent in movements of attack, strong sympathetic feeling in the stretching out of the arms as if to embrace the object, etc. According to the hypothesis of evolution these phenomena find a natural explanation in the fact that the involuntary expressive movements were originally purposively adapted voluntary movements.¹³

[118] It is only in the course of psychological development that differentiation between feeling and will sets in. A greater and greater opposition develops between the two ways in which the inner movement finds vent. We see here distinctly the psychological significance of the law of the conservation of energy. For the more energy the individual expends on the one kind of reaction, the less it can expend on the other. This truth is aptly expressed in Saxo’s well-known story of the different effect that the report of Regner Lodbrog’s murder had on his sons: the one in whom the expressive movement was weakest had the greatest energy of action. It is a well-known thing that sentimentality has a weakening effect on the will. In harmoniously developed natures such an opposition

¹²Psychologie. Kbhvn. 1856. p. 150 f.

¹³*Darwin*: Expression of the Emotions. Chap. 1. — *Spencer*: Principles of Psychology. II. p. 545 ff. — *Wundt*: Physiologische Psychologie. II. p. 417.

does not come to the fore; feeling and will each find their proper domain and at the right moments give way to each other.

Feeling, like knowledge, has an immediately practical character. By a course of development similar to that in the other case, feeling is freed from practical impulse and can move freely (thus in aesthetic and religious moods) without immediate regard to outward action.

e. If any of the three kinds of elements of consciousness were to be regarded as the original form of the life of consciousness, it would surely have to be the will. In instinct, raised only a little above reflex movement, we have the primitive form of consciousness, and there the side of will is evidently the strongest: the intellectual and emotional elements acquire significance only as transitional links to action. The will also forms later, at all stages, the constant substratum. Activity is a fundamental property of the life of consciousness, and when we ascribe will to the conscious individual we merely emphasize this fundamental property in particular. In the narrower sense, as the capacity to choose between various given possibilities, will is only a result of psychical development, not originally co-operative. But if we take will in a wider sense, as all activity determined by feeling and knowledge, then the whole life of consciousness can be said to be gathered up in will as its fullest expression. Consciousness can, in accordance with the foregoing, be defined as arrested movement. But the transition to movement nevertheless always stands as the conclusion of everything that goes on in the world of consciousness. The development of the conscious individual goes from will to will. This development can take place very sporadically, move through one-sidednesses and oppositions; but there will always (if not in the individual, then in the race) be an obscure urge that leads beyond the scattered, one-sided and conflicting to inner harmony among the fundamental spiritual tendencies. Ethics and aesthetics are, psychologically speaking, fully within their rights when they find in such a harmony the highest type for a human life. [119]

Chapter 5

Sense-Perception and Thought

[120]

A. Sensation

1. When we consider the phenomena of cognition by themselves alone, we perform, as we saw in the preceding chapter, an abstraction, since elements of feeling and will are surely at work in every actually given state of consciousness. But such an abstraction is necessary; every inquiry must to a certain degree take its object out of the living and manifold connection in which it exists. And we must carry the abstraction even further. The phenomena of cognition themselves are in a high degree composite. We may, it is true, provisionally divide them into two groups, sense-perception and thought; but it will turn out precisely that these two groups do not lie outside each other, but stand in the most intimate interaction. Consciousness appears — at least in the forms in which we know it from immediate experience — never as a perfectly single series of inner phenomena. Manifold series of thoughts and memories, sensations and ideas, interlock with one another in each single experience, each single inference. But in order to see the relations and the laws clearly we must, so far as possible, think of consciousness as forming a single series. And not only that; we must begin by thinking of this series as consisting of sensations alone. The difference between sense-perception and thought rests on the difference between sensations and ideas: between the immediately given new elements and the residues or after-effects of such elements that emerge anew. As soon as the elements previously acquired cooperate in the reception of the new, we [121] have no pure sensation; then there is already a certain degree of thought involved in the sense-perception. We shall therefore try whether it is possible to conceive of a consciousness that consists merely of a series of perfectly pure and simple sensations, or of simple, immediately given elements arising in each single moment. Even if it were

possible that there could be sensations entirely without admixture of ideas, it would not follow that these sensations existed entirely independently of one another and were each of them absolutely simple. There are thus three main questions that present themselves to us and indicate the points of view from which we shall here occupy ourselves with the doctrine of sense-sensations: namely, the question of their independence of ideas, the question of their independence of one another, and the question of their simplicity and indissolubility.

It will easily be seen that if these questions are answered in the affirmative, the provisional characterization of consciousness on which we built in the earlier chapters will be incorrect. For we found it to be the characteristic mark of the life of consciousness that it appeared in a series of states which did not exist independently of one another, but determined one another in an inner way, this inner interaction finding its most typical expression in memory (see II, 5). It is thus a kind of *test* of the correctness of this characterization that we now proceed to.

2. By a purely psychological route we can convince ourselves of the *simplicity* of our sensations only to a certain degree. We can never be quite sure whether we really stand before something indissoluble. Where observational psychology leaves off, experimental sense-physiology takes over, and it has shown in some cases that the seemingly simple psychological phenomenon presupposes an intricate and composite physiological process. The possibility that psychological simplicity is only the result of a composition taking place below the threshold of consciousness can therefore not be denied.

The more indefinite the sensation, the less probable it is that it should be uncompounded. Of the vital or general sensations, accordingly, one will not readily assert it. A sensation such as, e.g., nausea (sickness) has something composite about it even for [122] immediate observation, as is seen from the fact that it has been classed now under the sensations of taste, now under the muscular sensations, while at other times the wish has been to separate it from both these kinds. Many of the sensations of taste and smell are so mixed with sensations of touch that they cannot be called pure or simple. Salt, sour and astringent taste, pungent and sharp smell, are properly combinations of sensations of taste or smell with sensations of touch. The enjoyment of many kinds of food (e.g. jelly and the like) is no doubt due chiefly to their effect on the fine skin of the mouth cavity, and is thus connected with the sense of touch rather than with the sense of taste. In the wider meaning in which we are accustomed to speak of the sensation of taste, the Shah of Persia was therefore right when he reproached the Europeans (who use knife

and fork) for not knowing that the sensations of taste began in the fingertips. Besides smell, sight too plays a part in taste, since in the dark one does not judge the taste of a thing so surely as when one can call on sight for help. An isolation is possible here only by artificial means, and it then often turns out that the taste-sense proper has nothing whatever to do with the apparent sensations of taste.

The sensation we have when we lift a burden from the ground is very composite. Contact, pressure and muscular effort fuse together into an indefinite whole, which becomes still more intricate from the fact that several different muscles are tensed, and in different degrees. And yet it may seem to us as if it were a single sensation that we had.

As regards sensations belonging to one definite sense, the matter might seem to stand more simply and clearly. Yet here too there can be dispute whether a sensation appears as simple or as composite in immediate apprehension itself. *Goethe*, e.g., held that only the sensations of yellow, blue and red were simple; he called them principal colors and held that the other colors were composed of them, since in violet he believed he could trace red and blue, in green yellow and blue, and in orange yellow and red.¹ On the other hand, in our day E. *Hering* asserts that green is an entirely simple sensation, and that yellow and blue can never be sensed together as elements in a composite color. He therefore sets up [123] a series of four principal colors: red, green, yellow and blue.² When such able observers differ so decidedly, it is a proof of how uncertain direct psychological apprehension must be at this boundary point. As regards the matter itself, there is evidently yet another possibility left, namely that all color sensations are simple. He who practices abstracting from memories and preconceived opinions, and concentrating his attention on a single narrow band of the color scale, will surely be able to obtain an entirely simple sensation of each single color; and if language had formed words enough, he would feel the need to express each of these color tones by its own independent word. There is the further point³ that the dispute would begin anew when it came to stating more precisely which shade of red, green, etc., is the real principal color. Different observers would each name his own shade and assert that the other shades had arisen by mixture of it with other colors. When the expressions of language point to certain principal colors, this is probably to be explained by the fact that there were certain prominent natural objects whose colors impressed themselves early and dominated apprehension (the green plants,

¹Farbenlehre'. 1ster Bd. § 60.

²Zur Lehre vom Lichtsinne. 2 Aufl. § 38. — This fourfold scheme had already been set up by *Leonardo da Vinci*.

³V. Krenchel: Om Grundfarver. Et kritisk Bidrag til Farvesansens Teori [On Fundamental Colors. A Critical Contribution to the Theory of the Color Sense]. Kbhvn. 1880. p. 11 f.

the red blood, etc.).⁴

To be carefully distinguished from the question of *principal colors* in the subjective-psychological respect is the question whether there are *fundamental colors* in the objective-physiological respect. There are certain colored rays which, mixed in suitable proportions, can produce all possible subjective color sensations. These colors, by whose objective composition the others can arise, have been called fundamental colors. They are three: red, green and violet. According to a theory put forward by celebrated investigators (*Young* and *Helmholtz*), there are in the optic nerve three different kinds of fibers, each corresponding to its own "fundamental color." At every stimulation of the retina all [124] three kinds of fibers are set in activity, and the kind of sensation therefore depends on which kind of fiber is most strongly stimulated. Every color sensation will thus — physiologically regarded — be of a composite nature; in order to obtain an entirely simple stimulation, the two other kinds of fibers must first be dulled by strong stimulation with the rays corresponding to them. If we want, e.g., an unmixed red color impression, it is obtained by first looking at a green surface.

This theory, however, is regarded by many as unwarranted and unnecessary for sense-physiology. It carries physical relations over without more ado into the physiological domain. "From the fact that objective red, green and violet are sufficient to produce all qualities of light, we evidently may not yet conclude that there also exist only three physiological processes underlying all sensation of light and color; on the contrary, we must conclude that, since the qualitative sensations produced by those three objective colors and their mixtures are very manifold, the physiological effects of the proportional mixture of the three fundamental colors are qualitatively of very different kinds."⁵ Dr. V. *Krenchel*, in his treatise cited above, has tried to show that the phenomena before us can very well be explained without assuming fundamental colors in the physiological sense, simply by supposing that to each kind of subjective color sensation there correspond different forms of motion in the brain molecules to which the sensations must be assumed to be attached.

On the other hand, there seems to be bound up with every *color impression* a *colorless impression of light*, which, when the strength of the stimulation is greatly decreased or increased, passes over into a distinctly colorless sensation (of white, gray or black). Only with a stimulation of middle strength does the color impression (the chromatic irritation) have the preponderance over the colorless impression (the achromatic irritation). This mixture of a chromatic and an achromatic process in the organ of sight is inferred partly

⁴ *Wundt*. I. p. 415.

⁵ *Wundt*. I. p. 451.

from the fact that the parts of the retina most remote from the place of distinctest vision (the yellow spot, *fovea centralis*) are totally color-blind, partly from the fact that every color sensation, with sufficient decrease or increase in the magnitude of the stimulation, [125] passes over into colorless sensation.⁶

As regards the sense of hearing, the sensations are no doubt always composite. Every tone has its timbre, i.e. consists of a combination of a fundamental tone and weaker overtones, different for each source of tone. The same tone therefore sounds different when it is produced by means of different instruments. But just as the practiced ear at a concert is able to distinguish the contributions of the single instruments to the resulting impression, so too specially gifted or trained organs of hearing can separate the partial tones in the clang, although in immediate sensation this stands as entirely simple. A single tone is therefore properly an abstraction, since we probably never apprehend tones that are wholly free of timbre. There is only a difference of degree between clang and consonance, conditioned by the weaker or stronger prominence of the overtones in relation to the fundamental tone.

This theory, set up by *Helmholtz* and generally accepted, shows how by way of physical and physiological experiment the composite nature of subjective sensations can be established. The sensation of clang is resolved into elements on whose varying relations to one another the character of the sensation depends. But if this is how it stands with one kind of sensation, should it not stand the same way with our other sensations? If here we lack the ability to trace the elements of sensation of which the sensation given to our consciousness is composed, this might well be connected with the fact that the sense of hearing on the whole stands so far above the other senses in the fineness with which it is able to apprehend differences and nuances. The study of the sensation of hearing has in any case shaken the principle of the simplicity of sensations and opened a new horizon where the psychological world accessible to us seemed to close.

There are also certain phenomena that point toward simpler psychical elements than those we find in our distinct sense-sensations. The sensations we receive while our attention is taken up in another direction, or in a sudden surprise, have no definite, [126] qualitative character. We start, notice that something has happened in or to us, but what it is, whether it is an impression of light, a blow, an electric stimulation, we do not know — in the first moment at least. So the more suddenly the sensations occur and the shorter they last, the less can they be referred to any of the definite qualities of sense.

⁶ *Wundt*. I. p. 453.

The same holds — as has been shown experimentally by *Fick* and *Wunderli*⁷ — when the impressions are very weak and very limited. Unless the impression falls precisely on the inner palm of the hand or on the face, one cannot tell whether a weak and limited stimulation comes from contact or from warmth. It seems, then, as *Fick* remarks, that in the distinctly appearing sensations of warmth and contact we must see the results of a combination by which various elementary sensations are united into groups that stand before consciousness as indissoluble unities.

It is all the more natural to conceive the matter in this way since the nerve process, whatever its nature may otherwise be, takes place in pulsating shocks or oscillations. It is the highest law of general nerve-physiology that a nerve process can never be released by a state of equilibrium, but only by sudden changes in the state of the nerve proceeding with a certain rapidity. A seemingly continuous nerve process (a tetanus) comes about only through a series of single alterations in the equilibrium following rapidly on one another. With this law the special relations in the individual sense organs also seem to agree, so far as they are known.⁸ Sensation as we know it must correspond to several such shocks, or to the different moments of the oscillations; there is thus bound together in one moment of consciousness, in the single momentary sensation, what physiologically regarded fills several moments of time. And since the structure [127] and mode of operation of the nervous system seem to be uniform throughout, only one way stands open to an explanation of the qualitative differences of the sense-sensations, namely to derive them from the different ways in which those elementary sensations corresponding to the single nerve pulsations or discharges of force are combined. Just as the different color qualities can find their physiological representation in different directions of motion of the same brain molecules, so one might go a step further and think of the difference of the *modalities of sensation*⁹ as physiologically represented by differences lying still deeper, in the form and direction of the processes that take place in the central and sense organs. Only thus will there be a possibility of understanding how the different senses can have arisen in the course of the development of organic beings. The further one goes from the higher animals down to the lowest, the fewer modalities one finds. The modality that remains as a starting point for the development of sense-perception is the sense of touch. From it the special senses or modalities *must*

⁷Hermann's Handbuch. III, 1. p. 164. — III, 2. p. 322.

⁸*Funke* in Hermann's Handbuch. III, 2. p. 328 ff. — The principal law of nerve-physiology referred to was demonstrated for electrical stimulations by *du Bois-Reymond* in 1845. L. *Hermann*: Allg. Nervenphysiologie (Handbuch. II, 1.). p. 50.

⁹*Modality* is a term proposed by *Helmholtz* for what *Fichte* calls a circle of qualities: the sum of the sense-sensations belonging to the same class (sight, hearing, taste, etc.).

have arisen by differentiation. It agrees with this that in all sentient organisms the special sense organs develop from the outer covering of the body.¹⁰

This hypothesis (first set up by *Spencer*) of the origin of our single sensations by combination of elementary sensations of greater simplicity thus opens a wide horizon for us. What interests us most here, however, is that in being led to this hypothesis we find, at the borders of the domain of distinct consciousness, the traces of a work which has no doubt gone on below the threshold of consciousness, but yet according to the same laws that rule within the life of consciousness (cf. chap. III). The general characterization of consciousness as comprehending and unifying proves to hold of the very limiting parts to which the resolution of composite states of consciousness leads. If, then, one wanted to assert that consciousness is only a sum of sensations, it would in any case have to be remarked that the terms of this sum are not absolutely simple, but have probably arisen [128] by synthesis of still simpler elements. It is thus not possible to point out psychical atoms.

3. In close connection with the question of the simplicity of sensations stands the question of their *independence* of one another. Here too sense-physiology gives us interesting information.

There seems to be a certain resistance to be overcome, both in the end organs of the sensory nerves and in the nerve centers, before the impression can produce its full effect; but once it is overcome, the effect persists for some time after the impression has ceased. It is the law of inertia, valid for all material phenomena, that appears here in nerve-physiology too. The different senses, however, do not behave alike in this respect. The greatest elasticity is possessed by the sense of touch. If one lays one's finger on a toothed wheel that is turned with a certain speed, one can have as many as nearly 1000 separate sensations in one second. If the speed is increased still further, on the other hand, only one continuous sensation arises. Next after the sense of touch in this respect comes the sense of hearing. That the latter is surpassed by the sense of touch is seen already from the fact that by laying one's hand on a musical instrument one can feel the vibrations of even fairly high tones as a whirring. In observation with one ear the crack of two electric sparks is still heard as independent of each other when the one springs across 0, 002 second before the other. In observation with both ears the limit lies higher up (0, 064 second). Electric shocks can still be distinguished when they come at a rate of 35 in a second; if the rapidity becomes still greater, only one single sensation arises. On the forehead 60 shocks in a second can still be distinguished. Lowest in respect of elasticity stands the sense of sight,¹¹ which is inferred from the nature of the end organs.

¹⁰Cf. *Wundt*. I. p. 280.

¹¹For the sensations of taste and smell no reliable observations are available. It cannot be definitely

[129] After looking at a clearly lit object an after-image remains when one closes the eye. The new impression thus does not find the place empty, but is combined with the after-effect of the preceding one. If a disk divided into equal, alternately white and black sectors is turned rapidly round, the impressions fuse and a continuous sensation of gray arises when there come about 24 stimulations in a second and the intensity of light is that of ordinary daylight. A glowing sulphur match turned rapidly round in the dark looks to us like a luminous circle. With slow rotation we distinctly bring out the single sensations; with increased rapidity a flickering arises, and with still greater rapidity the fusion into one sensation then takes place.¹²

The man born blind on whom Franz operated still found it very unpleasant several months after the operation to walk in busy streets. The many different things and the rapid movements of people, carriages, etc., confused his sight so much that in the end he saw nothing at all; the impression produced by the object last seen had not yet vanished when the next object was already calling forth a new one. Here, then, the single impressions did not wholly fuse, but called forth a chaos that made definite apprehension impossible. At every stage of consciousness there is a certain tempo that the sensations must observe if they are to be independent of one another.

The one term in the series of sensations thus does not stand absolutely independent over against the other; their existence depends on their distance from one another in time. We move here in a domain where it is difficult to find the right expressions. For, strictly speaking, one cannot say that the sensations fuse, since under the conditions stated they no longer come into being each by itself. One might therefore hold that [130] these phenomena are no concern of psychology, since psychology has to keep to what occurs within consciousness. But it is of great importance for psychology to learn that the independence of sensations is only conditional and relative. We have now seen that already at the boundary of consciousness a struggle is waged between the single elements, and that the significance each of them acquires depends on its relation to the other elements. The same will prove to hold when we examine the strength and the quality of sensations.

4. When we listen to a dying tone, we can follow it down to a minimum of *strength* that could not be apprehended by one who had not heard the tone struck. So too the

established that these sensations leave after-sensations, since one cannot with certainty exclude residues of the tasting and smelling substances in the organs. *Vintschgau* in Hermanns Handbuch. III, 2. p. 221. 284.

¹²*Fick*. Hermanns Handbuch. III, 1. p. 211 ff. — *Exner*. *ibid.* II, 2 p. 256–260. — The strong and lasting after-effects of light in the eye are an important support for the photochemical theory, according to which the decomposition of the so-called visual purple in the retina brought about by the rays of light is the real stimulus of the optic nerve. *Kühne*: Chemische Vorgänge in der Netzhaut. (Hermann's Handb. III, 1). p. 238. 261.

eye can follow the flight of a bird to a distance at which it could not be discovered by an eye that was only now to apprehend it. What we call the threshold of consciousness is thus considerably lowered in such cases: it lies further down when we come to it from above than when we discover it from below. The strength an impression must have in order to become an object of sensation thus depends on the preceding state of the sentient organism. It is easier for consciousness to hold fast a given weaker impression than to receive a wholly new one, though this may perhaps be stronger in itself. Here it is active adaptation, or practice, that makes up for what the impression lacks in strength. Prisoners who had long lived in dark rooms acquired the ability to distinguish the smallest objects and the faintest differences in the intensity of light. When Lavoisier, on the occasion of an academic prize question, wished to investigate the luminous intensity of different kinds of flame and noticed that his eye was not sharp enough, he shut himself up for six weeks in a room hung with black, and thereby became able to notice very small differences. When sensation is not trained in this way, a higher degree of strength in the impression is required for the threshold of sensation to be crossed.

By gentle and successive strengthening an impression can remain imperceptible even after it has reached a degree of strength at which it would otherwise call forth sensation. By very slow strengthening of an electric current a nerve acted on by it is at last destroyed without any sign being shown of sensation. By successive, very small [131] raisings or lowerings of the temperature, frogs have been boiled or made to stiffen with frost without making the least reflex movement. Sensations of warmth and cold arise only when the temperature of the skin undergoes a change that proceeds with a certain rapidity. One and the same impression can under different circumstances produce now a sensation of warmth, now a sensation of cold. If, e.g., we bring the hand into a vessel in which the temperature is the same as in the room, to whose temperature the hand was accustomed, we sense warmth, because the radiation of heat from the hand is hindered. If one fills a vessel with water of a temperature corresponding to that of the skin, a vessel to the right with water of higher, one to the left with water of lower temperature, then dips the right hand into the right vessel and the left hand into the left vessel, and immediately afterward both hands into the middle vessel: then the right hand feels cold, the left warmth in this vessel, where neither of them before noticed cold or warmth. In the sensation of temperature it depends merely on the skin (or more exactly the thermal apparatus) acquiring a temperature that lies sufficiently above or below its “zero-point temperature” (i.e. the temperature which at the place of the skin in question is sensed

neither as warmth nor as cold). So long as the current of heat passing through the skin from within outward does not rise or fall, no sensation of temperature arises, whatever the constant temperature may be.¹³

A certain contrast is thus required between the new impression and the one it replaces, for a sensation to arise. There must be a background in relation to which the new impression can assert itself. Every impression is measured, as it were, by what precedes it. Now if the new impression is weak in relation to the preceding one, it will be, so to speak, absorbed by the latter and not noticed. After being electrified for some minutes with a strong current one does not notice a weaker one that would otherwise [132] have been sensed. If one receives a strong current through the one hand and a new current through the other, the latter is not noticed. If one leg of a pair of compasses is set on a painful wound, the other on the skin outside it, and both places are pressed equally hard, one gets only one sensation, even if the distance is double that at which two sensations would have been felt within the wound.¹⁴ Already *Hippocrates* taught that when two pains arose at the same time, but not in the same place, the lesser was obscured by the greater. Shakespeare has King Lear say in the night of storm on the heath (Act 3, Scene 4):

“Thou findest it hard
that the storm presses right in to the skin?
for thee perhaps; but where a greater malady
has taken root, the lesser is scarcely felt.
— — — When the mind
is free, the body senses every pain;
but the storm in my soul has taken away
all other feeling from my senses
save that which beats here in the heart.”

In a strongly excited state of consciousness even impressions strong in themselves may be unable to win any place in consciousness. The ecstatic enthusiasm of the martyrs must have weakened the sensations of their sufferings. Likewise hypochondria and mental derangement can prevent painful sensations from arising; the fixed idea that takes possession of consciousness can be so strong that no other impression is able to attract attention.

¹³Cf. *Fick*: Anatomie und Physiologie der Sinnesorgane. Lahr. 1864. p. 54. — *Hering*: Der Temperatursinn. (Hermann. III, 2), p. 415 ff. — *Richet*: Recherches sur la sensibilité. p. 42 ff. 170 ff.

¹⁴*Richet*. p. 168. 222.

Fechner has sought a mathematical formula for the relation in which the effect of every impression is determined by the impressions previously received. The increase in the strength of the sensation that is brought about by growth in the strength of the impression is determined not by the absolute magnitude of this growth, but by the ratio in which it stands to the impression already present. The stars send out just as much light by day as by night; but by day their brightness is for us vanishingly small in relation to the sun's, while by night they have at most only the moon to compete with. For the sensation to grow by a certain degree, the impression must grow by the more degrees the stronger it already is. This *Fechner* expresses in his "fundamental formula": $d\gamma = [133] d\beta/\beta$, where γ denotes the sensation, β the impression. This formula can be converted into what *Fechner* calls the "measurement formula": $\gamma = \log. \beta$, which means that in order for the strength of the sensation, e.g., to rise from 1 to 2, and from 2 to 3, etc., the strength of the impression must rise from 10 to 100, and from 100 to 1000; or in other words, the strength of the impression must increase in geometric progression for the strength of the sensation to increase in arithmetic progression.

We disregard here *Fechner's* presupposition that the just noticeable increments in sensations are equal (cf. I, 8 d), as well as his own interpretation of the measurement formula as the expression of the fundamental relation between spirit and matter (cf. II, 9 d). The question here is merely whether his formulas are really confirmed by experience. *Fechner* himself at once admitted that *Weber's* law, expressed in the fundamental formula, holds only for adequate impressions, i.e. those neither too strong nor too weak. There is a definite limit to the receptivity of the sense organs, and when two unequally strong impressions exceed this limit, the difference between them will, in the nature of the case, be impossible to apprehend. The lower limit of the law's validity is caused by the fact that the internal state of the organ and of the nervous system comes to play a part with very weak impressions. In the case of the eye there operates here the so-called "intrinsic light," which arises without stimulation from outside, a kind of "normal hallucination."¹⁵ Nor do the other senses seem ever to be entirely without subjective sensations,¹⁶ which [134]

¹⁵"The black that we see in the dark and with closed eyes is a sensation of sight that takes place without external stimulation, and must not be confused with the not-seeing that holds in the case of the finger or the back of the head, nor be compared with not-hearing in the absence of external noise. The black we have in the closed eye is only the same sensation of light that we have at the sight of a black surface, which can pass through all gradations over into the strongest sensation of light; indeed, the inner black of the eye sometimes passes, even from purely inner causes, over into clear light, and contains, so to speak, interspersed luminous phenomena. — On closer attention one discovers in the black of the closed eye a kind of fine luminous dust, which in different persons and in different states of the eye is present in different quantity, and in morbid states can be heightened into vivid luminous phenomena." *Fechner*: *Elemente der Psychophysik*. I. p. 165 f.

¹⁶See G. E. *Müller*: *Zur Grundlegung der Psychophysik*. Berlin. 1878. p. 236–239. Cf. for the sense of

play a more important part only when the impressions from outside are very weak. But even for adequate impressions the correctness of Weber's law has been denied; it seems to have undisputed approximate validity only for impressions of middle strength in the domain of the sense of sight, the muscular sense, and perhaps the sense of hearing.¹⁷ In the work in which Fechner has answered his critics, the old man of genius also declares that just as surely as Weber's law, with the laws following from it, can within certain limits and in several domains of sensation provide a useful starting point for judging the relations of measure of sensation on the basis of its dependence on the impression, just as surely it cannot provide anything of the kind beyond certain limits and applied to any domain whatever. The question then is only whether it is possible to point out special reasons for the deviations from the law that take place in several domains of sense.¹⁸

But whether or not it proves possible to carry through a mathematical formula, all experience indicates that the strength of sensations is determined by their mutual relation. Every emerging sensation has a struggle to go through with other possibilities of sensation, and on the outcome of this struggle depends the strength with which it appears in consciousness.

5. From a purely psychological point of view no sharp line can be drawn between the intensity of the sensation and its *quality*. The contrast between light and dark, between white, gray and black, is for subjective sensation just as much qualitative as the contrast between red, yellow and blue. Just as there are countless nuances and transitions between white and black, so too there is continuous transition between the color tones among themselves. Just as we place gray between white and black, so we place, e.g., orange between red and yellow; and just as white passes over through a connected series of nuances into gray, so red into orange.

[135] Not from a psychological, but from a purely physical point of view the sensations of light and dark stand in contrast to the color sensations. Physically regarded, namely, the sensation of black is characterized by the eye's receiving no rays of light; black is thus here something entirely negative, in contrast to which gray and white express the weaker or stronger action of light. But one has no right to force this physical point of view on psychology. It was the feeling of this that drove *Goethe* to so bitter and unjust a polemic against the physical theory of light. In our day *E. Hering* and *Aug. Classen* have maintained that, psychologically and physiologically regarded, *Goethe* was right. The mere fact that black can become the content of a sensation at all is enough to decide

hearing *Hensen*: Physiologie des Gehörs. (Hermann. III, 2). p. 121 f.

¹⁷G. E. Müller. p. 225.

¹⁸In Sachen der Psychophysik. Leipzig 1877. p. 212.

the matter; if the physical view exhausted the relation, black would denote the same as non-sensation (and so non-consciousness). Yet everyone senses something peculiar in itself when he has the sensation of black. Not to speak of how deeply the contrast between light and dark reaches into the life of feeling!

An eminent physiologist, to be sure, says in his polemic against Hering: "I find in my consciousness only a quantitative change of one and the same sensation when I gradually diminish the illumination of a white sheet of paper in the field of vision until at last it looks black, or when in some other way I let all the gradations follow one another from gray to black. I find this process entirely analogous to the one in which a definite clang is heard with decreasing intensity until at last it vanishes into complete silence."¹⁹ It would be interesting to know whether an observer who knew nothing at all of the results of physical optics would judge in the same way. Psychologically regarded, *Hering* is fully justified in demanding that the opposite procedure also be applied, namely to start from black as a positive starting point and, with increasing illumination, to pass over to gray and white. The question is, on the whole, whether we do not sense all differences of strength as qualitative. The difference between a weaker and a stronger tone, a weaker and a stronger pressure, etc., is for our sensation of the same kind as that [136] between two colors. He who follows the gradations from red to orange with strained attention, so that he skips nothing, will likewise be able to find in his consciousness that it is one and the same sensation whose quantity merely changes. Let us suppose that, just as there are eyes that are able to apprehend only the difference between light and dark, there were eyes that sensed only the difference between red and orange: would it not, from the standpoint of such a consciousness, be justified to call this difference a mere difference of strength?

Goethe's great mistake was that he wanted to force the psychological point of view on physics. He wanted to derive all colors from combinations of white and black. With increasing illumination all colors pass over into white, with decreasing into black. But this shows only that intensity and quality stand in a definite interaction physically regarded as well, not that one has a right to derive all nuances of color from white and black. Physiologically too the difference between the series of color tones and the degrees of intensity of light must be maintained, inasmuch as different processes seem to underlie them. The yellow spot (*fovea centralis*) on the retina is the proper place for the reception of color impressions; already a few millimeters from it not all color tones can be apprehended, but only yellow and blue. Still further out toward the periphery

¹⁹A. Fick: Die Lehre von der Lichtempfindung. (Hermann. III, 1. p. 205.

of the retina no color tones are apprehended, but only the difference between light and dark. In total color-blindness what normally holds only for the outermost parts of the retina holds for the whole retina (the yellow spot included). From this it is concluded that the processes in the elements of the retina that correspond to the sensation of colorless light must be independent of those that correspond to the sensation of color.²⁰ The lowest animals in which organs for the perception of light are thought to have been shown can no doubt (like human beings with congenital cataract) sense only the difference between light and dark.²¹ In the first years of childhood the color sense is very [137] little developed; only glaring color tones attract attention.²² But because the physical difference between strength and quality also becomes of importance for physiology, it does not follow that physiology can place the sensations of white and black (or the corresponding physiological processes) outside the domain of qualitative phenomena. Even if the organ of sight receives no stimulation from outside, a process can nevertheless take place in it to which there corresponds a sensation just as positive as any other. We have already remarked that inner irritations seem constantly to take place in the sense organs, which come forward the more, the weaker the external stimulations are. Here, then, a positive process takes place, and it is no doubt this process which, by its contrast with the process arising from stimulation by rays of light, corresponds to the sensation of black or dark. The purest black is apprehended only beside the purest white and in contrast to it. During a stay in a completely dark room, on the other hand, the sensations of the resting eye acquire a fairly considerable brightness and become almost as much akin to white as to black. The sensation of black proper therefore arises only under the influence of external impressions of light, namely as a contrast sensation to the sensation awakened by them.²³

As it is with the sensation of black, so it is in general with the sensation of quiet and rest.²⁴ If there were absolute quiet outside us and within us, we should never get the sensation of rest. It is only through the contrast with motion and change that the sensation of rest comes into being. As a sensation it is just as positive as the sensation of a change or a motion: it is precisely the sensation of the change itself from the state of motion to the state of rest. When this transition is made suddenly, it can act very strongly. A sudden pause in a noisy piece of music startles us. The miller wakes when [138] the mill stops, the sleeping traveler when the carriage halts. An entirely continuous

²⁰Fick. p. 206–207. — Wundt. I. p. 432 f.

²¹Emmert: Die Organe des Sehens in den verschiedenen Thierkreisen. Bern 1872. p. 14 ff.

²²Vierordt: Physiologie des Kinderalters. p. 154. — W. Preyer: Die Seele des Kindes. Leipzig 1882. p. 6.

²³Hering: Zur Lehre vom Lichtsinne. 2 Aufl. Wien 1878. p. 62 ff. — Cf. above p. 133.

²⁴Cf. G. H. Schneider: Die Empfindung der Ruhe. Zürich. 1876.

motion would be sensed as a state of rest, and would in general be sensed only when it came into contrast with interrupted and irregular motion.

However it may stand, then, with the relation between the strength and the quality of sensations, the same fundamental law holds in both respects. The peculiarities of sensations that are qualitative in the special sense are also determined by their relation to one another. The *law of contrast* holds both for quality and for strength. Just as light is apprehended most sharply in contrast to dark, so red comes out most definitely beside blue-green, blue beside orange, green beside purple, yellow beside indigo, violet beside yellow-green. It is the complementary colors (i.e. those whose rays when mixed produce the sensation of white light) that stand in strongest contrast to one another and thereby make one another appear in the most pronounced way. On the other hand, red, e.g., looks different when it is viewed beside other colors: it here always approaches the complementary color of the color in question. A gray strip on a colored ground, when covered with thin transparent paper, takes on a tinge of the complementary color of the ground. Besides this *simultaneous contrast*, the contrast effect between two successive impressions (*successive contrast*) also plays a great part. When the eye has dwelt a while on a definite color impression, it is thereby disposed to an especially pronounced apprehension of the complementary color. This is seen from the fact that when the eye, after a strong red color impression, is directed to a white surface, it sees blue-green. By the union of simultaneous and successive contrast the most pronounced sensation of a color arises. Without our noticing it, these contrast effects take part in all our sensations of color. We seldom keep our gaze fixed on a single point; even a quite brief fixation costs great effort. After-images from the one point of the field of vision therefore gain influence on the apprehension of others. The most intricate combinations of successive and simultaneous contrast come in. If we keep our gaze firmly fixed on a single point, the after-images coincide with the objects and make them indistinct; we feel dazzled, [139] and an irresistible impulse stirs to change the position of the eye.²⁵

We see how little the single sensation can be isolated: both simultaneous and preceding impressions determine it. *Helmholtz* goes so far as to explain the phenomena of contrast by false inferences. This expression, which he himself moreover declares to be half figurative, does not fit, since — as *Wundt* has aptly remarked (II. p. 457) — it presupposes that we possess an absolute standard of *correct* quality of sensation, whereas

²⁵*Helmholtz*: *Physiol. Optik*. p. 389. — *Panum*: *Sandserne og de vilkaarlige Bevægelser* [The Senses and the Voluntary Movements]. p. 209 f. — Examples can also be taken from the sensation of taste. The taste of cheese heightens, in the opinion of many, the taste for wine. After bitter or salty sensations of taste, water tastes sweetish.

the truth is that *every* determination of quality is relative. The determination of a normal or typical nuance that we fix for each color has only practical significance. It is difficult to find linguistic expression for phenomena that, like those brought forward here, lie at the border of the life of consciousness; the psychological expressions denote higher and more intricate processes of consciousness. But here in the border region we nevertheless find processes that form an elementary analogy to the higher psychical activities. This naturally does not exclude that these same processes must also be explained physiologically. According to the fundamental conception at which we arrived in chap. II, every psychological phenomenon should, on the contrary, be viewed from both sides. Physiologically the dispute here can only concern whether the end organs or the central organs play the larger part; probably both cooperate.

6. The doctrine of sensations thus confirms the general characterization of consciousness that we have given earlier (II, 5). It is impossible to resolve consciousness into a series of single and independent sensations, each with its own strength and quality. From the very hour of each single sensation's birth, its existence and its properties are determined by its relation to other sensations. A sensation that stands in no relation to any other sensation we do not know: it would not be an element in any consciousness, and so could under no conditions be known.

[140] Entirely uniform impressions and states therefore do not become objects of consciousness. The *pressure* of the air we do not notice, except when it varies. We take part in the earth's so rapid motion without knowing it, because it is constant. We do not notice that the blood vessels of the retina cast shadows on the retina itself, because the retina has always been accustomed to this; these shadows are noticed, however, when one artificially lets them fall on parts of the retina that are accustomed to stronger impressions of light.²⁶ A substance has a taste only when its taste differs from that of the saliva. To the saliva itself the tongue is accustomed, and is thereby dulled to its faint, saltish taste.²⁷ A sound always ringing alike, or a color always uniform, that surrounded us from the beginning of life to its end, we should not sense, if we had no other impressions against whose background they could stand out. If, as Pythagoras taught, there were a harmony of the spheres, produced by the motion of the stars in their orbits, ordinary mortals (and with others psychology cannot concern itself) would certainly not hear it. In recent years attention has been drawn to the remarkable circumstance that the writers of antiquity (the Vedas and Homer, e.g.) name only the radiant and bright *colors*, and from this it has even been sought to conclude that the human race in those

²⁶Helmholtz: Physiologische Optik. p. 161,

²⁷Henle: Ueber den Geschmacksinn. (Anthropologische Vorträge. 2tes Heft). p. 18.

times still had only the ability to apprehend the contrast between light and dark, not the differences of the color tones. This improbable theory (which is already refuted by the fact that the Egyptians depicted the blue sky) is unnecessary, since the fact pointed out is far more naturally explained by the law of contrast or of relation. The blue sky, which spread out before the gaze the whole day through, must have made less impression than the splendid play of colors of the dawn and the evening glow; red and yellow fruits must have awakened stronger sensations than the everlasting green. "Even our peasants as a rule hardly show such admiration for the color of the sky and the meadow as we town-dwellers, who only seldom have occasion to enjoy in full draughts the blue and the green in nature, and whose color sense, through the strengthening rest in the colorless rays of the walls, the pavements and the papers, has acquired an increased receptivity."²⁸ [141]

The *law of relation* agrees with the principal proposition of nerve-physiology, that not a constant state, but only a change setting in with a certain suddenness, awakens the nerve process. The preceding state in the organism and in consciousness thus forms the background for the following one. The contrast effect in the narrower sense is only a particularly striking example of what to a certain degree does and must take place in every sensation. We overlook it so easily only partly when the degree of contrast is slight, partly when two sensations lie at so great a distance from each other that we do not put them together: we have then forgotten the intermediate terms. The sharply and definitely appearing sensations are built up out of a multitude of small sensations of difference.

The difference or relation can be either simultaneous or successive. But the successive relation is the more primitive. Simultaneous sensations have a tendency to fuse (especially in the domain of the senses of touch, hearing and smell); and since, on the other hand, our attention, like our bodily eye, does not keep still but wanders from point to point, a composite impression too will naturally be apprehended successively. There is the further point that successive apprehension is more distinct than simultaneous. Small differences of weight are more easily apprehended by successive weighing in the same hand than by simultaneous weighing in both hands; the temperature of two liquids is better compared by dipping the same hand successively into both than by dipping both hands simultaneously, each into its own liquid.²⁹ Very faint shadows are apprehended only when the source of light is moved.³⁰ Finally, it can be said outright to follow from

²⁸V. *Krenchel*: Farvesansen og dens Tilstand i Oldtiden [The Color Sense and Its Condition in Antiquity]. (Tidsskr. for popul. Fremst. af Naturvidenskaben. 1879). p. 153.

²⁹E. H. *Weber*: Tastsinn und Gemeingefühl. (Wagners physiol. Handwörterbuch. III, 2). p. 544 ff.

³⁰*Fechner*: Elemente der Psychophysik. I. p. 174.

the general physiology of the sensations that successive differences must be apprehended more easily and more strongly than simultaneously existing ones. "Supposing that the apprehension of difference rests directly on change in the nerve process or life process, [142] the intensity of the former must grow with the intensity of the latter... Of simultaneous states, however, one will not say that on account of the simultaneous difference they indicate a *change* in the life process of the same kind and degree as successive states; for they can do so only, first, when each of them forms a contrast to preceding states, that is, when they have a successive character, and secondly, when they act modifyingly on new changes. The simultaneous states are, in their mutual relation, to be regarded as *indirect* changes."³¹

No distinction, then, can be carried through between absolute and relative sensation, or between sensations and differences of sensation. This consequence of the psychology and physiology of sensation is, however, still disputed, even by an investigator like *Fechner*, who has done so much to maintain the importance of the law of relation. "Certainly," says Fechner (In *Sachen der Psychophysik*. p. 114 f.), "since we never have sensations of a certain kind or strength without preceding or simultaneous ones of another kind or strength, no experimental proof can be given that there would be sensations at all if these did not precede them or go along with them; only I find neither any theoretical reason nor any reason from experience that could forbid assuming it; therefore I believe — but the opposite view too can rest only on a belief — that if a child woke for the first time into an entirely uniform brightness, so that all other sense-impressions were as far as possible kept away (which, to be sure, is not completely possible), it would nevertheless sense the brightness of the light." Fechner has rightly seen that one must go back to the first sensation, that is, to the beginning of consciousness, in order to get beyond the law of relation. Yet his example does not denote any absolute beginning; for the child that wakes to the brightness of the light has had general sensations, sensations of movement and perhaps other sensations before it gets the sensation of light; this sensation thus gets those more indefinite sensations as its background, and thereby appears with a [143] characteristic mark different from all subsequent sensations of light. Perhaps it was even prepared by processes in the organ of sight awakened from within, so that the threshold had already been crossed. The impression of light, moreover (even if it were absolutely uniform in itself), does not act alike on the organ of sight in two successive moments; in the first moment there will set in a bewilderment which only gradually gives place to seeing proper; — but in this successive transition the law of relation asserts itself, since

³¹G. H. *Schneider*: Warum bemerken wir mässig bewegte Dinge leichter als ruhende? (*Vierteljahrsschr. für wissensch. Philos.* II.). p. 411.

the state at each moment is determined by the preceding one. So much is certain, that the more a psychical state approaches absolute unity, or rather simplicity, the more we approach the border of consciousness. The phenomena of mysticism, hypnotism and other abnormal states³² establish this. In consequence of the law of relation, then, we must assert that so long as consciousness still exists, every nuance and every rhythm cannot have vanished. (Cf. with this III, 7 and 9).

An important consequence of the law of relation is that it becomes impossible to distinguish between the matter and the form of our knowledge in the way Kant attempted. He started from the presupposition that “that in which the sensations are ordered and placed in certain forms cannot itself be sensation,” and therefore set up the forms, as the expression of the receptivity of consciousness, in contrast to the mere sensations as the matter received. But there is no entirely unformed matter in our consciousness, since there is no entirely pure sensation, uninfluenced by other sensations. There are [144] only differences of degree in this respect. Only by way of abstraction do we approach mere sensation, the unformed elements of matter. One of Kant’s most acute disciples, *Salomon Maimon*, had already shown this. “Sensation is a modification of the faculty of knowledge which sets in merely through being acted upon (without spontaneity); but this is a *mere idea, which we constantly approach by the diminution of consciousness*.”³³ Just as there is no absolute matter in our consciousness, so there is no absolute passivity either. The mutual interaction between the sensations is at the same time the most elementary activity of consciousness itself. We see already here the germ of what at a higher stage of development appears as memory and thought. *Elementary memory* shows itself in the preservation of earlier impressions, so that they can modify the later ones. An *elementary process of comparison* we have in the way in which every new sensation is referred to other simultaneous or preceding sensations and assessed by them.

The moment of activity that thus comes forward already in this elementary domain shows itself still more distinctly when we consider the significance of movement for sense-perception.

³²»Ich habe von einer intelligenten und mir nahe stehenden Kranken, welche zeitweise, in Folge einer durch trockenen Katarrh bei Emphysem bewirkten venösen Blutüberfüllung, beziehungsweiser Blutstauung im Gehirn, sehr benommen war und ganz weg zu sein schien, einen Zustand schildern hören, in den sie dabei wohl gerieth, und der sehr merkwürdig war. Es war dabei von einem Sich-in-sich-Unterscheiden, ja überhaupt von einem weitem Unterscheiden gar nicht die Rede, aber doch wusste, richtiger gesagt, fühlte oder empfand sich die Kranke irgendwie noch als etwas, das sie nicht näher definiren konnte. Es sei eben ein dunkles Etwas gewesen, ein Gefühl etwa von dunkler Existenz eines runden Klumpens, dem nichts gegenüber stand, nicht einmal das Nichts.« *Koch*: Ueber das Unterscheidungsvermögen. (Zeitschr. für Philosophie u. philos. Kritik. Neue Folge. 79ter Bd.). p. 78. — On analysis of this description one will easily see that the state was not entirely simple.

³³Versuch über die Transscendentalphilosophie. Berlin 1790. p. 168.

7. We have so far kept to the sensations, or the elements of sensation, as merely given and received. We must now correct the account given of the fundamental laws of sense-sensations by showing that we are not passive in the origin of the sense-sensation. In the first place, the organism is itself a little world with a certain independence over against the great one. From the activities in its own interior come (in the general sensations) important impressions. Nutrition, the circulation of the blood and respiration go their way to a certain degree independently of what goes on outside; and these inner processes awaken stronger or weaker sensations. In the second place, the organism does not wait for the impressions to come to it from the surrounding world, but makes movements that bring sensations with them. Even before actual sensations arise, movements are, in [145] the opinion of some, made as a consequence of the nerve centers' wealth of potential energy. These movements, psychologically regarded spontaneous or unmotivated, can themselves become the cause of peculiar sensations: sensations of energy and muscular sensations.

The change or transition that every act of consciousness presupposes or consists in can thus be of either an active or a passive nature. Impressions do not merely come to us, but we ourselves also produce impressions. And the active changes perhaps even precede the passive in time, since consciousness itself first arises and develops when the reflex and instinctive movements undergo an arrest. (Cf. IV, 4–6).

The sensations of movement are of different kinds. Psychologically regarded they fall into two main groups: sensation of effort (sensation of energy) and sensation of the state of the muscle at the moment. The first kind of sensation is bound up with an activity proceeding from the seat of conscious life. The second can have causes that are not connected with such an activity; for to it belong not only the sensation of fatigue, but also the sensation accompanying spasmodic contraction of muscles that are not subject to the influence of the higher nerve centers (cramp in the calves; colic; labor pains; according to Weber the contraction of the muscles of the empty stomach calls forth a part of the sensation we call hunger).

As regards the sensation of energy, it has been much disputed to what physiological process it is bound. Some have held that the motor impulse proceeding from the brain became an object of sensation already in the brain itself, since sensory fibers from the motor centers³⁴ immediately bring every movement initiated to consciousness. We should then have here a kind of sensations not due to impulses from outside the brain itself. Already Joh. Müller³⁵ adduced in favor of such a view that we can adjust and measure

³⁴Whether these are found in the cerebrum or only in the ganglia lying below it.

³⁵Handbuch der Physiologie. Koblenz 1840. II. p. 500.

out in advance the degree of effort that is necessary for a certain movement (lifting a weight, climbing a height, etc.), and that the certainty of powerlessness is different [146] from the sensation of fatigue. Even if one embraces this view, one need not deny the cooperation of impressions that derive from the muscle itself and its surroundings. From such impressions many derive all sensation of energy and muscular sensation, and this explanation has in the last few years gained support especially through Sachs's discovery of sensory nerves that run from the interior of the muscles to the higher centers.³⁶ On this view, then, the sensation of energy would arise only when word comes back from the muscle to the brain of the degree of contraction that has taken place; the innervation itself would not come to consciousness. *Lotze* had already shown the probability of this view by an apt observation. When a limb has gone to sleep, so that its nerve trunk has become unable to conduct owing to compression, we do indeed sense the limb as a passive and heavy mass, but cannot move it, and despite all effort of will we get no sensation of movement either.³⁷

Several facts show that the central sensation of energy is in any case imperfect and needs support and correction. It can, as pathological cases establish, be replaced by other senses. Patients who had lost the muscular sense in the arm could still carry objects when they had their eyes fixed on them; as soon as they turned their gaze away, they dropped what they were carrying.³⁸ The sensation of energy is, on the other hand, not enough in itself when it is not supported by other sensations. A person who had entirely lost the sense of touch asserted, while her eyes were closed, that she had clenched her hand, raised her arm and the like, without this being in fact the case. With open eyes, on the other hand, she could close and open her hand very *well* and surely, etc.³⁹ Finally, there are experiments showing that the sense of energy functions equally well whether [147] the muscle is set in motion by the will or by electrical stimulation; here, then, all reason for assuming a special central consciousness of innervation seems to disappear.⁴⁰

The sensation in itself tells us nothing about whether its physiological source is central or peripheral. Whichever of the two assumptions one embraces, it is at any rate certain that the sensation of energy is bound up with muscular movement. From the examples adduced we learn only that in our sensation we have no absolutely sure criterion of what goes on in reality. In elementary form we come up here against a

³⁶C. Sachs: Physiologische und anatomische Untersuchungen über die sensiblen Nerven der Muskeln. (Reichert und du Bois-Reymond: Archiv für Anatomie, Physiologie und wissensch. Medicin). 1874.

³⁷Medicinische Psychologie. p. 310.

³⁸Maudsley: Physiol. de l'esprit. Trad. de l'angl. par A. Herzen. p. 455.

³⁹Exner: Physiol. der Grosshirnrinde. (Hermann. II, 2). p. 248.

⁴⁰Cf. Sachs, op. cit., p. 185.

problem that will occupy us more fully later. We believe that we have an immediate sensation of how the impulse of the will flows over into the muscle; but if the theory of the peripheral origin of the sensation of energy is correct,⁴¹ it is not the impulse of the will itself, but its effect, of which we become conscious.

8. In almost all sense-sensations the movements of the organism play an important part, so that sensations of energy or muscular sensations prove on closer examination to be elements in states that on superficial estimate we take to be entirely simple. In taste the movement of the tongue is of importance; solid parts of the food are pressed against the hard palate and only thereby become tastable. The sensation of smell arises only when the air is drawn up through the nose. If one holds back the air, all sensation of smell ceases, even if one is in a strongly scented atmosphere. In hearing we move the body, or at any rate the head, until we find the position in which the sound is heard most strongly. In attentive listening the muscles of the eardrum seem to contract. It is, however, especially in sight and the sense of touch that movement is of great importance. The eye must be accommodated to the distance of the object, which happens by contraction of the fine muscles by which the surfaces of the lens are made more convex; this change takes place with a certain effort. In every definite position of the eye some of the eye [148] muscles are actively shortened, others passively stretched; with every position of the eye, therefore, a certain sensation of energy or muscular sensation is bound up. We move the eye, or perhaps the whole head, until the impression of light we want to apprehend falls on the place of distinctest vision (the yellow spot). The fineness of the sense of touch at the different places of the body stands in a definite relation to the mobility of the part of the body in question; it is (according to *Weber*) greatest on the tongue, lips and fingers, least on the chest and back. It is by the help of the movable parts that the sense of touch becomes of such great importance for living beings; by them active experimenting becomes possible. Touch and sight, the senses most bound up with movement, are at the same time our most important means of orientation with regard to the relations of the surrounding world.

We are thus not cast out as a purely passive prey to the impressions of the surrounding world. In the spontaneous and reflex movements that go on before the awakening of consciousness, an active nature already manifests itself. The impressions from outside, moreover, soon call forth movements that serve to hold them fast and follow them. An active turning toward the impression takes place here, as when the infant follows or seeks the light with its head or with its eyes.⁴² An involuntary seeking and adapting conditions

⁴¹Cf. the misgivings expressed by *Sigm. Mayer*: *Specielle Nervenphysiologie* (Hermann II, 1) p. 220–222.

⁴²A male child born in the seventh month of pregnancy I saw, on the second day of life, late in the

the character of the sense-sensations. How primitive this first form of attention is can be seen from the fact that a pigeon deprived of the cerebrum turned its head after a light that was moved. Attention here, then, still bears the stamp of reflex movement. A conscious concentration of attention presupposes a certain development of the ability to remember and to imagine independently of purely momentary impressions, and is therefore found only at a higher stage than the one we are considering here.

The sensation of attention is closely akin to the sensation of energy or muscular sensation, and is perhaps connected with the fact that a stronger or weaker contraction [149] takes place in the muscles belonging to the organ in question. "When we turn attention from one domain of sense to another, we have at the same time a definite sensation of the altered direction, not describable, but easily reproduced by anyone from experience, which we can designate as the sensation of a differently localized tension. We feel a tension directed forward in the eye, one directed sideways in the ears; it grows with the attention... In a corresponding way the sensation is localized with regard to the different sense organs when we want to smell, taste, feel something exactly."⁴³

Even direct, subjective sensation therefore refutes Condillac's theory that attention is only an exclusive sensation.⁴⁴ When a sensation wholly occupies us and is about to crowd out everything else in our consciousness, it also takes possession of our activity. An exclusive sensation thus presupposes attention; it is not one with it. Besides: how does a sensation become exclusive? Impressions can come to us from several sides at the same time. The eye, e.g., apprehends impressions from several points of light at the same time. Several senses can moreover act at once. As purely passive, sense-perception would yield a chaotic multiplicity of sensations in each single moment. But out of this multiplicity, in each little moment, a single one is taken out, which becomes the center. Reflexively and instinctively attention moves from the one impression to the other and endeavors to form an ordered series of sensations. As we have seen above, succession is apprehended more easily and earlier than simultaneity; this seems to be connected with the great importance of movement for sense-perception.⁴⁵ — The motive for attention's

evening in the twilight, repeatedly turn its head, which had been turned away from the window, toward the window and the light, even when its position was changed. It was undoubtedly seeking the light." *Kussmaul*: Untersuchungen über das Seelenleben des neugeborenen Menschen. p. 26. — Only later, according to *Preyer* (*Die Seele des Kindes*. p. 31) on the 23rd day of life, do the child's eyes follow the light without movement of the head.

⁴³*Fechner*: Elemente der Psychophysik. II. p. 475.

⁴⁴*Logique*. II, 7.

⁴⁵*Condillac*, despite his own definition, acknowledges the independence of the element of attention when he says (*loc. cit.*): «Lorsqu'une campagne s'offre à ma vue, je vois tout d'un premier coup d'œil, et je ne discerne rien encore. Pour démêler differens objets et me faire une idée distincte de leur forme et de leur situation, il faut que j'arrête *mes regards* au chacun d'eux... Ce regard est *une action par laquelle mon*

[150] leaving one impression and turning to another must be sought in a sensation of fatigue or a feeling of dullness. (Cf. above p. 138). In every such transition an *elementary choice* takes place.

B. Idea

1. In the interaction of sensations and in involuntary attention, the unity and activity of consciousness still manifest themselves only in a purely elementary way; we there stood before phenomena so simple that we were even short of psychological terms for them. We now pass beyond the domain of pure sensations, as we bring forward the fact that new impressions can be determined and altered not merely by simultaneous and immediately preceding sensations, but also by sensations that lie further back in time. This happens through the new impression's reawakening earlier sensations. The presupposition for this is that the impressions repeat themselves. A consciousness that passed *over* from A to B, from B to C, and so on, always to new impressions, would not get beyond the elementary stage described in the foregoing. But no consciousness is so constituted. The beings endowed with consciousness have their definite conditions of life, which not only make existence possible but also narrow the horizon, since the series of impressions cannot run on to infinity, but unfolds with a certain rhythm. Both within and outside the organism the same circumstances recur with more or less regularity. Life pulsates, and external nature likewise presents a rhythmical alternation. The most elementary real consciousness would be one in which there was found only a rhythmical alternation of pleasure and displeasure, with simple sensations of sense and of movement attached to it. When, now, the series of sensations had come, e.g., to D, A would enter again — at least as A₂, since the sensation perhaps did not have quite the same strength [151] or quality. It will then have a tendency to reproduce the same state as took place before B. It gains by this repetition, as it profits from the traces that A₁ has left behind. It is the law of exercise, valid for all organic life, that is at work here. Every function goes more easily through repetition and exercise. This holds especially for the functions of the nervous system; indeed, all practice of compound movements is really exercise of the central nervous system. The nerve-paths become, so to speak, more passable the more often they are traveled. The repeated sensation thereby acquires a mark of its own, and stands with a peculiar character over against other sensations, which are, so to speak, measured only against one another, but not against themselves.

œil *tend* à l'objet... par cette raison je lui donne le nom d'attention; et il m'est évident que cette *direction* de l'organe est toute la part que le corps peut avoir à l'attention.« —

Reproduction here does not yet become a real and definite recollection. For the reawakened state fuses immediately with the new one, does not come forward independently, but serves only to take from the new impression its stamp of strangeness. It is a kind of *instinctive or automatic recognition* that takes place here. On such recognition rests all perception (*Perception*), in so far as it is different from mere sensation. Perception thus arises through the fusion of new elements of consciousness with earlier ones of a similar kind. To perception there therefore belongs an involuntary classification, a referring of new impressions to older ones, an involuntary and primitive judgment, which of course is not at all worked out in conceptual fashion. "This is red!" means nothing more than: "this impression of color is of the same kind as a certain one received earlier". In so simple a form as we have here described it for the sake of clarity, perception hardly occurs in any accessible example. We can, on the other hand, think of the difference there is between the first and the second reading of a book, or the first and the second time one hears a piece of music. The partial tones in a sound are heard more easily the more vivid the memory is of the sensation of the tones taken singly; if this memory has quite disappeared, the single tones fuse entirely together for the ear.⁴⁶ When no more than 10 seconds passed between two sensations of weight, E. H. *Weber* could distinguish [152] between 29 and 30 loth; but after an interval of half a minute he could perceive only the difference between 24 and 30 loth. With the real sensations there unite the second time — without our becoming aware of it — the after-effects of the first time, and thereby everything stands clearer and more familiar in the immediate apprehension itself. Infants and operated persons born blind must therefore learn to see colors; the first impressions of color can in the nature of the case be referred only to other kinds of sensations, not to sensations of the same kind. Well known is the story of the blind man to whom one tried to describe the color scarlet, and who at last exclaimed: "It must be something like the blast of a trumpet!" As with every entirely new impression, the element of feeling will therefore also have the preponderance; only through repetition can knowledge awaken. Repetition is not the same as continuation, but presupposes an interruption, whereby the contrast with the intervening sensations (B, C, D) comes to play a part by bringing out the likeness (of A2 with A) — thus a kind of contrast-effect between likeness and difference.

In the account of the person born blind who was operated on by Franz, one sees clearly both the predominance of the element of feeling at the beginning and the significance of repetition and of difference. As soon as the sensitivity to light had decreased so much

⁴⁶*Helmholtz*: Die Lehre von den Tonempfindungen. Braunschweig 1863. p. 110.

that the patient could look at an object without feeling pain long enough to form an idea of it, the following experiment, among others, was made. Differently colored silk ribbons on a black ground were used to show him the colors. "The patient knew the various colors, with the exception of yellow and green, which he frequently confused, but which he distinguished when they were presented *simultaneously*. He could name each single color correctly when *several at once* were shown him. Gray pleased him most; red, orange and yellow were painful; violet and brown were not painful, but unpleasant; the last-named he called hideous."⁴⁷ — As with newborn children and operated persons born blind, so it goes with dogs that have been deprived of larger parts of the surface of the cerebrum, so that they have become "soul-blind" and "soul-deaf", i.e. have lost the [153] ability to combine the new sense-sensations with corresponding memory-images and thus to arrive at recognition and apprehension. (Cf. II, 4). *Hermann Munk* remarks on this: "One can say that by our intervention the dog has, as regards the sense of sight, been set back to the state of earliest youth, the state in which the puppy finds itself when its eyes have just been opened. Just as the puppy learns to see, i.e. to know what is seen, so a dog must *learn* anew to see, except that its developed power of movement, the advanced development of its other senses, etc., can shorten the time of learning. . . . It is oriented first, and very soon, about the things most important for its existence. One need only once or twice have pressed its head down into the pail, so that the water touched its muzzle, and in future it always seeks out the pail by itself when it is thirsty. It goes the same way with the food-bowl. Then it gradually learns to know people, and likewise the objects in its surroundings, the largest first, the smaller later."⁴⁸

The more frequently the repetition takes place, the more easily recognition proceeds; the more ready consciousness is for the new impressions. The rubrics have been laid out in order.

Just as we called the immediate interaction among sensations *elementary remembering and comparison*, so we may call the fusion of new sensations with the remains or traces of earlier sensations *bound remembering and comparison*, because the earlier states do not come forward anew freely and independently, side by side with the sensations aroused by momentary impressions. This bound remembering is the subjective element in perception. The more earlier elements of consciousness the new impression releases into fusion with itself, the greater the part played by subjective interpretation in relation to what is immediately given. In all perception there is an objective and a subjective pole; but the relation between them varies endlessly.

⁴⁷Philosophical Transactions. London 1841. I. p. 64.

⁴⁸H. Munk: Ueber die Functionen der Grosshirnrinde. Berlin 1881. p. 30. — Cf. above, II, 4d. e.

The Danish word *Iagttagelse* means both apprehension by means of sensation (*Perception*, *Wahrnehmung*) and voluntary, inquiring and attentive apprehension (*Beobachtung*). Here we take it in the first sense. Psychological usage, however, is unsteady. *Sibbern*, e.g., defines perception as immediate sensation, and so seems not to distinguish between sensation and perception. Yet he adds that perception is “an immediate *consequence* of the affection, or of the object’s stepping forth before, or coming up into or for, consciousness in consequence of the influence” (*Psychologie*. 1856. § 50). This “coming up into or for consciousness” is precisely what I designate by sensation, and according to *Sibbern* perception is only a *consequence* of this; so that even according to *Sibbern* sensation and perception do not coincide. While *Sibbern* almost merges perception with sensation, *Helmholtz* puts a greater opposition between the two expressions. “Sensations”, he says (*Die Lehre von den Tonempfindungen*. Braunschweig 1863. p. 101), “we call the impressions on our senses in so far as they come to our consciousness only as states of our body (specifically of our nervous apparatus); perceptions we call them in so far as we form from them the idea of external objects.” With this usage we should come to lack a word for the simple phenomenon that a sensation is recognized, or referred to others of the same kind. This simple psychological operation does not yet presuppose a consciousness of the source of the impression as something existing outside us. Perception in the sense of the recognition of a sensation applies just as well to the domain of internal as to that of external nature. — As regards the word impression, I use it in two senses, which the context in each particular place will suffice to keep apart. Either it designates the emerging sensation as something given, something received with preponderant passivity, without, however, containing any hint of how the sensation comes to be in our consciousness; for of that we know nothing whatever. Or it designates the physical force that affects the sense-organ, thus the first link in the chain of effects whose last link we know as sensation; it is in this sense that impression is set in opposition to sensation [154] (see, e.g., V A, 4).

2. What repeats itself and is thereby perceived is, now, not merely single impressions, but series of impressions in a certain order. The repetition then concerns also the order in which the impressions arise; this order will imprint itself, the elements of consciousness corresponding to the impressions will enter into a fixed connection, so that when the first (A) has been reawakened and fused with the new impression (A2), the elements connected with A (B, C, D) will also have a tendency to emerge again, and it then depends on whether this tendency is favored by the following series of impressions. In any case that older series will always be used more or less for the interpretation and judging of

the new series, and will be able to fuse with it in various degrees. The subjective element in perception thus consists not merely of the earlier elements of consciousness with which the new impressions agree, but also of such elements of consciousness as have entered into fixed succession with those earlier elements of consciousness. In this way the effects of a new impression can branch out widely on all sides in consciousness. It does not then stop, e.g., at the recognition of the color red; since this color has frequently occurred together with certain forms, certain sensations of touch, etc., it will have a tendency to draw this whole series forth after it. When, now, only the first link (A2) is nevertheless really given, the other links (b, c, d) will not be able fully to fuse with the given impression. They therefore stand, when they come forward at all, more freely and independently over against it. They become, in other words, *free memory-images or ideas*.

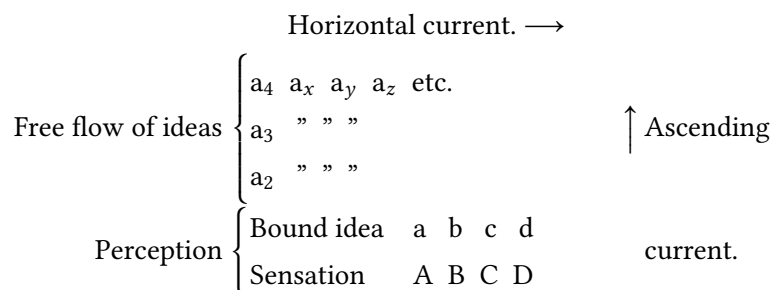
By another route, too, we can come from the bound remembering in perception to the free remembering in the idea. When A2 has reawakened A1, this latter will, in spite of its fusion with A2, have a tendency to reawaken other elements of consciousness (Ax Ay etc.), to which it stands in the same relation as A2 to itself. The presupposition is only that consciousness has had a sufficient number of similar experiences, and that these experiences, besides their agreement with one another, present marked differences [156] which hinder immediate fusion. This strengthening of the subjective pole in perception gradually removes it from the immediate impression, which now acts only as a motive that sets many hidden forces in motion. What thus gathers about the subjective pole will then be able to come forward as an independent element, not fused with, but in a certain opposition to, the immediately given impression. The perception of a round figure can, e.g., call up ideas of other round figures, so that these successively move further away from the one perceived.

At higher stages of development there forms by these routes an independent *circle of ideas*, a world of memory, which stands with a certain independence over against the momentary sensations and perceptions. In the reading of a book, e.g., the pure sensations are entirely subordinate; the paper and the black signs hardly come to our consciousness, but vanish before the ideas and feelings which they — at first through many intermediate links — set in motion.

In a consciousness that has reached the stage of development here indicated, two different currents take place. The one, which might be called the *ascending*, is called forth by the sensations we receive at each moment. However little we may perhaps heed these sensations, each of them nevertheless has a tendency to set certain definite series

of ideas in motion. I find myself, e.g., in a large hall, adorned with works of art, filled with talking people and resounding with music. Among these different impressions a struggle is waged over whose series of ideas is to win. Shall I hear the music, listen to the conversation, or lose myself in the contemplation of a painting? As my glance falls on a painting, a definite feeling is now awakened and a definite, coherent train of thought called forth. A current has then arisen which, in contrast to the ascending one, we may call the *horizontal*. But the other impressions storm on nonetheless, awaken faint beginnings of series of ideas, and it then depends on whether any of these series can become strong enough to initiate a new, dominant horizontal current. The following schema can illustrate this relation.

[157]



When we are to follow a piece of music, the point is to hinder the formation of a horizontal current, so that the mind can be open and fresh to receive the tone-impressions that ever press in anew. Rigoirist musicians will even have no title on a piece of music, so that it shall not call up a dominant series of images — In this example there now reigns a definite (rhythmical and melodic) connection among the immediate sensations (A, B, C ...). On a walk that is taken for the refreshment and rest of the mind, we rock ourselves, on the contrary, in disconnected and changing impressions (the light, the clouds, the trees, the people, etc.) and let each one initiate its ascending current until this is relieved by a new one. Such a state approaches the dream, which, as we know, is among other things often characterized by a constant change of motives of ideas, since the “ascending” currents do not permit any steady “horizontal” current to form. (See III, 8). The total contrast to this is reflection, self-absorption, where the many impressions vanish before a single, often narrow, clearly lit current of ideas.

3. Between the two currents, the free course of ideas and the series of real perceptions, and likewise between the two elements in perception, the sensation and the bound idea, there reigns an inverse relation. They strive to inhibit and displace one another. The more energy the one element lays claim to, the less, in the nature of the case, remains for the other. Both elements and both currents are present in every state of

consciousness, but with different strength. When they are equally strong, a rhythmical
 [158] alternation takes place between them, so that now the sense-sensation, now the idea has the preponderance.⁴⁹ They contend for attention; but an equilibrium between them would presuppose that they could both come before consciousness with equal clearness, which is impossible, since consciousness at each moment, like the place of clearest vision on the retina, concentrates itself on a single side. Wearied of the one current, consciousness seeks rest in the other. Here too there holds a law of contrast-effect.

Spencer finds in this opposition the explanation of the fact that we apprehend the tints of color in a landscape more freshly when we look at it with the head downward. "When objects are looked at in the usual way, consciousness is mostly occupied in interpreting the impressions made on the retina, is filled with the ideas aroused by these, which make up the knowledge of the objects as this or that; and so long as they are looked at in this way, it is impossible to prevent acts of recognition from taking place and claiming a part of the attention. But when the eyes are placed in an inverted position, or otherwise so placed that all the relations of form and arrangement by which the objects are recognized recede, the ideas connected with recognition do not emerge, and consciousness continues in so much the higher degree to be occupied with the sensations of color that compose the image received." (*Principles of Psychology*. II. p. 249).

The recognition of the composite nature of perception marks a great advance. The older psychology distinguished sharply between sense-perception and thinking as two entirely different functions (Plato), or conceived sense-perception as obscure thinking (Leibnitz), or thinking as transformed sense-perception (Condillac). *Kant* first showed the position of perception in relation to sense-sensation on the one side and thinking
 [159] proper on the other. Although he is still strongly influenced by the old, Platonic dualism between sense-perception and thinking,⁵⁰ he nevertheless has a clear eye for the fact that perception arises only through an activity whereby the given impression is taken up into consciousness ("apprehended") and worked over by it. It is by the forming power of consciousness ("the synthesis of the imagination") that a coherent image⁵¹ is formed out of the manifold elements given in the sensations. Through this forming power (*Kant's*

⁴⁹Cf. *Spencer*: *Principles of Psychology*. I p. 475. — II p. 251. — *Kant* had already seen the inverse relation between sense-sensation and perception (*Anthropologie* § 19). Later it has been stressed by *Fries* (*Psychische Anthropologie*. I p. 96. II p. 30) and *William Hamilton* (*Discussions on Philosophy*. p. 63). *Spencer* has presented the relation most exactly in an interesting chapter (vol. II. part. VI. chap. XVIII).

⁵⁰Even in his "Apologie der Sinnlichkeit" (*Anthropologie* § 8) he says of sense-perception that "in itself it belongs to the rabble, because it does not think". — The dualism manifests itself in his sharp distinction between lower and higher faculties of knowledge, especially between intuition and understanding.

⁵¹As to the definite forms (the forms of space and time) in which this image woven of sensations appears, reference must be made to the next section.

“productive imagination”) the “outermost ends” of our knowledge, sense-perception and understanding, hang together.⁵²

Only with the development of a coherent circle of ideas does the peculiarity of the life of consciousness come clearly forward. So long as we keep to the series of sensations, it looks as if consciousness were only the work of the moment. The connection is here elementary and externally mechanical. But in recognition and in the steady circle of ideas the unity of consciousness has a clearer expression. The stream of images that at each moment passes through consciousness is for it what the blood is for the organism: an inner medium that makes it independent of what the moments bring. If it stops, life ceases.

4. The question here naturally presses itself upon us, how the free course of ideas separates itself from the real perceptions. We cannot ascribe to consciousness any original knowledge of this difference. There is, to be sure, as a rule a difference of degree in strength between an idea of memory and a perception; but this difference of degree can be very small, indeed can disappear entirely. The first time a present impression calls up the image of earlier experiences, we cannot in any case know what the difference in strength means. It perhaps conditions a greater attention to the real impression than to [160] the memory-image; but the latter could for all that just as well stand as something real. It must be with incipient consciousness as with dream-consciousness: everything that presents itself is provisionally taken at face value, and there is no reason to refer the content of consciousness to two different spheres, the world of possibility and fantasy on the one side, the world of reality and perception on the other. On the contrary, this opposition is discovered only through experiences which are for a great part bitter. We must run our heads against reality a good many times before it dawns on us where its limits lie.

Let us start from such an exclusive sensation as *Condillac* spoke of. Every new element of consciousness that is to make itself felt beside it must share the energy and the interest. It will have a preponderant tendency to fuse with the given state when this is at all possible, and the least possible change will in any case take place. But from this it will in turn follow that the present sensation or perception will cast its strong and clear light over the less strong elements that emerge, that is, over the memory-images it awakens and which are firmly attached to it. Through the connection with the impression of reality by which they have been recalled, the ideas of memory will then also receive the stamp of reality, even if they do not appear so definitely as that impression. In this

⁵²Kritik der reinen Vernunft. 1 Udg. p. 119–125. (Kehrbachs Udgave p. 130–134).

way what is given is involuntarily supplemented and extended, so long as no definite contradicting experiences are known.

This power that the present impression exercises is due not merely to the strength of the sensation itself, but also to the active adherence of all the organs of sense and movement. The primitive impulse to movement is, in the nature of the case, directed especially in the direction that the given sensations indicate. These thereby acquire an effect that reaches far beyond what their strength and validity would in and of themselves bring with them. *Bain* has therefore rightly pointed to the original impulse to movement as an important element in the sanguineness with which the life of consciousness [161] begins.⁵³ We stand, as it were, ready to set off at the first signal that comes, and once we have set off, much is needed to change the direction. So long as we are not stopped, we suspect no danger. The animal follows the scent of its prey, and only when the trap snaps shut over it does it turn out that one and the same chief impression does not always bring with it the same series of circumstances.⁵⁴ The child puts everything it gets hold of into its mouth and begins to suck, but often it thereby suffers bitter disappointments. It learns that there are other things and relations than it had suspected in its tender philosophy. It is thrust back after its first sanguine assault upon reality.

There is greater reason here to speak of “unconscious inferences” than in the case of the immediate sense-sensations. Formulated logically, it is a positive inference in the second figure that incipient consciousness constantly draws, and by which it comes to have experiences that often confirm in painful fashion Aristotle’s warning against such a way of inferring. A is C, B is also C — from this it is inferred that A is C. But because a finger or a comforter has something in common with a nipple, it cannot for all that take the place of the latter in every respect. Not every smiling face announces food or [162] play soon to follow. Because the sheep gladly took the leaves that were held out to it, it does not follow that the bird does the same. A little boy even stated outright the claim

⁵³In his earlier presentations *Bain* laid the chief weight on spontaneous movement as the cause of belief (*belief*). Later he speaks more unclearly of an original credulity (*primitive credulity*), which is gradually checked by the contradiction of experience. See his interesting presentation in “Emotions and Will”. 3 ed. p. 511–518. — Even if his earlier view took too little account of the intellectual side, the later presentation nevertheless suffers from the essential defect that the “original credulity” receives no many-sided psychological explanation. Such an explanation might well lie in the power of attention to concentrate all the activity at its disposal upon its object.

⁵⁴Another example from the animal world I take from Lyell’s “Travels in North America” (Deutsch von Wolff. Halle 1846. p. 66). At the Lehigh Summit mines the coal is conveyed down in wagons which by their own weight run down rails on the mountainside. To draw the empty wagons up again, mules are used, which are then driven down again each evening, comfortably enough consuming their fodder on the way. When now these mules come into other service, they are very willing to draw heavy loads uphill, but refuse obstinately to take a wagon downhill, halting at the least slope, and there is no moving them from the spot.

that snow was sugar: “for snow is white, and sugar is white too”. The sight of the white calls up the memory of the sweet, and for the time being this union of white and sweet reigned in consciousness; experience now mercilessly tears this connection apart, the consequence of which is that the idea of the sweet acquires a peculiar stamp, withdraws as it were into a special corner of consciousness, whither a whole series of other ideas with which things have gone the same way are gradually shown (the idea of play in relation to the smiling face, of the ability to eat grass in relation to being an animal, etc.) — in other words: the first foundation is laid for the opposition between possibility and reality. Only now do the *free ideas* enter into a definite relation of opposition to sensation and perception. The original sanguineness is broken.

This now has an influence also on the immediate state itself under the impression of reality. The ideas awakened by that impression no longer fuse so closely with it. A certain *doubt* and unrest make themselves felt; the state is not so closed and uniform as at first. The idea of what has earlier been found to accompany or follow the given impression does not now become an object of full assurance. When the child has learned that satiety does not always follow hunger, there enters into its consciousness an opposition between the present feeling of displeasure and the idea of the satisfaction of the want. Formerly the two fused together; the transition from the one state to the other was continuous. Now, on the other hand, there is something vibrating, as it were, about the ideas connected with the present perception. They do not play the same part as that perception and do not produce the same effect. In so far as the connection nevertheless holds fast, hope stirs, since the ideas of memory still constantly have a tendency to approach reality, even if this tendency meets resistance. Only when this resistance becomes considerable is the sanguineness still lying in expectation also subdued at certain points, and *hope and memory* become opposites. The possibilities no longer lie over the real perception, waiting for it to give place to them, but stand as pale shadows with which the idea of [163] full presence no longer connects itself. Their tendency to take on the stamp of reality has been completely overcome, and with this the first step is often taken toward their complete disappearance. A higher stage of consciousness is required to preserve ideas that can no longer be realized and acquire practical significance.

In this process it is of course also of significance whether the perception (B) corresponding to the awakened idea used formerly to precede or to follow the present perception. If A used to precede B, it will, when B calls it up, have a tendency to stand as memory; if A used to follow B, it will, when called up again, more easily stand as expectation. One might think that every element of consciousness has two poles: by the

one it is attached to the preceding, by the other to the following element of consciousness (B thus hanging together with A by α , with C by γ). If it is now C that recalls B, B will place itself *before* it in the series, because in this case it is the pole γ that first emerges in consciousness. If, on the other hand, it is A that recalls B, B will place itself *after* A, because the pole α first emerges. The hungry (A) child is calmed by being taken up, because this is the introduction (α) to satiety (B); B places itself as of its own accord after A in the series.

This theory,⁵⁵ which of course uses figurative expressions, brings out elements of importance, but only of supporting, secondary significance. The chief cause of the distinction between hope and memory is the same as the cause of the definite opposition between perception and idea: experience and the disappointments it brings with it. After the ideas have been separated from the perceptions, there is now a new distinction to be practiced, that between memory and expectation, and it is supported, but not exclusively effected, by the definite place that the perceptions usually occupy in relation to one
[164] another. This is seen already from the fact that originally the forward pole of the idea always emerges first: consciousness is originally practical, strives forward, and notices only what can be means to its ends. Backward reproduction is always more unnatural to us, and at the lowest stage of consciousness it probably does not occur at all. Here the original sanguineness comes forward again; life strives forward, and it is only by being stopped that it is led to look back. As soon as it comes about that a perception (C) awakens the idea of its predecessor (B), this will nevertheless provisionally be placed not as predecessor but as successor, and expectation will arise. Only when experience has exercised its purifying influence can the difference between α and γ become of significance.

We see from this presentation how little it is in reality possible to carry through the abstraction of the elements of knowledge from the other kinds of elements of consciousness. Yet we here leave aside many questions that might be raised concerning the influence of feeling and will on the course of development here described; they belong in later sections of our inquiry.

That the free world of ideas is gradually separated off, through disappointed expectations, in opposition to reality and to the expectations attached to reality, has not always been clearly seen. According to *Plato* the transition from expectation (or impulse, Eros) to memory takes place through an evenly ascending development, in the course of

⁵⁵It is found in outline in *Robert Zimmerman*: Philos. Propädeutik. 3 Aufl. p. 223, worked out in *Taine*: De l'intelligence. Livre III. Chap. 7 & 9. Cf. also *James Sully*; Belief, its Varieties and its Conditions. (Sensation and Intuition. London 1874. p. 89 f).

which the primitive objects of impulse recede before ideal ideas, until expectation comes to rest in contemplation.⁵⁶ That modern psychology sees the matter otherwise is due to the more penetrating experience that the development of life everywhere proceeds through a struggle with opposites. — The examples we have used in the foregoing were taken from primitive and elementary stages.⁵⁷ But it will easily be seen that the same process repeats itself every time experience exercises its criticism of sanguine and [165] preconceived assumptions and hopes. Here is an ordeal by fire that all theoretical and practical endeavor must pass through. —

One element is still lacking to give memory its full clearness. We have distinguished between elementary remembering, bound remembering and free remembering, and have tried to show how free remembering emancipates itself from perception and expectation. With this free remembering there can further be connected a definite consciousness that the idea has its origin in an earlier time. The idea of time and its process of development we shall come to examine more closely only in the next section. Here it need only be pointed out that this definite referring of the idea to a certain point of time marks a chief difference between memory and free *fantasy*. Fantasy alters the relations and combinations of ideas and produces new orderings and groups, while memory in the narrower sense follows the series of real perceptions step by step. Here too the criticism of experience is the most essential cause of the distinction; for real memories have more practical significance than the products of fantasy, which cannot so easily lead to controlling experiments.

5. Already in the first sections of our inquiries we found in memory, and in the intimate way — standing alone within the circle of our experiences — in which various elements are there connected into unity, a typical expression of the nature of consciousness. My remembering falls in the present moment, but what I remember belongs in an earlier moment. In the unity which embraces and comprehends the various sensations and ideas and makes interaction between them possible lies the germ of the concept of the ego or the self. The difficulties that have been found in this concept derive for a great part from the fact that the ego has been sought as something entirely simple, which could therefore be given in a certain definite state, a certain definite sensation or idea. If [166] one starts from the presupposition that the ego itself must appear as a single element of

⁵⁶Plato's *Phaedrus* and *Symposium*.

⁵⁷Children do not understand the difference between poetry and history, fairy tale and reality. They know nothing as yet of criticism, and wonder that one does not know about everything. — Cf. *Egger*: *Sur le développement de l'intelligence et du langage chez les enfants*. Paris 1879. p. 60 f. (In his eighth year Felix was given a book of fairy tales and wondered that his parents seemed to doubt the truth of its contents. — He learned a little history, and it interested him very much; but he did not like there being anything that people doubted).

consciousness in opposition to other elements of consciousness, it is no wonder that one seeks in vain. Thus *Hume* says, in attempting to prove that the idea of the self contradicts experience: "If any sensation gives rise to the idea of self, it must continue invariably the same through the whole course of our lives, since the self is supposed to exist after that manner. But there is no constant and invariable sensation. Pain and pleasure, grief and joy, passions and sense-sensations succeed each other, and never all exist at the same time. The idea of the self cannot therefore be derived from any of these sensations, or from any other; and consequently there is no such idea. ... For my part, when I enter most intimately into what I call myself, I always stumble on some particular sensation of heat or cold, light or shade, love or hatred, pain or pleasure. I never can catch *myself* without a sensation,⁵⁸ and never can observe anything but the sensation."⁵⁹ In this *Hume* is quite certainly right. But he seeks in the wrong place. The nature of the ego manifests itself in the connection of sensations, ideas and feelings, and in the forms and laws of this connection, that is, in memory and comparison, from their purely elementary and automatic forms up to the highest and clearest forms they are able to take on. *Hume* cannot see the forest for the trees. His polemic is justified against the spiritualistic view that makes "the soul" a simple substance behind and separated from the single elements of consciousness. But he sins against psychological experience itself when he mockingly declares that, setting aside some few metaphysicians, the rest of mankind are nothing but bundles or collections of sensations (*perceptions*), which succeed each other with an inconceivable rapidity and are in a perpetual flux. He overlooks the inner bond [167] between these elements of consciousness, by which they become precisely elements in one and the same consciousness and not in several. And yet he would naturally have to come to ask what holds the elements of consciousness together and makes a "bundle" of them. There must surely be a uniting force here, but for *Hume* it receded entirely behind the single members of the bundle. He even went so far as to ascribe independent or substantial existence to the single perceptions. It therefore became for him an insoluble riddle how there can be a connection between the perceptions when each of them stands by itself as independent. "I must confess" — thus he ends the cited chapter of the "Treatise" — "that this difficulty is too hard for my understanding." It is also certain enough that once the single elements of consciousness have been set up as entirely independent, it is impossible to build a bridge between them. *Hume* of course did not overlook the actual connections between ideas; on the contrary, he lays

⁵⁸The *expression* *Hume* uses here (*perception*) designates with him both sensation (*impression*) and idea (*idea*).

⁵⁹Treatise on human nature. Vol. I. Part. IV. Sect. 6.

strong weight on them; he also shows how the easy transition that such connections bring about causes us to pass imperceptibly from the one idea to the other. It is, after all, precisely the actual connection in spite of the presupposed independence that occasions the great problem.

The more recent English school has not entirely worked its way out of this difficulty. *Stuart Mill* halts at the same paradox as Hume. "A memory of a sensation, even if it is not referred to any definite date, nevertheless involves the suggestion and belief that a sensation, of which it is a copy or representation, existed in the past; and an expectation involves the belief that a sensation or other state of consciousness, to which it points, will exist in the future.... We are faced with the alternative of either believing that the mind, or ego, is something different from any series of sensations⁶⁰ or possibilities of sensation, or of accepting the paradox that something which is supposed to be a mere series of sensations can be aware of itself as a series".⁶¹ Mill rightly stresses that memory [168] presupposes a self.⁶² But he is still, like Hume, caught in the idea of consciousness as a mere series or bundle, and therefore the connection and the unity stand for him as a paradox. When all psychological experiences point to connection, it is surely most reasonable to take this as fundamental, as *Kant* did in setting up the concept of synthesis as the fundamental one in the domain of the life of consciousness. If there is any substantiality in the spiritual domain, it must not be sought in the single elements, but in the unity that makes their mutual relation possible, through which each of them first is what it is, and which reveals itself most clearly in memory.

This is not to say that Hume's demand for constant elements of consciousness as the condition of the unity of the self is entirely unjustified. The synthesis or unity is in itself purely formal; it gets its individual and practical shape from the content it comprehends. If no sensations at all were repeated, no memory would take place either, and the life of consciousness would dissolve into momentary flashes. (Cf. V B, 1). Of a person who hastens from the one impression to the other, always seeking something new, we also say that he does not get time to collect himself or to be himself. There must be dominant sensations, ideas and feelings, if there is to come true unity and connection into consciousness and a peculiar self is to be formed. Through a sudden change, especially in the life of feeling, we can become strangers to ourselves. The fundamental feeling that

⁶⁰Mill's expression is "feelings", which is more comprehensive than the expression sensation in the sense in which I use this word.

⁶¹Examination of Sir William Hamilton's Philosophy. 2 ed. p. 212 f. — Cf. my work "Den engelske Filosofi i vor Tid" [English Philosophy in Our Time], p. 43 f.

⁶²See especially his notes to *James Mill's* "Analysis of the human mind". London 1869. I. p. 229. 339 f. — II. p. 174 f.

animates us is the real side of the ego, just as the synthetic unity of memory is the formal side. If the real connection is shaken, the *formal* one is only a weak bond that in the end is torn apart. An insane person in a state of partial derangement often describes his earlier state as insane, or declares himself dead, or speaks of himself as a third person. Total
 [169] change in the general sensation, in connection with hallucinations, seems to be the cause of this want of ability to recognize oneself.⁶³ The strongest feeling and the most coherent and comprehensive group of ideas act attractively and determiningly on everything else that emerges in consciousness. It is therefore no wonder that consciousness becomes a stranger to itself when they undergo a radical change. In the remarkable cases of so-called “double consciousness” two different existences are even lived, each with its own circle of experiences and memories. The patient can sometimes read and write in the one state, but not in the other. When the dream of one night, as is sometimes the case, is continued the next, there is already an approach to such a double existence. Yet there are hardly any examples of the continuity between such different existences in the same individual being entirely broken. A lady who suffered from “double consciousness” could nevertheless make herself familiar with the fact that there was a difference between her “old” and “new” state. In most examples one sees memories and feelings pass over from the one state to the other. A zealous abstainer (*teatotaler*), who in his somnambulistic state was brought to the conviction that he was drunk, nevertheless kept obstinately maintaining the impossibility of his having drunk “gin and water”. A young woman who had been on the point of drowning, and after being brought back to life had a fit after which she lost for a long time her memory, the power of speech and several of the senses, nevertheless still hung together by dim recollections with her earlier state; she could not bear to see water, not even in a painting, while on the other hand the presence
 [170] of a young man who was the object of her affection always exercised a calming influence on her.⁶⁴ — A *real* double consciousness would be the beginning of the dissolution of the life of consciousness.

⁶³See *Griesinger*: Die Pathologie und Therapie der psychischen Krankheiten. 2 Aufl. p. 334. 241. — Quite recently the French physician *Krishaber* has described cases where the power of reflection was entirely unimpaired, but where all inner and outer sensations took on another character. When the patient had grown used to the new way of sensing, he declared himself to be another; in the period of transition he doubted his own existence. Cf. *Revue philosophique*. I. p. 289 f.

⁶⁴The examples are found in *Brierre de Boismont*: Des Hallucinations. 3e éd. Paris 1862. p. 339 ff. *Carpenter*: Mental physiology, p. 459 ff. — A French medico-legal writer, *Fodéré*, even goes so far in holding fast the connection between the normal and the somnambulistic state that he maintains the imputability of sleepwalkers' actions: »Il me semble, qu'un homme qui aurait fait une mauvaise action durant son sommeil ne serait pas tout à fait excusable, puisque, d'après le plus grand nombre des observations, il n'aurait fait qu'exécuter les projets dont il se serait occupé durant la veille.« Cf. against this *Tardieu*: Etude médico-légale sur la folie. 2 éd. Paris 1880. p. 236 ff.

The very fact that there is something else in consciousness than the strongest feeling or mass of ideas shows again that the self is not exhausted by it, and that *Herbart* is not right when he lets the one group of ideas perceive (apperceive) the other. The ego is just as much in the weak as in the strong feeling and mass of ideas, and is determined just as much by the effect of the strong group on the weak as by that of the weak on the strong. The ego is precisely that which by its unity makes it possible for such an interaction among the elements of consciousness to take place.

Meanwhile we must grant *Hume* and *Stuart Mill* that there is a difficulty here; but it lies at another point than where they found it. The difficulty lies in the fact that memory itself, and with it the unity of the life of consciousness, develops successively and is never absolute.

We have, it is true, tried to find an analogy to memory already within the circle of mere sensations; but we saw that the more we approached the limit of consciousness, the more elementary the connection became, the more multiplicity gained the upper hand over unity. Could this not lead to the view, held by some physiologists (*Vierordt* and *Preyer*), that the unity of consciousness is not original? One of the most important laws of development is, after all, that we first have a growth out from scattered starting points, and that these sporadic beginnings only gradually unite into a common stream. Thus one might think that psychical life begins with scattered and independent sensations, [171] which are only gradually comprehended together and brought into mutual connection.⁶⁵ *Preyer* even goes so far as to speak of several different egos in the infant (a cerebrum-ego, a spinal-cord-ego, and besides these an ego for each of the central sense-organs), which later fuse together.⁶⁶ One cannot express more strongly the fragmentary and sporadic character of the earliest manifestations of the life of consciousness in comparison with the later ones. It is certainly also both useless and unjustified to assume a hidden unity where the experiences point in the direction of incoherence. Psychologically regarded, there is nothing else to answer to the question raised than that we here stand at a limit, where understanding fails. We have repeatedly pointed out that entirely single states are psychologically unintelligible. A flash of sensation that is not remembered is as if it did not exist; it may be a preparation for the clearer appearance of later flashes; but so could unconscious states. A coherent consciousness seems just as well able to form itself even if those sporadic sensations did not precede. Synthesis can never come about through a mere pushing together of single elements. It is precisely in this that spiritual connection differs from bodily connection.

⁶⁵*Vierordt*: *Physiol. des Kindesalters*, p. 157. 169.

⁶⁶*Die Seele des Kindes*, p. 368 f.

Later on, too, we meet the sporadic in the life of consciousness, and, as F. C. *Sibbern* has shown,⁶⁷ precisely in especially fruitful periods. Various promptings, ideas and impulses stir in chaotic fashion, especially in times of ferment and transition, and it is only through long work and reflection that order and harmony are brought into this material pressing forward from various sides. And the sporadic is never entirely overcome. The unity of consciousness is always a struggling unity. Life, after all, consists precisely in the constant taking up of new material that is to be worked over. But at such stages dispersion does not mean absolute separation and isolation. This is seen already from the fact that the sporadic and chaotic itself is *felt*, becomes an object of consciousness, precisely through its opposition to the connection reigning in other [172] regions and periods of the life of consciousness. This could not happen if there were not a unity of consciousness embracing both the sporadic and what has been worked together, any more than two colors contrast when each is apprehended by a consciousness of its own.

6. Sensations arise, as we have mentioned earlier (III, 1. 11–12), psychologically regarded, out of nothing. They are the primitive elements out of which the life of consciousness is woven, but whose explanation cannot be found within this web, if it can be found at all. Ideas are, in their simplest form, renewed or recalled sensations. But since we do after all distinguish between idea and continued sensation, so that there lies an interval between the original sensation and the idea in which it is renewed, the question arises whence the ideas come. Do they too rise out of a psychological nothing, or do they have an existence beneath consciousness? — What can be answered hypothetically to this question in connection with the general problem of the relation between the unconscious and the conscious, we already know (see III). We cannot avoid ascribing to ideas we have had a closer relation to our consciousness than to such as have never appeared in it. To this is added the remarkable circumstance that ideas, when they emerge again, may have undergone changes, may have been arranged in new groups or fused together. It is both the preservation of ideas, although they are not always in consciousness, and their working over, without consciousness taking part in it, that justify ascribing a positive validity to the concept of unconscious psychical activity, in spite of all the difficulties that are bound up with carrying through a law of the conservation of energy in the spiritual domain. When one speaks of traces, residua or dispositions that remain after sensations and ideas have vanished from consciousness, the assumption lying at the base is precisely that there must be in the spiritual domain an analogy to the conservation

⁶⁷Filosofisk Arkiv. p. 265.

of energy in material nature. The difference between material and spiritual nature is here this: whereas in the former one can point out potential energy as a definite given state of equilibrium, something corresponding is not possible in the spiritual domain. [173] Purely physiologically there is no real difficulty here, for it is in itself easy to understand that there can arise a brain process of the same kind as that which took place on an earlier occasion, and that this can happen all the more easily the more frequently it has been repeated and practiced. One can imagine that through repetition the molecules concerned are arranged in such a way that they are brought out of equilibrium with greater ease. On the other hand we lack any image to make clear what spiritual tension (potential energy) means.

The reason we here return to this consideration is the way in which the theory of the independence of ideas, criticized in the foregoing, has led to a conception of the preservation of the idea. Some psychologists (especially of *Herbart's* school⁶⁸) assume that the faithful and steady preservation of the idea is the rule, and forgetting the exception. The problem would thus not be remembering, but forgetting. Every idea is then assumed to have a tendency to self-preservation, to assert its existence in consciousness, a tendency that is checked only by other ideas coming forward with the same tendency. The single idea can therefore easily be displaced from consciousness by other ideas; but as soon as these vanish, it comes forward again of itself, without needing any help or any motive, just as a compressed watch-spring flies out at once as soon as the pressure is removed. A distinction is therefore made here between two kinds of reproduction: the immediate, by which an idea comes forward again by its own force as soon as the hindrances are cleared away, and the mediate, by which an idea is drawn forth again in consciousness by means of its connection with other ideas. An immediate reproduction takes place, e.g., after sleep, in that the ideas of the preceding day present themselves at once, as soon as the inhibiting influence of sleepiness has fallen away. Even if an idea seems to be entirely forgotten, one may not therefore regard it as entirely vanished; it lies below the threshold of consciousness and can on a given occasion come forward again. Memories that one did not oneself suspect one still possessed may arise when one sees again the surroundings in [174] which their content was lived through. In old age and shortly before death, memories of childhood can be awakened and long-forgotten ideas assert themselves. A great sorrow awakens what has vanished to a melancholy but clear memory. It is also known from a number of remarkable case histories how in a state of fever and other pathological states things and words are uttered which the patient in a healthy state would not be able to

⁶⁸Cf. however already *Fries: Neue Kritik*. (Heidelberg 1807). I. p. 96.

remember — which can only be explained by earlier overlooked ideas finding occasion, under the abnormal conditions, to press forward.⁶⁹

[175] That we naturally may not say of any idea whatever that it has entirely vanished from consciousness follows of itself. We do not know all the threads by which the single, apparently accidental and isolated idea can be attached to other elements of consciousness. The close connection between conscious and unconscious psychical life makes it possible that what stands as isolated, so long as we keep to clear consciousness, may have its subterranean connection. Precisely for this reason our self-knowledge is most often so purely empirical; we frequently discover an actual connection between certain ideas and feelings without being able to find the intermediate links. Thoughts emerge, feelings assert themselves, whose previous history we do not know, although it must after all be attached to the circle of our experiences. Here we can often discover something in ourselves that we did not suspect.

But it does not follow from this that the single ideas have a kind of individual immortality. The independence of the single idea is grounded only in its being a peculiar form of the activity of ideation, in its marking one of the paths that psychical activity enters upon under certain definite conditions. When the life of consciousness has gained new strength through sleep, it is quite natural that it should first and most easily enter upon the accustomed ways. Our habitual train of thought is for the life of our consciousness what the type is for the organism: the form in which development moves under usual conditions, and from which it lets itself be removed only with a certain resistance. The chief weight is thus to be laid not on the single ideas as such, but on the general activity of consciousness. No single idea is preserved as such and recalled except when the conditions of the course of the life of consciousness as a whole bring it

⁶⁹Some special examples are cited in *Mynster*: Om Hukommelsen [On Memory]. Kbhvn. 1849. p. 17 f. 21 f. — A young lady in a somnambulistic state maintained that a rare plant, whose Latin name she gave, and which was to be found in the botanical garden, would cure her. The plant was found there, but could not, for the rest, cure her. That she knew its name could only be explained by her having heard it in her childhood from her father, who was versed in natural history. — A peasant in an illness uttered some Greek words that he had heard many years before during a short stay at the parsonage. — Erasmus of Rotterdam tells of an Italian who spoke good German during a fit of raving brought on by a painful illness, while otherwise he did not understand this language. The only possible explanation is that in his youth he must have learned German, but later forgotten it. — A Venetian lady who had been brought up in French later forgot this language for Italian; but in her old age she could speak French, the language of her childhood. — See also *Carpenter*: Mental physiology. p. 436–439. — Through the taking of opium, too, long-forgotten recollections can suddenly emerge. De *Quincy* tells of this (En Opiumsæders Bekendelser [Confessions of an Opium-Eater]. Oversat af Hazeland. Kristiania 1878. p. 161): “The minutest incidents of childhood, or forgotten things of later years, could suddenly emerge. Remembered them in the ordinary sense I surely did not; for had I been told of them when waking, I should not have been able to acknowledge them as belonging to my past life. But placed as they stood before me in my dreams, like revelations, clothed in every little accompanying circumstance and every corresponding feeling, I recognized them at once.”

with them. There is here, as it were, an inner psychical world-order that bears the single manifestations of life. The single ideas may very well help to determine the further development of the life of consciousness without being themselves preserved. It is thus in reality about the conditions of the *preservation* of ideas that psychology must ask. The Herbartian psychology introduces an anarchy into the life of consciousness by ascribing to the single ideas an imperishable existence. Consciousness is not merely a stage on which the ideas wage their struggle for existence; it is itself at work in and through the single ideas.

Our spiritual states, our thoughts, feelings and resolutions are very far from being imperishable. It is precisely here that we have the bitterest experiences of the perishability [176] of everything that has developed, when, as the poet says, “even the paradises of fantasy, the spirit’s inner possession, blow away.” It must therefore be regarded as a victory when memory is preserved, and it becomes of interest to dwell on the conditions and the laws to which experience shows us this victory to be attached.

7. a. The *free* memory-image is different not merely from immediate perception and from the immediate after-effects of sensations (after-images, after-sounds, etc.), but also from certain ideas that emerge from within without any external occasion, the so-called hallucinations. Hallucinations present themselves, like perceptions, with the stamp of reality, and it can often be difficult, if not impossible, for the individual himself to separate them from real perceptions. Many differences of degree make themselves felt here. While the insane person firmly holds fast to the reality of his hallucinations, there are cases in which the individual knows clearly that what he sees is only phantoms. A man of whom *Bonnet*⁷⁰ tells saw before him from time to time, without the slightest external occasion, men, women, birds, carriages, buildings, etc. These figures moved, grew larger or smaller. Sometimes all the walls of his rooms seemed to him hung with landscape paintings. All this made just as vivid an impression on him as if the objects themselves had been present. And yet he did not take these apparitions for realities; in any case he always knew how “to correct his first judgments”. They were for him a play at which he was a spectator, without knowing what scene the next moment would bring. — By this *stamp* of involuntariness and externality (exteriority) the hallucination, like the *dream-image*, stands close to sense-perception. The dream-image is no doubt due in part to causes similar to those of the hallucination, namely the stimulation of the sensory brain-centers from without (especially by the altered state of the blood),⁷¹

⁷⁰Analytischer Versuch über die Seelenkräfte. Aus dem Französ. übers. von Schütz. 2ter Bd. Bremen und Leipzig 1771. Kap. 23. p. 68.

⁷¹Cf. *Wundt*: *Physiol. Psychol.* I. p. 178–181.

[177] thus an automatic irritation, by which the brain is affected in the way it is normally, and in the waking state, affected only by impressions from the surrounding world. Yet in dream-images the real impressions received in sleep play a very essential part (see III, 8), so that they are rather to be put alongside *illusions* than hallucinations. By illusion is understood a wrong subjective interpretation of an objective impression, as when a white towel in moonlight is taken for a figure clad in white, or bits of wreckage on a beach for human beings. According to the schema we used earlier, illusion is psychologically characterized by the bound idea gaining the upper hand over the element of sensation in perception (a over A, so that a draws b c d along with it).⁷²

The hallucination has this in common with the memory-image, that it can sometimes be called up at will. An English painter possessed the ability to imagine persons whom he was to paint, and who had sat for him only half an hour, with even more vivid colors and forms than in reality, so that he could paint from this idea. From time to time he raised his eyes from the painting to look at the imaginary person and compare the picture with it. It ended, to be sure, with his losing the ability to distinguish imaginary persons from real ones, and having to spend thirty years in an asylum for the insane. But the ability to call up images with the stamp of reality can also occur without any pathological character.⁷³ Goethe was thus able, with closed eyes and bowed head, to see a flower out of which ever new flowers unfolded, as long as he wished.

Although memory-images, like most hallucinations, can arise automatically, they are nevertheless for the most part called up by their connection with other ideas or with given sense-perceptions. While the hallucination therefore stands as something foreign, psychologically unmotivated, it is as a rule possible to trace the motives for the emergence of a memory-image in consciousness under the given circumstances. The chief forms under which this motivation appears, the laws of the association of ideas (*Ideassociation*), we shall go into more closely in the next paragraph (8).

As regards strength and vividness, there is only a difference of degree between hallucinations and memory-images. In this respect great individual differences make themselves felt.⁷⁴ While some people have very vivid and colored memory-images, with

⁷²An illusion on the basis of a hallucination is depicted by *Benevenuto Cellini* in his *Life* (2nd Book, 12th Chap. — Goethe's transl. I. p. 323). In his underground prison he had long ardently wished and prayed to see the sun, at least in a dream-vision. One morning he at last had a vision of the sun. But the disk of the sun soon changed into a golden disk, in the middle of which Christ on the cross, and a little later Mary with the Child, appeared as work in relief. As Goethe already remarks, he behaved toward the visionary sun as a plastic metalworker. Had his fantasy been accustomed to work in another direction, he would have got other things out of the same vision.

⁷³*Brierre de Boismont*: Des hallucinations. p. 27. 462.

⁷⁴Cf. *Fechner*: Elemente der Psychophysik. II. p. 469–491. — *Galton*: Statistics of mental imagery. (Mind 1880).

others these are colorless and indefinite in their outline. With some the memory-image approaches real perception or hallucination. They see things in the idea just as clear and vivid as they appear in reality, so that the memory of the sun can even have a dazzling effect. Others, on the contrary, declare that in memory they cannot see individual objects, but have only general and uncertain ideas of what was seen; their memories are not so distinct as to deserve the *name* of images. — Not only original endowment, but also age and practice are of significance here. Memory-images are most vivid in childhood and youth. A child said, after it had been absorbed in the memory of a person it was very fond of: “Now I have dreamed.” Poets, naturalists and travelers have more vivid memory-images than abstract men of science. There is on the whole an opposition between imagination and abstract thinking of the same kind as that between the free course of ideas and real perceptions, and as that between the two *elements* of perception (the sensation and the bound idea). (See V A, 2–3).

Those who do not possess individual and vivid memory-images may therefore very [179] well have a good and sure memory. They remember that they have lived through something, although in memory they do not see it before them. Very strong and uncompound sensations are no doubt all referred to remembering in this way. A cannon shot, a sudden blow or a flash of light one remembers more indirectly than directly. Even colorless and abstract memories can well be exact and fully sufficient to form the basis for the description of what was lived through and for the appraisal of it.

Some psychologists, such as *Bain* and *Lotze*, deny that ideas, like sensations, each have a certain strength of their own. They derive the so-called strength or vividness of ideas from the feeling and interest attached to them. The strongest and most vivid idea is thus the one connected with the greatest emotion.⁷⁵ Against this, however, speaks the fact that memory-images, without being especially attached to a feeling, can approach real perceptions in distinctness and firmness. The difference between the full individuality and splendor of color of some memory-images and the misty indefiniteness of others cannot be better designated than as a difference in strength and life. When *Lotze* says that the idea of a bright luster is no greater work of the activity of ideation than the idea of a dull glimmer, and that the idea of thunder requires no greater effort than the idea of a sound, he seems to confuse the idea that emerges involuntarily (if one will: the idea in itself) with the idea that one calls up and holds fast with voluntary attention. It can even at times cost greater effort to imagine a faint glimmer than a strong light. But we are speaking here of the idea merely as such, apart from the special difficulties to which its

⁷⁵*Bain* in a note to the 2nd edition of *James Mill's Analysis*. I. p. 116 ff. — *Lotze*: *Grundzüge der Psychologie*. 1881. p. 18 f.

arising and its being held fast may be subject.

[180] Just as illusion and the dream-image lie between sense-perception and hallucination, and just as very strong and vivid memory-images approach hallucination, so on the other side we have an intermediate link between sense-perception and the memory-image in what *Fechner* has called “memory-after-image”.⁷⁶ This is a memory-image that one calls up immediately after a sense-impression, while the after-effect is still making itself felt. Even those who otherwise do not have vivid and colored memory-images can produce them in this way, at least for a moment. Their power of memory thus needs, as it were, a helping hand from the real sensation. Here too many individual differences make themselves felt. With some the after-image itself, with others the concentration of attention on the after-image, plays the greatest part. *Fechner*, after looking at an object in ordinary daylight, got a complementary after-image; but on concentrating the activity of memory on it, there arose a memory-after-image with the object’s own color and without complementary after-effect. — An observation I have made shows that the after-image itself can be called up anew. One morning, without thinking further about it, I had looked at an oval photograph surrounded by a white border. When I closed my eyes, a light oval emerged with dark surroundings. This negative after-image vanished after a little, but a whitish patch of mist remained at the place in the field of vision of the closed eye where it had vanished, and when I concentrated my attention on this mist, the after-image arose again. This was thus really a memory-after-image of an after-image. The further the memory-image lies in time from the immediate sense-perception, the more difficult it is for it to take on a vivid character.

As regards the chief conditions of the preservation and arising of memory-images, there are especially three things to stress: the circumstances in which the original experiences were lived through, the circumstances of the recall, and finally the nature of the memories themselves.⁷⁷

[181] **b.** Since memory has its physiological expression in the organism’s ability to preserve the traces of impressions received, it follows of itself that the fresher and more energetic the life-process as a whole, the better will one also be able to learn, i.e. the more lasting and deeper will be the traces that sense-perceptions leave behind. Therefore childhood and youth are the proper time for learning, and what is learned then is preserved more easily than what is experienced at a later age. In old age, therefore, what one lived through in childhood is better remembered, while the experiences of later years and

⁷⁶Anfr. Skr. p. 491 ff.

⁷⁷Cf. as regards what follows: *Spencer*: Principles of Psychology. I. Part. II. chap. 5–6. — *Carpenter*: Mental physiology. p. 441 ff. — *Ribot*: Les maladies de la mémoire. Paris 1881.

of the last days pass into oblivion. "The old man's spectacles are ground for seeing near". His brain process lacks the energy to preserve the new impressions. This *quicker* dissolution of what was *later* acquired is a general physiological law (cf. also III, 4).

What is lived through and experienced in an especially vigorous and elated state is preserved better than what is appropriated in a slack and depressed state. The heightened vitality benefits also the newly received elements of consciousness. In cases of apoplexy and epilepsy the same relation sometimes shows itself as in old age: while older memories are preserved, the most recent are effaced. Exhausted by fatigue, one is *not* able to gather up material for memory.

Time and repetition are needed if memories are to be founded. What one has appropriated in haste as a rule also vanishes in haste. Actors who have learned a part at short notice do not remember it so well as one they have really studied. Drilling for examinations (*what* the English call *cramming*) does not lead to such thorough results as real study. Here belongs also the remarkable fact that in pathological weakening of memory the words first forgotten are those that express concrete and individual objects, while designations for abstract concepts and relations are better preserved. Most frequently, therefore, proper names, and nouns in general, are forgotten; later verbs, adjectives and pronouns. *Kussmaul*⁷⁸ derives this from the fact that, while we can easily imagine persons and things without words, abstract concepts and relations first gain firmness for our consciousness with the help of words. Far more numerous processes [182] and combinations in the cell-network of the cerebral cortex must therefore no doubt be needed to produce an abstract than to produce a concrete concept, so that the organic bonds that connect the abstract concept with its name are far more numerous than in the case of the concrete concept.

It might seem to conflict with this that one remembers so little from the very first years of childhood, when the brain must surely be freshest. Memories, after all, seldom go further back than the third or fourth year. *Preyer*, by way of explaining this, has aptly remarked that the experiences of the very first years of childhood are very different from those of the later years.⁷⁹ What the child learns in its first years — to hold itself upright, to walk, to talk — is like a closed course by itself; when it has gone through it, the way is paved to the great circle of experience common to all adults. There is thus lacking continuity and agreement between the experiences of the first years and the later ones, and therefore, as a rule, interest in keeping fresh the memory of those older experiences.

Yet one would also be able to point out other contributing causes. The first sense-

⁷⁸Die Störungen der Sprache. p. 163–165.

⁷⁹Die Seele des Kindes. p. 226.

perceptions still have a chaotic and sporadic stamp, are little ordered and organized; and a certain definiteness and order is, as will be developed later, a condition for the preservation of experiences in memory. In the consciousness of the infant the “ascending” currents reign, just as in dream-consciousness. Dreams too, after all, are seldom remembered. The cerebrum, to which the activity of memory is attached, plays on the whole a not very active part at this stage (see III, 4). The impressions have a greater tendency to discharge themselves in reflex movements than to deposit themselves as memories.

c. Just as a sound and fresh brain is required for gathering up material for memory, so it is also a condition for the recall of this material that there be sufficient energy in the organism, and especially in the brain. An English physician, Sir *Henry Holland*, through overexertion during a visit to the mines in the Harz, suddenly forgot his German, and it came back only when he had rested and refreshed himself. In an elated mood, and [183] in general in a strong emotion, memories emerge that are not at one’s service under ordinary circumstances (cf. 6). The taking of opium and similar means can have the same effect. In some way or other there must, besides, be something in the organic state and in the reigning fundamental mood that corresponds to the state and mood that were present when the experience was lived through; though at times precisely the contrast can here be of great significance.

It depends, besides, on what sensations are pressing in at the moment. The stronger these are, the more difficult it is for the memory-images to unfold (see 2). The more akin they are to the memory-images, the more they will hinder the latter’s free appearance. We can have the memory of colors or tones while we touch an object, but hardly the memory of red while we sense blue, or of other tones than those we are hearing at the moment.

d. Not everything that we live through is equally suited to being remembered. The more simple and uncompound it is, the more easily it vanishes. What is many-sidedly marked and articulated is better remembered. Feelings and moods are therefore remembered, as we shall see in a later connection, only with the help of the ideas to which they are attached, and we recall the fluctuations and transitions in emotion more easily than the feeling itself. The general sensations, the dimmest and least articulated of our sensations, cannot easily be recalled either. We can remember that we have felt hunger or thirst under these or those circumstances, but of the hunger or thirst itself no memory-image forms. Taste and smell, too, cannot really be remembered. The higher senses, on the other hand — touch, hearing and sight — yield clear and vivid

memory-images, and for seeing people the world of memory is no doubt even peopled chiefly by visual ideas. We remember relations better than we remember the single members that stand in the relation, the form better than the content. Of the forms, again, those are best remembered that are most distinctly marked. Thus the spatial relation is remembered better than the temporal relation, and this better than the general relation of opposition and difference. Because of the easier preservation of the spatial relation, it was made the basis in the so-called art of memory, as a frame (*memoria localis*) that could enclose and support all material for memory.

e. Memory can be of different kinds. — Already *Plato* and *Aristotle* distinguished [184] between memory and recollection, so that the former is the mere mechanical preservation of experience, the latter the ability freely to recall it. Aristotle remarks that some people are more gifted in the direction of memory, others in the direction of recollection. In Danish literature the same opposition has been set up between *Hukommelse* and *Erindring*⁸⁰ by *Treschow*, *Sibbern* and *S. Kierkegaard*. According to this usage, which is not observed in everyday speech, one can remember much without recollecting it. Recollection is more akin partly to poetic reproduction, partly to logical thinking, while memory keeps to the external connection in which what was lived through and learned happened to occur. Many features and details may have vanished from memory without recollection having suffered anything thereby.

The ability to preserve (memory) is with most people restricted to particular domains, where in return it can often express itself with a remarkable power. An Englishman by the name of *Baxton*, who had not even learned to write his name, could reckon the hardest problems quickly and surely. Once he undertook to multiply 39 figures by one another, and finished this calculation in two and a half months; the result consisted of 78 figures, and the whole operation took place in his head. A Danish memory artist, *Carl Otto*, could repeat a number consisting of 162 digits after merely having heard it named once. He let himself be examined in all directions within the number and could, e.g., say without hesitation what the 121st digit was. (See *O. Borchsenius: Fra Fyrreerne* [From the Forties]. I. p. 115). A man of science remembers best what belongs to his subject. The botanist *Brown* knew 25,000 names of plants, *Knuth* 20,000; when Knuth tried to impress more on his mind, others were in return lost.⁸¹ I have heard a man of science tell that when he learned English at an advanced age, it was at the expense of his French; “the bag was full”. Such phenomena confirm the assumption, often put forward, that a [185] certain sum of energy stands at the disposal of every individual. This is of course not

⁸⁰ A corresponding distinction is made in German between »Gedächtniss« and »Erinnerung«.

⁸¹ *du Bois-Reymond: Ueber die Uebung*. Berlin 1881. p. 28.

constant; it varies with age and practice; but it has a maximum. Within this maximum the point is to make the best possible distribution, and as regards memory this will be determined by the individual's essential interest and activity.

8. a. When we give ourselves over to the course of our ideas, the emerging images seem to come "of themselves", like sensations. We have in any case a feeling that we produce them as little as we produce the immediate impressions. Both we must take as they are and as they come. Especially when there is great difference and opposition between the emerging ideas and their predecessors, they stand as inexplicable. The effect here seems to stand in no relation to the cause. If there is added to this the suddenness with which ideas sometimes present themselves, it is no wonder that even many who assume no break in the causal series in the physical world nevertheless think that the psychical world is not subject to fixed laws. We have already seen (see III) that the world of consciousness is not a closed whole; it is understood only by assuming an interaction between conscious and unconscious activity. All the conditions for the arising of a psychical state are not given in the conscious life of ideas and feelings; unconscious dispositions and instincts, inherited or self-acquired, often play the most essential part, and the observer comes to know them only by their effects. The laws for the interaction of conscious ideas are therefore only guiding threads that can guide us in our attempts to understand the changes of the phenomena of consciousness — empirical rules by which we order the chaos of experiences. But in so far as we are able to find these rules confirmed by closer inquiry into psychological phenomena, the justified presupposition that the inquirer brings to inner as well as to outer nature is also confirmed. And in so far as a phenomenon cannot be satisfactorily explained by these rules, we conclude only that there must either be laws we do not know, or that the connection must be so intricate that it cannot be traced back to simple points of view.

[186]

b. The laws that hold for the connection of ideas we have already come upon in the foregoing. We have seen how a sensation fuses with the traces of earlier sensations (A with a), and how a fixed and repeated connection between sensations (*ABCD*) also brings with it a fixed connection between the corresponding ideas (*a b c d*), so that when one of them is reawakened it will have a tendency to draw the others after it. In the interaction of sensations, in the arising of sense-perception and in the liberation of memory-images, the same laws are at work that hold for the connection of free ideas. The difference is that here we know the single members as independent elements of consciousness before the connection. In the connection between sensations and bound ideas we know only the product. It is only analysis that demonstrates the composite character of sense-perception.

Yet associations of ideas too can become so fixed and steady that it is forgotten out of what elements they arose. Some of the greatest mysteries in the psychological respect are due to such ingrown associations of ideas, whose origin and history have been forgotten. The theory of *inseparable association* is the chief weapon of the more recent English school against the appeal to a priori and original forms and ideas of consciousness. This theory starts from the correct presupposition that what stands before us as a unity and as necessarily belonging together may nonetheless well have arisen through the fusion of independent elements. It therefore demands a more deep-going and longer continued psychological analysis than dogmatizing psychology engages in. This applies especially to connections that have not formed in the individual's own consciousness, but are an inheritance from earlier generations, either through innate organization or through tradition and language. That associations of ideas can be inherited is shown, e.g., by the fact that young hunting dogs that are out hunting for the first time and have never yet heard a shot rush forward eagerly when the first shot sounds, even if they see no game fall. The connection between the sound of a shot and the spring after prey must, in the time since the invention of gunpowder, have become hereditarily fixed in the dog's brain, or have become a so-called instinct. Birds in uninhabited lands are easier [187] to come near than birds that for generations have grown used to human enemies.⁸² The fundamental psychical phenomena can in such circumstances be found only with comprehensive regard to sociological and physiological conditions. But even then it is the simple psychological laws that furnish the principles for understanding the influence of the historical and physical circumstances pointed out upon the life of consciousness.

If a certain idea (a) is given, it will be able to call up other ideas in two directions. It will either awaken ideas of a similar kind and content to itself ($a_2 a_3 a_4 \dots$), or ideas whose objects usually occur in connection with its own object (a b c d corresponding to A B C D). The two chief rules are thus the *laws of similarity* and of *external connection* (of contact, of contiguity). What is akin in nature belongs together, and what occurs in connection belongs together for consciousness. We shall consider briefly the most essential relations that make themselves felt here.

I) *Association of ideas by similarity.*

1) The first relation to be named here is identity, *complete sameness*. It is the relation that makes itself felt in sense-perception, when the sensation awakens a bound idea and fuses with it. Here, then, lies the starting point for all the influence an impression

⁸²*Sigm. Exner*: Physiologie der Grosshirnrinde. (Hermann's Handbuch. II, 2). p. 282 f. — Cf. above III, 5. — See besides the chapter on instinct in Darwin's "Arternes Oprindelse" [The Origin of Species] and Spencer's Principles of Psychology. I. p. 423. 432 ff.

can exercise in consciousness. For whatever states and further-reaching effects it may otherwise be able to call forth, the first condition is still that an instinctive recognition take place, i.e. that the impression have a point of contact and a point of attachment in consciousness. This point of attachment (a) then becomes the starting point for further effects. Once recognition has taken place, various relations of similarity can then make themselves felt.

2) The next simplest connection by similarity consists in recognition's leading to the idea of an earlier experience of the *same* kind. If I have recognized a man, there naturally
[188] comes before me the image of him from the last time I saw him. At the portrait of a person we come to think of the person himself. A step further off lies already the idea of people who resemble the one I see, i.e. who have features in common with him, or who resemble him in character and fate (as when at Napoleon I come to think of Alexander and Caesar, or at the Bourbons of the Stuarts).

3) Through the last examples cited we are naturally led to remote relations of similarity, analogies, parallels, *metaphors* and allegories, which play a great part, especially at primitive stages of consciousness. We have already mentioned earlier that all the expressions of language for spiritual phenomena are transferred from analogous material experiences (see I, 3). There is, according to *Max Müller*, a period in the development of the race that may be called the mythological, because "all thoughts that went beyond the narrower horizon of our everyday life had to be expressed by metaphors, which had not yet become what they are for us, namely mere expressions made familiar by use and tradition, but were felt and understood half in their original, half in their modified character." From roots that meant to shine, to glitter, there were thus formed names for sun, moon, stars, the eyes of man, gold, silver, play, joy, happiness and love. *Max Müller* distinguishes between *radical* and *poetical metaphors*. It is a root metaphor when a root that means to shine is used to form expressions not only for sun and fire, but also for spring, the light of morning, the clearness of thought and the hymn rising with glad rejoicing. The poetical metaphor arises through the transfer of a word whose meaning is already fixed to designate another object, as when the sunbeams were called the hands and feet of the sun, the rain-clouds cows with heavy udders, the lightning an arrow or a serpent, etc.⁸³ — Even to this day the poet finds by this route a connection where the prosaic eye is blind. In the power of rhythm over the dancing couples he sees, e.g., a symbol of the world-law that the globes follow in the heavens (*Schiller*: Der Tanz). — In
[189] recording a series of associations of ideas from my own experience it has struck me how

⁸³*Max Müller*: Vorlesungen über die Wissenschaft der Sprache. Deutsch von Böttiger. II. p. 339 f.

easily and quickly the metaphorical meaning presented itself, even where it did not lie quite near (e.g. with the words swelling, fog, gild).

4) The idea of a *whole* calls up the idea of its *parts*, and conversely.⁸⁴ The whole and the parts have, after all, certain features in common, which furnish motives for a reproduction. — By looking at the embers in a stove one may, e.g., come to think of a smithy. — The mad Lear wants to comfort the blind Gloster in his misfortune; the comfort turns into a sermon, during which, in the manner of Puritan preachers, he holds his hat in his hand; from the hat he comes in turn to think of a possible stratagem: to cover the horses' hooves with felt, so as to fall silently upon his enemies.⁸⁵ — In complete dementia (*démence*), where an almost complete dissolution of the circle of ideas prevails, since not even fixed ideas prevail any longer, the association of ideas is determined only by the like sound of the words (assonances).⁸⁶ Even in the normal state every word awakens certain associations by its sound alone.⁸⁷

5) Ideas of *qualities and actions* call up the ideas of the *things or persons* that possess the qualities or perform the actions, and conversely. The word treason awakens the ideas of Ephialtes, Judas, etc. With the idea of a gambler there naturally comes forward the [190] idea of pallor (perhaps under the influence of reminiscences from novels).

6) The idea of the *cause* calls up the idea of the *effect*, and conversely. We refer this relation to the relation of similarity, since — as will appear in a later section — the concept of cause is, on closer analysis, traced back to the principle that like conditions bring like consequences, so that the effect is really only a continuation or prolongation of the nature of the cause. The planetary motions lead us naturally to think of the law of gravitation. — When Lear sees the mad Thomas, he asks at once: "Have his daughters brought him to this pass?" His idea goes back from phenomenon to cause — the only cause he can conceive of. — The idea of an end leads to the idea of the means, which is, after all, the cause of the end's realization.

⁸⁴Wundt refers this relation to association by external connection (Physiol. Psychol. II. p. 300). It seems to me that *Sibbern* with greater right refers it to association by similarity (Psychologi. 1856. p. 230).

⁸⁵

Be content; we came weeping
into the world, you know well — —
I will preach you a sermon; listen!
— — That hat-crown is good. — A splendid stratagem
to shoe a troop of cavalry horses with felt!
I will try it, though; and when I then
have come upon your sons-in-law from behind:
cut down! cut down!

(King Lear. Act. IV. Sc. 6).

⁸⁶Griesinger: Pathol. u. Therapie der psych. Krankheiten.

⁸⁷Tegnér: Språkets makt öfver tanken [The Power of Language over Thought]. Stockholm 1880. p. 25 ff.

II) *Association of ideas by external connection (contiguity).*

1) Sensations that constantly occur together also produce connected ideas. In this way *the idea of a single object* is formed. Certain visual ideas (yellow color), olfactory ideas, tactile ideas (smoothness) and gustatory ideas are combined into one idea (of an apple). What has occurred together in our experience in time and space will on the whole naturally occur together in our ideas, even if it does not shape itself into a closed whole. The idea of a man leads naturally to the idea of his house, his friends, etc., the idea of the gambler to the idea of the green table, the idea of a wreck to the idea of a beach, etc. There is, moreover, a natural connection between an event and the time and the place at which it took place.

2) A special kind of association by external connection is that between the *thing* and the *sign* of the thing. An emotion and its external expression are naturally associated in our ideas. If, e.g., one looks up the word terror in a dictionary, the imagination involuntarily forms the picture of a start, a blanching, or the like. The Greek word for flight later came to mean fear. Language is such a sign, originally perhaps called forth as [191] an involuntary outburst at the observation of a phenomenon that was characteristic or that affected the individual's weal and woe, but thereafter employed as a means of mutual understanding between individuals. "Just as the subject itself, in its state still poor in ideas and in self-determined forms of utterance, had a tendency to return with the same idea to the same utterance (sound), so the sound also became common to others by being repeated as a response, and through the impulse to imitation, which is active precisely at this stage of life and before the sound had yet acquired significative determinations; but thereby there was at once formed a connection, common to several, between this sound and the idea affecting the mind ... The means of communication and understanding had been found for this idea."⁸⁸ Why it is precisely sound that becomes the universal sign for all sensations and feelings can perhaps be explained by the fact that it commands the greatest wealth of nuances for expressing feelings. Already in the animal world the cry is at once the involuntary effect of what strongly affects the individual and a signal to other individuals. The cry of pain warns, and mating calls bring the sexes together. According to *Darwin*, the capacity to produce tones developed among man's ancestors in the course of sexual selection, and was thereby bound up with the very strongest emotions: burning love, jealousy and triumph. This capacity would thus have arisen before the capacity for articulate speech.⁸⁹ The various sense-sensations have this in

⁸⁸ *Madvig: Om Sprogets Væsen, Udvikling og Liv* [On the Nature, Development and Life of Language]. (Universitets-program). Kbhvn. 1842. p. 9 f.

⁸⁹ *Expression of the Emotions*. London 1872. p. 87.

common, that each in its own way affects feeling; it is no wonder, then, that they are all in the end translated into the language of feeling.

c. The two chief laws of the association of ideas, the law of similarity and the law of contiguity, it has sometimes been attempted to reduce to a single one. In English psychology the prevailing tendency for a time was to take external connection as the basis, and to explain all connection within consciousness by habitual connection in time and space. So already in the founder of English psychology, *Thomas Hobbes* (*Human Nature*, 1640), and later, e.g., in *James Mill*. This is the principle of the extreme “Association- [192] psychology”.⁹⁰ It conceives consciousness as a series or bundle of sensations and ideas, and so cannot consistently admit any connections other than those which rest on external coincidence. Association by similarity then becomes only a special case of association by external connection, since, namely, similar and related experiences will, by the nature of things, often occur simultaneously or in immediate succession. The artificiality of this view is easily seen. It conflicts, moreover, with the experience that it is often the relation of similarity itself which first brings us to put together things remote from one another in time and space. Caesar, Alexander and Napoleon certainly occur together readily in our imagining; but that is just because a comparison between them has so often been made. The steps of a mathematical proof follow one another with ease in our memory once thought has linked them together.

It would be more justified to trace all association of ideas back to the relation of similarity. Recognition is, as we have seen, a condition of sense-perception itself. It is further a presupposition for the relation of external connection being able to produce associations of ideas. For A to awaken the ideas of B, C, D, together with which it usually arises in consciousness, A must itself first, as it were, establish its identity. A must therefore call up a, and it is only then a which in turn brings b, c and d with it. The relation of similarity is thus the innermost kernel of every association of ideas; external connection can act only on the presupposition of the internal.

But because the relation of similarity thus underlies association by external connection, the latter does not lose its independent significance. Mere recognition and identification would not carry the life of ideas further. External connection is the acquisitive, material-absorbing principle; the material absorbed is then gradually ordered according to the principle of similarity. Thus two laws work together in every association [193] of ideas, a centrifugal and a centripetal tendency. The two laws or tendencies appear in

⁹⁰The expression association of ideas was introduced by *Locke*, who, however, designates by it only certain individual peculiarities, and does not, as *Hobbes* and the later school do, make the “association of ideas” the sole governing psychological principle.

different degrees in different individuals according to their nature and gifts. Some strive to gather a manifold material of ideas and observations; the aspiration of others, on the contrary, is to gain as simple and clear points of view as possible, for which reason they concentrate their interest on the common and the typical. Historical learning and specialization in the special sciences point in the one direction; mathematical and philosophical inquiry in the other. It is only artistic genius that is able to unite the particular and the typical at once.⁹¹

Yet there is a psychological standpoint from which the two laws can be united under one and the same fundamental law.⁹² However many different sensations and ideas may occur simultaneously or in immediate succession in our consciousness, they do not remain entirely separate; they work together to produce a total state or total activity. The various sense-sensations unite in the perception of a definite object; the various ideas are combined into more or less definite products of thought. Now when one of these sensations or ideas is renewed and draws the others along with it, what is really operative is the tendency to reproduce *the common state or activity* to which all the ideas belonged. The deepest ground of all association of ideas would thus be to be sought in the unity [194] with which every mental state and activity appears, and which stamps all the single elements with a common mark. Seen from this standpoint, then, the association between the parts and the whole would be the typical form of all association. — Schematically this can be expressed thus: $x = (a_x b_x c_x)$. a_x calls up b_x and c_x because they are parts of the same whole. Here similarity and external connection work inseparably together.

This consideration takes us a step further. For what gives each single mental state its peculiar character is not merely the particular combination of sensations and ideas, but it is above all the *mood* that prevails during it, determines it and is determined by it. As with immediate sensations, so also in the course of ideas, interest and the *attention* determined by interest play an essential part. We are never entirely passive in the associations of ideas, any more than in our sensations (A, 7–8). The connection between our ideas is therefore conditioned at every single moment not only by relations of similarity and connection, but also by the prevailing feeling. An approximation to pure, absolute validity of the stated laws of the association of ideas we have only in

⁹¹The two most eminent English psychologists of the present day (*Bain* and *Spencer*) conceive the two laws as working together in every association of ideas, but differ from each other in that *Spencer* regards the relation of similarity as the most deeply operative, the really decisive one, in relation to which every other motive of association is secondary, while *Bain*, conversely, takes association by external connection as the basis and regards similarity as a presupposition of subordinate importance. Cf. *Bain's* latest contribution: *Mind*. April 1881. p. 267.

⁹²Cf. *Fries*: *Neue Kritik der Vernunft*. I. p. 114 f. — *William Hamilton*: *Lectures on Metaphysics*. I. p. 233. (See “Den engelske Filosofi i vor Tid” [English Philosophy in Our Time]. p. 114).

moments when the mood is roughly neutral, or rather, when it is so adapted in a certain respect that in a given domain it contributes just its part to allowing the ideas to order themselves according to their kinship and connection. Originally, definite practical ends and interests weigh down the scale in favor of definite series of ideas. A kind of selection thus takes place among the possible ideas, and here the way stands open for many unconscious influences, which assert themselves even where we think we are following the stream of our thoughts and not directing it. Through interest we come back to impulse, instinct and temperament, hidden sources which are often known only through their effects. The connection between feeling and knowledge (see on this VI E) lies deeper than the connection of the ideas among themselves. If all associations of ideas rested on the individual's actual experiences and their combination according to the laws of the association of ideas, consciousness in each person would be far more clear and transparent than it actually is. This is not to be understood as if it were only the lack of intelligible connection that had to be explained by influence from the side of [195] feeling. Feeling can, on the other hand, also bring about a firmer connection than there would otherwise be. The strongest feeling is that with which man embraces his ideal or practical ends; this feeling drives him to seek the means for the realization of these ends, and thereby lays the foundation for a firm connection among a whole circle of ideas.⁹³

The closer investigation of the influence of feeling and will on knowledge must, however, be postponed to a following section (VI E). Here it need only be added that when association by *contrast* has sometimes been set up as a special case of the association of ideas, the phenomena belonging here can naturally be explained by the influence of feeling, in so far as they do not quite simply become intelligible by the laws of similarity or of connection. It is characteristic of the life of feeling to move through opposites; it is determined from first to last by the great contrast between pleasure and pain, which is stronger than any contrast-effect in the domain of sensations. A strong tension in one direction is readily followed by a slackening, if not by a tendency to turn interest in the opposite direction, just as the eye saturated with one color seeks the contrasting color. This could explain the urge to pass from the idea of light to the idea of darkness, from the idea of the great to the idea of the small. Yet it is not in all cases necessary to go back to feeling's propensity for states of contrast; often the explanation lies in a relation of

⁹³Hobbes already stressed the fixing and ordering power which the thought of an end exerts on the association of ideas. (Human nature. Chap. 4. Leviathan cap. 3). In more recent times William Hamilton has set up the law of interest (law of preference) as supplementing the laws of similarity and contiguity. (Cf. *Mansel*: Metaphysics. Edingburgh 1875. p. 241 f.). In Wundt's theory of apperception (the concentration of consciousness) as an act of will (Physiol. Psychol. II. p. 205 ff.) the law of interest is combined in an acute way with the concept of apperception set up by Leibnitz and Kant. Cf. already Fries: "Neue Kritik der Vernunft" and "Psychische Anthropologie".

[196] similarity or a relation of connection.⁹⁴ Contrasts fall under the same common concept, like two poles that move away from one and the same center, each in its own direction. Dwarf and giant both deviate from the ordinary standard. Further, the course of life brings with it that opposites succeed one another, touch one another and pass over into one another, as day succeeds night, joy sorrow. Here, then, external connection can act associatively.

d. Even if one cannot agree with those psychologists who hold that forgetting is what is difficult to explain, whereas remembering is self-explanatory, one cannot, conversely, maintain that ideas are forgotten merely because there are no special reasons for their being remembered. Forgetting can be just as great an art as remembering, which is also why Themistocles, on Simonides's offer to teach him the art of memory, is said to have answered that he would rather learn to forget: "for I remember even what I do not wish to remember, but I cannot forget what I wish to forget." What is indifferent or of little significance disappears as of itself; but precisely the painful ideas are as a rule bound up with such far-reaching experiences and circumstances that the involuntary course of ideas does not wash them away. Besides, there may lie in the individual's very nature a tendency to hold on with a certain stubbornness to the painful ideas. In other circumstances, naturally, the task may be precisely to forget ideas with which pleasure is connected. — Here we shall only briefly consider the ways in which an idea can be more or less completely displaced from consciousness. These are the *laws of forgetting* in contrast to those of remembering.

1) A refractory idea naturally cannot be counteracted quite directly.⁹⁵ The art of forgetting (or, as it has also been called, of abstraction) can consist only in displacing certain ideas with the help of others. Whoever wants to forget must seek strong and great [197] series of ideas in which his thought can be absorbed. Which he seeks (amusements or penances, work or fantasies) will depend on his character and on what is at his disposal in a mental respect. — The capacity for self-education depends in great part on whether one can practice the art of forgetting. Fortunately, as what follows shows, nature comes to the aid of art.

2) In many cases an idea is from the first very closely bound up with another of such strength and significance that the latter easily obscures or displaces the former. When one points at something for a small child and then withdraws one's hand, the child's eye for the most part follows the hand instead of fixing itself on the object. But when

⁹⁴Cf. *James Mill*: Analysis of the phenomena of the human mind. 2 ed. London 1869. I. p. 113 f.

⁹⁵"Oblivion is like the heather; it grows only of itself, and all the world's protection and care adds not an inch to its growth." J. P. *Jakobsen* (Marie Grubbe).

the object for one reason or another is able to attract attention, the child does not care where the hand goes. This is the history of all true education (see III, 4): authority leads the ward to a truth which gradually acquires independent validity and significance, so that the original authority is forgotten, just as the scaffolding is taken away when the house is built.

3) In other cases the first idea does not disappear entirely, but enters as a subordinate element into the one it awakens. When we read, the letter-signs awaken a multitude of ideas and feelings, but the signs themselves do not on that account disappear entirely from our consciousness. In transferred and figurative designations the original meaning often still glimmers in the background. When, e.g., there is talk of the fire of enthusiasm, there still remains, for those who are not all too accustomed to rhetoric, a gleam of the idea of real fire.

4) An idea can thus be displaced either by an entirely foreign idea, arbitrarily awakened from other regions of consciousness ($a < x$); or by an idea which it itself awakens ($a < b$); or it can enter as a subordinate element into the victorious idea (ba). But there is yet a fourth possibility. It can preserve itself in its independence over against the other idea to which it attaches itself, and yet enter into so intimate a connection with it that a new idea arises which is determined by both of those ideas, though neither of them is definitely noticed in the new idea. The schema for this is: $ab = c$. Here there thus shows itself a kind of psychical chemistry⁹⁶: in chemical composition, after all, the product has [198] other properties than the substances of which it is composed. The history of compound words furnishes examples of this. At the word *Vindu* [window] only the linguist now thinks that it comes from *vind-øje* [wind-eye]⁹⁷; but there must have been a time when the etymology asserted itself immediately in the idea. In the relation between the agglutinative and isolating languages on the one side and the inflectional languages on the other, this transition stands out clearly, if indeed *Max Müller* is right that agglutinative languages maintain the consciousness of their roots and do not let them be affected by phonetic weakening. “Even if they are no longer conscious of the original meaning of their endings, they still feel distinctly the difference between the significant root and the modifying elements. It is otherwise in the inflectional languages. There the various elements that are joined to the word-form can fuse with one another.”⁹⁸ — The idea of

⁹⁶*Hartley* first called attention to this phenomenon in his “Observations of man”. (Cf. “Den engelske Filosofi i vor Tid”. p. 19–20).

⁹⁷*Tegnér*: Om språkets makt öfver Tanken. p. 49. — According to *Molbech*, the first part of the word is rather the Old Swedish *wind*, loft, than *Vind*, *ventus*.

⁹⁸*Vorlesungen über die Sprache*. I. p. 281. — Whether the transition from agglutination or isolation to inflection is always an advance is another question. At one time it may be good to concentrate interest, at another to disperse it. In the expression “amavi” three ideas act as a collected mass; in the expression “I

distance seems to us to be simple and immediate, and yet it is certainly (as we shall see in the next section) the product of sensations which no longer assert themselves individually. The same holds of every apprehension of a whole that we have gained by laborious work with the details; the whole stands as the object of immediate intuition, of an “intuitive knowledge”, in which all discursive elements and processes have vanished. We seem to know the thing “in itself”, apart from all relations and comparisons. And yet in reality we know what the thing is only because we have investigated all the relations [199] in which it stands to other things, and compared it with the things with which it has points of similarity.

Such a psychical synthesis can take place more or less mechanically. It may be the linguistic expression that goes first and draws the thought after it, as seems to be the case in the development of language at the transition from agglutination to inflection.⁹⁹ But it may also be a fusion of the ideas themselves that takes place, and which is essentially conditioned by the need for unity and intimate connection among the elements of consciousness which lies in the very nature of consciousness. In the phenomena of psychical chemistry there appears — even more clearly than in the other forms of association of ideas — the general law of synthesis, which we have traced at every stage of the development of conscious life. Seen from this side, forgetting is a necessary condition of remembering and thinking. We could neither remember nor think if we could not abbreviate our experiences by eliminating intermediate links and forming new, compound unities. The material would overwhelm us. *Ribot* has therefore aptly remarked that remembering and forgetting work together like assimilation and disassimilation in the organic process of life.¹⁰⁰

9. The laws of the association of ideas must by the nature of the case be determinative for *the formation of compound ideas*. This is namely the place to point out that we have hitherto used the word idea with a certain indefiniteness. We introduced it originally (see 1–2) to designate reproduced sensations. But the sensations each have their definite content, which appears to us as something simple (see on this A, 2); the corresponding ideas must therefore also have a simple content (color, light, etc.). Yet in the foregoing we have also spoken of ideas of objects, persons and relations. The question then arises how such compound ideas are formed out of those single ideas.

[200] The idea of an individual object or an individual relation is formed chiefly by a

have loved” the emphasis will on each occasion be laid on one of the three ideas.

⁹⁹*Tegnér* p. 54: »Flexionen har sin väsentliga källa ej i tankan utan i *tungan*; den uppkommer ej därigenom att man tänker snabbare utan därigenom att man talar snabbare.«

¹⁰⁰*Les maladies de la mémoire*. p. 46.

combination of single ideas according to the law of external connection. (Cf. 8 b). When a series of single sensations has frequently arisen, the corresponding single ideas also enter into a firm connection, so firm that we are inclined to assume that there exists as its counterpart a certain mystical unity, the thing itself, distinct from those of its properties that are given us in our sensations and ideas. Yet such total *or* individual ideas, as we might call them to distinguish them from the single ideas, are by no means entirely fixed and unchangeable. They constantly change elements, and are not at one moment what they were at the preceding one. My idea of the table at which I sit is formed by a combination of various single ideas (a certain color, a definite degree of hardness, form, position, etc.). But each single time I have seen the table I have seen it in a different way; point of view, lighting, position have never been quite the same; now one element, now another (now color and hardness, now form and position) has, owing to special circumstances, attracted my attention. When I nevertheless ascribe to myself an idea of the table, and when I recognize it by means of this idea, then my idea, if it is to fit all the single experiences of the table, can contain only certain common points or features that recur every time I have the table before me. Only when we have observed an object or a relation a single time can we have an entirely individual image of it. If several observations have taken place, the differences will more or less assert themselves; we imagine, e.g., our home now in one lighting, now in another, and there can arise a certain struggle among the elements to determine the character of the idea, and this all the sooner, the richer and more many-sided our experiences are.

Since we are unable to hold one and the same memory-image fast for any length of time (which is why even what we call dwelling on a memory is in reality a continual letting go and recalling of the image), one and the same individual idea will have a [201] tendency to change its form every time it emerges anew. The elements that prevail at one moment will, by virtue of the law of contrast, give way to others that now have greater freshness. The question then is: do we really have ideas of individual objects and relations that are more than mere repetitions of the single experiences and observations? Do we have *common or typical ideas* that hold for all the single observations we have made?

The difficulty of this question lies in the fact that every real and complete idea of an object must give us the object with all its features and properties; now if these vary in each single experience, then I have not one individual idea of the object, but many. We therefore distinguish between a *concrete* individual idea (of *the table in this* lighting, seen from *this* side, etc.) and a *typical individual idea* (of this table as opposed to other tables).

The psychological difficulty that lies in this was noticed long ago. Yet it was not the individual ideas but the *general ideas* that drew attention to the difficulty. It was not observed that our ideas of single objects and relations which occur more than once in our experience are just as much abstract ideas as the general ideas proper. My idea of a table is related to the various tables I have seen as my idea of this particular table is related to my various observations of this table. The general idea arises by a continuation of the same process by which the typical individual idea is formed. Just as the concrete individual ideas struggle to determine the typical individual idea, so the various individual ideas struggle to determine the general idea. If I want to imagine a triangle, it becomes now an isosceles, now an equilateral triangle, etc. The common element is not enough to constitute a real idea. Just as little as one can eat fruit in general, but always only apples or pears, etc., just so little can one imagine fruit in general. — But then do we, psychologically speaking, really have general ideas?

[202] *Berkeley* was the first to call urgent attention to this psychological difficulty (in the Introduction to his “Principles of human knowledge”). The current theory of abstraction ascribes to us without further ado a faculty of “extracting” common properties and laws and of forming from them new, “abstract” ideas. Now *Berkeley* flatly denied that he had such ideas, although it was adduced by other philosophers, e.g. *Locke*, as one of man’s chief advantages over the animal that he was able to form them. “It is impossible for me,” says *Berkeley*, “to form the abstract idea of a motion without a moving body, a motion that is to be neither swift nor slow, curvilinear nor rectilinear, and the same holds with regard to every other abstract general idea.” Every idea concerns something quite individual and single. Only in the sense that we can let a concrete individual idea serve as an example or representative of a whole series of individual ideas are there typical or general ideas. The generality of an idea thus means nothing other than its usefulness for being set up as an example or representative.

In this *Berkeley* has certainly pointed to the decisive point. But it must still be asked by what psychological process an idea comes to be set up as a type in this way.

In the formation of the idea of a single object the law of external connection is at work. In the formation of the typical individual idea and of the general idea the law of similarity is at work. The single experiences and the single individuals have common, pervasive elements, which stand in contrast to those peculiar to each single experience, each single individual. Let $A = ax$, $B = bx$, $C = cx$ be three experiences; then, according to the law of similarity, the single idea x will stand differently before consciousness than a , b and c . The light of recognition falls more strongly on it. x will form a fixed and bright

kernel, around which a, b and c will move as dimmer and looser elements. The error of the old theory of abstraction was the assumption that x could be taken out and imagined by itself. x must be imagined either together with a or with b or with c; which of these possibilities occurs depends on the individual circumstances.

The simplest psychological explanation of this matter might seem to be the one that makes the interaction of the ideas themselves decisive. With repetition x will by the nature of the case come far more frequently than a, b and c, and will therefore attain greater strength and firmness. a, b and c, on the other hand, will inhibit and obscure one another. The common elements keep themselves up, while the struggle goes on among the other elements at the threshold of consciousness. In this way, then, the individual features would be worn off and only the general retained. — But this explanation would lead us back to the old theory of abstraction. The process would have to end with the common elements alone being left, which conflicts with the individual character of all distinct ideas. [203]

Often, to be sure, we behave so passively that the involuntary play of ideas itself takes care of the separation between the clear kernel and the dim periphery. Many confused and accidental ideas can approximately be said to be formed in this way. But for the most part we ourselves have an interest in the course the ideas take. We have a definite practical end whose means we seek, or a definite theoretical problem whose solution we seek. Our attention will therefore concentrate on the elements that point in the direction of what we seek, and will not take more account of the other elements than the apprehension of the image of the idea itself makes necessary. And this concentration will strengthen and bring out these elements, just as attention in general strengthens sensations and memories. This will not least be the case where the points of similarity are to be held fast in spite of strong differences. For a, b and c will also draw attention to themselves, and only when the goal and motive of the movement of thought is strong enough are obstructing associations of ideas kept away. In the idea ax we thus take special note of x. The general theorems about the triangle we prove by means of any triangle whatever, taking care not to take account of the special peculiarity of the triangle drawn (whether it is right-angled or obtuse-angled, etc.). General and typical ideas therefore exist in the sense that we are able to concentrate attention on certain elements of the individual idea, so that a weaker light falls on the other elements. We therefore form only those general ideas that we have an interest in forming. Such individual ideas as exhibit in a particularly striking way the elements decisive for us we choose as representatives. How comprehensive and correct the general idea becomes depends on the wealth of [204]

experiences and the exactness of comparison. The following example, in which child psychology and folk psychology meet, will be illuminating with regard to the way in which general ideas are involuntarily formed.

A small child who had learned the letters saw a book with Greek on one side, Latin on the other. It then exclaimed: “that is Greek, but that is letters.”¹⁰¹ Under the concept letter it thus included only the signs it knew. Here the general idea was closed and did not yet possess sufficient power of expansion. Peoples in their childhood have fared similarly. They regarded only their own language as real speech; the languages of foreign peoples seemed to them mumbling or stammering, childish babbling or animal bellowing. It is well known that the Greek word for all non-Hellenes, barbarians, really means those who speak roughly and inarticulately. Corresponding words and ideas have been shown among the ancient Indians, Hebrews and Arabs; indeed, even the Herero people (a Kaffir tribe) regard only their own language as fluent, all others as stammering.¹⁰² — It is only through closer acquaintance that the general idea is widened. Thus *Strabo* (in the 1st century A.D.) says, apropos of the word barbarian: “With extended intercourse and dealings with the barbarians it was seen that it was not a matter of *defects* in the [205] organs of speech, but of *peculiarities* of the languages.” (Quoted in Geiger.) — While linguistic differences thus stood out so strongly for the Greeks that they were long inclined to restrict the concept of language to their own language, they had, conversely, a striking tendency to rediscover their own gods in foreign deities — a tendency which, especially in the imperial age, “so completely dominated the pious believers that it let them perceive only what was really or apparently alike in the various religions, and made them completely blind to the sharpest and most glaring contrasts.”¹⁰³

Just as the naive consciousness, as regards the single sensations and observations, must pass through a purgatory of disappointments before it has learned by practice the difference between observation, memory and expectation (see 4), so too the involuntarily formed general ideas, which by the nature of the case are built on the nearest-lying cases, are purified and sifted by advancing experience. New experiences are always measured by those one has already had, and on which one has built one’s general idea. In the first example cited, the new phenomenon (the foreign language) was found unworthy of coming under the previously given general idea. In the second case one was precisely

¹⁰¹*Egger*: Observations et réflexions sur le développement de l’intelligence et du langage chez les enfants, Paris 1879. p. 22.

¹⁰²*L. Geiger*. Ursprung und Entwicklung der menschlichen Sprache und Vernunft. I. Stuttgart 1868. p. 300 ff. — In Greek authors ἄγλωσσος (speechless) occurs in the same meaning as ἑτερόγλωσσος (speaking another language). Thus, e.g., in *Sophocles*’s “Women of Trachis” ἄγλωσσος as the opposite of ἑλλην (v. 1060)! — The word ἄθεος (atheist) has a similar history.

¹⁰³*Friedländer*: Sittengeschichte Roms. 5te Aufl. Leipzig 1881. III. p. 501.

too quick to classify under given rubrics (likewise when the whale is counted among the fishes and the bat among the birds, or when the Bushman calls the European traveler's wagon "the white man's big animal"). — In both forms there appears the sanguine presupposition of being sufficiently equipped to understand and judge any phenomena whatever that may occur.

10. Not even such primitive general ideas as those just discussed could have formed and fixed themselves in consciousness without the help of *language*. To association by similarity between the homogeneous elements in observations and ideas there is thus added association by contiguity between the idea and its sign (8 b).

Even if language is from the first essentially a means of communicating finished ideas and not a means of forming and retaining ideas, and even if associations of ideas [206] can develop to a certain degree without the help of language, there nevertheless comes a point in mental development at which language becomes a necessary means of getting further.

As proof of the independence of the process of ideation one might appeal to the fact that infants have ideas and think before they learn language, — to the fact that the faculty of speech can be lost in morbid cases without the understanding having suffered at all, — and finally to the fact that certain languages lack expressions for ideas which nevertheless cannot have been lacking in the consciousness of the peoples concerned.

Even if one cannot ascribe a language to the infant, it nevertheless does not lack signs and marks that can serve as means of fixing ideas, by entering into association by contiguity with them. It is instinct and feeling that rule in the child at this stage. They find vent in involuntary outbursts, and these, when the child can notice them itself, will soon become a kind of sign, whereby the observation or idea that calls forth the feeling acquires a stamp that distinguishes it from others. The very first formation of ideas is thus not entirely without support, even if it is not the word that furnishes the support. — The child does begin its life with a cry, but this cry it cannot itself hear. It is not long, however, before it finds great pleasure in the tones it can itself produce. Out of such tones, at first involuntarily produced, the child forms its first language. Already when it is half a year old it is often able to answer in this language what is said to it. Then comes a period in which it uses and alters this instinctive language with greater freedom, so that imitation of the language it hears around it also asserts itself.¹⁰⁴ At this transitional stage language is entirely individual for each single child. Only at a third stage does the

¹⁰⁴This characterization I base on a record of a two-year-old child's vocabulary. — Cf. also *Egger* op. cit. p. 20. *Preyer*: Die Seele des Kindes. p. 259 ff. — In the list given by H. *Trier* in "Vor Ungdom" [Our Youth] 1881 p. 207, imitation of sounds (onomatopoeia) and of adults' speech stands out more than in my list.

[207] historically developed language gain power over the child, and now the association of ideas also goes more easily and more briskly.

It is no wonder that people can do without language after they have been able to make use of it for a whole life. The ideas must then naturally be able to be strong enough to do without the help of the symbol, at least for a time. They have, moreover, facial expressions and actions by which the ideas can be fixed. A remarkable example of the independence of ideas from the word, once they have been formed, is a 70-year-old woman whom *Kussmaul* mentions.¹⁰⁵ As a result of apoplexy she had lost the ability to speak and write; she could only utter exclamations when she became agitated. Yet she could read, and she showed great energy and intelligence in regaining the right of disposal over her fortune, which had been taken from her in the mistaken belief that she was insane.

When a language lacks an expression for an idea which the people must nevertheless be assumed to have had — as, e.g., the Samoyeds, according to *Castrén*, have no expression for gratitude — this can be explained in several ways. It may be that the feeling exists as an instinctive, involuntary stirring which has not become an object of consciousness; it is a trait of savage nations that they are children of the moment in their kindly as in their egoistic feelings. *Castrén* adds, too, that the Samoyed will go to his death for the one who has given him a present. “I have,” says the famous traveler, “also begun to believe that the word to thank was invented by someone who wanted to get off his obligations as cheaply as possible.” — More plausible, though, is perhaps the explanation that the Samoyeds have not formed quite the same general idea as we have, so that the elements of what we call gratitude are found expressed in the designation for a related virtue. We have, after all, e.g., no single word to express the Latin *pietas*.¹⁰⁶

[208] The primitive functions of cognition, sensation and observation, do not need the definite mark of language. Already the memory-image may need the word when it is not entirely fresh and vivid. The more memory and idea renew reality for us and approach intuition (cf. 7 a), the more independent of the word they become. But with paler and especially with typical and general ideas the word is an essential help. In some people thinking is to such a degree an inner speaking that they become hoarse from strenuous thinking. Thinking has therefore been called “a process of speech running its course imperceptibly in the central organs”, which stands in the same relation to

¹⁰⁵Die Störungen der Sprache. p. 22.

¹⁰⁶*Tegnér*: Språkets makt öfver tanken. p. 100–101. — The Hawaiians are thus said to have only one word for love, friendship, gratitude, benevolence and esteem; all these feelings they designate by the word *aloha*. *Max Müller*: Vorlesungen. II. p. 295.

actual speaking as the will stands to actual movement.¹⁰⁷ This agrees with the fact that our word-ideas consist chiefly in motor ideas, corresponding to an activity in the nerve centers governing the muscles of articulation, while the auditory and written form of words asserts itself only in a subordinate way, and in some people not at all.¹⁰⁸ When lively thinking leads to speaking aloud, this is thus all the more natural, since with every idea an innervation of the muscles of articulation seems to take place.

With the general idea the word becomes of special significance. For intuitability is here entirely lacking, and the very need for intuitability contains a temptation, since it can lead to confusing the typical with the individual. It is only with the help of the word that the bright and fixed kernel can avoid being mixed up with the dim and shifting periphery. The word is, as it were, a substitute for the impossible intuition, and it is only by misuse that it also replaces the general idea itself.

Deaf-mutes who have not learned finger-language nevertheless express objects and experiences in the most vivid and individual way by facial expressions and imitative movements. But precisely this individual and concrete character of their designations makes it impossible for them to form clear and definite general ideas; these are not properly freed from the individual ideas. Thus they designate meat and food by pointing to their own body, red by touching their lips. Thereby the lack of hearing and speech [209] cuts so deeply into the mental development of deaf-mutes.¹⁰⁹

The abbreviation or condensation of ideas which the general idea makes possible is secured and fixed by the word. If the word is a symbol, then the general idea is already a symbol too. The idea of an animal is a symbol for a series of ideas (mammal, bird, fish, etc.); the idea of multiplication is a symbol for a certain process of addition, the idea of a power a symbol for a certain process of multiplication, etc. The general idea takes a certain series of ideas for granted, but precisely thereby frees consciousness from now running through this series. While observation and the memory-image still have the character of intuition (*cognitio intuitiva*), the general idea is in itself blind and becomes seeing only with the help of the word (*cogitatio coeca vel etiam symbolica*).¹¹⁰ Once such a symbol has been gained, it in turn makes possible further-reaching general ideas and

¹⁰⁷L. Geiger op. cit. I. p. 58.

¹⁰⁸Stricker: Studien über die Sprachvorstellungen. Wien 1880. p. 20. 33.

¹⁰⁹Oehlwein: Die natürliche Zeichensprache der Taubstummen und ihre psychische Bedeutung. Weimar 1867. (Quoted in Preyer: Die Seele des Kindes. p. 406).

¹¹⁰Leibnitz. Meditationes de cognitione, veritate et ideis. (Op. philos. ed. Erdmann p. 80). — Leibnitz already called attention to the fact that with symbols what matters is not agreement between their original meaning and the content of the ideas, but the connection and relation of the symbolic signs among themselves. The single signs can be chosen arbitrarily, provided only their use and connection are not arbitrary. Dialogus de connexionione inter res et verba. (Op. Phil. p. 77).

combinations of them. We then get symbols for symbols, as in mathematics letter-signs for number-signs, or as paper money symbolizes silver and gold coin, which in turn symbolizes the immediate necessities of life.

11. Through the definite delimitation made possible by the word-symbol, the transition from the general idea to the *concept* in turn becomes possible. The concept states the fixed characteristic marks that hold for all the phenomena to which it is to be applied. The concept has in common with the general idea that it needs an individual idea, a concrete image, as its psychological representative; it differs from the general idea in
 [210] standing as the conclusion of a more or less arbitrary search for an exact determination of the universally valid characteristic marks within a given circle of phenomena. The psychological peculiarity of the concept as opposed to the mere general idea thus lies in this, that it now presents itself to consciousness as a task and an ideal to make its ideas precise and to ground them. General ideas are wild-growing concepts, which are now brought under the garden shears. This presupposes a turning point, a special awakening of attention and interest. And such a turning point presupposes in turn that so rich and provisionally ordered a content has been involuntarily formed in consciousness that consciousness comes to stand more freely over against its single elements. The naive, primitive consciousness feels no need to form concepts, but goes passively from disappointment to disappointment and forms only general ideas that act instinctively. The formation of concepts presupposes an ability to look away from what lies immediately at hand and to take account of *more remote elements* and relations. Here, then, there must be exercised on the one hand a self-control that keeps down the involuntarily intruding ideas when they are not relevant to the matter, and on the other hand a positive striving to call up and gather everything *that* belongs under the problem set.

It was *Socrates* who first put forward the demand for definite and conscious concept-formation, supported by examination of all the relevant cases. He thereby became the founder of logic. But his real aim was practical: definite concept-formation was necessary for self-knowledge and for knowledge of the difference between good and evil. The close connection between thinking and will stands out clearly here.

We stand here at the boundary between psychology and logic. Logic investigates the validity of concepts, the ideal demands made on concept-formation. Psychology has to do only with the psychical activities on which the formation of concepts rests. It therefore takes no account of the validity of concepts, but of the conditions of their origin in the soul. The transition from the general idea to the concept coincides with the transition from *association of ideas* to *thinking*. It is this transition, then, that must be

illuminated somewhat more closely.

a) Just as the will (in the narrower sense) presupposes the original spontaneous and reflex movement and can already be said to be present in it as a germ, so thinking is a special development and elaboration of an activity which already manifests itself in sensation, sense-perception and involuntary association of ideas. In the interaction and mutual combination of sensations, in automatic recognition and in the grouping of ideas according to the laws stated above (8), we have activities that can rightly be called elementary thinking. Here recognition and association by similarity especially stand out as what is central and most characteristic of conscious life. Thinking differs from the association of ideas only in that it tests, measures and makes precise the relation of similarity. It is thus a critical activity; it seeks to dam in the associative process, which runs its course involuntarily and often beyond all bounds. But the goal is the same, for thinking too seeks what is common; it wants to establish the valid principles and the prevailing laws. It only distinguishes more definitely than the association of ideas left to itself between identity, homogeneity and analogy. [211]

The motive for seeking an exact determination of the relation of similarity is here again a disappointment, a disturbance of the equilibrium of consciousness. The involuntarily produced associations by similarity must be shaken in their accustomed dominion by conflicting experiences. The original sanguine tendency (see 4) leads consciousness to cling to the first automatic generalizations. Throughout the whole history of human knowledge this tenacious holding on to general points of view once gained can be traced. Precisely because the need for unity and connection is, psychologically speaking, so deeply grounded in the nature of human consciousness, it is intelligible that it so often seizes and holds fast, with blind and ruthless energy, ideas that offer a simple and clear ordering of the state of consciousness hitherto given. That is why *Bacon*, the philosopher of empiricism, maintained that what the human understanding needed was not wings but leaden weights to check its leaps and its flight (Nov. Org. I, 104). Such leaden weights are experiences that conflict with what has hitherto been accepted. A king of Siam would not believe that water could take on solid form, so that one could walk and drive upon it. [212] If he was now convinced of it by observation, the equilibrium of his consciousness was disturbed, and the task was to reconcile the new phenomenon with what is otherwise known of water. He who for the first time is astonished that a stick put into water looks as if it were bent has thereby acquired a problem that can lead to the study of the laws of refraction. It is the differences that awaken doubt; they shake the previously given unity and connection; but thereby they also awaken and nourish. Difference and change are,

as has often been remarked, a fundamental condition of consciousness. But equilibrium is always sought to be restored anew by the striving to find a deeper-lying similarity and unity that can embrace the new phenomena as well as the old. Doubt and criticism are thus natural fruits of widened experience, and they are justified where they are borne by faith in a deeper-lying truth than the one hitherto found (*alte dubitat, qvi altius credit*). A great example is modern astronomy, which broke down the ancient and medieval world-view and set up in its place a magnificent unity in the universe, the same forces that act on the earth's surface guiding the course of the globes in infinite space.

b) The striving by which consciousness seeks to restore its equilibrium and connection, disturbed by new differences, is a striving for *grounding*. If a phenomenon A is given, and we, on going further, find B under conditions where we should expect A to repeat itself, astonishment is awakened, and we are satisfied only when we find in hitherto unknown or unnoticed phenomena (C) a bridge between A and B ($A = C = B$). If we are again puzzled at the transition from A to C, we should have to seek a D, such that $A = D = C$, etc. Grounding is found only where similarity overcomes and embraces the given differences.

The extreme "Association-psychology" (see 8 c), which would trace all association of ideas back to external connection, is therefore incapable of setting up a logic in the sense of a doctrine of the norms of thinking. And yet *Stuart Mill* founded his famous "inductive logic" on the theory that inference from particular to particular is the primitive form of [213] inference, from which all other has developed, and to which it in turn points back. The foundation on which logic rests thus becomes accidental, becomes a mere "association of ideas", a facticity that is not itself grounded. And yet Mill himself expressly demands more than mere associations of ideas.¹¹¹ — Only when, in accordance with what was developed above (8), one regards the relation of similarity as fundamental, can one find within psychology the salient point from which the logical principles and norms can develop. It is by similarity, by unity, that we measure everything else; similarity is what brings light and clarity into the chaos of ideas.¹¹² That inference from particular to particular does not in reality exist. When "a burnt child dreads the fire", i.e. when the sight of fire awakens the memory of the pain that touching fire has previously caused,

¹¹¹See especially III, 21, 1: There are many things which generate *belief besides evidence*. A mere strong association of ideas often causes a belief so intense as to be unshakeable by experience or argument. Evidence is not that which the *mind does or must* yield to, but that which it *ought* to yield to.

¹¹²In his later writings (the book on Hamilton's philosophy and the notes to James Mill) Stuart Mill has recognized association by similarity as the fundamental one; but this recognition did not lead him to any change in his logical theory. — Cf. further on this question: "Den engelske Filosofi i vor Tid." p. 26–27. 32. 37 f. (In the last-cited place, p. 37 line 3 from the bottom, should read: "Sammenfattelsen er Betingelsen for Sammenligning" [Comprehension is the condition of comparison].)

this is a justified train of thought only if this fire (c) which the child now sees is related to a certain possible sensation (x) as the fire (a) which the child once saw before is related to the pain (b) it felt that time.¹¹³ But whether this logical proportion is correct does not depend on external, accidental connection. It depends in the first place on whether c is essentially of the same kind as a (whether c is $= a_c$); only then must x also be of the same kind as b (be b_x). It is possible that the child is mistaken — that, e.g., what it starts back from is a shiny metal plate; but only continued and exact *comparison* can decide that. The principle of similarity or identity thus underlies it clearly enough.

Without going more closely into what it is the business of logic to set forth, we [214] must nevertheless briefly bring forward the forms of thought and principles that assert themselves when instinctive association is clarified into logical thinking in the school of experience and doubt. In other words, the task is to show how *logical necessity* develops out of merely psychological necessity.

Let the relation between A and B be the object of our consideration. Here there are four possible cases: 1) A can be connected with B (which we express by AB); 2) A can be found without B (which is expressed by $A (\div B)$); 3) B can be found without A (which is expressed by $B (\div A)$); 4) finally, both A and B can be absent from our experience (which is expressed by $\div A \div B$). — Now if, e.g., the first and the fourth case (AB and $\div A \div B$) are given, it follows that $A = B$. The presupposition naturally is that we have convinced ourselves that the 2nd and 3rd cases (A without B and B without A) do not occur in experience. — If the first, third and fourth cases, but not the second, occur in experience, then A must be AB, i.e. A presupposes B, but B does not necessarily presuppose A. In this way one infers, e.g., that man (A) is a rational being (AB), in such a way that the existence of other rational beings is not excluded, since reason (B) may also be ascribed to the animals (so that B can occur without A). He, on the other hand, who finds the third possibility excluded in experience, so that only AB and $\div A \div B$ are given, must infer that man is the only rational being. — If all four cases are given in experience, it is impossible to demonstrate any law, any definite relation between A and B.¹¹⁴

The experiences are each given by themselves, but the combination and working-up of what is given in experience is carried out according to a law which consciousness must acknowledge as its own law. *Induction* itself, by which it dawns on us what experience contains and signifies, presupposes the activity of consciousness just as much as *deduction*, which is the reverse process. If I know that $A = B$, I can infer from this that [215]

¹¹³Cf. *Spencer*: Princ. of psychology. II. p. 101 f.

¹¹⁴*Jevons*; Principles of Science. 2 ed. London and New York 1877. p. 134–136. Cf. G. *Boole*: An investigation of the laws of thought. London 1874. p. 81.

only the first and the fourth of the stated possibilities can be found in experience.

Here the difference stands out clearly between *logical thinking* and *the immediate course of ideas*. The former fixes its gaze on definite relations and seeks to exhaust and measure them. Every investigation aims at making a choice among the possibilities that the nature of the case offers. A logical compulsion takes the place of the psychological necessity by which the ideas order themselves according to the laws of the association of ideas. Psychologically speaking, every association of ideas is *possible* that accords with the laws of similarity and contiguity. Logically, the question is whether the transition from one idea to the other is *justified* and *necessary*. More than the four combinations just cited *cannot* be given as regards A and B; if they are *all* given, no necessary inference can be drawn; if certain definite combinations are given, then there also follows inevitably the path the train of thought must take if it is not to become entangled in self-contradiction. For logical necessity differs from merely psychological necessity in being a *demand* addressed to consciousness, an ideal that is set up but is not always fulfilled. Psychologically, inconsistency and self-contradiction can be just as intelligible as clear, consistent thinking.

Logic is a doctrine of art, psychology a doctrine of nature. But art grows out of nature and develops nature further. The demand that logic makes is grounded in the very nature of consciousness and of the activity of ideation. This shows itself when we consider what laws consciousness follows in the logical working-up of experience.

When we set up A B and A ($\div$ B) above as two different cases, the presupposition underlying this was that A cannot be anything but B or $\div$ B, but must be one of the two. Here we come to the logical principle which states that between two mutually exclusive determinations there is no third (*principium exclusi medii inter duo contradictoria*), the principle of exclusion or, as it has been called most recently, the *principle of duality*.

[216] This principle, on closer consideration, includes in turn two still simpler principles. — The first is the principle of *self-contradiction*: A cannot at once be B and $\div$ B, or: B is not $\div$ B. — This principle plays a part in all thinking. It is from the first contradiction that sets thought in motion, and all our reflection is in a way a hunt for contradictions, for eliminating the cases that conflict with the presuppositions (e.g. A ($\div$ B), when A = B). On the other hand, everything may be assumed that does not lead to self-contradiction, and everything must be assumed whose negation leads to self-contradiction. A discussion is a trial of strength in consciousness between the various associations of ideas, a systematically conducted process of elimination.¹¹⁵

¹¹⁵Cf. Spencer's essay "Sandhedens Kendemærke" (Test of truth) in "Mindre Afhandlinger af Herbert Spencer" [Minor Essays by Herbert Spencer]. Danish transl. Kbhvn. 1877. p. 148.

In the second place, the principle of duality presupposes that every thing is what it is. This is the principle of *identity* ($A = A$), the most abstract of all principles. Every concept must, wherever and whenever I apply it, be identical with itself, preserve its essence under all forms and relations. This most abstract and formal principle is the inner bond that makes transitions of thought and inferences possible. Whether the proposition $A = B$ holds we cannot always directly experience. They perhaps stand so far from each other that they cannot be immediately compared. The task then is to find a connecting middle term. If $A = C$ and $C = B$, then $A = B$ must also hold. For when $C = B$, B must everywhere be substitutable for C — thus also in $A = C$. The inference thus rests on substitution by virtue of identity.

Aristotelian logic, which has been dominant right up to the most recent times, did not yet grasp this simple and fundamental point of view. It took the relation between the whole and its parts as the basis of the theory of inference. When A is a part of C , and C is a part of B , then A must be a part of B — that was the model inference (inference in the first figure) of Aristotelian logic. But alongside this form of inference several others were set up, whose validity and scope were determined by laborious and intricate rules. [217] One had seen the difference and collision between psychology and logic, but had not found the connecting and common element. Its warnings against invalid inferences set up boundary posts, but these boundaries are not traced back to a clear and general principle. A single example, already hinted at above (4), will be illuminating at once with regard to the relation between psychology and logic and with regard to the relation between Aristotelian logic and the logic of identity.

For the immediate course of ideas any similarity and any contiguity is enough to bring about a connection between ideas. The mythologies, dreams, children's fancies, the deliria of the insane, and the transitions between the different meanings of one and the same word seem to show that there is no impossible association of ideas. Only under compulsion does the immediate course of ideas submit to logical control. When A is C and B is C , it promptly decides that A is B — as when the child asserts that snow is sugar. *Aristotle* (Analyt. pr. I, 5) had already pointed out that one cannot infer positively in the 2nd figure: from the fact that A and B both belong to the same whole (C), nothing can be inferred as to their mutual relation.¹¹⁶ Aristotelian logic first sets up a form of inference and afterwards determines the limit of its application. *The newer logic of identity*, on the

¹¹⁶Cf. S. Heegaard: Den formelle Logik systematisk fremstillet [Formal Logic Systematically Presented]. Kbhvn. 1871. p. 78 f. — Erasmus Montanus's inferences, by which he turns his mother into a stone and the parish clerk into a cock, are positive inferences in the 2nd figure. He has thus not studied his "Logica" properly.

other hand, whose principle was stated by *Leibnitz*, and which has been developed in our day by English investigators (*Boole* and *Jevons*), takes the principle of identity as its basis, whereby the unjustified inferences are cut off from the first: When $A = C$ and $B = C$, then $A = B$ as well. The various “figures” and the various warnings against their misuse thereby fall away. The concept snow and the concept white do not coincide, nor do the concepts sugar and white; substitution is therefore impossible. The newer logic thus rests on the principle of substitution, while the older logic rests on the principle of [218] subsumption. The logic of identity throws a clear light on the fundamental activity of thinking, and thereby on the inner connection between logic and psychology.

c) Thinking is distinguished from the immediate course of ideas not only by more exact generalization and by the striving for grounding, but also by seeking *new differences*. Just as little as thinking relies on the exactness of the agreements that immediately present themselves, just so little does it rest content with the completeness of the given differences. This is just as important a point in the process of thought as those cited hitherto; for logical inference itself presupposes a completeness in the given experience. A more comprehensive point of view would perhaps in many cases discover combinations that one had not regarded as given. What matters, then, is to widen the horizon of experience, to cast the net that seeks for facts out over deeper grounds. But the motive for seeking can lie only in the connection of what is already given. When this has gaps or discrepancies, the need for filling in and clearing up asserts itself. If the full explanation does not lie in the given elements, then other elements must be sought, through whose combination with the given ones the explanation can be reached. It is Herbart’s merit to have shown that the ground cannot be anything single, but always consists in a relation among several factors.¹¹⁷

The new nuances that are sought contribute to a closer determination of what is already given in consciousness. The given ideas are seen in new lights, new relations. This process has its expression in *judgment*. Psychologically speaking, it is more correct to conceive judgment as a nuancing, a making precise or an analysis of a given idea than as a combination of different independent concepts. In a judgment such as: The [219] man is good, the concepts “man” and “good” are not fetched from different regions of consciousness in order only then to be combined. There stands from the first before consciousness a total action, an event, to which both the man and his good sides or

¹¹⁷Cf. already Tschirnhausen’s criticism of *Spinoza*’s metaphysics. (My book “*Spinozas Liv og Lære*” [*Spinoza’s Life and Teaching*], p. 69 f.). — Later this consideration was urged by *Fries* against *Reinhold*, who opened the era of German speculation with the demand that all scientific propositions should be derived from one single principle. (See *Fries*’s work: “*Reinhold, Fichte und Schelling*”. Leipzig. 1803. p. 11 f.)

actions belong, and the mental operation carried out in the judgment is a division, a distinguishing bringing-out of the elements that were given in unresolved unity for the immediate apprehension. — *Hegel* has already aptly pointed to the etymology of the German word *Urtheil*, by which it is signified that the unity of the concept is the first and precedes the distinction between the parts of its content.¹¹⁸ In the course of the development of language we are pointed to a time when the root-words each by itself expressed an event, a relation, or in other words were “sentences in the larval state”, so that, e.g., the root that meant to strike designated the agent, the action and the patient all at once.¹¹⁹ Only when attention was directed to each single one of these parts of the fundamental idea could the need for a distinct designation of them arise. The form of judgment as a combination of subject and predicate is thus the expression of the fact that the unity lying in the fundamental idea is not to be abolished by the distinguishing bringing-out, but is to persist in and through it. — At later stages of the development of language, where — as in the modern European languages, especially in English — the inflectional language is in turn replaced by the isolating language¹²⁰, a *process* takes place of the same kind as that which expresses itself in every judgment. What was before comprised in one word is now distributed over two or three (*amavi* — *j’ai aimé*). — The child’s first words are likewise judgments or sentences “in the larval state”. *Vov-vov* [bow-wow] means both “the dog”, “there is a dog”, and “the dog barks”. Preyer’s child uttered its first judgment in the 23rd month. “The child was drinking milk that was too hot for it, quickly *put the cup down* and said loudly and firmly, with great seriousness: [220] ‘hot!’ This one word was to mean: ‘The drink was too hot.’ ”¹²¹

The thinking search for and setting-up of differences thus leads partly to new observations, partly to the analysis of ideas already given. By both paths the multiplicity of the content of consciousness is increased.

d) As regards material, thinking is dependent on the course of ideas, and it cannot produce anything new out of itself. It shapes and seeks, but only by the laws of the course of ideas itself do the new forms and ideas arise. While from this point of view thinking is only the correct association of ideas, the contrast between thinking and the association of ideas comes out in this, that the former has the stamp of *activity*, the latter that of *passivity*. Thinking is a matter of will, as already remarked above.

¹¹⁸Encyclopädie der philos. Wissenschaften. § 166. — Cf. also *Wundt*: Logik. I. Stuttgart 1880. p. 49 ff, 135 ff. Wundt develops this conception of judgment in a most instructive way.

¹¹⁹L. *Geiger*: Ursprung und Entwicklung der menschlichen Sprache und Vernunft. I. p. 205.

¹²⁰*Tegnér*: Om Språkets makt öfver tanken, p. 54 f. — Cf. *Wundt*: op. cit. p. 36 f.

¹²¹Die Seele des Kindes. p. 310. — Cf. p. 279: »Tuhl bedeutet: 1) Mein Stuhl fehlt; 2) Der Stuhl ist zerbrochen; 3) Ich möchte auf den Stuhl gehoben werden; 4) Hier ist ein Stuhl.«

Certain ideas are to be inhibited and others strengthened; this happens according to the laws of remembering and forgetting; but the active interest in the solution of definite problems pushes accidents aside. The willing that manifests itself in thinking is no lawless arbitrariness; it aims precisely at being freed from the accidental and becoming bound by a *valid law*.

Thinking is not content with what the involuntary streams of experiences and ideas bring along. It uses this as material or as motive. Many discoveries are owed, in a sense, to chance¹²², in so far as it was a coincidence of circumstances that awakened attention; but once this has happened, the role of chance ends. Thinking is not content with passive observation, but undertakes experiments and constructions by which it demands of experience definite answers to definite questions.

[221] That we ascribe the character of activity to thinking must not be understood as though we always consciously directed our ideas toward a certain object. Thinking, when it is free and energetic, runs its course with such *speed* that we forget ourselves. When we really think, we “fall into thought”, are overwhelmed by our thoughts. But will is not the same as self-reflection and self-constraint either. We can quite well forget ourselves entirely and yet guard against accidental and unjustified associations of ideas, avoid straying onto side-paths and into inconsistencies. Practice is needed for thinking as for all other activity. And before practice has been gained there is resistance to overcome, so that a special exertion must take place. In the effort to hold a train of thought fast the element of will comes out clearly; when the practiced thinker surrenders himself to the course of his thoughts, the element of will is present in a hidden way, unites its energy with the energy of the connection of thought itself, and comes out specially only when difficulties and resistance arise.

12. While the development from the concrete individual idea through the typical individual idea to the general idea and the concept leads from mere association of ideas to thinking proper, another process of development can take place which does not lead away from the concrete individual ideas, but whose result is precisely the formation of new concrete individual ideas. It is *the imagination or fantasy* that develops here, from the same root as thinking, but in another direction.

To understand this process of development it must be held fast that the concrete individual idea itself is of a compound nature, is the product of a psychical synthesis. Herein lies the possibility that the same elements can also be put together in other ways, occur in other combinations. If only some of the elements are exchanged for others, the

¹²² *Jevons*: Principles of Science. p. 529–532. (Accidental Discovery).

whole individual idea can acquire a different stamp. Now this happens involuntarily in all our remembering. Single features are effaced and replaced by others without our noticing it. Precisely when we hold the essential features fast with certainty, such a rearrangement and shifting of subordinate features can take place without attracting attention, if we have no opportunity to compare the memory-image with reality. The dream, both the dream proper and the waking dream, goes further still and transforms the very elements that rule and determine the individual idea, thus producing ideas of individual persons, things and events that have never presented themselves in our [222] experience. We can dream of persons who stand before us quite clearly and vividly, and whom we have nevertheless never seen.

This *free power of combination*¹²³ we make use of daily when we try to get into a matter whose full and entire connection we do not know. When we understand a half-sung ballad, we supplement the scattered elements given us into an individual whole. The *discoverer* of a new mechanism combines given elements, whose laws of action he knows, into a whole and a connection that has no complete counterpart in experience. The scientific discoverer likewise tumbles the elements of experience about, trying their possible combinations in order to find the one that agrees best with other experiences. During all this a series of individual ideas forms in his consciousness, which are successively rejected until that one presents itself which best accommodates and arranges the given elements. The admirable thing about scientific genius is the freedom of mind with which it is able to look away from experience and let various possibilities tumble about and unfold their consequences for the imagination — in order thereby to find a new reality that was not accessible to direct experience. *Kepler* stressed this freedom of mind as a significant trait in Copernicus's gifts.¹²⁴

This free combination has its psychological motive in the feeling of disharmony, of being overwhelmed by the chaos of the isolated elements of ideas. What drives thinking proper forward is outright self-contradiction, the spots on the bright surface of similarity and inner identity. Thinking is analytical; it seeks to make differences into disguised similarities. Free combination lets the differences stand as they are, but arranges the manifold elements in a new harmony. Where it works in the service of scientific [223] inquiry, it nevertheless always needs thinking proper as a corrective. The formal or abstract sciences (logic and mathematics) therefore form the basis and corrective for

¹²³It is hardly found at the lowest stage of human existence. The imagination of savages is reminiscent, not constructive. — *Spencer*: Princ. of Sociol. § 39 and 47.

¹²⁴Copernicus vir maximo ingenio et, quod in hoc exercitio magni momenti est, *animo libero*. (*Reuschle*: Kepler und die Astronomie. Frankfurt 1871. p. 119).

the real or concrete ones (natural science and history). The development of scientific knowledge aims at finding so deep-lying a unity and similarity that all differences can be accommodated within it, and on the other hand at ordering all differences in such clear and definite forms that the laws of similarity come forth as of themselves.

As long as fantasy stands in the service of knowledge, it is only a detour which the process of ideation takes where the straight path is inaccessible. Sometimes this detour cannot be walked to its end; it may nevertheless be justified when its general direction, the bends it makes, point toward an agreement with experience. Knowledge then ends in a *hypothesis*. But there is also a use of free combination in which such a return to given experience is not possible, and where what is striven for is precisely an independent and new creation, just as the dream involuntarily produces such a one. *Artistic fantasy* forms new individual ideas with more or less consciousness. It can bear most resemblance now to the involuntary combinations of the dream, now to the scientific search for the solution of a problem. In the poet of genius the combination often takes place unconsciously, so that an image finished in its main features suddenly emerges in consciousness. The joining together of the single elements has taken place below the threshold of consciousness. Or the combination is the work of the moment, as in improvisation, where the given theme acts through the movement it at once awakens in the life of ideas and feelings, just as in a lively conversation one remark draws the other after it. "L'improvisation est pour moi comme une conversation animée", Mme de Staël has Corinne say. After first having praised "the glory and happiness of Italy", which her hearers had given her as a subject, Corinne feels, from the expression on the stranger's face, an urge to strike up more serious tones and "to speak of happiness with less certainty" (Corinne II. c. 2). In contrast both to the instinctive creation that does [224] not know what it is doing and to the free unfolding of images according to the mood of the moment, there stands the energetic labor of forming a refractory material into a new form. The poet tolerates the self-contradictory and the incoherent just as little as the inquirer does. But for the poet the greatest contradiction lies in the elements' not shaping themselves into an individual figure. Every man receives his impressions, lives through his moods; but that they can be used as stones for the erection of a new building dawns on few. Most let the single experiences stand as they are. The poet's fantasy, on the other hand, goes on working with them until they shape themselves into an image, and *only* then does he feel free and clear. Just as *Newton* found an inner agreement between the motion of the moon and the fall of a stone, so the poet discovers a whole event, a piece of world history, where the ordinary eye sees only scattered features of

everyday life.

Even where the poet rises highest, he works with elements of experience and according to the general laws of the association of ideas. Every poet is in so far a realist. When *Lavater* wanted to depict the entry of the Antichrist, *Goethe* rediscovered in his description, step by step and point by point, the entry of the Electors into Frankfurt at the coronation of Joseph the Second, “so that even the tassels on the heads of the isabella-colored horses were not missing” (*Aus meinem Leben. Fünftes Buch*). It is well enough known how *Goethe* used real events and persons in his works. Of his “Tasso” he says himself (in a conversation with *Eckermann*): “I had Tasso’s life, and I had my own life; and in joining together two such strange figures with their peculiarities, the image of Tasso arose for me, to whom I opposed Antonio as a prosaic contrast, for whom I did not lack models either. Court, life and love relations were, moreover, to be found in Weimar just as well as in Ferrara, and I can rightly say of my presentation that it is bone of my bone and flesh of my flesh.” “I received,” he continues, “in my inner self impressions of a sensuous, lively, colorful and manifold kind, such as a lively imagination offered them to me; and as a poet I had nothing else to do but artistically to round off and develop such intuitions and impressions within me and, by a lively presentation, to let them come [225] forth in such a way that others received the same impressions when they heard or read my presentation.” *Schiller* got the idea for “Lied von der Glocke” through frequent visits to the bell foundry at Rudolstadt, and he did not neglect, as a basis for the composition, to acquaint himself with the technique. *Walter Scott*, on a visit to a Scottish mountain glen that he wanted to depict in one of his novels, noted down even the names of the most inconspicuous flowers that grew there. The main features of the Sistine Madonna have been rediscovered in a portrait of a Roman woman whom *Raphael* also painted.

The artist everywhere seeks the individual, while the inquirer seeks what is common to the individual phenomena. The concrete individual idea, which for the thinker is only an example or symbol, often a disturbing symbol, is for the artist the highest thing. But neither of them lets the given be what it is. All art differs from a mere impression or duplication of reality in bearing the stamp of the spirit whose work it is. This stamp does not rest on the artist’s personality coming forward immediately before the beholder; it rests on his having made a definite *choice*, conscious or unconscious to himself, of what he will give and how he will give it. The element of will thus comes forward here, just as in the psychology of thinking. And just as little as there does this element of will signify arbitrariness. It is the expression of the striving, led by a ruling feeling, that brings the artist to his *work*. How this feeling develops does not concern the psychology

of knowledge, any more than does the question on what the value of the work of art for the beholder depends.

In the choice of material and mode of treatment lies the idealistic element in all art, an element from which so-called realism only apparently frees itself. It is the open side of art, on which it stands in connection with the stirrings of spiritual life in other domains. However deep the movements that arise here, what matters artistically is what result they lead to. The poet may, e.g., be a thinker and a man of science; but poetically what counts is whether his work becomes more than abstract thoughts translated into verse — whether image, feeling and thought really fuse together from first to last. It is [226] an unpsychological idealism that would individualize what has lost its living connection with concrete, individual reality. On the other hand, in that idealistic element which is present in all artistic production there may lie the germ of an *idealization* of what is actually given. *Goethe* found it apt when *Merck* characterized his striving, in contrast to that of the brothers *Stollberg*, in the following way: “Your endeavor, your unshakeable direction, is to give the real a poetic form; the others seek to realize the so-called poetic, the imaginative, and that yields nothing but foolish stuff.” (Am meinem Leben. 18tes Buch). Through the poetic form a light can be cast on the real which the real does not have in itself or at all times, and the poet’s ruling feeling discovers a world-order in which his ideals find their satisfaction. In this way art seeks to spin further on a thread that can already be traced in nature; it becomes an ideal continuation of natural development. Thus, e.g., the visual arts complete the subordination of vegetative life to animal life which appears clearly on comparing higher and lower forms of organic life.¹²⁵ Precisely through its idealism art comes anew into kinship with nature; for nature too makes its choice (qualitative and sexual selection), directly or indirectly, whereby the transition takes place from lower to higher forms. A conception of artistic striving that would restrict it to giving a mosaic of single observations of real things would come into conflict with nature itself, which is constantly engaged in “idealizing”, inasmuch as forms and characteristic marks are, in the struggle for life, always stamped more distinctively, in close connection with the whole sphere within which the struggle is waged.

¹²⁵J. C. *Schiødte*: Det vegetative og det animale i den dyriske og menneskelige Form [The Vegetative and the Animal in the Animal and Human Form]. (Nordisk tidsskrift, udg. af den Letterstedtske Forening. 1878. p. 345).

C. Perception of Time and Space

1. Already in the preliminary characterization of the life of consciousness it was mentioned that psychical phenomena appear in *the form of time*. Change, transition, alternation — and an inner connection through all alternation — these were the most important characteristic marks of consciousness. But in this the form of time is already given. Psychology must therefore stop at it as something originally given, a psychological ultimate, which is implied in all phenomena of consciousness and which cannot itself become an object of explanation. [227]

It is another matter when we speak of the *idea of time*, the idea of the relation of time. This idea, like every other, has its psychological history of development. Psychical states may go on succeeding one another without there therefore arising in consciousness an idea of this succession. The more consciousness approaches to being a series of different, mutually independent sensations and ideas — which, as mentioned earlier, is the same as an approach to the dissolution of the life of consciousness — the less possibility there is of an idea of time. The moments change, but each moment lays full claim to consciousness, without there being energy to spare for what lies before or what comes after. Not only in its dissolution, but also in its beginning, consciousness approaches to being a series (cf. B, 5), which is why the idea of time can hardly be traced in the child before the third year. We have seen in an earlier connection how expectation and free recollection only gradually arise as special states in the struggle of experiences with the exclusive sensation and the original sanguine disposition (B, 4). This account of the development of memory we now complete by adding an important moment in its nature, namely the referring of the content of memory to definite points of time.

The simplest form of consciousness would be one in which two states (a and b) succeeded each other without intermediate links. Now so long as a and b each by itself occupy consciousness while they are present, no idea of time can arise; a is forgotten when b arises, and conversely. There must be something which so encompasses a and b that their alternation can appear as a different filling-out of one and the same schema. This common encompassing element can be nothing other than a sensation or feeling which remains constant while a and b alternate, and which therefore furnishes the fixed background in contrast to which the change, the succession, can stand out clearly. Besides the alternating a and b we must therefore have a third, relatively unchanging element x, which can make an effect of contrast possible. We are reminded here that the unity of the self, of consciousness, is borne not only by the formal connection and interaction among everything in consciousness, but also by a dominant feeling (B, 5). [228]

This fundamental feeling, which in large part, and in the case of the lowest conscious beings exclusively, is determined by the general or vital sensations, is thus a necessary presupposition for the rise of the idea of time.

2. The idea of time gains greater clearness when intermediate links are inserted between a and b, so that consciousness, in order to get from a to b, must pass through m and n, constantly with x as the background for the whole series of alternating states. The presupposition is, however, that m and n do not themselves appear with such great strength and interest that they obscure a and b, but that a and b continue to stand as the main points. Let a, for example, be the sensation of hunger, b the sensation of satiety, so that m and n designate the means by which hunger can be replaced by satiety (the sight of prey, capture, etc.). A fixed connection a-m-n-b will thereby be formed, and a rhythmic alternation will take place, with which consciousness gradually becomes familiar and which it easily surveys. The more manifold the states become with higher development, the more necessary it is that there be certain points which constantly recur, like elevated places in the succession, from which the rest can be surveyed and measured.

For the idea of time presupposes two things: 1) *consciousness of the change*, the succession; this arises through contrast with a constant sensation; — 2) an order which makes *a certain measuring and grouping* possible in the series of changes; this arises through sensations which recur according to a rhythmic law.

A single constant sensation or feeling would not make an idea of time possible. The more we become absorbed in a single thought, the more we are, as it were, removed from time, which is also why the mystics call eternity a “standing now.” On the other hand, [229] the mere succession of sensations would not make an idea of time possible either; there would be lacking something that could lead to surveying and measuring the succession.

The more rhythmic series form themselves, and the more practice consciousness gains in surveying them, the more clearly does the idea of the time-series stand out, in a certain contrast to the sensations that fill it. Just as many parts of time as are filled by the series a-m-n-b may under other circumstances be filled by the series c-p-q-d, and so on. With this the conditions are present for the formation of a general idea of time. The limits of the time-series are pushed ever farther out, both backward and forward, the further experience and practice in handling the idea of time advance. We try in vain to imagine a first beginning and a last end. Homer is right that rational beings are those that “look before and after”; we do this involuntarily, even when we speak of the beginning or end of time. It does not follow from this that we can have an idea

of an infinite time. We merely cannot cut the time-series short. Our imagination may grow tired of continuing the succession in idea, of adding link to link; but reflecting consciousness recognizes that it is only fatigue that compels it to stop. That time is infinite can thus have the correct negative meaning that it cannot end. — Infinite time is therefore not the same as eternity. Succession and change conflict with the concept of eternity. The popular view conceives eternity as a continuation or a forerunner of time. The view of the philosophy of religion, on the other hand, has always maintained that time and eternity cannot stand in this external relation to each other; for it, eternity is an inner concentration of what, in the form of time, is spread out in an endless series of succession. Thus *Master Eckart* says: “The day that passed a thousand years ago is no farther from eternity than the hour in which I stand here; and the day that is to come in a thousand years is no farther from eternity than the hour in which I am now speaking. — He who in the midst of time has set his heart on eternity, in him is the fullness of time.”¹²⁶ The concept of eternity in the philosophy of religion brings out *the value of the content of time* as independent of the differences of time. It comes into conflict with the psychological idea of time when it confuses this determination of value with an actual annulment of the differences of time, which is psychologically unthinkable. [230]

3. But even the limited idea of time is not without difficulty for psychological understanding. That general schema or pattern, which we think of as filled out in different ways, we cannot imagine in and for itself. It shares the lot of all general ideas, that of needing an individual representation. But the time that we can immediately imagine, or encompass in a single moment, is very short. Now when the idea is nevertheless to have as its content a time-series stretching far beyond the present moments, this content must undergo an abbreviation. Quite clearly I imagine only the transition from one moment to another; the idea of spans of time that include myriads of moments can come about only by a symbolic route. The vanished days, weeks, months and years — not to speak of longer periods — pass before our consciousness, so to speak, with diminished dimensions. Were we to remember past time as clearly as we imagine the minutes that make up what we call the present time, memory would become an impossibility. We therefore always imply a different scale for the past than for the present and the nearest future. With the more distant future it goes with us just as with the past. Only when we really want to put ourselves back vividly into the past do we seek to expand the

¹²⁶— Augustine defined time as “the distension of the mind” (*distentio animi*. Confess. XI, 26). Distension is the opposite of concentration. The mystical and speculative philosophy of religion postulates an absolute concentration, in which all “distension,” all alternation and becoming, is annulled; here psychology is not able to follow it. Cf. *Spinozas Liv og Lære* [Spinoza’s Life and Teaching]. p. 142–144.

differences of time in memory, though never so much that the extent of the present time is reached. It goes, then, with remembered time as with the strength of remembered sensations (B, 7 a); both the duration and the strength of our original experiences are remembered only indirectly. — On this symbolic element in the idea of time, once it has [231] reached a further development, depends the fact that it is formed only comparatively late, and long remains within certain limits. It first reaches its full clearness when it becomes possible to present it in a distinct symbolic form, to fix the fleeting rhythms of ideas in a resting image. And this happens only when the perception of space comes to its aid. It is only through the form of space that an intuition of time becomes possible. We then conceive time as a straight line, infinitely prolonged in both directions.

The idea of time is a typical individual idea. Wherever we observe time, it is the same time whose parts we have before us. It is like a river that we observe at different stretches. It often hides itself from us, as when we sleep, are unconscious, or for other reasons do not know “where the time has gone.” But as soon as attention is aroused again, we reconstruct the missing course of time. Time is thus related to the particular times not only as a general concept is related to the individual cases included under it, but also as our dominant idea of an individual is related to our various observations of this individual.

4. So long as the idea of time rests only on the alternation of our inner states, the *estimation of time* is very uncertain. In particular, the estimation of the content of what is experienced plays in and diverts attention from the succession itself, its stages and landmarks. Even a long time filled with monotonous experiences can pass quickly when what is experienced lays claim to our interest. The seven years went for Jacob as a few days, because he loved Rachel. If we experience many important events in a very short time, the time may seem to us to have been very long, since we involuntarily infer from the weightiness and richness of content of the experiences to a long extent of time. This is especially so when what is experienced has brought about a far-reaching change in our fate; what lies before the turning-point is pushed back in time, even if the turning-point lies very near the present.

If we disregard entirely the significance of what is experienced, the estimation of time is determined merely by the quantity of the features experienced. Time that is divided [232] up and studded with salient points seems longer to us than time that has flowed by uniformly.¹²⁷ Here every individual has his own scale, taken from the speed with which

¹²⁷Here perhaps the answer might be sought to a question which a contributor to the first volume of the *Revue philosophique* raises: why the content of a recollection which is itself remembered seems to us to lie farther back in time than when we first had the recollection. The first moment of recollection forms, in

his ideas are accustomed to move. If he is now compelled to fall into a much slower tempo, through the impressions slackening their pace, he feels unsatisfied, hampered, impatient. A feeling of boredom and weariness arises. A much quicker succession of impressions than usual can have the same effect, since such a quantity of ideas streams into consciousness that it cannot order and master them, so that it soon becomes unable to keep up. — When the time of waiting seems very long, it is not merely because it is not filled with experiences, but also because the idea of what is expected constantly comes up and must constantly be pushed back again. A painful feeling is thereby aroused, which gives rise to an unfavorable judgment about the time.

In states that are especially favorable to recall in memory (cf. B, 7 c), such manifold events can unfold before the imagination that it seems to us as if long times had been unrolled before us. Persons who have been on the point of drowning have in a single second seen their whole past life pass by in memory, often with distinct appearance of the smallest details. The taking of opium can bring about the same thing. *De Quincey* describes how time, as it were, swelled up for him. “I often seemed to have lived 80 or 100 years in a single night; indeed it was sometimes as if a millennium must have passed from the one day to the next, or at least a span of time which stretched far beyond the limits of a single human being’s experience.”¹²⁸ A dream which could be ascertained to have lasted only 2 or 3 minutes represented a procession of bloody figures which seemed to the dreamer to last five hours.¹²⁹ This also explains how visions can unroll [233] time and eternity for the ecstatic seer, although not so much time elapses as it takes for the hourglass to empty.

The estimation of time thus stands in a certain relation to the organic state, just as memory does. If the measure of time originally depends on the alternation of our sensations and ideas, and if the vital or general sensations, as the most constant sensations, form the background necessary for consciousness of succession and change, one must necessarily conclude that the manner in which the organic functions take place in the individual organism will condition the estimation of time. Many animals seem to possess an instinctive estimation of time (*unconscious chronometry*), which is explained by regular organic changes in the body.¹³⁰ Of no less importance must be the individual’s habitual active movements. These stand out more distinctly and more objectively, and therefore furnish more definite points of support. Beings whose vital functions and

relation to the later one, a station, an intermediate link, which serves for division and thereby makes the perspective more distinct.

¹²⁸ *En Opiumæders Bekendelser* [Confessions of an Opium-Eater]. p. 161.

¹²⁹ *Brierre de Boismont*: Des hallucinations, 3 ed. p. 267.

¹³⁰ *Carpenter*: Mental physiology. p. 347 ff.

active movements take place quickly must therefore be supposed to have a finer scale for time than beings in whom the corresponding activities take place with greater slowness. A gnat's wings make (according to Marey) 330 strokes a second; if these wing-strokes are noticed by the gnat as we notice the movements of our arms, one and the same time must seem far longer to the gnat than it seems to us.

Experimental investigations point in the same direction as these general experiences and inferences. In Vierordt's investigations of the ability to estimate differences of time it turned out that with very small spans of time there was a tendency to overestimate, with larger spans a tendency to underestimate, the length of time. Later investigations by *Wundt* led to the result that at a difference of time of 0.72, or nearly $\frac{3}{4}$, of a second the actual difference of time and the reproduced one coincide. This, then, is the length of time that we most easily apprehend and retain in memory and use as a scale. What lies below it we overestimate, what lies above it we underestimate. Remarkably, the value stated [234] agrees with the time we need to form simple associations of ideas; the course of ideas thus proves to be our original clock. And further, the same value agrees approximately with the time the leg needs to swing in brisk walking. It is thus a psychical constant, valid for both mental and bodily movement.¹³¹

The need to put an objective measure of time in place of the subjective one, whose uncertainty was bound to be easily noticed, made itself felt early. The great, regularly recurring phenomena of nature furnished a good schema. The movements of the sun and moon, day and night, morning, noon and evening served as a basis. For finer division the sand in an hourglass was used, the water in a clepsydra, or else a burning candle. But only with the pendulum and the chronoscope has greater exactness become possible. *Wheatstone* measured the speed of an electric spark and found it to be 1/115,200 of a second. With Siemens's chronoscope even 1/1,000,000 of a second can be determined.¹³²

This exactness we can imagine continued to ever higher degrees. But a conclusion is as little thinkable here as with the length of time. We measure time by means of uniform movements in nature. But this very uniformity must then be proved — and here we thus come to move in a circle. An absolutely uniform time is an ideal; it is the demand that every estimation of time whatever shall again be subjected to control. Every scale that has been tried in order to proceed from complete uniformity has proved to vary. Only in the symbolic presentation of time as a line is absolute uniformity present. But there, precisely, the idealizing abstraction has done its work.

¹³¹ *Wundt*: *Physiol. Psychol.* II. p. 286. — *Julius Kollert*: *Untersuchungen über den Zeitsinn.* (In *Wundt's »Philosophische Studien«*. 1 Bd. 1 Heft. Leipzig 1881).

¹³² *Jevons*: *Principles of Science.* 4 ed. p. 307 f. — *A. Paulsen*: *Naturkræfterne.* III. p. 129.

5. That the form of time is present from the beginning of consciousness there can be no *doubt*. The psychological investigation of time therefore concerns only the idea of time and the estimation of time. On the other hand, it is a matter of dispute whether *the form of space* is also original. That in any case it cannot stand in the intimate [235] relation to consciousness that the form of time does follows from the general character of consciousness. The states of consciousness follow one another in time; but no meaning can be attached to their spreading out in space. What appears in the form of space can then only be objects for consciousness, not consciousness itself. Already in this it lies that the form of time, psychologically regarded, is more original than the form of space; the latter does not seem to be absolutely necessary for consciousness. The question now is whether experience points to the form of space being in reality a psychical product, formed according to the general psychological laws. There would then be, or have been, a stage in the development of consciousness in which sensations and ideas appeared only with a certain strength and a certain quality, without their content being shaped into extended images.

It is difficult for us to think ourselves into a consciousness without any perception of space. We think, after all, constantly in images. Even our moods and feelings we cannot apprehend and express except by means of intuitions and images. From this one might conclude that the form of space must be original; the symbol might even seem to lie nearer to us than what is symbolized, so that the intuition of space would possess greater originality than the intuition of time. An acute thinker, *Albert Lange*, has in fact drawn this conclusion. "The empirical observation of our inner state cannot," he says (*Logische Studien*. Iserlohn 1877. p. 139), "be accomplished in the mere form of time. We always find a plurality of sensations simultaneously, which can only be combined into a synthesis in the form of a spatial image. ... All our empirical ideas of time rest on ideas of space. A line forms our idea of the course of time. Movements in space give us occasion to measure time. Should one not conclude from this that the idea of time is secondary in relation to the idea of space?" — To this it must first be remarked that because the higher and clearer development of the idea of time and of the measurement of time presupposes the idea of space (see 4), it is not thereby said that the latter is also the more original. *Lange* infers too much from the necessity of symbolism. Thought must after all be more original than the word, although it needs the word for its full clearness and exactness. However intimately feeling is bound to its expression, it would [236] still be to stand the matter on its head to take feeling for secondary in relation to its expression. It is, after all, only at the higher stages of the life of consciousness that a

wide circle of symbols stands at its service. Only with two of our senses, sight and touch, do sensations arrange themselves for us in the form of space; with hearing, smell and taste localization is not originally present; there we have to do only with the strength and quality of the sensations. The proper, definite intuition of space is, as we shall see in what follows, bound up with the sensations of sight. Visual images play so great a part in our world of ideas that we abstract from them only with difficulty. The more we fix our attention on tones, and on impressions of taste and smell, the more we shall approach a consciousness endowed merely with the form of time, and see the possibility that an inner manifoldness may be given in consciousness without its being combined into unity in the form of space.

This shows itself still more clearly in the states of feeling. These are certainly most often accompanied by local sensations (in the chest, in the heart, e.g.); but inner observation makes it easy to distinguish between the feeling proper and its organic consequences. Pleasure and pain, joy and sorrow, hope and fear can stir in us without being directly symbolized. There is indeed in every feeling something inexpressible, incommensurable with every expression. The feelings drive us to seek forms and expressions, but they are not themselves ordered under the form of space. When, in a despondent mood, a gleam of hope emerges, we do not imagine the hope as above, or to the right or left of, the despondency, as we do when we simultaneously have sensations of different colors. Different elements of consciousness can thus assert themselves simultaneously without being ordered in the form of space. The obscurity and difficulty of immediate psychological observation is connected in part with this (see I, 8 a). We lack a form of intuition for simultaneous inner *phenomena*, while we possess one for *simultaneous outer phenomena*.

6. We perceive space in three *dimensions*: up and down, right and left, forward and back. This threefoldness can be reduced to a twofoldness: distance (or depth) and extension in surface.

[237] That the *perception of distance* cannot be due to any single sensation is easily seen. Every sensation presupposes, physiologically regarded, that a physical impression comes to the sense organ. But distance itself makes no physical impression on us. We measure distance by a line from the object to ourselves; but no impression can inform us of the existence of this line. It designates the direction of the impressions, but itself calls forth no sensation. We perceive distance only when we can in one way or another measure out this line; but such a measuring-out is not any single sensation; it is a process of comparison, which either already presupposes a certain idea of space or else can build

only on the strength and kind of the sensations we get by moving ourselves, or parts of our body, toward the object.

Such a measuring-out by means of movement we do in fact perform in every perception of an object. The eye adjusts itself involuntarily so that the impression of light strikes the yellow spot on the retina. The two eyes are set so that the visual axes meet at the point that attracts attention. We reach for the object or move toward it in order to feel it. We thereby get sensations of movement which stand in a definite connection with the position of the object in relation to us. With the sight or the touch of the object there now combines, through association and practice, the idea of the sensations of movement that are a necessary presupposition for the most distinct perception of the object. More or less clearly, distance means for us the greater or smaller series of sensations of movement that we would have if we moved from our standpoint to the object. In agreement with this is the fact that the perception of distance is clear and distinct only with near objects, and that among distant objects it is the more distinct the more familiar they are to us. We have the most distinct perception of space of what our own hands have immediately measured. Greater distances we understand (like greater periods of time) only symbolically, by regarding them as a sum of small, immediately measurable distances.

The sense of touch and the sensations of movement bound up with its activity are the most important basis of the perception of distance. The proper measure of the object we get by active feeling of it. The distances we perceive in this way we always imply in visual perception. Distant objects, which to sight look smaller, we nevertheless estimate immediately according to the size they would have for touch. Only secondarily does the size of known objects for sight in turn become a means of estimating their distance. [238]

This theory, which was first put forward by *Berkeley* (Theory of vision. Dublin 1709),¹³³ is confirmed in part by observations of newborn children and of those born blind who have been operated on. Although the child turns toward the light quite early, it hardly has a perception of distance. The movements of the two eyes are as a rule not coordinated, so that the visual axes do not from the first necessarily intersect at the point that is the object of perception, which would be a necessary presupposition if the perception of distance rested on an innate mechanism coming into action at once.¹³⁴

¹³³ *Berkeley*, however, does not yet lay such great weight on the sensations of movement as has been done in the account given here. It is especially *Bain* and *Spencer* who have brought out the importance of the sensations of movement for the perception of space. — *Helmholtz* has expressed a similar view in his famous »*Physiologische Optik*«.

¹³⁴ *Descartes* already held this view, which in our day is maintained by the so-called *nativism*. Yet *Descartes* grants that in the immediate perception of distance, which seems to be simple, there lies a comparison as presupposition (*ratiocinatio involuta, similis illi, qva Geometræ per duas stationes diversas*

Only gradually (in the course of the first three months) does practice in the coordination of the eyes arise, so that squinting no longer occurs so frequently. And even after the adjustment has been brought about, the sure perception of distance is lacking, which shows itself in the child reaching for things it cannot reach. Even in the 2nd and 3rd years of life the estimation of distance is imperfect.¹³⁵

Nor even at a later age is the cooperation of the two eyes entirely harmonious. With practice one can see double images of objects. This happens most easily when one directs [239] the eyes at a point out in the background of the field of vision, but fixes attention on an object lying in a straight line in front of it; this is then seen double.

It is not quite certain whether it goes with newborn animals as with newborn human beings. *Spalding's* experiments with newly hatched chickens and newborn pigs prove that these animals are at once able to find their food with great sureness. The chickens dart off after a grain or an insect,¹³⁶ and the pigs after their mother's udder. A pig that was set on a chair ten minutes after the bandage it had been given over its eyes immediately after birth had been taken off seemed to measure the distance from the ground, knelt, and jumped down. Here, then, a perception of distance seems to assert itself at once, although one must surely beware of reading too much into it, since an immediate instinct might be supposed to assert itself, which under the guidance of the visual impression led the animal in the direction of the food, and something similar might be the case with the jump. It is therefore hardly justified to posit a perception of distance prior to experience and practice, even if this course of experience is here, as in so many other fields, far shorter for the animal than for man.

The experiences of those born blind who have been operated on also present features that can be interpreted according to the theory founded by Berkeley. The man born blind who was operated on by *Cheselden* (1728) was, when he had gained his sight, so unable to estimate distances that he believed that all objects "touched his eyes" (as he himself put it), just as what he felt touched his hand. The one operated on by *Franz* (1840) took a cube for a square, a sphere for a disk, and a pyramid for a triangle; it was only by feeling them that he learned to understand what they were. All things seemed to him perfectly flat. The man born blind who was operated on by *Dufour* (1876) could [240] not judge distances without the help of his hands. With outstretched hands he walked

loca inaccessa dimetiuntur. Dioptrica. Cap. 5. § 13). Here, then, the theory of "unconscious inferences" is already intimated, and that not in an empiricist but in a nativist.

¹³⁵ *Preyer*: Die Seele des Kindes. p. 24–29; 38 f; 112.

¹³⁶ *Adam Smith* already («On the external senses» in his »Essays en philosophical subjects«, edited by Dugald Stewart) remarks that this sure ability of orientation immediately at birth is found only in birds that build their nests on the flat ground, not in those that build in trees and in other less accessible places.

toward the shiny door-handle to which his attention had been drawn, but stopped two steps from it, and missed several times before he succeeded in grasping it. — Should one say, apropos of the last example, that he who moves toward an object must have an idea of its distance, this is an unjustified inference. He need only adjust his movements so that the guiding visual sensation does not lose but successively gains in strength, and for this no idea of distance is required. There is a game in which one is to find an object by music, in such a way that the music sounds the louder the nearer one comes to the object, the fainter the farther one moves away from it. Here the seekers are obviously not guided by perception of distance; the movement is adjusted immediately to the strength of the music.

Most of the operated persons born blind mentioned in the literature could already before the operation distinguish between light and darkness and be guided in their movements by the impressions of light. But it does not follow from this that the impressions of light themselves immediately brought with them a perception of distance.

7. a. That the idea of distance or depth arises through a psychological process is granted even by investigators who regard the perception of *extension in surface* as original and innate. But even if it is the case that the perception of distance is due *merely* to association and practice, the question of the originality of the perception of space would thereby only have been pushed a step back, so long as we have not settled it with regard to the perception of height and breadth. One might, after all, maintain that we become able to measure the distance between an object and ourselves only because we already have the idea of space as regards lines and surfaces.

With regard to the perception of extension in surface, too, attempts have been made to explain the perception of space by association of visual sensations with sensations of touch and movement. What is immediately perceived would then here too be sensations with a certain strength and quality, and the perception of space would arise as a certain ordering of the sensations. It is natural to assume, and observations of newborn children confirm it, that what is first perceived by sight is light and colors. The clear and bright, [241] yet not too dazzling, impressions are sought out and sought to be held fast; only later is the form of objects perceived.¹³⁷ It is owing to observations and experiments by means of touch and movement that the particular objects mark themselves off. The language of sight becomes perfectly distinct only by the help of sensations of movement and of touch. Those born blind who have been operated on must therefore gradually learn to know by their appearance the same objects with which they were very familiar by their tangible

¹³⁷Preyer. p. 36. 41.

properties. The question then arises whether the blind man, who remains restricted to sensations of touch and movement, really has any intuition of space like that of the seeing. We, the seeing, think at once, when we think of space, of a visible surface at some distance from us; but what can the blind man really imagine by space? — Here the real paradox comes out in the assumption of space as a psychological product. For if, by the exclusion of visual images, we keep back only something that is not of the same kind as visible space, we would not be able to say what space is in and for itself, since no definition could be set up that fits both the space of sight and that of touch. According to *Berkeley* there is no more natural connection between visual space and tactile space than between the word and the thing it designates (Theory of vision § 121–143).

There is in fact a definite observation on record in this direction. *Ernst Platner* writes in his »Philosophische Aphorismen« (Erster Theil. Leipzig 1793, p. 446 f.): “As regards the sightless idea of space and extension, the observation and examination of a man born blind, which I undertook in the year 1785 and continued for three whole weeks, has convinced me that the sense of touch by itself alone is wholly ignorant of everything that belongs to intuition and space, and knows nothing of a being-outside-one-another in space, and, to put it briefly, that man without sight perceives nothing at all of the external world but the existence of something acting, which is different from the feeling self acted upon by it, and for the rest merely the numerical difference of the impressions or things. Time serves the man born blind in place of space. Nearness and distance [242] mean for him nothing but the shorter or longer time, the smaller or greater number of sensations, which he needs in order to get from one sensation to another. That the man born blind makes use of the language of sight [just as the color-blind have learned to use the names of colors] can deceive, and at the beginning did deceive me too. But in reality he knows nothing whatever of things that are outside one another, and if the objects and the parts of the body touched by them did not make impressions of different kinds on his nerves of touch, he would (this I noticed especially clearly) regard everything external as one and the same thing, which affected him more strongly when he laid his hand on a surface than when he put his finger on it, more weakly when he merely stroked with his hand or passed lightly over a surface. On his own body he distinguishes head and feet not by the distance, but merely by the difference, perceptible to him with incredible fineness, in the sensations he has of these two parts, and for the rest by time. Likewise he distinguishes the figure of bodies merely by the character of the impressions of touch, the cube, e.g., affecting the sense of touch differently with its corners and edges than the sphere does.”

We have an approximation to the perception of space of the man born blind when we try to find our way in a dark room. Only we then have the advantage that visual space lies ready-made and can be called upon to help in interpreting the sensations of touch and movement. Something similar holds when we concentrate our attention on the tongue and — with the greatest possible abstraction from visual ideas — observe what perception of space it gives us. For the tongue is like one born blind, and yet is excellently oriented in its surroundings.

b. Here, however, a possibility still shows itself for maintaining the originality of the perception of space. The sensations of movement are, it is true, always successive. But through the sense of touch we can get several impressions simultaneously. Likewise several impressions of light can be received simultaneously on the retina. Might there not lie in this the possibility of an immediate perception of the impressions as ordered in space? It might even seem necessary to assume this. For the sensation of a color means, after all, the sensation of a colored surface; if, namely, the colored thing were only a mathematical point, it would make no impression. Even if larger objects can be perceived [243] only by movement of the organs of sight and touch, one would still think that small objects could be perceived immediately, without a successive process. There must surely be an immediate difference between the impression of a krone and the impression of an øre. The perception of small surfaces would then be the minimum to which *nativism*, the view that maintains the originality of the perception of space, would restrict itself, the last position from which it could not be dislodged.¹³⁸

It naturally cannot be denied that we can get, and constantly do get, a multitude of simultaneous impressions of light and touch, and that we immediately perceive them as ordered in space. But it is not certain that the beginning consciousness has the same

¹³⁸Cf. *Stumpf*: Der psychologische Ursprung der Raumvorstellung, Leipzig 1873. p. 56–71. — Professor *Mahaffy* of Dublin reports, in a letter printed in *Mind* 1881. p. 278 f., on a conversation with the man born blind who was operated on by Franz. This man, who was a physician in Kingstown, stated that immediately after gaining his sight he saw and distinguished figures, and that outlines and figures corresponded to what he had expected from the sense of touch. Mahaffy regards this communication as the testimony of a competent judge to the originality of the ability to distinguish figures by sight alone. But neither this interpretation nor the communication itself agrees with Dr. Franz's report, which was printed shortly after the operation in »Philosophical Transactions« (1841). As stated above, the patient took a cube for a square and a sphere for a disk. When Franz asked him to describe the impressions the objects had made on him, he said that he had at once noticed a difference between the cube and the sphere, but was not able to form the idea of a square or a disk from them *before he had perceived a sensation in the tips of his fingers*, as if he were actually touching the objects. (Phil. Transact. 1841. I. p. 65). He thus really had to carry out a translation from the language of the one sense into that of the other before he could recognize the figures. There remains only the fact that he at once noticed a difference between the figures, which speaks in favor of nativism. But here too the experience of the sense of touch may have helped him. He did not stand before the figures as a consciousness without any experiences, i.e. a beginning consciousness, would stand.

perception. A manifoldness of simultaneous impressions will from the first rather act as
 [244] a collected mass, give one single chaotic sensation. Just as quality and strength cannot from the first be distinguished, so too the relation of space could not from the first be perceived by itself, but would, so to speak, hide itself in a higher degree of strength, or, to use Bain's expression, a greater massiveness of the impression. Platner's account quoted above points in this direction, since it shows that it was only qualitative differences in the impressions that led the blind man to assume a manifoldness outside himself. One can here also refer to Weber's experiment, which proved that when the whole hand is dipped in warm water, the degree of heat seems to be higher than when only the finger is dipped. Since no innate idea of the difference between the fingers and the hand can be assumed, that difference in strength must from the first have been the only one that stood out.

A purely passive perception of simultaneously given impressions can only be quite momentary. Activity is aroused at once; the eye moves over the surface or the hand feels it. At once, then, succession begins to take the place of simultaneity, the discursive that of the intuitive. We perceive moving objects more quickly and easily than resting ones, and if they do not move themselves, we put ourselves in motion in relation to them. Two successive impressions on the skin can be distinguished at a smaller distance from each other than two simultaneous ones. The lowest animals and newborn children do not notice simultaneous differences in their surroundings, while they can perceive successive differences (changes).¹³⁹ By movement, then, is discovered and perceived what would otherwise go unnoticed. The first chaotic sensation is thus analyzed more closely by the help of a series of succeeding sensations, in which the sensations of movement, in the nature of the case, take a prominent place.

One might even maintain that successive and discursive perception is a necessary presupposition for real intuition of space. Space means a relation. That something is in space means that it occupies a certain position in relation to other things, and that its
 [245] particular parts occupy certain positions in relation to one another. Instead of speaking of space *en bloc*, as a complete unity, one therefore does better to use the more elementary determination *position*. The perception of space is then seen to rest on a comparison or a combination. It cannot therefore be present at the first immediate observation.

c. It is not a matter of indifference which place on the body is struck by the impression. Since we must learn to know our own body through successive experiences, just like the world surrounding us, one cannot assume an original knowledge or sensation of

¹³⁹G. H. Schneider: Warum bemerken wir mässig bewegte Dinge leichter als ruhende? (Vierteljahrsschr. für wiss. Phil. II. p. 412).

the place at which an impression strikes us. Now if the impressions nevertheless act differently at the different places, this difference can only appear to consciousness as a certain nuance, a qualitative subsidiary determination of the impression. The peculiar stamp the impression receives by striking a single definite place Lotze has called its *local sign*. The conditions are different at every place of the skin and of the retina; there must therefore be a manifoldness of local signs. In the case of sight, these local signs might consist either in the impulses of movement, different for each point, which tend to turn the eye so that the impression of light falls on the yellow spot (Lotze), or in the different quality of the sensations on the different parts of the retina (*Wundt*). In the case of the sense of touch, Lotze finds the local signs given by the different secondary sensations which one and the same touch brings with it at different places on the skin, on account of the skin's differing density and tension and the different substratum it encloses.¹⁴⁰

These local signs can from the first assert themselves only in successive perception. Consciousness cannot sense them all at once; only after having run through their series can it assign to each of the sensations furnished with local signs its definite place. In order to fulfil their function, the local signs must therefore make up a coherent system. Stimulation of a single place on the skin or the retina can then lead to an orientation, a [246] local determination, without the whole series of local signs needing to be gone through anew; for the idea of the series as a given whole serves as an abbreviated symbol of the original successive experiences.

8. With all this the perception of space proper has not been reached. We have obtained a coherent group of sensations of movement, local signs, and impressions of light and touch, connected according to the laws of the association of ideas. But the ordering of all this in such a way that there results an intuition of an image whose particular parts are *outside* one another, and which is itself *outside* us (or rather outside the image of our organism), is not thereby given; for all those sensations were qualitative and intensive, not extensive.

Here it becomes apparent that all theories of the perception of space come into difficulty. The so-called nativistic theory regards the perception of space as given with the very first impressions.¹⁴¹ According to it, any psychological explanation of the

¹⁴⁰ Lotze has given several presentations of his theory of local signs, most recently in *Grundzüge der Psychologie* (1881). For Wundt, see *Physiol. Psychol.* II. p. 25. 28. 163.

¹⁴¹ The most important representatives of nativism at present are, on the psychological side, *Stumpf* (*Der psychol. Urspr. der Raumvorst.*), on the physiological side, *Hering* (whose latest work is: »Der Raumsinn und die Bewegungen des Auges«. *Hermann's Handbuch.* III, 1). Hering's general standpoint is indicated in the following statement: "That the complete exclusion of the motives of experience is unthinkable is acknowledged by all... Where the boundary between the innate and the acquired lies, we do not know. But it seems inadmissible to assume that the spatial properties of sensations and their localization are

development of the perception of space is both superfluous and impossible; the perception of space must here be taken as something given from the first. The experiences cited above (6. 7 a and b) seem, however, to speak against this theory. The *genetic* theory therefore regards the perception of space as a psychological product, arisen through the combination of intensive and qualitative sensations in accordance with the general psychological laws of the association of ideas. Now, however, an unexplained remainder [247] is left over, since the psychological product has other properties than the elements from which it is supposed to have arisen, and that precisely the property with which the problem is concerned, namely the extensive form itself. *Herbart* and *Spencer* have indeed tried to explain the transition from the perception of successive impressions to the intuition of what is simultaneous and extended in this way: that when it is a matter of indifference from which end we begin the reproduction of a series of impressions, an intuition of space forms itself; for while time has only one direction, space has several. In the end the whole series will then stand before us at once, wherever we begin. But by this we arrive only at *simultaneity* (coexistence), not at space. We can, e.g., repeat a series of tones forward and backward without the tones arranging themselves for us in space. — Here, therefore, a transformation must be assumed to have taken place, a psychical synthesis analogous to the chemical syntheses by which compound substances arise with quite other properties than the elements possess (cf. B, 8 d 4). In this synthesis, in the seeing human being, the sensations of sight play the most important part and become the dominant language into which are translated the contributions to the perception of space that other senses can make. Whether a corresponding synthesis also takes place in the case of the blind, such that the sense of touch becomes the dominant one, seems doubtful according to *Platner's* account.

Even if it should be the case that tangible and visible minima are perceived as extended, a synthesis would still have to be posited. For our images of space are continuous, uninterruptedly coherent; but neither the retina nor the surface of the skin furnishes a basis for such a continuity. The organs of sight and touch give the image of a mosaic, not of an unbroken coherence. On the retina there is even a place quite insensitive to impressions of light, “the blind spot,” where the optic nerve enters the eye; but in our visual image there is no corresponding gap; we therefore involuntarily fill out the hole in the series of sensations. Every theory which does not hold that the perception of space is given at one stroke from the first must therefore, in one form or another, appeal to the

merely the product of experience, partly because many newborn animals directly prove the contrary, and it cannot be assumed that the newborn human being is in this respect more than in degree different from the animals; partly because visual sensations without any spatial properties are something inconceivable.”

constructive capacity of consciousness.

Such an appeal is not without its drawbacks. For a psychological view which finds [248] the very nature of consciousness expressed in synthesis, in the gathering into unity and coherence (II, 5. — IV, 5), there might seem to be something quite reasonable in this fundamental form asserting itself also in the particular psychical processes. But it is one thing to hit upon the right characterization of consciousness as a whole, another to apply the characteristic mark thus found as a *deus ex machina* to special psychological problems. The way is easily opened to arbitrariness; into such a psychical synthesis one can, after all, put whatever one likes. Lotze therefore surely proceeds most cautiously when he declares that the purpose of his theory of local signs is not to explain the intuition of space itself, but only the motives and aids the soul uses when it forms its images of space. That the soul forms images of space out of certain of its sensations Lotze regards, on the other hand, as a factual capacity or impulse that cannot be an object of further explanation.

The *genetic* theory too, then, must acknowledge an original element in the perception of space. This original element points back to the general property of the activity of consciousness; what was not explained by the special sensations was, after all, precisely the unity and continuity of the space intuited. But unity and continuity there could be, even if the sensations were combined into other forms of intuition than our space. Here something always remains that must be regarded as merely factual or historical.

The genetic theory stands in complete opposition to the nativistic only when it assumes that all the conditions for the development of the intuition of space are given with the experiences the single individual can have in the course of its development. But this view is improbable on account of the unexplained remainder that is left when we compare the elementary sensations with the fully developed intuition of space. If in the synthesis to which the intuition of space owes its definitive rise we see the manifestation of a constructive capacity acting instinctively, then the question of how this capacity came to be directs us beyond the single individual to the natural context within which it arises. The experiences which in the life of the single individual cannot lead to the goal [249] indicated may, in the course of the development of the race, have gradually adapted the organization in such a way that inherited dispositions supplement what is insufficient in individual experiences. The hypothesis of evolution, which was first applied in this field by *Herbert Spencer* (1855), opens the prospect of following the problem further back than was possible so long as one kept to the experiences of individual life.

How near the genetic and the nativistic theories can in this way come to each other

may be seen from the fact that, while on the one side it is granted that what is immediately given in the perception of space may be vanishingly small in relation to what is added by association, and that in the original sensations there is really given only a possibility of a definite intuition of space (*Stumpf*), on the other side it is intimated that everything may be so arranged in the organization that the time between the first impression of light on the retina and the rise of the idea of space becomes vanishingly small (Wundt).¹⁴²

9. Whether one adheres to a nativistic or a genetic theory, it is in any case a necessary presupposition for the rise of the intuition of space that a definite organic basis be present. The dispute of the theories concerns only (or ought to concern only) whether the functions conditioned by the organization come into force at once or need some time of preparation and practice.

The more precise demonstration of the organic structures that are of importance for the rise of the intuition of space is the task of the physiology of the senses to give.¹⁴³

[250] Besides what has been indicated in this respect in the foregoing, we shall here only point to the importance of the central mechanism by which a close connection is made possible between the sense-impressions and the muscular movements of the sense organs. In the case of the sense of touch the optic thalami, in the case of sight the corpora quadrigemina, seem to be the reflex centers by which this connection, and with it the physiological expression of the psychological synthesis, is brought about; though the cerebrum probably also plays a part.

The apparatus that have here been made ready are perhaps (cf. 6) in some animals able to function immediately after birth, so that the experiences necessary for the perception of space can be had at once and easily. In man, as it seems, several months pass before this point is reached.

10. Hitherto we have spoken of the form of space itself, the capacity to intuit images of space. As regards the idea of space as a general idea, or rather a *typical* individual idea, it develops in a way corresponding to the idea of time (1–3), as attention is directed

¹⁴²*Stumpf*: Der psychologische Ursprung der Raumvorstellung. p. 114 ff. 184. — *Wundt*: Physiol. Psychol. II. p. 164. Cf. also *Spencer*: Principles of Psychology. II. p. 203 ff.

¹⁴³On this see Panum: *Sanserne og de vilkaarlige Bevægelser* [The Senses and the Voluntary Movements]. p. 234–238, and Wundt's very detailed presentation (II. chaps. 11 and 13). — In the preceding investigation only the most general psychological points of view have been taken into account. Whoever wishes to be introduced to the special psychological and physiological questions which the theory of the perception of space has led to discussing is referred to Wundt's work, where the whole material is found collected and critically worked up. — The question whether the geometrical axioms are grounded in the capacity of intuition determined by our organization, or whether with the necessary practice we should be able to imagine and intuit space of another character, belongs to the theory of knowledge, not to psychology. Purely logically such a possibility naturally cannot be denied; but in fact we are surely bound to our space, characterized by the axioms of Euclidean geometry, so long as our organization has not undergone a change through adaptation to other conditions of existence.

to the schema common to all particular images of space and to its capacity for extension. From the first the idea of space is limited. The man born blind who was operated on by *Cheselden* could not imagine lines in space beyond the limits of his field of vision. He knew that the room he was in was only a part of the house; but he could not comprehend that the whole house could look larger than the room. He thus still lacked the ability to use the individual idea in a symbolic way. When this ability develops, it is discovered that no limit can be posited for the division or the extension of space, any more than for that of time.

[251]

D. The Perception of the Real

1. Sensations, ideas, concepts, time and space are forms under which the content of consciousness appears and orders itself. We have followed this ordering from its simplest stage, in the grouping and interaction of sensations, to its highest stage, in thinking directed toward definite problems. The laws in force were everywhere the same. The motive for the advance from the involuntary ordering in the play of sensations and ideas to scientific thinking and artistic imagination lay in the criticism that advancing experience necessarily brought with it. New differences and contrasts demand new activity, in order that connection and unity may be maintained and chaos overcome. The logical principles and the aesthetic laws, which it is not the business of psychology to set forth, nevertheless develop according to psychological laws of nature, and thereby prove their intimate connection with human nature; they do not suddenly come forward as something forced upon it from outside.

But here the question arises how it comes about that consciousness, in the content of its sensations and ideas and in the connection among them brought about by thinking, sees a *reality* independent of itself. Because our knowledge develops according to definite psychological laws of nature, it is not thereby said that it leads to a reality. The fantasies of the insane and the images of the dream obey, after all, the same laws, which is why we have so often been able to use them for orientation. How then does it dawn on consciousness that in its content it has a reality and not a dream? Can any definite psychological activity — observation or act of thought — be indicated by which we are led to the conviction that we stand before a reality?

It is not a question that first arises at an advanced standpoint in the development of consciousness. From the first beginning of consciousness it is really posed without

interruption. The individual is not allowed to order his ideas as he pleases. There goes on a constant education through disappointments, which may be hard and painful. (Cf. [252] V B, 4.) The first disappointment lays the first foundation for the contrast between the possible and the real, or between dream and reality. — Here we return to this point in order to carry the train of thought further.

2. The impulse to movement which stirs so early in conscious beings leads them involuntarily to intervene in the surrounding world. But they soon learn that their movements are not allowed to proceed unhindered. At certain points they meet a resistance, and in the sensation of resistance, in the sensation of hampered movement, the individual meets something foreign, something which — whatever else it may be — is not itself. It tries perhaps again and again to overcome the resistance; yet it never quite succeeds, so long as life lasts. New barriers are set up in place of those overcome.

Seen from one side, *every* sense-sensation is a sensation of resistance. Every physical impression acts only by reaching the outer surface of the organism, in particular those parts of it where receptivity to the kind of impression in question is especially developed (thus the eye for rays of light, the ear for waves of air, etc.). It is an old thought that the sense of touch is the universal sense, and that the other senses have developed out of it. But precisely on account of the fineness of the special senses they do not play so great a part in the development of the firm conviction of a reality. They cooperate, but presuppose a firm kernel about which they can gather, and this firm kernel is given in the sensation of resistance to our movement. A being that possessed merely sight, hearing, smell and taste would lead, as it were, an ethereal existence, would be placed outside tangible reality. The very word *Genstand* [object] means, after all, resistance; an object is what stands against us.

No single observation whatever can be relied on, however. We have seen that observation is always of a composite nature, since new sensations immediately fuse in it with memories. Here lies the source of a multitude of *illusions*, which arise through false interpretation of the given sensations (B 7, a). Observation is an attempt at interpretation; but how is the correctness of the interpretation guaranteed? Sensations may call forth incorrect memories. And besides, the brain can, through processes in the interior of the [253] organism, be put into a state similar to that when impressions are propagated to it from the surrounding world. Then *hallucinations* arise, false sensations and observations, which one is often unable to distinguish from normal sense-observations. Not only sight, but the other senses too can suffer under such abnormal states. It is remarkable, but in keeping with what was noted above about the relation between the sensation of

resistance and the other senses, that while hallucinations of sight and hearing do not always bring with them hallucinations of touch and resistance, the latter, conversely, seem as a rule to bring with them hallucinations of sight or hearing.¹⁴⁴ False sensations of resistance thus give the most forceful illusion of a reality. They therefore also act in a more harrowing and destructive way on mental health.¹⁴⁵

But the question is precisely how, in the particular case, a distinction can be made between mental health and mental illness. When all the sense-sensations join together, and when the hallucinating person moreover rejects with great acuteness all the objections that can be directed against the imagined world into which he has lived his way, how can it then be decided who is right?

Particular sense-observations as such cannot decide the matter. Each of them may rest on illusions or hallucinations. The only decision that is possible is reached by looking to the connection among the different sense-observations. The particular points at which the sensation of resistance (to keep to it) occurs are not isolated, but prove to stand in connection with one another. The task is then set for the individual of ordering his ideas according to this connection. If he does not find the right ordering, he will run up against contradictions, and in the end he will suffer a practical disappointment or pain. If he perhaps found the right ordering among the ideas as they presented themselves, but nevertheless came into conflict with new experiences, then doubt is raised about the correctness of the observations themselves. If doubt is not raised, and the interaction with the surrounding world is continued, destruction is certain. That is why the child [254] and the insane are withdrawn from the struggle of life: they are not able to correct their ideas according to experience.

The real is what we perceive as real — what, in spite of all resistance on our part, we must in the end nevertheless let stand as it is — what we cannot but acknowledge. This “cannot but” is a negative and subjective criterion, and there can be no question here of any other. For the dreamer his dream is reality; if he wakes, he discovers that the dream was only an illusory reality, which is conditioned by a more comprehensive reality and finds its explanation within it. As far as we can go within the dream without running up against glaring contradictions and conflicting experiences, so far we believe in the reality of the dream. But there comes a point where the thread breaks. Even the most systematic dream is only a fragment in comparison with the totality into which advancing experience leads. In this way all those of our ideas that have no root in reality are corrected; sooner or later their limitation will show itself, and it is discovered that

¹⁴⁴*Brierre de Boismont*: Des hallucinations. 3 éd. p. 597 f.

¹⁴⁵Cf. E. *Kraepelin*; Ueber Trugwahrnehmungen. (Vierteljahrsschr. für wiss. Phil. V. p. 365).

there is more in heaven and on earth than we thought of in our philosophy.

By this route the knowledge of the single individual finds its completion in a total picture of the reality of which he is a member. But the individual's own powers can do little here. He cannot correct his own illusions when he stands alone. Just as little can the single people and the single age. The total pictures which the single individuals, peoples and ages form in turn carry on a great struggle among themselves, and through this struggle the world-view of the race slowly approaches reality. The psychology of the individual here leads over partly to the psychology of peoples, partly to the history of the sciences.

Whether the goal can in itself be reached is not thereby decided. But before we touch on the great problem that arises here, we must bring forward an important concept to which the foregoing development has led us, and whose psychological basis it is the place here to point out.

3. Reality, according to what has just been developed, shows itself in the fixed [255] connection among observations. Of particular observations, of the reality of particular things or events, we can therefore never be so firmly convinced as of connected series of things and events. The more "method" there is in the "madness," the more dangerous it therefore is.

The fixed bond by which things and events are joined into the connection of reality we call the *causal relation*. The real, we may say, is that which is effect and has effect. Without having effect it would not exist for us, would not meet us, would not stop us on our way as a non-ego. Without being and having effect it would not fill one definite place in the series of things and events. That we know reality therefore means the same as that we apply the concept of cause to our observations.

According to the popular view of the causal relation, the cause is one thing and the effect another thing. The difficulty that might lie in the fact that things which exist each by itself nevertheless have so much to do with one another as cause and effect must have is easily solved from this standpoint by ascribing to the thing one calls the cause a producing or compelling power. One personifies causes, attributes to them something analogous to a personal exertion of will.

David Hume was the first to carry through a criticism of the popular concept of cause. What does one really mean — he asked — when one says that one thing is the cause of another? If the answer is that the cause *produces* the effect, what then does one understand by "production"? Can one give any explanation of this concept without its becoming the same as — causation, so that one is thus moving in a circle? — Let us take

a thing by itself alone. Can one by any reasoning whatever infer what effect it will have? All our distinct ideas can be held apart from one another, and it is easy for us to imagine an object as non-existent at this moment and existent at the next, without needing for this any idea of a cause or a producing principle. When we consider things each by itself, there is none of them that *necessarily* presupposes any of the others. — Nor by the route of experience can we arrive at the concept of cause. We do not discover an inner bond [256] between things and events. We see the one phenomenon exist together with or follow after the other, but we do not see it *follow* from the other. The producing or causing itself we do not see. — The fixed connection that we posit between two things which we call cause and effect thus does not belong to the objects themselves. Necessity exists only in consciousness, not in things. But what then can it be that thus binds our ideas together? This is in itself just as mysterious as the connection between the objects. The only possible explanation is that constant experiences produce a habit, an instinct, an inclination to pass from one idea over to other ideas with which it is usually connected. This subjective compulsion, which we experience in the course of our ideas, we then transfer anthropomorphically to nature.¹⁴⁶

Hume dissolves the popular view of the concept of cause by carrying through its presuppositions. His criticism is built on the cause being one thing and the effect a quite different thing. This isolation of the particular members of the causal relation stands in close connection with his psychological theory, which regards consciousness as a sum or series of independent ideas (see above B, 5).

Instead of saying with Hume that we cannot see in a thing, or infer from its concept, that it is the cause or effect of another thing, we must conversely maintain that we know a thing at all only in so far as it is cause or effect. Hume has not hit upon the real basis of the popular view. Things are always given to us as members of a connection. The popular view takes them out of this connection, in which they live and move and have their being — and then it certainly becomes wonderful that they have anything to do with one another. Hume says that a thing continues to be what it is whether or not it is a cause. But the thing that is a cause, i.e. from which a change proceeds, must surely be in a different inner state than when no change is determined by it. It cannot bring about any change without itself being changed. Instead of beginning with the idea of [257] independent things and wondering at the causal relation, it would be at least as justified to begin with the causal relation and wonder at the independent things.

Hume saw clearly the connection between the theory of the concept of cause and the

¹⁴⁶Treatise on human nature. Book I. Part 3. Sect 3–14.

psychology of knowledge. “The uniting principle among our internal perceptions,” he says (I, 3, 14), “is as unintelligible as that among external objects.” If Hume’s psychology is corrected, the theory of the concept of cause will also undergo a corresponding correction. *Kant* attempted both, after Hume’s criticism itself had awakened him from his dogmatic slumber. Kant’s theory of psychical synthesis as the typical characteristic mark of consciousness lies at the basis of his theory of the causal relation.

The objects of experience form — such is Kant’s train of thought — the connection, fixed and universally valid for all thinking beings, which we call nature, only through the particular sensations and observations being ordered in a definite way. This ordering in turn becomes possible only through the combining or synthetic activity of consciousness. The laws of this activity therefore determine our conception of nature. In them are given principles according to which we seek a definite place for each observation, until there forms a connection that is continuous and yet at the same time marked out in the finest nuances.

Kant thus agrees with Hume that all necessity is in the end subjective. But he does not regard it as mere habit or instinct; for him it stands as the expression of a law of the understanding grounded in the nature of consciousness. Of this law of the understanding we naturally become conscious only when we reflect on the involuntary thinking we employ in all our experience; but it proves to be closely connected with the fundamental peculiarity of consciousness, to be a special expression of it.

Kant’s theory comes out clearly in a statement such as the following. “That something happens is a perception which belongs to a possible experience, which becomes actual through my regarding the phenomenon as determined as to its place in nature, and thus regarding it as an object which can always be found according to a rule in the connection of perceptions. But this rule, according to which something is determined with respect [258] to the succession in time, is: that in what precedes, one can meet with the condition under which the event always (i.e. necessarily) follows. The principle of sufficient reason is therefore the ground of possible experience, namely of the objective knowledge of phenomena with respect to their relation in the sequence of time. — — If, on the other hand, I posit what precedes, and the event does not necessarily follow upon it, then I regard it only as a subjective play of my imaginings, and, if I nevertheless represent it to myself as something objective, I call it a mere dream.”¹⁴⁷

The imperfection of Kant’s theory is that it often looks as if the principle of causality were an independent, originally given form of consciousness. From a bent for scholastic

¹⁴⁷Kritik der reinen Vernunft. ed Kehrbach. p. 189–190.

formalism he has let himself be led into setting up a whole series of coordinate forms in which the original synthesis of consciousness was supposed to show itself. If we keep to a statement such as the one quoted, the theory amounts merely to this: that the concept of cause arises through the application of the law of sufficient reason to what is given.

In an earlier connection (B 11, b) we have seen how the law of sufficient reason is connected with elementary thinking. Here it is only a matter of stating the relation between ground and *cause*. We ask about grounds also when we have to do merely with our own thoughts and seek nothing other than what is *thinkable*. There is talk of cause when the question concerns the *existence* of *real* phenomena. The cause is then that in the series of phenomena which satisfies our asking for the ground of the existence of these or those phenomena. It goes here, then, as with all grounding: a difference, a contrast, is to be reconciled with the rest of the content of consciousness. The equilibrium, the connection, has been disturbed by a new experience and is now to be restored. The concept of cause thus does not lie ready-made in us; in the particular case we do not know whether the cause can be found. Purely empirically, the universal validity of the law of causality can to that extent be denied — a sign that it cannot derive from experience. The concept of cause designates a demand we set up; it contains the *postulate* that everything in our experiences shall stand in mutual connection. To that extent Hume is right that it is an instinct that guides us; only he did not see the origin of this [259] instinct in the very nature of our consciousness. Consciousness anticipates experience, sets up an ideal which it follows in its treatment of experience. This ideal would certainly die away if it were arbitrary and were not confirmed by experience. But the concept of cause is not merely a postulate; it is also a *result*. It is no foreign power to which our sensations and observations are subjected when we apply the law of causality to them. The receptive and the active side of our nature, the predominantly passive taking-in of experiences and the predominantly active working-up of them, play more and more into each other's hands. It might, regarded purely abstractly, be supposed that nature never presented quite similar cases. The ideal satisfaction of the postulate of causality rests on the presupposition that like conditions are followed by like consequences. But if reality now consisted of nothing but differences, if it varied infinitely, then the concept of cause would find no application, or rather would never have arisen at all. It would in any case soon die of atrophy, like organs that are not used.

Reality consists of a multitude of mutually intersecting series of phenomena. If we want to understand a phenomenon, the task is to find the series to which the phenomenon belongs. Were there only one single series of phenomena, there would be no difficulty.

Now there are not only many series, but they intersect one another in manifold ways. From this it comes that the law of causality stands as an ideal, because the circumstances of the past never quite repeat themselves. We can learn from the past only when the future to a certain degree resembles it. The sanguine tendency in the human mind can lead to overestimating both the similarities and the differences. In the first case it builds bold inferences on few and superficial similarities. In the second case it holds, even where the circumstances are essentially alike, that it is after all “another matter”; that is why it is, e.g., so difficult to learn from history. But this does not shake the general validity of the law of causality; on the contrary, that validity is confirmed by the fact that that sanguine tendency can be corrected.

[260] A thought-experiment such as that of a reality of absolute differences is, however, surely impossible to carry through. *Stuart Mill* held that by sufficient exercise of one’s power of abstraction one would be able to imagine a world where events followed one another in a chance manner (*at random*), without definite law.¹⁴⁸ At every moment, then, something wholly new would enter which had no connection with anything preceding. Even in our dreams and our wildest fantasies there is nevertheless a certain connection, however unstable and arbitrary. The law of causality, and through it the general nature of consciousness, necessarily asserts itself. *Stuart Mill*’s thought-experiment we can carry through only in so far as we can imagine the dissolution of consciousness itself. What he postulates as “thinkable” cannot be distinguished from the “*Ideenjagd*” of the insane, the last embers before the fire of consciousness is wholly extinguished.

Just as the concept of cause is an ideal, so reality or nature itself is something coming to be for us. There is no original harmony between our thinking consciousness and nature. But our consciousness, in accordance with its fundamental peculiarity, constantly seeks greater connection — i.e. seeks nature, reality; and since the history of the sciences shows us that this striving succeeds at least in some fields, nature may be said to come to meet our striving. But a conclusion of this process of development is in itself unthinkable. The connection can always be brought to encompass more phenomena, and within the connection new nuances and transitions can always be sought out.

4. The complete development of the concept of cause belongs to the philosophical theory of knowledge, not to psychology. In the foregoing we have in part overstepped the boundary between these two disciplines. We now return to the psychological field in order to point out that the determination of the concept of cause which Hume and Kant agreed in striking out — the anthropomorphic conception of every causal relation

¹⁴⁸Logic. III, 21, 1.

in nature as an activity of will analogous to the human — contains a hint as to *the* [261] *psychological origin of the concept of cause*. For with regard both to *motive* and to *form* the concept of cause proceeds originally from a practical interest. Only at an advanced stage of development does man take an interest in the connection of nature apart from whether it can serve his own purposes. It is the impulse of self-preservation that from the first leads to knowledge of the surrounding world: necessity teaches thinking, just as it teaches spinning. Knowledge that cannot immediately be made useful is not esteemed. When a missionary told an Indian about the creation, the latter answered: “My father, my grandfather and my great-grandfather used only to look at the earth to see whether it gave grass and water for their horses. They never troubled themselves about what went on in the heavens.”¹⁴⁹ And later on, too, knowledge and learning stand as means by which man adapts nature to his purposes. Thus *Baco*, the prophet of modern natural science and of industrialism, says: “Human knowledge and power coincide, for ignorance of the cause makes the effect impossible. For nature is conquered only by obeying her. And what in inquiry stands as cause appears in operation as rule.”¹⁵⁰ The cause is thus originally the means, a detour that man is compelled to take in order to reach his goal. It is only in pure instinct that the dim impulse immediately gives rise to the action. Reason (in the widest sense of the word) raises itself above instinct in that there joins itself to the impulse an idea of what must first take place in order that the impulse may be satisfied. In this idea of unavoidable intermediate links lie the germs of the concept of necessity and of the concept of cause; and when this idea acquires a more comprehensive content and becomes an object of independent interest, the concept of cause emancipates itself from the concept of purpose.

In close connection with this originally practical motivation of the formation of the concept of cause stands the original form under which it is conceived. The concept of cause, on account of its intimate connection with the nature of consciousness, asserts itself at all stages of human development; but that in which the cause is sought may [262] be extremely various. What at the different stages of mental development is regarded as good and sound causes depends on the whole standpoint one occupies. When the Australian native sees one of his tribe die without having been struck by a shot or other external injury, he concludes that there must be sorcery at play, and in order to discover who by his magic power has killed his comrade, he follows the direction in which the first insect that comes along moves away from the place of the death; the one whom he

¹⁴⁹ *Lubbock*: Origin of civilisation. 3 ed. p. 370.

¹⁵⁰ *Novum Organon*. I, 2.

then meets must be the murderer.¹⁵¹ On his presupposition this is good, rational thinking. That similar presuppositions in part still hold sway to this day may be seen from the fact that millions of people in the most civilized countries derive the so-called spiritistic phenomena from the influence of spirits.¹⁵² So long as the gods of mythology, or similar beings, stand as realities for consciousness, they furnish excellent and easily applicable means for satisfying the causal impulse. This impulse is, moreover, quickly sated at such a standpoint. When imagination has taken a couple of steps back in the series, it needs rest and posits a conclusion. The Greeks regarded the gods as the originators of the phenomena in nature (in any case of all important or conspicuous phenomena). Where did the gods in turn come from? This question *Hesiod* answers in his *Theogony* by describing how the dynasties of gods gradually developed out of chaos. But where chaos came from he did not ask, although he expressly declares it to have come into being. ("First of all chaos *came into being*," v. 116.) — From the mythological to the scientific explanation of nature there is a continuous series of stages. The anthropomorphic form is broken up gradually as more and more intermediate links become necessary, and the more the conviction arises of the constancy of these intermediate links and their independence of personal caprice. The scientific concept of cause is characterized by its finding the explanation of a natural phenomenon in tracing it back to a circle of

[263] other natural phenomena. Nature is explained by itself, not by something outside it. The medically trained observer of a sudden death gathers information about the man's constitution, way of life, descent, etc. He perhaps finds at the post-mortem an obstruction (embolism) in one of the arteries, and now explains the death to himself either as an effect of the supply of blood to the brain having been cut off, or by the heart having stopped. In this way he forms the picture of a connected process in which link is joined to link; the mysterious and sudden phenomenon then stands as the natural conclusion of this process. The law of causality is thus closely connected with the law of continuity. The scientific causal series too must in each particular case stop somewhere (as in the example just given, at the obstruction of the artery), either from fatigue or because there is no way of getting further. Thus the modern hypothesis of evolution stops at the primeval nebula, without therefore regarding it as an absolute first. For the conviction arises, through reflection on the nature of the concept of cause, that the series of causes — like the time-series and space — has only relative limits, and can always be thought of as continued further. The path of thought is prolonged before our gaze with every

¹⁵¹Fr. Müller: Allgemeine Ethnographie. 2 Ausg. Wien 1879. p. 214.

¹⁵²Cf. Girard de Rialle: La Mythologie comparée. I. Paris 1878. Chap. 14: Le Fétichisme chez les peuples civilisés.

advance. What seemed to be the outermost proves to be a transition to something still more remote.

5. But even if a conclusion of this process could be thought, what *validity* then has knowledge? This again is a question for the theory of knowledge; but it arises naturally at the close of the psychology of knowledge, and we will not pass it by without answering it as well as we can.

All our sensations and ideas are subject to *the law of relativity or the law of relation*. In the case of sensations this has been shown in particular in an earlier section (A, 6). As regards ideas, it follows from the development we have given of the ideas of time, space and cause. These ideas formed series in which every member is supported and determined by the other members. Every totality is in turn enclosed by a higher totality. In its most abstract form the law of relation shows itself in the fact that all grounding requires a plurality of ideas, and that thinking always moves forward through the choice and [264] rejection of possibilities (B, 11).¹⁵³ Knowledge is conditioned by the reciprocal relation between the similarities and differences of our ideas, or, to use Spencer's expressions, which recall the general laws of evolution, integrations and differentiations. Yet there is one relation which lies deeper than all others, and which becomes of special importance for the question of the validity of knowledge, namely the relation between consciousness and object, *subject* and *object*.

The popular answer to the question of what conditions the validity of our knowledge is: "our knowledge is correct when it agrees with the object." But how can we convince ourselves of this? "The object" we know only through our sensations and ideas! — On the basis of our *sensations* we ascribe certain properties to objects (light and darkness, colors, sound, heat, cold, smell and taste, etc.). But these properties do not belong to the objects themselves; they are a language in which we express them according to the way in which they affect our organism. Purely physically, color is only vibrations which — perhaps in an exceedingly fine substance, the ether — are propagated from the objects to us; sound is waves of air, etc. Were there no eyes and no brain, light as we sense it would not exist. We thus do not really sense things; but our sensations correspond to the state into which our brain comes when effects from the objects are propagated to it. Even the sensation of resistance does not help us beyond this; we always measure the resistance

¹⁵³Just as there has been speculation about the possibility of more than three dimensions in space, so one could also, purely formally, set up the possibility of a thinking that moved forward through a division into three instead of through a division into two. Instead of resting on the principle that A must be either B or ($\div$ B), thinking would then function according to the principle that A must be either B, ($\div$ B) or C (which is neither B nor ($\div$ B)). Such a thing is as impossible for us to *think* as it is impossible for us to *imagine* a fourth dimension. — Cf. *Boole: An investigation into the laws of thought*. p. 50 f.

by our own exertion of force; what it is in itself we cannot sense. There remain, then, the relations of space themselves, in which the objects (among which our organism itself, [265] the brain included, must here be reckoned) appear. But the relations of space we know only through the intuition of space, and the intuition of space is a psychological activity. Whether we follow the nativistic or the genetic theory, the intuition of space belongs to the subjective forms in and through which objects are given to us, and apart from which we know nothing about them. — In sense-observations *ideas* at once play in as well; it is by virtue of psychological laws that we form the idea of the particular object as a whole. Our knowledge is and remains, however logically we proceed according to the law of sufficient reason, a play of ideas. And just as impossible as it is to perceive anything about objects except through sensations, just so impossible is it to think anything about objects except by means of ideas and concepts. In order for that popular criterion of the validity of our knowledge to be usable, we would have to be able to go behind our consciousness and compare the object with the image or the concept of it that we have in our consciousness; but this is impossible, because it is self-contradictory.

In everyday life we do not stop at this difficulty, because we are constantly occupied with letting our ideas and sensations mutually control one another. When we say that we correct our concepts by comparison with “reality,” “reality” does not mean anything independent of consciousness, but more definite and more comprehensive observations than those we had hitherto had. But now we take the sum of all actual and possible sensations and ideas and ask how they are related to reality. Here we stand at the limit of knowledge. We cannot positively refute *Fichte* and *Stuart Mill* when they teach that the non-ego, the sum of all barriers to our action and all objects of our knowledge, is the product of a mental activity unconscious to ourselves. We cannot, then, if we want to push the matter to its extreme, know whether life is not a dream, a great, coherent and consistent dream.

It does not help to exclaim that such a doctrine is bottomless skepticism. If one will not, with *materialism*, hold fast with blind dogmatic defiance to the absolute validity of the content of sensations, although this content exists only in and for consciousness, [266] then one must, with *speculative idealism*, teach that appearance presupposes essence, and that behind the world of phenomena there rises an ideal world, which is the true reality. But however high one may wish to climb in the airy realm of speculation, the same fundamental relation between subject and object will come into force. *Kant* posited an ideal world (*noumena* in contrast to *phaenomena*) as the true reality. But this ideal world itself exists only for a consciousness. Therefore we cannot even form any concept

of an absolute knowledge, still less possess one. For an absolute knowledge the subject and object of knowledge would have to coincide; it would have to be a knowledge that itself produced its objects, and that knew no barrier, no resistance. But to it our logical and psychological laws, and thus our concept of knowledge, would not apply; and in spite of the many speculative and theosophical attempts, it has not yet been possible to point out and carry through other (“higher”) psychological and logical laws.

But just as materialism and speculative idealism sin against the law of relation, and in an illusory leap go beyond the limits of knowledge by positing an absolute reality, so subjective *idealism* is wrong in assuming an absolute dream. Dream too is a concept of relation, which falls away when it does not have a reality as its contrast. Fichte’s pure ego is just as dogmatic as the pure being of the dogmatists. What we call the reality of waking life is a greater connection than the reality of the dream, and if it is itself a dream, it must be so in relation to a still more comprehensive reality.

When it proves impossible to apply the popular definition of truth as the agreement of knowledge with the object, since the object itself presupposes our knowledge, we must seek the criterion of truth *within* the world of consciousness and not *outside* it. And it can then be nothing other than the inner harmony and consistency among all thoughts and experiences (cf. 2). If we cannot get outside the dream, we can still, as *Calderon* demands in *Life Is a Dream*, “live rightly in the dream”; that is to say, here, we can constantly extend the realm of our experiences and thoughts and found a deeper and firmer harmony among them. The law of relativity thus by no means deprives our concepts of their validity. That “everything is relative” does not mean that it is all equally true, but that all truth holds in quite definite relations — that all truth is a harmony. If there is a higher truth than that which human knowledge can reach, then all the truth we can know is a part of it. By using the means and the scale that are given us in the nature and organization of our mind, we can therefore really advance in the knowledge of objective truth. [267]

Chapter 6

The Psychology of Feeling

[268]

A. Sense-Sensation and Feeling

1. When we set feeling in contrast to knowledge, we do not thereby mean, as was already explained earlier (IV), to assert any contrast between different faculties or powers in the soul. The psychological distinctions concern only the elements of which psychical states prove, on closer consideration, to be composed, and it has been shown earlier with what right we distinguish between elements of feeling and elements of knowledge in every psychical state. It proved impossible to derive all other forms of conscious life from a pure state of feeling; even if the elements of feeling were altogether predominant at the primitive stages, closer consideration nevertheless established the presence of elements of knowledge. The task now is to show by what laws and along what paths the higher forms of the life of feeling are developed out of the primitive feelings attached to the immediate sense-sensations.

Some have wished to deny altogether that any such development takes place. Just as within the domain of knowledge there has been an inclination to maintain a sharp contrast between sense-perception and thinking, so it has been regarded as a degradation of the higher, ideal feelings to be akin to the primitive feelings. Here an ethical valuation is unjustifiably allowed to determine the psychological view. As an example of this tendency we may refer to the work, able in other respects, by *Nahlow*: »Das Gefühlsleben« (1862). This psychologist, who belongs to the Herbartian school, distinguishes between [269] the way in which the sense-sensations determine our general organic condition and the way in which inner movements and stirrings in our circle of ideas affect us. Only the latter kind of affection will he call feeling. The sense-sensation has, to be sure, its peculiar "tone," but this concerns only the bodily state, not the soul. One sees here how

a spiritualistic theory of the relation between soul and body can interfere in a special psychological question. According to Nahlowsky, physical pain is a sensation transmitted from the body to the soul, while psychical pain is a real feeling, an expression of the soul's own inner state during the interaction of the ideas with one another. Sensations are therefore explained by the relation between soul and body, feelings by the relation of the ideas to one another.

From the standpoint we have taken up, it is impossible to draw a definite boundary between physical and psychical pain. All pain is psychical, since psychical life is known to us only as conscious life. The psychological difference can consist only in this, that so-called physical pain, i.e. that which is due to immediate sense-sensations, is less composite, contains fewer elements, than so-called psychical pain. Toothache is a single, primitive feeling, while sorrow and remorse are feelings that presuppose ideas and memories. On the other hand, the higher feelings, just as well as the lower, have their corresponding physiological process. The difference can consist only in this, that the central processes, those taking place within the brain, play a greater part in the higher feelings than in the lower, which are determined predominantly by the single impression.

It thereby becomes possible that the "tone" of the sense-sensation, or the way in which it immediately attunes us, may psychologically be a germ out of which higher feelings develop. Before we go further into this development, we shall consider somewhat more closely the relation between the sense-sensations and the feelings of pleasure and displeasure, or pain, connected with them.

2. Feeling stands out clearly as an element distinct from the sense-sensation proper in certain experiences which establish that the pain of an impression needs a longer time for
[270] its arising than the sensation itself, and that sensation can arise without corresponding feeling and vice versa.

According to *Beau*, when one strikes a corn with a stick, one to two seconds pass between the sensation of touch and the feeling of pain. E. H. *Weber* found that when one dips the hand into very cold or very hot water, one first has a very lively sensation; this then decreases, but only to increase again at once and become pain. He sees an example of the same thing in the fact that when one starts at a sudden sound (e.g. at the sudden blare of kettledrums and trumpets after a pause in the music), a measurable time passes between the impression and the startled movement; and since the propagation of the impression through the sensory and motor nerves takes no noticeable time, he explains the phenomenon by saying that the activity of the brain is a condition of the arising of feeling. The same slowness in the arising of the feeling of pain in the sensations has been

established for electrical stimulation and for pinching the skin with a pair of tweezers, as well as under pathological conditions.¹

In a discussion of the relation between feeling and knowledge carried on between *Horwicz* and *Wundt* in the »Vierteljahrsschrift für wissenschaftliche Philosophie« (3rd and 4th volumes), the first-named investigator appealed to sudden and violent blows or knocks as examples in which the feeling of pain arose before the sensation. With very strong impressions this may perhaps be so; but with impressions of moderate strength one can easily observe that Beau and Weber are right. I experienced this very clearly once when, with my hands behind my back, I stepped a couple of paces backward, so that I touched a hot stove which I had not believed so near; I then noticed quite definitely the sensation of touch before the feeling of pain.

Pain must both spread and have a certain duration in order to be felt. *Richet* even says that without memory and without radiation pain would not be pain. (See above, [271] p. 114.) It is therefore not so simple in its nature as the sense-sensation; it presupposes the overcoming of a certain resistance in the central nervous organs.

In certain cases the feeling of pain is abolished while the sense-sensation is unimpaired. After section of the gray substance of the spinal cord, the part of the body lying behind the cut can be maltreated in every way, cut, crushed, burned, without any sign of pain being shown. Anaesthetics, cold, intoxication and hypnotic sleep have the same influence. A chloroformed patient who had a leg amputated did indeed feel the operation, but it seemed to him as though it were being performed on a wooden leg. After the battle of Eylau the operations were almost painless, because there were 10° of frost. Hypnotized persons feel no pain when their teeth are drawn. Just as there can thus be analgesia (freedom from pain) without anaesthesia (or apselaphesia, lack of the sensation of touch), so there can also be apselaphesia without analgesia (*anaesthesia dolorosa*). On section of the posterior columns of the spinal cord the sensations of touch in the hind part of the body disappear, while the feeling of pain persists.²

The feeling of pain can be of different character. It can run through a whole scale, from mere tingling or "falling asleep," pricking, formication (*fourmillement*), and prickling, to pain proper. The pains themselves differ in their immediate appearance. The feeling of fatigue is different from the feeling of displeasure in faintness and dizziness. There are burning, cutting, pressing, gripping and boring pains. In the opinion of some, these

¹E. H. *Weber* in *Wagner's Physiol. Handwörterbuch*. III, 2. p. 565–571. — *Richet*: *Recherches expérimentales et cliniques sur la sensibilité*. Paris 1877. p. 290–293. — *Funke*: *Tastsinn und Gemeingefühle*. (Hermann's *Handbuch*. III, 2. p. 298–300).

²C. *Lange*: *Rygmarvens Patologi*. p. 11. 92 f. 111. — *Richet*. p. 118 f. 258 ff. — *Preyer*: *Die Entdeckung des Hypnotismus*. p. 44.

differences are not to be regarded as differences in kind, but rest on the different strength, extent and duration of the pain. Here too the more central character of pain in relation to the sense-sensation comes out. It is only at its peaks that pain appears as altogether simple, while the sense-sensation from the first appears predominantly with the stamp [272] of simplicity (but cf. on this V. A, 2). But those differences in the immediate appearance of pain contain at the same time a new proof that in every state of feeling there are also elements of knowledge. The difference between cutting and boring pain is just as much of an intellectual character as the difference between the sensation of yellow and the sensation of blue.

As regards the feeling of pleasure, it was already maintained by *Aristippus* that it presents no differences in kind. Pleasure was not different from pleasure; only the circumstances, the causes, were different. But *Aristippus* pushed this so far as to make the feeling of pleasure something altogether simple, setting up the purely momentary pleasure, unmixed with all remembering and looking forward, as the end of life. Such altogether simple and instantaneous feelings do not exist. *Aristippus*'s moral principle would presuppose a dissolution of consciousness into flashes; it would then be a matter of possessing such virtuosity that one could move from flash to flash, and at each moment be wholly concentrated in the single flash. But as soon as *Aristippus* maintains that insight and practice are necessary for attaining such virtuosity, he admits that the one moment has something to do with the other. And even if the general character of the feeling of pleasure is the same through all its forms, it nevertheless acquires in each particular case certain definite nuances in consequence of the way in which it is called forth. Thus the feeling of pleasure in muscular movement has other elements in it than the feeling of pleasure in taste or smell. It is on this composite, nuanced character of even the simplest feelings of pleasure that the possibility rests of pleasure being developed and transformed through connection with elements of idea and thought. *Aristippus* therefore also consistently ranked physical pleasure highest, because it comes nearest to being altogether momentary and simple.

3. In so-called physical pleasure and pain, then, there are already elements of knowledge besides the elements of feeling proper, although the latter decidedly predominate. We shall now briefly consider the relation between these two kinds of elements in the domains of the different senses. It will appear that the domains of sense can be arranged in a series, such that at one end the elements of feeling have a decided preponderance [273] over the elements of knowledge, while at the other end a more even development of both kinds appears.

a. Characteristic of the general sensation is the lack of definite and local characterization of the particular sensations. These vanish in a general feeling of well-being or indisposition, which constitutes, as it were, the result at which the brain arrives from the reports it receives from the various parts of the organism. We have here a feeling of our existence as a whole, of the general course of the life-process; this feeling, attached to the general sensations, we shall call the *feeling of life*. The quality and quantity of the blood, the liveliness of the circulation, the tension of the vessels (tonus), the more abundant or more sparing secretions of the glands, the slackness or tautness of the muscles (both the involuntary and the voluntary), the brisk or labored course of respiration, the normal or abnormal course of the digestive process — all this helps to determine it, without any of the factors mentioned needing to stand out by itself. The general sensations form a chaos, which receives its stamp from the contrast between well-being and ill-being, and whose special nuances are, in the nature of the case, determined by one organ or another playing a particularly predominant part, without there always arising on that account an express consciousness of it as the source of the sensation. On the contrary, it is characteristic of the general sensations that they often “radiate” (irradiate) or are projected to places quite other than those where the cause really lies. In this little-articulated state of conscious life, the state of the organ predominant at the given time becomes determinative for the general fundamental mood.

This fundamental mood can be characterized only by certain general features, which stand in close connection with the easy and free, or hindered and heavy, course of the organic life-process. Thus the feeling of freedom, security and strength stands in contrast to the feeling of inner resistance, unrest, anxiety and faintness. In the contrast between the *feeling of strength* and the *feeling of faintness*, the muscular sensation plainly plays an important part. Even if we do not consciously stretch our muscles, they are nevertheless always in a certain degree of tension; resting muscles (in sitting, lying, etc.) are not wholly slack, but partly contracted; the muscles of mastication always involuntarily draw the lower jaw up against the upper; the *upper* eyelid is always held up, etc. During [274] sleep this “latent innervation,” as it has been called, is diminished; the posture of the body then conforms more to the laws of gravity; — and yet there is still a difference between the posture of the dead and that of the sleeping.³ The feeling of lightness and of *freedom* is connected especially with the course of respiration and of the nutritive functions. Hindered respiration produces a feeling of painful unrest and anxiety. If the first cry of the child is called forth by the hunger for air that sets in as a consequence of

³C. Lange: Rygmarvens Patologi. p. 344 ff.

the interruption of the placental circulation, life begins with anxiety. A physician from the West Indies who consulted Charles Bell reported that “when he fell asleep, just as will and consciousness ceased, the involuntary movements also stopped, with a sensation of death, during which he as a rule woke up in convulsions. His medical friends have sat up and watched over him, and they found that when sleep took hold of him, the respiration became slower and weaker. The heart and pulse also slacken and cease to beat when sleep comes over him, and a moment later he wakes up in terror.” Incubus or nightmare seems (according to *Laycock*) to be caused by the sleep of the respiratory centers.⁴ Many abdominal diseases bring with them the same feeling. It seems to the patient “as though nature had stopped its activity in him.”⁵ With nervous pains in the pit of the stomach (cardialgia) there may be connected — perhaps on account of disturbances of the circulation — a terrible feeling of anxiety and powerlessness, which makes these pains some of the very most dreadful sufferings.⁶

In their first stages the feelings of *hunger* or of *thirst* also have the vague character of the general feeling, expressing themselves as an indisposition, an unrest. With this, however, definite local sensations soon connect themselves: in the case of hunger, of pressure and gnawing in the stomach; in the case of thirst, of dryness and burning in the tongue and throat. — We must here disregard the way in which these feelings appear [275] when definite ideas of their significance attach themselves to them, and when the habit has been formed of taking nourishment at fixed times.⁷

b. The sensations of touch and of *movement* stand so near to the general sensations that they are often absorbed into these without standing out independently, and, as with the general sensations, the strength of the impression plays a greater part than its quality. But when the strength does not exceed a certain degree, these sensations are nevertheless fine enough to be connected with feelings of pleasure or displeasure which have a certain independence in relation to the general feeling of organic well-being or ill-being. In active movement a peculiar sense of comfort can be felt, and one kind or form of activity can be preferred to another, just as one color is preferred to another. Likewise there is a comfort in touching soft and smooth surfaces and a displeasure in touching rough and hard ones, which may already be said to have a certain aesthetic character.

⁴*Laycock*: On the reflex function of the brain. (British and Foreign Medical Review. 1845. 19er Bind). p. 306.

⁵*Richet*: Recherches sur la sensibilité. p. 312.

⁶*C. Lange* p. 152 ff.

⁷This “psychical factor” (*Ranke*: *Physiol. des Menschen*. 3 Aufl. p. 220) no doubt also plays a part in the sensation of cold, which is why the adult suffers more from cold than the child, in whom no further ideas are yet connected with it (*Perez*: *Les trois premières années de l’enfant*. Paris 1878. p. 8 f).

c. *Taste* still stands near the general sensation. It is closely bound up with the function of nutrition, as a kind of gauger and rejecter of what is to be taken up and worked over by that function. The feelings of comfort or of disgust which according to some are due to the part of the organs of taste lying on the last third of the tongue have quite the same stamp as the general feeling of life determined by the general sensation. But alongside this, the differences of quality acquire a definite significance here. Even newborn children seem able to distinguish between the different qualities of taste. To each of these qualities (sweet, bitter, sour and salt) certain nuances of feeling attach. In their simplicity these are indescribable; but that they are present is seen from the fact that expressions from the domain of taste are used to designate higher states of feeling. Only for the salt taste there seems to be no analogy in the domain of the higher life of [276] feeling; on the other hand we have sweet memories, bitter experiences and sour looks (biting into the sour apple).

d. The *sensations of smell* also present qualitative differences, but these have not attracted so much attention that language has formed separate words for them. In animals smell plays a predominant part among the senses; by it the prey is tracked, danger is avoided, and the sexes are brought together. This deeply penetrating significance smell has not quite lost in man. It is to the process of respiration what taste is to nutrition, and it also works together with the sensation of taste as the guardian of the alimentary canal. Quite apart from associations of ideas and idiosyncrasies, the sensations of smell moreover have a great, often violent, influence on the mood. But since this sense nevertheless stands almost as a relic in man, at any rate in civilized man, it is easily loosened from its close connection with the feeling of life and acquires its own scale of feeling.

e. The higher senses, *sight and hearing*, seem almost wholly emancipated from the immediate connection with the feeling of life. And yet they too are originally only outposts for it. Smell and taste make it possible to carry out a preliminary examination, so that nothing shall get into the alimentary canal that could be harmful to the continuance of life; smell and the sense of touch give warning that the enemy or the prey is approaching. In a similar way sight and hearing stand from the first in the service of instinct.⁸ Just as the sensation of taste is followed by an impulse to swallow, so the sight of a grain or an insect immediately awakens in the newly hatched chick the impulse to peck it up, or the sound of the hen's cluck makes it dart off toward the source of the sound. One can eat

⁸Through this connection with instinct, sight and hearing, as well as touch, are able to call forth certain elementary stirrings of what, under the influence of definite ideas, becomes hope and fear. See below (B, 2 d).

and enjoy with the eyes.

What places the higher senses more freely over against the feeling of life is, in the first place, their definitely marked scale of qualities. As long as differences of strength [277] play the chief part, the sensations intervene more directly in the feeling of life itself. This is seen especially at the peaks of pleasure and pain, even where purely intellectual and aesthetic feelings are in question. The special forms of tone and nuances of color awaken a finer play of feelings than impressions which by their strength affect the very processes to which the continuance of life is bound. Next, it is also of importance that impressions of light and sound on the whole do not belong to the stronger influences on our body, and that there are means in the sense-organs for damping impressions that are too violent.

In the developed consciousness so many secondary ideas attach themselves to colors and tones that it becomes difficult to discover what effect on feeling the elementary sensations produce in and for themselves. Practically, we use colors and tones as means of orienting ourselves; we do not think so much of them themselves as of what they signify. Their immediate effect is as a rule unconscious to us, and only when the mood they awaken comes into a certain contrast with other moods do we become aware of it.⁹ In order to feel such effects fully in their peculiarity, *Goethe* used to look through colored glass and so make himself one with the color, see the whole world green, yellow, etc. His remarks on the feeling-tones of colors are still classic. H. C. Ørsted, in his fine treatise on this subject, follows *Goethe* for the most part.¹⁰

The connection with the feeling of life itself is seen most clearly when we keep to the differences of strength. A lively sense of nature has, as Ørsted remarks, always connected light and life, darkness and death, with one another. In the teaching of Zoroaster this connection was even made the basis of the whole view of the world. Light is a condition of life for all organic life. Quite apart from all associations of ideas, the contrast of feeling [278] attached to the contrast between light and darkness is therefore easy to understand. But the comfort in light and the displeasure in darkness are not merely part of the general feeling of life; light is the condition of the functioning of the organ of sight, and the joy in light is the expression of this function unfolding in a healthy and normal way, just as the displeasure in darkness is the expression of a hindered impulse to activity.

The eye needs *colors* just as much as light. "Let one recall," says *Goethe*, "the refreshing

⁹»Von einem geistreichen Franzosen wird erzählt: Il prétendoit que son ton de conversation avec Madame étoit changé depuis qu'elle avoit changé en cramoisi le meuble de son cabinet qui étoit bleu«. *Goethe*: Farbenlehre § 762.

¹⁰To Capitler af det Skjønnes Naturlære [Two Chapters of the Natural Theory of the Beautiful]. Kbhvn. 1845. — This point is also interestingly treated by *Nahlowsky* and *Wundt*.

feeling that arises when on a dark day the sun shines on a single part of the landscape and the colors become visible out there. That healing power was ascribed to colored precious stones may perhaps have arisen from a deep feeling of this inexpressible comfort.” — With regard to their effects on feeling, Goethe divides the colors into two series: the positive (the plus side) and the negative (the minus side). The positive series, which consists of yellow, orange and vermillion, has a rousing, lively and striving effect; the negative series, which consists of blue, violet and purple, calls forth a restless, soft and yearning mood. Yellow and blue in particular appear as opposites to each other. *Yellow* stands nearest to white. In its highest purity it has the stamp of clearness, and has a cheering and warming effect. This is experienced especially when one looks at a landscape through yellow glass on a dark winter day. “The eye rejoices, the heart expands, the mind is cheered; an immediate warmth seems to stream toward us.” *Blue* has the stamp of darkness. It gives a feeling of cold, recalls black, and shows us objects in a mournful light. — The two series meet on the one side in red, on *the other side* in green. In *red* Goethe sees the highest of all color phenomena. It gives an impression both of seriousness and dignity and of beauty and grace, the former in its darker, the latter in its lighter nuances. A well-lit landscape seen through a purple-red glass appears in a terrible light: “such must be the tone of color spread over heaven and earth on the Day of Judgment.” *Green* stands midway between yellow and blue, while red unites both effects. The impression of green is rest and simplicity. »Man will nicht weiter und kann nicht weiter.«

These elementary moods are of course altered by different lighting. Diminished intensity of light gives greater *seriousness* and damps the energy of the positive series. Likewise it follows of itself that the elementary moods can enter into combinations and [279] modifications when the colors are combined in different ways. Here we have wished only to use Goethe’s account in order to characterize these simple nuances of feeling.

Corresponding to the contrast between light and darkness, the contrast between *sound* and *silence* is at work in the domain of hearing. There is a natural pleasure in all sound merely because it sets our organs of hearing in activity. The ear-splitting music of children and savages satisfies only this need for vigorous functioning of these organs. — Corresponding to the positive and negative series of color-nuances, the contrast between high tones and deep tones has been set up. The former have a cheering and rousing effect, the latter a depressing effect, or they call forth seriousness and longing. The timbre of the instruments has then been arranged according to the same relation of contrast. Here too, then, it is seriousness and elation, energy and rest, that mark the chief nuances of

the elementary feelings.¹¹

That there can be no complete contrast between the feelings attached to the differences of quality and those attached to the differences of strength is seen already from the fact that all differences of strength, psychologically regarded, appear as qualitative differences (see V A, 5). And if Goethe's account is correct, the great fundamental contrast in feeling, motivated by light and darkness, is a frame into which the feeling-effects of the particular colors fit. That feeling thereby undergoes alterations follows of itself. But this alteration may well be connected with purely quantitative relations. For darkness to awaken real displeasure, it must appear overwhelming and dominating. As a subordinate factor in a landscape it has the effect of setting off and enlivening, and it is then a question whether one is right in saying that an element of displeasure is present in the [280] impression. What caused the displeasure was, after all, the hindrance of the activity of seeing; but now this activity is precisely furthered by the dark shadow. The higher senses are distinguished precisely by being able in this way to apprehend combinations in which what, in and for itself and alone, would awaken dislike can serve to heighten liking. Something of this is already present in the lower senses, since, e.g., small quantities of bitter and sour substances heighten the agreeableness of a taste. But only the higher senses command such a wealth of combinations that new and peculiar nuances of feeling, partly independent of the fluctuations of the feeling of life, can develop.

What holds for the quality of sensations holds also for their *form*. By the very way in which the particular tones and colors are combined, liking and dislike can be awakened. Here belongs the feeling of pleasure in symmetry, in definite relations of form (the golden section), in rhythm and harmony. These feelings too detach themselves more or less from the general feeling of life.

4. The feelings attached to immediate sense-sensations thus form a graded series from the feeling of life to the finely wrought nuances of feeling in the qualitative sensations of the higher senses. This graded series marks the natural course of development of the elementary feelings. Before the definite appearance of special organs and functions, in the individual and in the race, feeling can only be a chaotic mass, a summary expression of the course of life. But it is necessary for the very continuance of life that the feeling of life be differentiated, take on special forms. In order to maintain its own existence, the individual must be able to feel the significance of the existence of other things. The

¹¹Cf. *Nahlowsky*: Das Gefühlsleben. p. 142 ff. — *Wundt*. I. p. 471 ff. — As *Panum* has remarked, the physical and the physiological approach here lead to different results. If one takes account of wavelength and frequency of vibration, red will correspond to a deep tone, violet to a high one; according to the physiological impression, on the other hand, red corresponds to a high, violet to a deep tone. (*Sanserne og de vilkaarlige Bevægelser* [The Senses and the Voluntary Movements]. p. 198 ff.).

special development of the sense-organs makes this possible, but it becomes actual only through a simultaneous development of higher, intellectual functions.

As regards the general relation between feeling and sense-sensation, we can now set up the following result. In respect of strength they stand in inverse relation to each other, such that the stronger the element of feeling becomes, the more the properly sensing or cognizing element vanishes. The sense-sensations that awaken the strongest pleasure and displeasure teach us least about relations outside us, however great their practical significance may otherwise be as warning or enticing. But where the qualitative character of the sensation can assert itself at a strength suited to the sense-organ, there feeling is formed and specified together with the sensation. What it loses in violence it gains in richness and manifold nuance, as well as in independence of the immediate struggle to exist. [281]

The same sum of energy that in the feeling of life is concentrated on the one question of being or not being, on organic weal and woe, is distributed in the qualitative feelings and led out into several different streams.

B. The Development of Feeling through Knowledge

1. The feelings attached to immediate sensations acquire a peculiar character through the qualitative character of the sensations. The mere fluctuation between pleasure and displeasure is developed into a series of states of feeling, each of which receives its individual stamp by being attached to a definite content of sensation. Already here, then, one may speak of the development of feeling through knowledge; for the sense-sensations, in so far as we can distinguish them from the feelings of pleasure and displeasure, belong to the domain of knowledge. But we come to the most decisive relation between knowledge and feeling only when we examine the influence of ideas on the feelings. As we have seen earlier, the pure sensation is only an abstraction; there will always be connected with the impression of the moment more or fewer, stronger or weaker residues or memories of earlier sensations. The point at which the circle of ideas and the associations of ideas gain influence on feeling must therefore lie at the beginning of conscious life, even if this influence stands out more clearly only later.

Since we start from the assumption that feelings of pleasure and displeasure are present in the most primitive psychical states and are given prior to any *definite* and distinct sense-sensation, we cannot use the common definition of feeling as the effect of sensations and ideas on consciousness. As a primitive element of consciousness, feeling

is already given before sensation and ideation can exert any influence; this influence [282] only develops and modifies what is already given beforehand. If we have nevertheless (IV, 7 c) declared ourselves against the assumption that all conscious life has developed out of pure states of feeling, this was partly because it is not probable that feelings occur altogether without sense-sensations (at any rate general sensations and sensations of movement), partly because an intellectual element is already given in the differences of strength of feeling and in the fluctuation between pleasure and displeasure. It is thus only by abstraction that we speak of mere feeling without any elements of knowledge. Yet it is justified theoretically to take such an abstraction as a basis, since it facilitates insight into the laws that hold by presenting the relation in a still simpler form than actual experience can.

2. a. When the feeling of *displeasure* or of *pain* is connected with the idea of its cause, it passes over into *aversion* or *anger*. Before such an idea can arise, the feeling as yet has no content, no definite direction. The violent movement that pain causes in the newborn child is a general discharge, but is not directed against any definite object.

“It was” — says *Darwin* (Biographical Sketch of an Infant. Mind 1877. p. 287 f.) — “difficult to decide how early anger was felt. When the boy was eight days old, he wrinkled the skin around his eyes before he screamed; but this may derive from pain or distress, not from anger. At about ten weeks old he was once given rather cold milk, and a slight frown came on his forehead while he drank, so that he looked like a grown person who is vexed at having to do something he does not like. But when he was four months old, one could see clearly that he fell into a great passion. The blood rushed strongly to his face and scalp. A small cause was sufficient. Thus, when he was seven months old, he screamed with rage because a lemon slipped away from him. At seven months old, when he had been given a wrong plaything, he pushed it away and struck at it. I believe that the blow was an instinctive sign of anger, just as a young crocodile, just crept out of the egg, snaps its jaws; but I do not believe that he thought he was [283] doing the plaything harm. At two years and three months old he was very skillful at throwing books, sticks, etc. at anyone who offended him.” — A child of three months which had been given bitter medicine from a blue cup afterwards always turned its head away when it saw this cup, made a wry face and cried.

The presupposition of aversion and anger, then, is that a fixed connection forms in consciousness between the feeling of displeasure and a certain observation or idea. The content of this idea will in the nature of the case be something that plays an essential part in the arising of the feeling of displeasure, even if it is not the real cause (as the blue

cup, e.g., was not). This feeling can arise both toward lifeless things and toward personal beings; everything that hinders and pains is its object. With further development, and on the presupposition of an ability to enter into the feeling of another individual, anger becomes pleasure at the personal cause of the pain coming to feel pain itself (what *Bain* has aptly called *malevolent sympathy*). There arise *desire for revenge* and hatred, while mere aversion and anger in themselves lead only to removing ourselves from the object or removing it from us.

b. By a similar metamorphosis the feeling of pleasure becomes *joy* and *love*. The idea of that which stands in essential connection with the feeling of pleasure fuses with it and determines it in a certain direction. There arises an involuntary need to hold fast and protect what awakens pleasure. Joy is this need seen from the intellectual side; it denotes pleasure in dwelling on the object; love denotes the active side, the impulse to an action that can secure the object or at any rate secure it to us. At higher stages of development there arises *the sympathy of love*, pleasure in the pleasure of others.

c. From this account it is seen that aversion and joy, anger and love, cannot be separated from impulse or *desire*. Every pleasure or displeasure sets the organism more or less in motion. The form and direction of this movement are determined by the original structure of the organism. While it is often an ineffectual, if not directly harmful, process of discharge for the energy roused to activity, in other cases (in the so-called instinctive actions) it is a purposive approach to or turning away from the object. An impulse arises when this involuntary initiation of a movement asserts itself in consciousness with a certain idea of the goal to which it leads. If the movement proceeds easily and immediately, no impulse arises; then the whole takes place below the threshold of consciousness as a simple reflex movement. The movement must meet a certain resistance, which, however, need not be so strong as to call forth downright displeasure. There is a certain unrest (what *Locke* called *uneasiness*) in all impulse; but this lies simply in the fact that impulse goes beyond the present and resting state, in that it either holds fast the cause of pleasure or removes that of displeasure. The stronger the resistance becomes, the more the unrest passes over into displeasure, in its simplest form displeasure at hindered movement. With this there soon follows displeasure at the object of pleasure not being able to be held fast, or at the means of removing the cause of displeasure not being procurable. In this way impulse becomes more and more composed of feelings of pleasure and displeasure, and thereby gradually moves more definitely away both from the simple reflex movements and from the instinctive movements called forth by immediate sense-sensations. Impulse then acquires a content of ideas, is connected [284]

with the thought of what hinders or furthers its purpose. — If one wishes to distinguish between impulse and desire, it will be most natural to regard desire as impulse guided by distinct ideas. — If satisfaction is delayed very long or fails altogether, impulse, if it has a deep ground in the nature of the individual, passes over into strong pain.

d. Impulse is originally sanguine expectation. A rising idea is at first not distinguished from a real observation; they have, to be sure, as a rule different strength, but of the significance of this difference there is no innate knowledge; only experience, and that here means disappointment, impresses the difference between the possible and the actual. (Cf. V B, 5.) Now when the idea of disappointment asserts itself with greater or lesser strength alongside the idea of satisfaction, so that the mind dwells now on the one, now on the other, there arise *hope* or *fear*. Let α be a feeling of displeasure, a the idea of that [285] which can remedy it; or α a feeling of pleasure and a the idea of what can preserve and heighten it. Let b be an idea by which a is favored, c another by which a is cancelled. Both b and c stand in connection with a and are therefore called forth by it according to the laws of the association of ideas. As long as neither b nor c is given as actual experience, consciousness will pass now from a to b, now from a to c. The question now is what influence these alternating transitions between the ideas have on feeling.

It has been shown (A, 2) that feeling on the whole arises more slowly than sensations. By observation one will easily convince oneself that feeling also arises and moves with greater slowness than ideas. It takes longer to pass over from joy to sorrow than from the idea of something joyful to the idea of something sorrowful. Even in the sanguine, thought and fancy move with greater rapidity than mood. Now when consciousness passes over from the one of the viewpoints indicated (ab) to the other (ac), the idea c will have a tendency to awaken a new mood (γ); but since the mood (β) awakened by the first idea (b) still persists, the two moods will meet each other and enter into a combination. It goes here as with the waves that beat against the shore; the newly arriving one takes up into itself the backwash of the preceding one. There thus arises a mixed mood: when b β keeps the upper hand, hope, — when c γ gains the upper hand, fear. Both feelings presuppose a certain play of possibilities.

The moods of hope and fear appear in countless gradations and nuances, according to the relation of the possibilities to one another. The greater the possibility of attaining the goal is assumed to be, the more will hope approach *confident expectation*, in which the mind rests in the idea of the bright future with no other unrest than that which is inseparable from the consciousness that the present is to give place to a future, — the smaller, the more fear approaches *despair* or *resignation*. If the possibilities are regarded

as equal, so that fancy is attracted equally strongly from both sides, the mind feels itself divided. Two different moods strive to spread in consciousness, but neither of them can gain the mastery. Hence arises the mood of *doubt*, whose chief peculiarity is the painful unrest that can awaken so violent a need for decision that it seems a matter of [286] indifference in what direction the decision goes, if only the pain of dividedness ceases.¹² People who are tormented by sudden promptings or fixed ideas sometimes suffer so much from them that they follow the prompting to murder or suicide merely in order to have peace.¹³

As we have here described the feelings of hope and fear, they presuppose a certain development of the life of ideas, a certain ability to weigh possibilities. But just as there is a more *elementary form of hope* in the original sanguine tendency, which only gradually is checked by experience, so there is also an elementary fear at sudden and strong impressions.¹⁴ This kind of fear is akin to astonishment at the new and unexpected and at first not to be distinguished from it. The unknown and unexpected can awaken both feelings, and both feelings find their expression in one's starting, standing motionless or shrinking back. The lower animals contract at a touch just like the higher ones and just as men still involuntarily do. We are startled by a sudden sound, a rustling in the leaves or an unexpected address, without accounting to ourselves for what affects us. It is one of the first feelings that can be traced in infants. It has something of the character of instinct, and the will cannot always check it. In so far as it is due merely to the unknown, it falls away with the development of knowledge. One of the most beneficial effects of science has been to stop up false sources of fear. But on the other hand the development of the life of ideas and of reflection has favored fear by leading to and giving practice in [287] occupation with possibilities. The play of free ideas makes possible a dividedness and cleavage in the mind which the unreflective state does not know, however much it may suffer from the instinctive terror of the unknown. Here too, however, scientific insight can exert a beneficial influence, in that by its criticism it restricts the possibilities and in its knowledge of laws builds a firm ground for the conviction as to what cannot become

¹²Othello (Act III. Sc. 3) says to Iago, when the latter has awakened doubt in him about Desdemona's fidelity:

You have set me on the rack!
Far better to be wholly deceived
than only to know a little.

¹³Cf. *Ideler*: Biographien Geisteskranker. Berlin 1841. p. 134. *Dagonet*: Nouveau traité des maladies mentales. Paris 1876. p. 311.

¹⁴This contrast between hope and fear can be followed still further back, namely to the feelings, conditioned by certain general sensations, of lightness and freedom on the one side, anxiety and oppression on the other. Cf. A, 3 a.

an object of knowledge.

e. When two conflicting feelings press forward at once and simultaneously assert themselves with equal strength, the unbearable dividedness mentioned above arises. This, however, will be a rare case and not of long duration. When *Shakespeare* has King Claudius describe his mood at his marriage to his brother's widow as a complete *equilibrium* of joy and sorrow,¹⁵ it was surely the intention of *the great poet-psychologist* to mark him thereby as a hypocrite, who betrays himself by the unnaturalness of the state he attributes to himself. Where the one feeling does not displace the other or reduce it to a subordinate factor (cf. A, 3 d), they will relieve each other rhythmically. *Plato* describes in this way the mood of the disciples of Socrates during their last conversation with their teacher. "I found myself," he has *Phaedo* say, "in a truly strange state, in an unusual mixture of joy [at the content of the conversation] and sorrow, when I considered that he was now soon to die. And all those present were in almost the same mood; now they laughed, now they wept." Such alternation is the natural state where different motives assert themselves. It will not be able to last, however, since the mind seeks equilibrium and converts the successive differences into simultaneous ones; the two feelings will thereby fuse into a new feeling, sorrow and joy, e.g., into wistful melancholy. In gentler natures this conversion will take place more quickly than in more passionate ones. *Homer* [288] represents Andromache "smiling through tears" (δεκρύνειν γαλάσσασα), when Hector hands her their little son in order himself to hasten out to the battle.

Sibbern has therefore rightly distinguished between the mixture or alternation of different or even conflicting states of feeling and mixed feelings in the proper sense. To a mixed feeling it belongs that the difference between the elementary feelings is no longer noticed, both working to form one single total feeling — "as when fear is overcome by boldness with regard to the very thing feared, or when strength in its struggle and exertion feels itself inflamed or furthered precisely by the hindrances or difficulties. A certain contentment with life, indeed blessedness, by virtue of overcome and subdued affliction or of some other effect of adversity, belongs here."¹⁶

3. We have analyzed some of the simplest states of feeling in order to discover what relation obtains between the elements of feeling and the ideas connected with them. We shall now see what this analysis teaches us about the law of the development of the

¹⁵*Shakespeare's Hamlet*. Act I. Sc. 2.

— with subdued joy, with one eye
wet with tears, one in the gleam of hope,
with delight at the funeral, dirge at the wedding,
while sorrow and pleasure held balance in the soul.

¹⁶Psychologie. Kbhvn. 1856. p. 380.

feelings.

In the older psychology there appears a tendency to see in knowledge the main thing and what is properly constitutive in consciousness. For *Plato* the immortal part of the soul was one with thought or reason; the feelings of pleasure and displeasure, like the sense-sensations, arose only through the soul's being enclosed in a material body. In modern idealistic philosophy down to *Hegel* and *Herbart* a similar view can be traced. *Kant* and *William Hamilton*, e.g., could conceive of spiritual beings with reason without feeling and will, but not the reverse. For them the essence of consciousness coincides with knowledge. *Consciousness is a knowledge*, says Hamilton.

It was therefore usual to apply without further ado to the feelings the laws found for the development of ideas. This has happened especially with the laws of the association of ideas. The natural growth of the life of thought rests on these laws; but do they also hold for the feelings among themselves? Some psychologists (among the older ones, e.g., *Spinoza* *Eth.* III, 14) assume this and hold that related feelings and feelings that have [289] earlier arisen together will later call each other forth. But it is a great question whether a feeling *in itself* has the power to call forth another, however great the kinship and likeness may be. Feeling *as such* rests in itself. One idea leads to another; therefore there can be a coherent system of thought. But the feelings lead beyond themselves only indirectly, namely with the help of thoughts and of the will, and are not connected by any dialectical coherence. It has therefore been wrongly objected to the expression "association of ideas" that it does not include the feelings, which after all are also members in the series of states of consciousness.¹⁷ There are, as we have seen, hardly any quite pure states of feeling without elements of knowledge; at any rate they must be relegated to the very first beginning of conscious life or to its cessation.

The question is: can the element of feeling of one mental state attract the element of feeling in another mental state, or does the transition always take place through the connection of the elements of knowledge?

1) A state of consciousness (A) consists in the connection of an element of feeling (α) with an element of knowledge (a). Now if there are other, related elements of knowledge (a_2, a_3, a_4 etc.), which a has a tendency to call forth, some of these will be able to enter into a connection with a, and through a with α . While α was earlier determined only by a, it is now determined by $a + a_2 + a_3 + a_4$. The mood is modified; the state A becomes A_4 . We thereby obtain no wholly new kind of feeling, but the given feeling spreads over a larger part of the content of consciousness. I love a human being — and thereby acquire

¹⁷ *Bouillier*: Du plaisir et de la douleur. 2 éd. p. 173 f.

interest in and love for everything that concerns him and is connected with him. Love is always an expansion of our being; it joins the lover and the beloved into a whole. But what we love is a part of the world, and since the world is a coherent whole, our feeling will extend further, when special causes do not prevent it. — It goes similarly with other feelings: it is according to this rule that the devil takes the whole hand once he has got the finger.

[290] 2) The original feeling undergoes a greater change when the new element of knowledge is not so immediately connected with the first. If we have $A = a + \alpha$, and a proves to be closely connected with b , then A becomes A_b , i.e. the feeling is still of the same kind, but more special. When, e.g., a certain quality (a) has great value in my eyes, and I discover, or believe I discover, that I myself (b) possess it, my admiration (A) becomes pride (A_b).

3) When, finally, the new idea (b) itself brings with it a feeling (β) which it would have awakened had it reigned alone, a new kind of feeling (B) arises. Instead of $A = a + \alpha$ we get $B = \alpha + (a + b) + \beta$: i.e. through a and b , α and β are connected. This is the schema for the development of hope, fear, wistful melancholy and similar composite feelings.

No sharp boundary can of course be drawn between these three cases, since already a_2 and a_3 etc. must call forth new nuances of feeling, which fuse with the feeling given beforehand.

The association of ideas is thus the channel through which the feelings mingle with one another. Through the relation of thoughts to new thoughts, the feelings pass over into new feelings. But since the movement of feeling proceeds more slowly than the movement of thought, it is no wonder that intellectual progress as a rule has the lead over the development of the life of feeling. Thought is the most mobile part of our being; feeling forms the foundation, to which the effects from the more mobile surface are propagated only gradually.

Although the laws of the association of ideas cannot be applied to the relations of the feelings, they are nevertheless, on account of the connection between feelings and ideas, of the greatest significance for the development of the life of feeling.

4. If this view¹⁸ is correct, it must be confirmed by the way in which feelings are recalled in memory; for the laws of the association of ideas are the laws of memory.

[291] Now it at once appears that ideas are more easily recalled than the feelings connected with them. Pictures and situations from our past life we can recall to ourselves, but only

¹⁸It is already hinted at by *Hume* (Treatise II, 1, 5. 8), more definitely by *Bain* (Emotions and Will. I, 5) and *Kirchmann* (Erläuterungen zur Kant's Anthropologie. p. 447).

most imperfectly the moods that animated us.¹⁹ The more nuances, the more definitely marked features and relations a mental state presents, the better can it be called back in memory. But nuances and relations presuppose comparison and belong to the region of knowledge. The smaller the part the elements of knowledge play in a state, the more imperfectly is it remembered. We thus recall the alternation and succession of feelings more easily than the single feeling itself in and for itself. In this respect it goes with the feelings as it goes with the immediate sense-sensations (V B, 7 a. d.); we can remember that we have had them without really recalling them. Special circumstances are needed for the state of feeling to be reproduced. Feelings are remembered with the help of the ideas to which they were originally attached, and together with which they once made up a certain state of consciousness. Only when one really immerses oneself in the memory, lives oneself into it, can the feeling be awakened. This follows directly from the slower movement of feeling; the thought comes back in a moment, but the unfolding of the feelings takes time.²⁰ There will always be a hindrance given in the feeling that [292] dominates us at the present moment; it will in any case modify the earlier feeling more or less, and a new feeling will arise which becomes the resultant of the two (according to the schema $a + (a + b) + \beta$). From this source stem many illusions that we form with regard to the past.

The feelings attached to the senses of sight and hearing and to free intuition and activity of thought are recalled more easily than those due to the lower senses, and especially than those due to the exercise of organic functions. The former therefore stand most at our free disposal, and external hindrances can less easily cut us off from them, which is of great significance, since the aesthetic, intellectual, moral and religious

¹⁹Longfellow has expressed this in the following fine lines:

Ah, our memory can take up again
all that belongs to the outward;
it builds up, as we would most have it,
every fairy palace of the vanished days.
Only ourselves it cannot draw in there!
Our heart's strange play of strings
never gets its first tuning back.

(The Golden Legend. Tr. Thor Lange)

²⁰In some cases of an almost pathological character, the memory can be connected with quite the same fresh feeling as the original experience. Littré relates from his own experience a remarkable example of such "automnésie affective." At ten years old he had lost a little sister under especially moving circumstances and grieved much over it. »Mais le chagrin d'un garçon ne dure pas beaucoup.« Yet he always kept the event in lively memory, although the fresh pain had vanished. Then, in his old age, he suddenly felt, without external occasion, the same pain again. »Tout à coup, sans que je ne le voulusse ni le cherchasse, par un phénomène d'automnésie affective, ce même événement s'est reproduit avec une peine présente non moindre, certes, que celle que j'éprouvais au moment même, et qui alla jusqu'à mouiller mes yeux de larmes«. This recurred several times in the course of several days, after which it ceased and gave place to the usual memory. (*Revue philos.* 1877. p. 660 f.).

feelings are of this kind.

5. So far we have spoken of feeling in general, and we have seen how the primitive feeling of pleasure or displeasure can be developed by being directed toward definite objects through the ideas that fuse with it. The further development of feeling now receives its decisive stamp according as it is regard for the individual himself or regard for something lying beyond the individual that comes to furnish the *center of gravity of feeling*.

At first this contrast, which in its full development becomes the contrast between egoism and sympathy (altruism), cannot stand out. In the incipient life of consciousness the ideas are little clear and definite; the idea of one's own ego therefore does not yet come into contrast with the idea of something outside the ego or of other egos. It is therefore psychologically meaningless to speak of an original egoism. It is another matter that no individual can feel joy or displeasure at anything that does not make itself felt [293] in him. Whatever he may attach his pleasure and his pain to — he can have no other pleasure and pain than his own. At the primitive stages it is as good as exclusively the feelings attached to sense-sensations (especially general sensations) that reign. The impulse to suck and to bring to the mouth everything that is grasped shows itself early; it can certainly be interpreted as a tendency to refer everything to oneself as center; but it must then be added that the child itself does not know this center.

A long process of development (described from another point of view in I, 4) is required before the consciousness of one's own ego as center arises. The child is unconscious of itself, feels no *shyness or bashfulness*, but stares calmly and fixedly at a new face, provided this has nothing about it that frightens. One could just as well speak of innate impudence as of innate egoism. Only when the child has the sensation of being observed and judged does it become bashful, turn its eyes away and blush when one looks at it. This feeling of being an object of observation and judgment contains the germ of feelings such as *humility*, *remorse* and *self-contempt*, which have in common that we feel ourselves to stand far below the standard that we ourselves or others apply to our person. These feelings presuppose such a development of the circle of ideas that a distinct idea of our person, of a demand that is to be fulfilled by us, and of the relation between the two can arise. — On the other hand, feelings such as *pride*, *feeling of power* and *self-satisfaction* form when our person in our own eyes possesses those advantages and reaches up to the height of the ideal in question.

The feelings that concern ourselves are the first to be awakened and the strongest that for the time being stir in us. The feelings of pleasure and displeasure, at first impersonal

so to speak (not concentrated or integrated), naturally gather about regard for self. This, in the nature of the case, comes forward so frequently that it grows together with feeling more firmly than any other experience and idea. The more regard for self excludes other resting points for feeling, the more there arises egoism proper, which makes everything a means to the self's feeling of pleasure.

6. How can it be explained at all that the individual can feel pleasure and displeasure [294] at something that does not intervene in its own existence? — This question has seemed so difficult to answer that some have even denied that what is asked about takes place. All sympathy has then been explained as hidden self-love. "Self-love," says La Rochefoucauld, "never comes to rest outside itself, and stops at foreign objects only as the bees do at the flowers, in order to extract from them what it can use." — Others have recognized that there is a pleasure and a displeasure at other things for their own sake, and have sought to explain the coming into being of such unselfish feeling according to general psychological laws. They even undertake to show that a psychological bridge can be built between absolute regard for self and absolute self-forgetfulness, between self-preservation and self-surrender. The most interesting and most original accounts in this direction have been given by *Spinoza* and *Hartley*. Later *James Mill* and *Stuart Mill* developed the same theory further.²¹

It is the laws of forgetting discussed earlier (V A, 8 d) that find application here. An idea that has been the occasion of another idea's arising may itself vanish, so that this other idea comes to work immediately and alone. An often used example of this is the independent value attached to money, although it is only a means of procuring certain goods. In the miser, the middle term, without which the value cannot be grounded and with whose help it originally arose, has passed entirely into oblivion. He loves money for its own sake, indeed even entirely renounces the goods it could procure. The feeling has shifted from the end to the means, or rather has made the means an end. In the same way, persons and things which at first became objects of joy and love only because they were causes of pleasure will be able to become immediate objects of these feelings. We love them for their own sake, having forgotten the original why.²² This will happen [295]

²¹Cf. also *Sibbern*: *Psykologisk Pathologi* [Psychological Pathology]. Kbhvn. 1828. p. 97–108.

²²*Paul Friedmann*, in his essay »The Genesis of Disinterested Benevolence« (Mind 1878), lays weight on the fact that people who live together do each other good turns even without expressly willing it. The feeling of having thus been a cause of joy to another awakens interest in him; one takes care that the benefit is not undone, because one feels one's *power* in thus helping others. The interest thus awakened can *then* come to embrace the whole person, and the original occasion is forgotten. — Aristotle already said something similar. "Benefactors seem to love those they *have* benefited more than these love their benefactors. ... The same we see in artists: every one loves his work more than he would be loved by it, *if* it were ensouled." Aristotle explains this phenomenon by the fact that the deed is a part or an expression

all the more easily the more other persons have in common with ourselves in nature, appearance, circumstances of life and interests. We then become accustomed not to separate ourselves from them; in everything that happens to them we involuntarily put ourselves in their place, suffer and feel with them. Involuntarily, by virtue of the law of similarity, interest glides from ourselves over to others. It is not only sympathy that can arise in this way, but also envy and ambition. These feelings too arise through one's vividly representing to oneself the feelings of other persons.

Sympathy presupposes that the common interests have the upper hand over the conflicting ones. It therefore also develops first in narrow circles and afterwards radiates out into larger circles. Each of the narrower circles (family, class, nation, sect) stands as an egoist in relation to more comprehensive circles. In the end sympathy must be able to extend to all living beings, to the whole of nature; it then finally acquires a religious character, becomes what Spinoza called "the intellectual love of God."²³

This view, however, surely overrates what can be attained in the course of a single individual's life. The feelings are, as we have seen, heavy masses to operate with; they do not shift so easily once they have found a center of gravity. More experiences than the single individual can have in his life are needed to complete such a metamorphosis, however aptly and naturally it is otherwise described.

[296] The theory moreover presupposes that there is a possibility of receiving help, or at any rate beneficial influence, from others. The individual thus does not stand isolated, not from the first sharply separated from other individuals. A society is presupposed within which the individual develops. The problem is then only pushed further back, and the question is whether the society in which relations are so arranged that a development such as the one described becomes possible can itself be conceived as having arisen through the union of individuals each of whom begins by himself with an unconditional impulse of self-preservation.

In so far, finally, as the theory lays weight on the motive lying in other individuals' likeness to the individual himself, it presupposes an original impulse of imitation or an instinct to feel and suffer with one's like. So *Spinoza*, who here hints at a theory later developed by *Adam Smith*. He who sees another burn himself involuntarily draws back his hand. The glad or sorrowful appearance of others at once attunes us in the same way. Some even hold that this ability to be attuned by the appearance of others is grounded in an inherited disposition. But in any case this ability or impulse then becomes an

of our being. Eth. Nicom. IX, 7.

²³Cf. »Spinozas Liv og Lære« [Spinoza's Life and Teaching]. p. 124–140. — »Den engelske Filosofi i vor Tid.« [English Philosophy in Our Time]. p. 57.

independent element alongside the impulse of self-preservation and requires its own explanation.

All this indicates that one must go further back than the individual impulse of self-preservation if one wishes to understand how individuals come to attach an independent value to something that lies beyond themselves.

7. The boundary of psychology lies where the question of the origin of the individual arises. Nor shall any attempt be made here to solve this problem. But it *will* be of interest to point out that what does not find its full explanation in what can be acquired in the course of the individual's life may become more intelligible by seeing the individual, with his whole nature, proceed from the race. However it may stand with psychological individuality (the center of pleasure and pain, of memory and self-consciousness), physiology shows us that the separation between individuals takes place gradually, and that the stage at which the mother organism and the new organism stand independently over against each other is preceded by a stage at which they make up one single living whole or organism.

Reproduction stands in the same relation to the race as nutrition and renewal (regeneration) to the individual organism. The series of living beings presents every possible transitional form between individual self-preservation and the production of new individuals from the substance of the mother organism. In plants and lower animals, single cells from one part or another of the organism can at once lead an independent organic life as soon as they are separated from the mother organism. The higher we go up in the series of beings, the more intricate become the conditions for such a development or prolongation of the organism's life in new organisms. But also in the beings that reproduce by seed and egg, and in which therefore the new organism arises from cells derived from two different organisms, the same fundamental relation holds, since egg cells and sperm cells have formed within the mother and father organisms and at their expense. Whatever the mode of reproduction, the organic individual nevertheless lives the first period of its life as part of another organism. According to the assumption now general, the egg is formed in the mother organism before the latter's birth, according to some even in the egg of the mother organism.²⁴ Although the origin of the individual is thus finally lost in the distance for physiology just as for psychology, we have nevertheless obtained the important information that it must be sought within earlier organisms, and that the separation between independent organisms is only the result of a process of development. The origin of the individual thereby falls under a more

²⁴*Ditlefsen*: *Menneskets Histologi* [Human Histology]. p. 244. — *Claude Bernard*: *Leçons sur les phénomènes de la vie*. p. 311.

general physiological point of view. — “Growth,” says v. Baer,²⁵ “is nutrition with the formation of new elements, thus a continued generation; and generation is nothing but the beginning of an individual growth. — Growth we find quite comprehensible; it is precisely the beginning that we should like to know. Above all we seek a really definite beginning, a sharp boundary between being and not-being. But do we then really find anywhere in nature such an absolute beginning?”

[298] Since self-preservation and reproduction thus pass over into each other, as *Plato* already taught (Symposion. Hejse’s transl. p. 73), we get here a physiological foundation for the transition between the feeling of pleasure in what concerns the individual himself and the feeling of pleasure in what lies beyond his own life-process. Sympathy seems here literally to grow out of self-preservation. Even after the physical connection between the mother organism and the new organism has been broken off at birth, *instinct* maintains a firm connection. The most wonderful instincts of living beings are precisely those that lead the one generation to prepare the way for the other. Many insects prepare food and safety for the larvae that are to develop from their eggs, and which they never get to see. In none of its forms does mother-love know any *why*; but it appears most obscurely and most instinctively where the mother does not have the child over against her as an independent organism. Only when this is the case can the general psychological laws begin to work, so that the maternal instinct can be developed into real *maternal feeling*. That instinctive care for beings of which the individual itself can have no idea is guided in its particulars by sense-sensations. Instinct consists in the union of a strong feeling with certain sense-sensations and involuntary motor impulses. When an observation of the child and an idea of it can take the place of the mere sense-sensation, the feeling acquires a higher form. The child’s caresses and smiles, its helpless situation, the feeling of belonging together which constant benefits call forth in the very one who confers them (see above, p. 294**), develop the originally blind and instinctive feeling into clearness and inwardness. The feeling now separates itself definitely from the general feeling of life and can enter into definite contrast to it. How strong maternal feeling is in comparison with the impulse of self-preservation is seen from the courage with which animals defend their young and the grief they feel at losing them. Swallows fly into burning houses to save their young. The mother whale does not leave her calf when it has been harpooned, as long as it is still alive. When sea-bears have had to abandon their young in flight, they [299] soon come back again, search for them, shed (according to *Brehm*) great tears, and swim about in their lamentation for several days along the shore. Many animals seek to draw

²⁵Entwicklungsgeschichte der Thiere. II. P. 4.

the attention of attackers from their young to themselves.

In animals and in the lowest races of men, mother-love fades when the child's helpless age is past. The child is left to itself when it no longer sucks. Where, on the other hand, there is a lasting relation between mother and child, the feeling will attain a still higher form by embracing not only the child's physical but also its spiritual individuality. When the vivid idea of an independent spiritual life in the child has formed, the psychological doubling in which sympathy consists is fully complete. Pleasure and pain are felt because another being feels pleasure and pain, just as the vibrations of one string make another vibrate in the same way.

In the foregoing only mother-love has been spoken of, not father-love, since the latter first shows itself with strength at higher stages. Mother-love is not only the strongest sympathetic feeling; it is also — when we regard the series of living beings as the expression of a great process of development — the one that reveals itself earliest, and which, by founding the most original human social relation, at the same time lays the foundation for all the means and forms by which sympathy can be developed further and higher. The relation between mother and child is the most primitive family and the most primitive human society. It makes a pure “state of nature,” an absolute individualism, impossible.²⁶ — In the animal world the male seldom takes part in the care of the offspring. In some species of fish the male takes over the whole work with great fidelity and watchfulness.²⁷ But otherwise the father is often even a danger and an enemy to his own offspring. A remarkable example at once of egoism on the part of the male and of self-sacrifice on the part of the mother is related by *Darwin* in his “Voyage round the World” (Danish transl. p. 213). The wild stallions on the Falkland Islands roam constantly from place to place and force the mares to accompany them, whether the young foals are able to follow or not. A man saw a stallion for a whole hour violently kick and bite [300] a mare until it forced her to leave the foal to its fate. Male egoism shows itself also in the human world, since here too at the lower stages the care of the children is left to the mother. Only where marriage takes on fixed forms, and *this happens*, as *Lucretius* already taught (V, 1008), only when *people live* in fixed dwellings, — only then can the relation of father become a source of the development of sympathetic feelings. Paternal tenderness then takes its place beside maternal tenderness.

A sign that the half-unconscious instinct in the development of sympathy within the human race²⁸ precedes the psychological development proper, which is determined by

²⁶Cf. »Grundlaget for den humane Etik« [The Foundation of Humane Ethics] p. 27 f. 58.

²⁷*Darwin*: Menneskets Afstamning og Parringsvalget [The Descent of Man, and Selection in Relation to Sex]. II. P. 19.

²⁸A detailed account of this course of development is found in *Herbert Spencer's* Psychology (Part 2),

the laws of the association of ideas, is that sympathy develops in those who give sooner than in those who receive. The love of parents for their children is as a rule greater than the love of children for their parents. During Sulla's proscriptions sons sometimes betrayed their fathers, never the reverse.

In the shelter of the bulwark that paternal and maternal sympathy builds, brotherly and comradely sympathy can now develop. It spreads also beyond the family by virtue of the psychological process described above (6). It reaches its consummation in the feeling that all men are brothers, of like nature and subject to like conditions.

8. There is yet another mighty feeling that grows out of a natural instinct and forms an important foundation for the development of sympathy. *The feeling of sexual love* is in its purely primitive form, like maternal feeling, a factor in the general feeling of life. Here too the first stirrings are connected with upheavals in the interior of the organism, by which the feeling of life acquires a character hitherto unknown. New, inexplicable presentiments and sensations arise. Something stirs in the individual that [301] drives it out beyond itself. But at the primitive stages the individual regards the object to which the impulse binds it only as a means. The feeling of sexual love is for the time being like a prolongation of the feeling of self. *Aristophanes* therefore explains it in Plato's »Symposion« by the gods' having cut human beings in two, which is why the two halves now go about full of longing and seek each other. The comparative physiology of reproduction in a way confirms this humorous explanation. It shows us various transitional forms between sexless and sexual reproduction. In the lowest forms of sexual reproduction the corresponding organs are still found in one and the same individual; such a hermaphrodite represents in itself the whole species, while in the higher beings the species requires two different individuals in order to be represented. In this double representation it is — as in the contrast between the mother and the child born — as though one ego were divided in two.

Here too it is only when the idea of the object works determinatively on the feeling that that *doubleness* becomes significant. At the lowest stages the object does not stand as another ego. An approach to it, however, is found already within *the animal world*. *The courtship of animals* is, as *Darwin* has shown in his well-known work on sexual selection, not so simple a thing as is generally supposed. An individual choice is often met with; beauty and other attractive qualities are taken into account, and often a *touching fidelity* is shown. With this, motives are already given which in the human world develop the

Sociology (Part 1) and in his »Data of Ethics«. — In our own literature we now have an account of the primitive family and social relations in *Claudius Wilkens's* work »Samfundslegemets Grundlove« [The Fundamental Laws of the Social Body]. Kbhvn. 1881. IV, 3–4.

feeling of sexual love from an unbridled, sensual craving, in which the individual seeks only its own, into inward devotion to and joy in another individual. Regarded purely egoistically, the sexual impulse is a deception; it looks as though it were the individual's own satisfaction that is at stake, and yet *it is the preservation of the race* that is secured. *Schopenhauer*, who sees the essence of the feeling of sexual love exhausted in the mere sexual impulse, *therefore preaches revolt* against it out of indignation at the deception that "the will to live" here practices. But this impulse, which in its lowest forms need not know the object that satisfies it, is purified and ennobled the more it is attached to and determined by the image of another, independent individual, an image that can awaken joy and admiration, not merely immediate desire. Instead of *coming forward with a demand* and with a word of power, the feeling can now be satisfied only by free [302] devotion. "Pleasure can be stolen, love must be a gift" (*Schiller*)²⁹. And that the means has become the end is seen from the fact that resignation is possible, that desire can be given up without the feeling ceasing.

9. In the instincts now discussed we have the helping hands that from the first lead man out beyond himself and bring him into relations under which the educative laws of the association of ideas can work. This was already emphasized by *Shaftesbury* in opposition to the individualistic explanation of sympathy. In our day this point of view has been asserted especially by *Spencer* and *Darwin*. — Once a feeling is attached to an idea, and this idea to other ideas, the way will be opened (according to 3) to the expansion and modification of the feeling. And as such a metamorphosis slowly takes place during the life of the race, it deposits its results in the organization that the new individuals receive. The laws of heredity make it possible that what earlier generations have acquired can become a capital with which later generations can begin. Together with heredity, tradition and upbringing are at work, in that the forms in which earlier generations have given expression to their sympathetic feelings meet the later generations as awakening and nourishing in a sympathetic direction. The sympathetic instinct unfolds differently in a Christian than in a Greek atmosphere, differently within modern humanity than within the medieval mixture of asceticism and barbarism. How much rests on tradition, how much on inheritance, and how much must be acquired and developed in the course of the individual life on the basis of the constantly working instincts, differs for every individual, every sex, every race and every age. Our analysis is here only qualitative, and only points out that these three forces are at work.

If one starts from the impulse of self-preservation as the original foundation, which

²⁹In »Mauprat« *George Sand* has described how brutal instinct can undergo a metamorphosis into ideal, human feeling.

is successively altered by the influence of experiences working through the laws of the association of ideas, one will be able to construct a dominant feeling of hatred, envy and malice just as well as a dominant feeling of sympathy. *Spinoza* already called attention to this. If it is asked, then, why in psychology we do not follow the development of a disinterested malice, but dwell instead on the development of disinterested love, the answer lies in the first place in the fact that the formal laws are the same in both processes of development; one need only, as it were, change the sign. But the most important reason is that disinterested malice (*malevolent sympathy*) does not find the nourishment in the conditions of life that the opposite feeling does. It can arise under certain individual and unhappy circumstances (cf. Shakespeare's *Richard the Third*, and Miss Wade in Dickens's "Little Dorrit"), but it cannot strike root in nature, because it is hostile to life, not life-preserving and life-furthering. Even if it finds favorable conditions in the conflict between the interests of individuals, families, races and confessions, historical development on the whole nevertheless tends to even out and cancel such disharmony, to lead the conflicting interests out into a common stream. Often the decisive help comes here from the sympathetic feelings grounded on organic instincts (the story of the rape of the Sabine women; *connubium* between patricians and plebeians; *Romeo and Juliet*).

The parallel mentioned between the psychological development of disinterested love and of disinterested malice explains why they sometimes so easily turn into each other. In both places ideal forces are in motion, and lukewarmness and narrowness have been left behind. It depends only on whether the railway train, with the speed it has attained, may come at a switch to rush off in another direction.

10. Even where instinct finds no direct satisfaction, it still exerts its mighty effect. The movement it causes in the feeling of life must have an outlet, and if this cannot be found in reality, it is sought in an ideal. For the ascetic and the mystic the ideal becomes "the heavenly Virgin," who takes the place of an earthly woman. *Plato* already (in the »Symposion«) described *Eros* with genius as the ideal impulse that is satisfied either in sexual love or in the striving for honor or power or knowledge. *Eros* is the great teacher who leads man to attach his heart to something that lies beyond his own ego. As a counterpart to this ideal effect of instinct there is no lack of examples of an uncanny mixture of sensuality and mysticism. In instinct itself there lies nothing but that the need for devotion has been awakened; it then depends on how this need is satisfied. That is why the age of transition from childhood to youth is so significant and fateful a time in every human being's life. The power of the obscure but strong feelings leads the individual out beyond his limitation, sets thought and fancy in motion, and awakens an

idealizing need. It is a time when everyone is a genius, even if he never is otherwise.³⁰ *Goethe* in his poem »Der Schäfer« has described with unsurpassable humor this all too often merely brief ideal flaring up, which “draws toward the distance,” but, after instinct has had its physical satisfaction, gives place to the normal prose.

The most significant crisis in the development of a feeling occurs when its object passes over from the sphere of sensation and observation to that of idea and memory. The source of all poetry, all morality and all religion lies where the sympathetic feeling no longer has its object immediately present. The immediate connection with the object is now replaced by a certain distance, and it depends on whether the feeling can bridge this distance, so that “its power is felt at a distance” (*Goethe’s* poem »Das Blümlein Wunderschön«).

Bain lays great weight on the significance that touch and embrace have in all forms of the feeling of tenderness, whether this is love of children and the helpless, or sexual love, or friendship. “Just as anger reaches its consummation and its satisfaction in striking someone to the ground, love is consummated and satisfied in an embrace ... [305] The pleasures of love begin and end with sensual touch.” The strength of this need finds its explanation in the fact that the sense of touch is the fundamental sense, the one from which, according to the hypothesis of evolution, all other senses have developed. Only this fundamental sensation can afford the impulse of love its highest satisfaction.³¹ All the greater is the pain it causes when separation occurs, so that the deep organic need cannot be satisfied. It then depends on whether an ideal connection can replace the physical one, whether a spiritual bond can become as strong as the bodily one. The transition from the physical to the ideal connection does not as a rule take place except through resignation. But this resignation can vanish altogether through the deeper and richer connection with the object which can take the place of the immediate and momentary connection. Where the object of feeling is great and manifold, sense-observation is even impossible. This is so already in the feeling for a single personality. A personality is never completely given in a single moment, a single situation; we have it completely only in the totality of its life, in its history. When we love a human being, we no doubt imagine him in some definite situation or other; but this stands only as example or type. (Cf. the theory of

³⁰*Cabanis* remarks (*Rapports du physique et du moral de l’homme*. V, 10): »J’ai vu nombre de fois la plus grande fécondité d’idées, la plus brillante imagination, une aptitude singulière à tous les arts, se développer tout à coup chez des filles de cet âge, mais s’éteindre bientôt par degrés, et faire place au bout de quelque temps à la médiocrité d’esprit la plus absolue. La même cause n’a souvent pas moins de puissance chez les jeunes garçons; souvent aussi les heureux effets n’en sont pas plus durables.«

³¹*Emotions and Will*. 3 ed. p. 126. — In this third edition of his excellent work, the best account of the psychology of feeling and will yet given, *Bain*, particularly as regards the explanation of the development of sympathy and of the feeling of tenderness, essentially adopts *Spencer’s* hypothesis of evolution.

individual ideas V B, 9.) The feeling therefore acquires an ideal character when it is to embrace the personality in its wholeness and unity, which it strives to do if it has got beyond the stage at which the object is only a means to the self-satisfaction of the feeling individual.

The feeling becomes still more ideal when it concerns a great, comprehensive whole (the family, the state, humanity) or is attached to what by its essence cannot be imagined as limited (the Godhead, nature). If a definite idea is to be connected with the feeling [306] here, this can only happen by way of symbolization. The history of religion shows us how deeply the need for symbols lies in the essence of feeling. Hence the need to fix the ideal and infinite in definite forms, so that the feeling may have a rallying point. On the other hand we also see how the fixed symbols can hamper and constrict the feeling, which is why the feeling ever anew bursts them and demands new forms for its satisfaction (the struggle of mysticism with dogmatics).

The development of the life of thought and fancy is thus a necessary presupposition for a higher development of sympathy. In order to feel with others, one must have had experiences of one's own, know from oneself what pleasure and pain are, and *what* calls forth these feelings. The range of sympathy is therefore determined by the experiences of particular individuals, nations and ages. Where circumstances, as among savage nations, compel hardness and contempt for torments, no fellow-feeling with the pain of others develops either. Love of mankind therefore presupposes a certain degree of civilization. Like hardness, asceticism too can hamper sympathy, *which is why the* principle of universal love of mankind, grounded in original Christianity, could first properly unfold after the ascetic tendency in the Church had been pushed back. — Next after having had one's own experiences of pleasure and pain, the task is to be able to hold these fast in memory and to be able to apply them to the understanding of the state of others. It depends on being accessible to lively impressions and possessing mobility enough to put oneself in fancy in the place of others. It has *rightly* been remarked that lack of sympathy often springs from lack of fancy and liveliness of mind, and not really from lack of feeling. It becomes especially difficult *to put* oneself in the place of others when their inner or outer circumstances *are very* different from ours. Different *language* (Greek [307] — barbarian), different color of skin (the Negro slaves)³², different rank and different faith have long offered hard resistance to the growth of sympathy in the human race.

³²It was a world-historical moment when *Abraham Lincoln*, on his entry into Richmond, took off his hat to a Negro who blessed him. — A lady who watched this scene from a window directly opposite turned away in unspeakable horror! (H. *Rasmussen*: *Abraham Lincoln's Liv og Gerning* [Abraham Lincoln's Life and Work]. Kbhvn. 1882. p. 116).

— Formal logical consistency itself can here become of great significance. Just as one can say with a certain right that egoism is unwise, since one often works for oneself by working for others, so one can — and surely with greater right — say that egoism is illogical when it restricts sympathy: it makes exceptions that do not accord with the nature of the case. In historical development there lies an inexorable logic which leads sympathy to victory not only over individual egoism, but also over family egoism and over national and confessional egoism. Impartial knowledge plays into the hands of the widest sympathy; both stop only at the natural limits. — Finally, the development of the life of knowledge also becomes significant for the form in which, and the means by which, sympathy is satisfied. By giving way to the sympathetic prompting of the moment one can just as well harm as benefit the object of sympathy. The task then is to be able to check such momentary emotion in order to take account of the lasting happiness of the objects, and this is impossible without the ability for circumspect and many-sided deliberation. Sympathy, like egoism, can show an aversion to limitation, a need for full discharge, and can so far be said itself to contain an egoistic element.

The relation between the intellectual and the emotional element in sympathy varies infinitely in particular cases. Now it is fancy's reflection of the nature and fate of the object, now the immediate unity in which we live and feel with it, that gives the feeling its peculiarity. Poetic sympathy is characterized by the preponderance of the first-named element, so that at times even a pure satisfaction of fancy is sought in busying oneself with the misfortunes and dark sides of life and depicting them. Writings that give a lively and correct description of individual and social sufferings can produce a certain repellent effect, because the element of feeling proper does not show itself. The intellectual element in sympathy is also the element of distance; it is what lets the object of sympathy recede somewhat in order to be apprehended all the more fully. It is therefore significant only where the foundation of feeling is strong and deep. The emotional element lies deeper than the intellectual; the latter has essentially the significance of purifying, expanding [308] and elevating.

11. We now return to the question of the possibility of disinterested sympathy. It is clear from the foregoing that the word interest may very well fit sympathy without its thereby becoming hidden egoism. The race can as little be separated from the individual as the individual from the race. In the instincts that underlie maternal feeling and the feeling of sexual love there is precisely this wonderful thing, that it is impossible to draw the line between what the race wills and what the individual wills. However ideal a character sympathy may assume, however exalted and comprehensive that may be to

which man attaches his pleasure, it is nevertheless just as much a part of him as he is a part of it. The pleasure and pain he feels in it are his own pleasure and pain; what other could he feel? But in "disinterested" love he does not feel it as his own in the sense that devotion to the object should be only a means to so much the greater self-enjoyment. Since he does not want to annihilate himself, the "disinterested" can only mean that he immediately shares the pleasure and pain of others and rests in it without demanding more for himself than precisely to be able to be animated in this way.

It is impossible to love something or someone without feeling joy and satisfaction in it. With every strong emotional movement there is connected a peculiar pleasure, whatever the character of the feeling. Even in sorrow there is, alongside the bitterness, a depth and a life in the mood, a strong movement of the powers of the soul, which has something captivating and alluring about it. All the sluices are opened, as it were. It is the pleasure or impulse toward weeping and grief of which Homer speaks. The strong emotion moreover brings with it a series of organic reflexes, and with this discharge a certain comfort is connected. It is thus in the spiritual and physical exaltation that [309] the ground of the attractiveness of sorrow must be sought.³³ What can thus take place with the feeling of sorrow holds more or less of all other feelings. Here, to be sure, lies the germ of an egoistic turn even within the sympathetic feelings. There can be a hysterical need to have one's feeling set in motion. Connected with this is the fact that the sympathetic feeling is so often satisfied without consideration and therefore without real benefit to the object of sympathy. The mark of unselfish sympathy is in this as in other respects the possibility of resignation. Pure and strong sympathy must even be able to deny itself.

Sympathy can be put to yet another test. Even if we are absorbed in the feeling for the object, we can still hold fast the wish that it should be precisely we ourselves who work and live for it. Now if circumstances turn out such that our service is a hindrance to the object's nature and worth being properly unfolded, or such that others can work for it better and more fully, then it depends on whether our sympathy is strong enough for us to be able to step back and resign ourselves to its not falling to us to further what we love.

Such resignation is possible and is the most glorious proof of the purity of sympathy. It was so far with full right that *Molinos* and *Fénelon* in theological form, as *Spinoza* before

³³This explanation, which has been given by W. *Hamilton* and *Bouillier*, seems to me the most probable one of this remarkable phenomenon. (Cf. *Bouillier*: *Du plaisir et de la douleur*. Chap. VII.) It is more natural than the explanation hinted at by *Spencer* (*Princ. of Psychol.* II. p. 590 f.), according to which the need to immerse oneself in sorrow springs from a *feeling* of suffering undeservedly, so that sorrow, by an effect of contrast, lets the consciousness of one's own worth come vividly forward.

them in philosophical form, maintained as the highest a love independent of all regard for self, of all reward and punishment. The Catholic Church condemned this doctrine, most likely chiefly because it did not wish to lose the means of education that regard for reward and punishment affords. But psychology cannot admit that disinterested love is a chimera; the chimerical begins only where the demand goes so far that the life and movement of feeling are to vanish for an absolutely simple, resting state. Such a state would, as we have often remarked, be one with the cessation of conscious life itself.

12. Akin to the higher forms of sympathy is *the ethical feeling*. But only akin; it does not — as *Adam Smith* and *Herbert Spencer* teach — wholly coincide with sympathy. [310] We have already seen that sympathy, left to itself, can work against itself, and that it must therefore undergo a refining in order to reach its goal. That an action springs from sympathy thus does not establish that it is ethical; it depends also on whether it furthers the progress of life and happiness in the world. Not sympathy alone, but the union of sympathy and wisdom is what leads to right actions. The immediate feeling of love becomes, through union with wisdom, the *feeling of justice*, the concept of justice being taken in the wide sense that both Greek and Christian ethics have developed, namely as the ordering and distributing principle which lets every benefit, every good imparted, flow to the right person, at the right time and place and in the right measure. — As sympathy undergoes this metamorphosis, its intellectual elements are expanded. It now embraces not only single individuals or circles of individuals, but bears within it the idea of an organic whole of human personalities, each of which lives its life out from a peculiar center, and each of which therefore has a claim to a peculiar form and direction of sympathy. As the view is thus widened, the individual feels himself only as one among many, as a single member of a great realm developing through the ages. What his impulse of self-preservation and his sympathetic impulse prompt him to is finally subordinated to regard for what the progress of this realm demands. This demand he perceives within himself as a law; and as this law comes into greater or lesser contrast with what stirs immediately in him in an egoistic or sympathetic direction, there arises the feeling of duty, which is conditioned by man's feeling himself bound by an ideal law. The feeling of duty need not, as *Kant* thought, contain an element of displeasure because the lower, sensuous promptings in our nature are pushed back before the majesty of the ethical law. There can be a relation, even a contrast, between a higher and a lower in us without any real feeling of displeasure arising thereby. The feeling of the truth and glory of the ideal can rouse our activity in such a way that the hindrances only make us feel our powers all the more definitely. But of course it is not always so. The hindrances can be so strong [311]

that the most painful contrast and dividedness arises in the mind. Feeling then stands against feeling, and the one feeling passes judgment on the other. In this painful feeling of the contrast between the acknowledged ideal and the imperfect reality *the ethical* feeling of remorse expresses itself, in which the individual turns in rejection against his own being and his own actions. Very often remorse is the first form of the feeling of duty, the birth-pangs of ethical character. Just as in the development of knowledge an original sanguineness shows itself which leads to hasty expectations and inferences and thereby to disappointments, so in the domain of feeling too a blind craving can *sweep* man along, so that the corrective must be developed afterwards with pain. The Greeks already had a sharp eye for how infatuation is followed by remorse.³⁴ The development of the ethical feeling makes it possible to forestall remorse (the ethical disappointment), in that scruple already asserts itself over against the possible action, not only over against the action performed. In *conscience* there make themselves known both the judging power of the ideal over what lies in the past and the ability to apply an ideal test to future conduct. It is the ethical memory. In it the experiences of the past fuse with everything that circumspect reflection teaches into one collected power, which can express itself with the immediacy and strength of instinct.

Not all remorse belongs to the ethical feeling. We have mentioned in passing that remorse can be felt already at the standpoint of the mere impulse of self-preservation (5). This will happen when we have acted against our dominant selfish interest. Remorse acquires an ethical character only when we feel ourselves in conflict with the demands of sympathy; and the more deeply and inwardly we apprehend the ethical law, the more we feel ourselves bound to it in our inmost being, the more violently and penetratingly does remorse work in us.

[312] The ethical feeling runs through a whole scale of forms of development. Historically regarded, it finds its full explanation neither in the impulse of self-preservation, nor in sympathy, nor in the influence of knowledge on feeling. It has developed in the shelter of educating powers, authorities. These powers have been educative partly with, partly against their will. They have often pursued their own ends and have nevertheless worked toward the development of ethical feeling by working on human nature through reward and punishment, checking certain feelings and nourishing others. At the lowest stage authority appears as overwhelming physical power; here it instills fear and trembling, thus purely egoistic feelings. When, on the other hand, at higher stages authority stands as the life-protecting and life-furthering power, fear is mixed with sympathy

³⁴Cf. the allegory in the 9th book of the *Iliad*, v. 502–507.

and admiration and thereby passes over into *reverence*. From this there is only a step to the ethical feeling proper, which presupposes a conviction independent of external authority.³⁵

The religious feeling is historically and psychologically closely connected with the ethical. It too has at its lowest stage the character of fear. That it is fear that from the first creates gods is confirmed by the fact, on which all anthropologists seem agreed, that men worshiped evil beings before good ones. Belief in gods is at the lowest stages one with belief in immortality, being belief in the spirits of the dead, and only those spirits which were considered able to work harm became objects of worship. Religious worship is here a simple acknowledgment of power. A higher stage is reached where the extraordinary, the incomprehensible and the wonderful awaken the religious feeling; this then acquires the character of admiration and reverence and begins to become disinterested in its nature. The religious feeling acquires a real ethical significance only when the great powers on which man feels himself dependent stand before him as bearers of ethical ends and values. This stage is reached in the higher national religions. The religious feeling has its lasting source, unassailable by any intellectual progress, in the great question of the connection of ethical endeavors with, and their significance for, the development of the world as a whole. All egoistic and personal stirrings of hope and fear it can separate [313] out, under the influence partly of insight into the fixed, lawful order of nature, partly of the demands of ideal ethics. But there then always remains the question of the relation between the ethical ideals and the factual reality of the world. The fact that everything we know and admire as true, beautiful and good has developed under the conditions given in nature already gives the idea of nature a religious character, and contains the motive for the idea of an ethical world order, according to which the inmost kernel of reality, the inmost force in the development of the world, is of the same nature as that which works its way forward in human ideals. But since what is ideal and valuable in the world of experience always appears as a *struggling* power, the religious feeling acquires the character of faith and presentiment, and that idea stands — theoretically regarded — as the last hypothesis or the last postulate, for whose validity no other pledge can be had than precisely those ideal values whose real grounding is postulated. Under what form the idea is expressed — to what speculations, symbols and dogmas it leads — it is not the business of psychology to investigate. Psychology has only to show how the religious feeling, just as well as the ethical feeling and all other ideal feelings, is determined in

³⁵I have given a more detailed investigation of the ethical feeling and its development in my work »Om Grundlaget for den humane Etik« [On the Foundation of Humane Ethics] (Kbhvn. 1876) and in two essays in the periodical »Vor Ungdom« [Our Youth] for 1880.

its development partly by the influence of knowledge, partly by the contrast between egoism and sympathy.

The relation between the ethical and the religious feeling is marked by the fact that in the former we feel our power of action spurred, in the latter our dependence. When religiousness and ethics come into conflict, it is thus the passive and the active pole of our nature that come into contrast with each other. This conflict can arise both within the consciousness of the single individual and within the race as a whole, in that the life of feeling of some individuals leads them more in an ethical, that of others more in a religious direction. But a definite and absolute separation between passivity and activity, dependence and power of action, is as little possible in the domain of feeling as in that of knowledge. In our receiving we are always active, in our activity receptive. The very powers we employ in our strongest exertion of will we nevertheless feel as something given to us. We feel that a nourishment streams to us from within, without which we can do nothing, and that all our activity really serves only to help forward [314] and unfold what is laid down in us by a quiet and unconscious growth. The ethical feeling is religious through the element of devotion and reverence which is inseparable from it, and the religious feeling is ethical when it is more than egoistic superstition and sentimental enthusiasm. It is the great merit of Kant and Fichte to have pointed to this inner connection. They thereby had a purifying and ennobling effect on both the religious and the ethical feeling, and have paved the way for a more perfect harmony between the two.

13. The disinterested feelings presuppose that idea and fancy are developed enough to hold fast the object of feeling as something that has its independent value. Only thereby do unselfish love and ethical and religious feeling come into being. But also apart from their object and content, idea and fancy acquire significance for feeling. The very activity of ideation and fancy can become a source of peculiar feelings.

Knowledge is from the first the servant of impulse. Thoughts are the scouts of the impulse of self-preservation. Knowledge is valued because it is a means to power. At this stage no real *intellectual feeling* arises. Even where it is not external power or external goods that are sought, but where knowledge is regarded as a means to spiritual freedom and independence, the feeling awakened by the activity of knowing is not purely intellectual. The intellectual feeling first comes into being where the inner relation between the ideas itself becomes the determining thing, quite apart from what inner or outer consequences knowledge brings with it for us. The presupposition is that the struggle for existence does not make its hard and inescapable demands.

Furthermore, such a host of ideas must have formed that they can in the main arrange themselves according to their own laws, without the immediate intervention of feelings and impulses. There then arises joy in agreement, consistency and coherence, and a feeling of displeasure at disharmony, contradiction and incoherence; and this pleasure or displeasure is felt not merely because it is our standard of truth that is upheld or violated, but because there is something immediately satisfying or paining in the harmony or disharmony itself. The joy in new facts and discoveries also belongs here. Even if they overturn hitherto accepted assumptions and thereby produce unrest and *doubt*, they [315] nevertheless open a view to more distant horizons than had hitherto been suspected.

Just as there are musical and poetic natures, so there are also intellectual natures. For these, self-contradiction, unclearness and lack of coherence are just as painful as false notes and mediocre verse are for the former.

The aesthetic feeling is in some of its forms akin to the intellectual. Pleasure in symmetry and rhythm, in the form of phenomena generally, finds its explanation in the easy and clear way in which the observations arrange themselves. The powers of knowledge here work without hindering resistance, subdue the material to themselves as in play. Already the higher sense-sensations (colors and tones) awaken, as we have seen, moods which detach themselves from the general feeling of life. This is still more the case with combinations and groups of colors and tones.

The feeling of beauty plays a part already within the animal world, since colors, tones, scent and rhythmic movement are used as enticements in *sexual selection*. In man too the feeling of sexual love rouses fancy to a freer and bolder life and opens the eye to colors and forms. *Plato* therefore called Eros the impulse toward the beautiful (8). At this stage beauty is still only a means, just as truth is as long as knowledge works in the service of the impulse of self-preservation. But through the psychological process described earlier, what is originally only a point of attachment for instinct can become an independent end. This aesthetic development goes in part hand in hand with the general development of sympathy. The two support each other; the capacity for unselfish devotion is common to both.

In the desire to adorn oneself (with feathers, pearls and pieces of bone, or by tattooing) an aesthetic feeling already expresses itself in the lowest men. The next step is joy in weapons and other implements, apart from their usefulness. They are worked with taste and adorned with pictures. Thereby *the means of waging the struggle for life* themselves become sources of the feeling of pleasure. And as it goes with the implements, so it goes with their use. When the pressing need is satisfied and one has rested after exertion,

[316] there arises an impulse to movement for the sake of movement itself. The beast of prey plays when it is not tormented by hunger, fatigue or dangers. The savage performs his war games and finds in the violent movements an outlet for his regained energy. In play, which comes forth in the use of "the luxury of powers," *Schiller* (Ueber die ästhetische Erziehung des Menschen. 27ter Brief) saw the germ of all art, a thought which *Herbert Spencer* has sought to carry through.

The aesthetic feeling has thus developed out of the impulses that lead to the preservation of the individual and of the race. It presupposes a *surplus of energy* which is not used up in the struggle of life and can therefore be employed in another way. But with this only the material is given. The particular way in which it is employed depends on intellectual development. For impressions to work aesthetically, there must be a certain articulation of the sense-organs, such that finer nuances are apprehended and not all sensations are absorbed into the general feeling of life. A free play of colors, forms, tones and movements must be able to unfold. But the direct impression is not enough. For it to work aesthetically, it must not merely assert itself by its strength, its kind and its form; its effect must also be able to branch out into consciousness by awakening a wealth of ideas and moods to clearer or dimmer appearance. In the aesthetic effect, therefore, a distinction can be made between the immediate, directly working element and the associations it calls forth.³⁶ In music *the direct factor* reigns, in poetry *the associative*; the visual arts stand between the two in this respect. And yet music has a decidedly intellectual character; what matters is not merely the strong, elementary impressions produced by sound and rhythm, which are dominant in the childhood of the art, but the contemplative enjoyment of the rising and falling, the conflict and harmony, of tones. The art of tones rests on the "unconscious application of original ideas of magnitude and relation, a hidden measuring and counting."³⁷ The visual arts approach music the

[317] more they work by the play of colors and the harmony of forms. But here the definite subject, the individual phenomena that are to be represented, make definite demands. Their figures must be recognizable and their situations understood. Associations of ideas and historical memories are thus necessary here for the immediate apprehension itself. Poetry, finally, works predominantly through the ideas and feelings it awakens; the direct factor is only a means. It gives what music cannot, the definite feeling with its full content of thought and fancy; and it gives what the visual arts cannot, the historical unfolding of characters and actions. — In all its various forms art gives access to the exercise and employment of powers that were originally taken up by the struggle for

³⁶*Fechner*: Vorschule der Aesthetik. Leipzig 1876. Kap. 9–13.

³⁷*Hanslick*: Vom Musikalisch-Schönen. 6te Aufl. p. 169.

existence. There therefore exists the most intimate interaction between art and real life; the ideal and the real working of the spiritual powers pass over into each other, prepare each other mutually.

It accords well with the theory of the origin of art from the struggle for existence that the sense for the beauty of art precedes the sense for the beauty of nature. Art stands nearer to man than nature; the former is his own work, which he cannot disown, while the latter can long stand as a foreign, hostile or indifferent power. Children and savages as a rule have no sense for the beauty of nature. Everything connected with man and his pursuits interests them, but nature only in so far as it serves the purposes of man. At the primitive, practical standpoint a beautiful region is the same as a fruitful one, i.e. a region fruitful in corn and grass. Peasants therefore wonder at the tourists' desire to visit barren heaths, dune country and mountains. A distinguished American traveler said to an Englishman: "Your country is very beautiful; in many parts one can walk whole miles without seeing a tree outside the hedges." The sense for the wild, sublime and romantic in nature is not much more than a hundred years old.³⁸ *Rousseau* was the one [318] who awakened the sense of nature, and especially the sense for wild, undisturbed nature, in wide circles. He discovered the mountain landscape, which hitherto had awakened horror and dread in most people. He taught men to turn their backs on human life and listen to the language of nature. Joy in nature stands as the highest stage of development of the aesthetic feeling and at the same time as one of the best examples of disinterested sympathy.

Two special aesthetic feelings, the feeling of the sublime and the feeling of the ludicrous, we shall later go into somewhat more closely, since they will furnish good examples for illustrating the general psychological laws of the life of feeling. The present work does not aim at a description and classification of all feelings, but at an investigation of the general psychology of feeling. Whoever wishes a description of the particular feelings will find one in Bain's »Emotions and Will«, in Nahlowsky's »Gefühlsleben«, in A. Horwicz's »Psychologische Analysen« and in Sibbern's »Psykologiske Patologi« [Psychological Pathology].

³⁸Cf. the interesting essay on the sense of nature in ancient and modern times in the 2nd volume of Friedländer's »Sittengeschichte Roms«. — To the objection that landscape painting already in the 16th and 17th centuries represents mountain scenery with sympathy and understanding, the author remarks that one cannot infer from single artists to the whole of their contemporaries.

C. The Physiology and Biology of Feeling

1. If it is true that it is only by abstraction that we distinguish between knowledge and feeling, and that every concrete state of consciousness consists of elements of knowledge and of feeling, there will be no reason to expect that knowledge and feeling should be bound up with the activity of different organs of the brain. The matter was bound to look otherwise as long as one started from the idea of different parts or faculties in the soul. *Plato* accordingly placed thought in the head, the feeling of courage and honor in the breast, and the sensuous impulses in the abdomen. *Aristotle* referred all feeling of pleasure and displeasure (as well as all sense-sensation) to the heart, whereas pure rational activity was in his opinion not bound up with any physical function. In modern times *Descartes*³⁹ and his school (*Willis*, *Malebranche*) referred feeling, with all other phenomena of consciousness, to the brain, and so on this point, as on so many others, were forerunners of modern physiology. But the old theory of the contrast between [319] brain and heart as a parallel to the contrast between understanding and feeling did not let itself be so easily displaced. It relied on what immediate experience seemed to teach. Every strong movement of feeling is bound up with sensations in the breast or abdomen; the heart beats harder, breathing is checked or intensified, digestion is modified; in short, emotion reaches deep into the functions of vegetative life. As late as *Bichat*⁴⁰ says: "Everything seems to prove that organic life is the goal in which the passions end and the center from which they go out." The brain is in his opinion the seat of knowledge and "is never affected by the passions, whose only seat is the organs of the inner life" (namely, now the liver, now the lung, heart, spleen or stomach). — *Charles Bell* and *Gall* seem to have been the first in our century to refer feeling to the brain, like the understanding and the other phenomena of consciousness. When *Gall* thought he could point out different parts of the brain as organs for the different special feelings (self-love, reverence, hope, benevolence, etc.), this lacked all physiological foundation and all psychological probability. As we have seen in an earlier section (II, 4), it has not yet been possible to demonstrate with certainty any localization of definite functions within the cerebrum, and even those who maintain such a localization (like *H. Munk*) state expressly that it is only sense-perception, not intelligence (the combining activity of ideas), that is localized. Since the development of feeling is so closely bound up with knowledge, there will be just as little reason to expect a special localization for it as for knowledge. In the different forms of feeling, as in the different forms of knowledge,

³⁹Les passions de l'âme. I, 33.

⁴⁰Recherches physiologiques sur la vie et la mort. Paris. An VIII. p. 71.

the same brain cells can be thought to work together, only in different degrees and in different combinations. Since, according to what has gone before, feeling seems to arise and unfold more slowly than knowledge, it must be assumed that it is represented physiologically by a greater spread of the nervous process in the mass of the brain.

Yet an elementary feeling can arise even when the higher organs of the brain are wanting. A rat that has lost its cerebrum, optic thalami and corpora striata starts when [320] one imitates the cry of a cat, entirely in the same way as it would do in an uninjured state. The elementary, instinctive feeling of fear (cf. B, 2 d) can thus, like the elementary sense-sensation, arise without the function of the cerebrum. The same perhaps also holds of certain morbid outbursts of feeling that stand in no connection with mental influences.⁴¹

But in any case there is this much justification in the old idea of the connection of feeling with the vegetative organs, that these do in reality play an important part in every feeling, even if they are not its physiological seat. Feeling makes stronger claims on the nerve centers than knowledge does, and the tension produced finds vent by being propagated to more or fewer parts of the rest of the organism. While the preponderance of the elements of knowledge shows itself in this, that as much energy as possible is concentrated in the brain, so that the rest of the organism is kept as quiet and passive as possible, the state determined by feeling has precisely a tendency to spread. From the brain the movement propagates itself to the heart, which in strong emotion may even stop altogether, so that death ensues. Strong and sudden fright, anger, grief or joy can in this way have a *fatal* effect. If the movement is less strong, the heart, after a short stoppage, begins to beat faster than before, and thereby sends a more vigorous stream of blood up to the brain, which in this way receives the recoil of its own movement. In warm-blooded animals this reaction of heart upon brain is stronger than in cold-blooded ones, and in higher animals than in lower. Man notices this reaction in the course of a couple of seconds. — Not only the heart, but other internal organs too are affected by the discharge of the strong movement in the brain. The more violent beating of the heart itself in fright and anxiety is explained by some from the fact that the arteries suddenly contract and thereby offer greater resistance to the stream of blood, so that the heart works harder. The sudden pallor in fright proves at any rate that such a contraction takes [321] place. In other cases the muscles of the vessels relax, so that a more abundant stream of blood is sent out (blushing). Emotion can also affect the lachrymal glands (grief), the intestines (fear), the liver (anger), the respiratory organs (fright), etc. Emotion can

⁴¹Cf. *Vulpian*: Physiologie du système nerveux. Paris 1866. p. 549, where such elementary feelings are referred to the pons Varolii (protubérance annulaire).

further bring about a heightening or a loss of the irritability of the motor centers (tetanus, paralysis, St. Vitus's dance); but can conversely also sometimes lead to recovery.⁴² — We disregard here the effect that feeling has on the voluntary movements, since that is most naturally referred to the psychology of the will.

The reason why it is first and foremost the centers of the vegetative processes that are affected by the strong tension produced in the state of feeling must be sought in the fact that these centers (which are situated in the medulla oblongata) are in constant activity as long as life runs its course. They therefore possess the greatest degree of susceptibility to impressions (the greatest irritability). Hence they respond quickly and easily to every influence conveyed to them as a consequence of the state of other centers.⁴³

The movement of feeling thus reinforces itself by passing through a circuit. Some have at times wished, especially from a spiritualistic standpoint, to distinguish sharply between feeling proper and the reaction from the internal organs. This latter moment certainly belongs also to the general feeling of life, and, as remarked earlier (A, 3 a), the state of those vegetative organs can call forth a feeling of a character similar to that awakened by external experiences and mental influences. But although the two stages in the development of a feeling are separated by a small interval of time, there is [322] nevertheless no reason to draw a sharp boundary between them psychologically. They at once fuse together, and only both together make up the feeling in its full peculiarity. On the other hand, it may be useful in medical, pedagogical and moral respects to keep the two stages apart. In different individuals they appear with different strength. In some the discharge to the internal organs is relatively weak, even in strong emotion; in others it is disproportionately strong. Often it serves to relieve and lighten, and not to reinforce, what goes on in the brain.

2. Still greater difficulties than the physiological localization of feeling are presented by the question of the nature of the organic process to which feeling corresponds. Since the nature of the nervous process is unknown, nothing more than a hypothesis can be reached here. Nevertheless it is of interest to bring up this question, since light is thereby thrown on the significance of feeling in the economy of life.

Aristotle conceived the feeling of pleasure as bound up with every natural and normal

⁴²*Claude Bernard*: Étude sur la physiologie du cœur. (Revue des deux mondes. 1865. Reprinted in "La science expérimentale"). — *Wundt*: *Physiol. Psychol.* II. p. 330. — *Ranke*: *Physiologie des Menschen*. 3. Aufl. p. 399. — *C. Lange*: *Rygmarvens Patologi* [The Pathology of the Spinal Cord]. p. 255. 391 ff. — *Darwin*: *Expression of Emotions*. passim.

⁴³Cf. *A. Freusberg*: Ueber die Erregung und Hemmung der Thätigkeit der nervösen Centralorgane. (Pflüger's Archiv für Physiol. 1875). p. 185. — See also *Spencer*: *Mindre Afhandlinger* [Minor Essays]. Danish transl. p. 98 f.

activity of life, and this conception is still the most common and the most probable.⁴⁴ In feeling we have the innermost state of the conscious individual, as it is determined by the influences it receives and the activity it itself exerts. It is then natural to see in the contrast between pleasure and displeasure — the fundamental contrast within the world of feeling — an expression of the contrast between advance or decline in the life-process itself. As a main rule, then, it could be laid down that pleasure indicates heightened activity of life, a higher and freer employment of energy. The normal functioning of the various organs, brain and nervous system as well as muscles and vegetative organs, is accompanied by the feeling of pleasure. If, on the other hand, greater demands are made than an organ can satisfy, or if, conversely, an organ does not get sufficient employment for its energy, displeasure or pain is felt. Since all function is bound up with the release of potential energy, the dissolution of the stored-up organic capital, one and the same [323] degree of activity may at different times be bound up with pleasure and with pain, all according to the energy that is at one's disposal.

A somewhat divergent conception has been put forward in the most recent times by *Fechner*. Since pleasure and displeasure are not only quantitatively but qualitatively different from each other, they must, according to Fechner, correspond to processes that differ not only in degree but in form. He therefore assumes that pleasure rests on inner agreement or harmony, displeasure on a disharmonious relation between the moments of the physical process to which the life of consciousness is bound.⁴⁵ — It is, as has been said, impossible to decide anything on this. Fechner's theory leads in essentials to the same result as the one founded by Aristotle, inasmuch as it must be presupposed that the harmony in question signifies an advance, the disharmony a decline, in the organic respect. If disharmony gains complete preponderance, life dissolves, just as when it is overwhelmed by impressions, overstrains its powers, or is kept out of all activity. Pleasure thus stands in any case as the expression of heightened life, pain as the expression of decline and as a warning of death.

It is not meant, of course, that in pleasure and displeasure a reflection or comparison asserts itself as to how far things are going forward or backward with us. Such reflection is in any case impossible in the simplest forms of feeling. Originally pleasure and displeasure announce themselves of their own accord, and only then, provided that knowledge is sufficiently developed, can one brood over their significance.

3. The theory stated is confirmed by the fact that impressions that cause displeasure

⁴⁴In *Léon Dumont*: *Théorie scientifique de la sensibilité*. Paris 1875. p. 26–63, a survey of the different theories is to be found.

⁴⁵*Vorschule der Ästhetik*. I. p. 12. 79. — II. p. 266.

and pain are as a rule also harmful. Of pain from blows and wounds, where the organism is downright mutilated, this is self-evident. Likewise with fatigue and with overwhelming sense-impressions. Bitter substances have a tendency to dissolve organic tissue; the agreeableness of sweet taste finds its explanation in the presence of sugar in most of the vegetable constituents of human food.⁴⁶

[324] On the other hand it might seem a serious objection to the Aristotelian theory that pleasure is sometimes bound up with what is harmful, displeasure with what is beneficial. This objection leads to a more precise statement of the theory. It is only the *partial* and *momentary* influence of the impression or activity that expresses itself in the feeling of pleasure or displeasure. A palatable poison brings about a momentary advance in one part of our organic life. Later, when it spreads in the organism, it unfortunately displays other properties, by which it threatens life. But the feeling of pleasure in the taste was not therefore a deception; a thermometer does not show us the degree of heat that will be present in a few hours, but the one that is here already.⁴⁷

The general rule of pleasure as a sign of advance, displeasure as a sign of decline, is hereby essentially restricted. We are sent from *Aristippus* to *Epicurus*: it is no longer by the pleasure or displeasure of the moment that life can be judged, but by the duration and final victory of pleasure. But it must still be required of the theory that it be able to explain why the pleasure or displeasure of the moment is no sure criterion. It is no satisfactory answer to say with Lotze that "the purposiveness (teleology) of nature extends, not necessarily, but only in some fortunate cases, to wholly unaccustomed events or morbidly perverted states." The question is: *why* then does purposiveness not extend further? It cannot, after all, be from a mere whim. And with this there is then naturally connected the question why purposiveness is present at all, so far as it is.

This is one of the points where the hypothesis of evolution brings in a clear light where before there was utter darkness.⁴⁸ We do not understand feeling as long as we keep it completely isolated from the will; it has also appeared in what has gone before that feeling and impulse cannot be separated. The feeling of pleasure leads to a striving to hold fast and appropriate what awakens pleasure; displeasure and pain lead to withdrawing from and defending oneself against what is their cause. This rule holds both at the
[325] standpoint of instinct and at that of conscious will. The difference concerns only the nature and cause of the pleasure and the pain. Now if a being were so organized that it felt pleasure in everything that was harmful to it and displeasure in everything that was

⁴⁶ *Grant Allen*: Physiological Æsthetics. London 1877. p. 69 ff.

⁴⁷ Cf. *Lotze*: Medicinische Psychologie. p. 237 ff.

⁴⁸ Cf. *Spencer*: Principles of Psychology. Part II. Chap. 9.

beneficial to it, it would not be able to live. Natural selection itself therefore leads to a certain harmony between the feelings and the conditions of life. But that this harmony is not complete follows of itself. To wholly foreign or very rare conditions the organization cannot be *adapted*. Pleasure in what is harmful is thus a sign of an imperfection in development, which will perhaps gradually be remedied. Such imperfections will, by the nature of the case, occur especially, on the one hand, with a sudden change in the conditions of life, on the other, with the transition to very complex and many-sided conditions. Both apply to man. His life of feeling is not yet adapted to the tasks and demands of social life. Civilization is barely a few thousand years old where it is oldest, and before it there lie perhaps myriads of years in which animal and barbarous impulses held sway. It is then no wonder that pleasure and pain cannot without further ado count as sure guides, and that it has even been possible to maintain the exact opposite of the main rule stated and to regard pleasure as a danger and misfortune, pain as beneficial. The single individual already often has the experience that pain has a “secret formative power”; in education pain is regarded as a good in so far as it warns and checks. In the development of the race, too, pain stands as the warning voice, though it is only the ascetic cast of thought that ascribes value to it in and for itself. Just as feeling is brought, through the struggle for existence, into a certain harmony with the conditions of life, so too it is the struggle for existence that teaches that we cannot rely on the feeling of the moment, but must seek a higher standard.

The general rule that pleasure indicates the advance of life, displeasure the decline of life, fits the higher, ideal feelings just as well as the lower ones. In sympathy and in the ethico-religious feelings the individual feels himself as part of a greater whole, as member of an organism whose pulse-beat he senses within himself. He does not separate his cause from that of the more comprehensive organism. What furthers its life thereby [326] furthers his too.

D. The Validity of the Law of Relation for the Feelings

1. If the distinction we make between knowledge and feeling is correct, there must be certain laws that belong only to the one kind of psychical elements, not to the other. This too has proved to be so, inasmuch as within the domain of the feelings we find nothing corresponding to the laws of the association of ideas. A feeling can enter into connection with one or more ideas; but a direct association of feelings there hardly is. — On the other hand, both knowledge and feeling are elements in every one of the states of

the life of consciousness; we must then expect that there are laws common to both kinds of elements, because they spring from the general nature of the life of consciousness. This too we find confirmed, inasmuch as the law of relation proves itself a fundamental psychological law by holding also in the domain of the feelings. Here it even comes forward in a more striking way.

At about the same time as *Hobbes* had stated the law of relation as a psychological principle (see II, 5), *Spinoza* demonstrated its validity as regards feeling. We feel, as he expresses it, pleasure in an advance in perfection or energy, displeasure in a decline; but it is not perfection in and for itself that awakens pleasure; for if man were born with it, he would feel no pleasure in it. Likewise with imperfection; only he who knows perfection is saddened by it. This is, as one sees, *Fechner's* law (IV A, 4) applied to feeling. As his predecessors in this domain *Fechner* himself cites *Bernouilli* and *Laplace*. These mathematicians taught that the value of a thing cannot be determined purely objectively, but becomes different according to the relation of things to the interests of each single individual. A distinction must be made between *fortune physique* and *fortune morale*. In physical fortune one disregards the circumstances of the individual to whose lot it falls. A sum of 1000 kroner signifies an equally great physical fortune for all. But for him [327] who already owns a million kroner, the mental fortune (i.e. the fortune that is really felt) from that increase becomes small in relation to what it is for him who owns nothing, or who is even in a debt that drives him to despair. Psychologically regarded it is only this "moral fortune" that we have to do with; that a fortune is merely "physical" can signify nothing else than that no real feeling of pleasure is awakened.

2. Just as we distinguish between a color in and for itself and its particular shades, which are in part determined by its relation to other color effects, so pleasure and pain too stand before us as fixed forms, although it is only through their contrast to each other that they are what they are. Every individual has, in the fundamental mood that governs his life, a practical standard, a level, above which his feelings rise only in single moments, and below which they fall only by exception. When the movements of feeling rise high above this "standard of life" or sink deep beneath it, there arise the so-called emotions or *affects*.

A feeling comes forth in its full strength only when it stands in complete contrast to another feeling. That we do not always notice the part contrast plays in our feelings — a part that is far greater than in sensations and ideas, because the life of feeling is through and through stamped by the great contrast between pleasure and displeasure — is surely to be explained by the fact that we as a rule turn toward the new feeling, which gets

its strength from the contrast with the vanishing feeling. Over the victor we forget the vanquished. Precisely because preceding pain gives joy greater life, the former is easily left altogether out of account⁴⁹; this also happens where it is not pain proper, but only a lesser degree of satisfaction, that forms the background of the new movement of feeling.

Just as contrasting colors not only set each other off but also easily pass over into each other, so a feeling also often prepares the way for its opposite. The transition from a strong feeling to its opposite takes place more easily than the transition from indifference to strong feeling. In the first case the source has, so to speak, been opened⁵⁰, and it is only a matter of the stream being led in another direction; in the latter case the living force must first be released. Often a feeling can be developed only by roundabout ways; thus piety and obedience toward authority as a rule precede the ethical feeling proper in its purity and independence. Even the great contrasts of feeling (pleasure — pain, love — hatred, hope — fear, reverence — contempt) prepare the way for each other. Satiety with the one member of the contrast begets a disposition to experience the other, especially at a stage of development where man immediately follows the promptings of the moment, or in strong nervous exaltation. (Cf. the famous “Himmelhoch jauchzend — zum Tode betrübt”). Thus there often comes, in the course of development of a mental illness, a point where extreme feelings of misfortune, disappointment and being misunderstood turn over into excessive joy in imagined loftiness and glory.⁵¹ That “les extrêmes se touchent” is seen nowhere better than in the domain of the life of feeling, where the sharpest and most important contrasts are at home. There are natures that do not get peace of mind until passion has raged itself out. The voice of conscience first becomes audible in many as a contrast to a violent stirring of wild impulses; temptation must thus, strange as it sounds, become very great for it to be possible to overcome it. Often conscience first awakens when a crime has been committed, and then drives to immediate self-denunciation.⁵²

The physiological basis for this peculiarity of the feelings must be sought in the conditions of life of the nervous system. The energy of the nerve organs is limited; when it is exhausted by lasting influence, they demand either rest or influence of another kind. That is why pains are intermittent; even when the cause continues to act, there still comes a point where the capacity to suffer is for the time being exhausted, and there

⁴⁹Cf. John XVI, 21: “A woman when she is in travail hath sorrow, because her hour is come: but as soon as she is delivered of the child, she remembereth no more the anguish, for joy that a man is born into the world.”

⁵⁰Cf. the first series of observations in Ludv. Feilberg’s book “Om størst Udbytte af Sjæleevner” [On the Greatest Yield from the Powers of the Mind].

⁵¹Examples in *Ideler*: Biographien Geisteskranker. Berlin 1841.

⁵²*Bischoff*: Merkwürdige Criminalrechtsfälle. 2ter Bd. Hannover 1835. p. 43 f.

[329] then comes a period of rest in which strength is gathered for new suffering.⁵³ Mental suffering and joy, too, express themselves in this rhythmical fashion; strong outbursts are followed by calmer moods, which can then again pass over into momentary affects. *Herbert Spencer* has drawn attention to the fact that the expressions of the movements of feeling in dance, poetry and music have a rhythmical character.⁵⁴

Those feelings of pleasure and displeasure are the most violent that are bound up with the most intermittent organic functions. The feelings bound up with the preservation of the individual and of the race can have the greatest degree of violence, because the deep-lying organic conditions of life on which they rest are subject to a natural rhythm. The special sense-functions (sight and hearing especially) go on more continuously and are therefore not subject to such great contrasts. (Cf. also A, 3 e). — For the rest, it must be remembered that a feeling can be very strong without being violent (IV, 7 a).

As finite beings we have only a limited capacity for pleasure as well as for pain. The profound idea of the ancients of the envy of the gods has a real basis here. Happiness leads by its own excess to unhappiness, when it consumes the potential energy of our nature. But in virtue of the same law of contrast we must set over against that idea the idea of the compassion of the gods, since pain too consumes itself.

This alternating play, which heightens the feelings of pleasure but also threatens their continuance, and indeed seems able to dissolve the life of the spirit into conflicting forces, has often been regarded as an imperfection, and the human spirit has, in contrast to it, formed an image of an ideal state in which the full feeling of blessedness is not broken by contrasts or changes. Even *Spinoza*, who, as long as he speaks as a psychologist, shows so full and correct an understanding of the law of relation, passes over at the end of his *Ethics* into a mystical speculation that abolishes all contrast, all change and transition. But it has never been possible to give this ideal state a definite positive content. Such a thought and its influence is psychologically intelligible only through its contrast with the sufferings of the present life. Its very genesis thus confirms the law of relation.

3. It has been mentioned earlier (VI A, 2 and B, 2 d) that feeling arises more slowly than knowledge. It is only with very strong and sudden impressions that this difference in time falls away. But the state that arises in such moments is characterized by an indissoluble fusion of elements of knowledge and of feeling. Such simple “shocks” as the elementary sensations from which, according to a hypothesis stated earlier (V A, 2),

⁵³Ch. *Richet*: *Recherches expérimentales et cliniques sur la sensibilité*. Paris 1877. p. 303–307.

⁵⁴First Principles. Part. II. Chap. 10: The rhythm of motion. § 86. — When hysterical patients alternately laugh and weep with violence, and when little children likewise pass suddenly from the one state to the other, this is, as *Darwin* (*Expression of Emotions*. p. 208) remarks, explicable by the great similarity between convulsive weeping and convulsive laughter.

all sense-sensations arise, can with equal right be counted as feeling and as knowledge. Something similar holds at a higher stage of all wonder and surprise, of every impression of the new and extraordinary. As a consequence of the strong contrast with the preceding state or with the rest of the content of our consciousness, our own new state produced by the influence plays at least as great a part as the content of the new observation or idea. This is, in the psychological respect, what is peculiar to all spiritual awakening. Owing to the fresh current of feeling, the new knowledge stands with an extraordinary clearness and in a particularly alluring light. That is why the ancients taught that wonder was the beginning of wisdom. But also only the beginning. For from this border phenomenon different roads may be taken. Either the element of feeling gains the upper hand to such a degree that unrest and enthusiasm make clear appropriation and deeper penetration impossible, or else it acts as an inner animating force and leads to many-sided and steady work in the service of the new idea.

In some cases wonder passes over into fear, disappointment and contempt, or into love and reverence, all according to the nature of what has awakened the wonder. Wonder thus stands as an introduction to very different psychological processes, and it was to that extent a correct thought that lay at the bottom of it when *Descartes* and *Malebranche*, [331] in their presentation of the feelings, let wonder open the series as a presupposition of each of them. Yet it is more natural, with Bain, to form a separate class of feelings, containing wonder and other “feelings of relation” (emotions of relativity), where the moment of relation and contrast that asserts itself in all our states of consciousness determines not only the strength and quality of the feeling, but its whole nature and content.

4. It is a necessary consequence of the law of relation that frequent repetition weakens the freshness and strength of feeling. The background against which it originally unfolded so vividly and energetically necessarily becomes more and more indistinct; a distribution of light and shadow sets in, so that the contrast gradually falls away. This process is only a form of the general adaptation that belongs to all life. Under all conditions the living strives to come into equilibrium with its conditions. For feeling it often has the effect that enthusiasm is succeeded by indifference and slackness and at last stands as an inexplicable memory. From this the pessimists fetch one of their most dangerous arguments. It is true that for sorrow and pain as well as for joy and inspiration adaptation has a damping effect; but it is not always that we therefore praise “the healing hand of time”. (Cf. B, 11).

S. *Kierkegaard* has taken this psychological law as his starting point for showing the

boundary between the aesthetic and the ethical conduct of life. Every awakening and enthusiasm is of an aesthetic kind; we are in an attitude of enjoyment as we are seized by the strong influence. The ego lets itself be carried away by the waves of feeling that break forth involuntarily. But in daily work, under the depressing and damping influence of repetition, it must be shown whether the feeling possesses any other strength than that momentary flaring up. That is why the possibility of repetition stands for S. Kierkegaard as the fundamental ethical problem.⁵⁵

[332] But while Kierkegaard poses the problem in a way that betrays his brilliant psychological eye, it is striking to see with what haste he turns his back on psychology. The problem, according to his assertion, cannot be solved with the help of psychology. "Repetition" — he says (p. 92) — "is a transcendence", which can hardly be understood otherwise than that it lies beyond the naturally human. He has overlooked that there is a psychological law of nature on which the ethical demand can rely — as indeed ethics, if it is not to hover in the air or be a constant appeal to the supernatural, must build on what is psychologically possible.

For it is only in so far as feeling is regarded as a purely passive state that it is dulled by repetition and habit. Active movements and skills, on the other hand, are perfected by repetition; here habit becomes practice. To the active side of our nature belongs not only the capacity for muscular movement, but also observation and thinking, attention and will. Feeling is damped by repetition precisely because the practiced movement can gradually take place unconsciously.⁵⁶ To that extent, then, a decisive contrast would here be established between the conditions of the feelings and those of the ideas. But now the section immediately preceding (B) has taught us how the development of knowledge benefits feeling. The purely elementary feelings, i.e. those called forth by single, definite sensuous impressions, cannot gain by repetition (any more than the simplest, immediate sense-sensations can). The ideal feelings, on the other hand, i.e. those bound up with and determined by a greater or smaller circle of ideas, can not only preserve their strength (if [333] not their violence), but also gain by repetition. The same sum of energy that is released

⁵⁵ "He who will only hope is cowardly; he who will only recollect is voluptuous; but he who wills repetition is a man.... When one has sailed round existence, it shall be seen whether one has the courage to understand that life is a repetition, and the desire to rejoice in it." — Gjentagelsen. Et Forsøg i den eksperimenterende Psykologi. Af Constantin Constantius [Repetition. An Essay in Experimenting Psychology. By Constantin Constantius]. Kbhvn. 1843. p. 5. — From another side the problem of repetition comes forward later in S. Kierkegaard, namely in his polemic against official Christianity as "an established order that has forgotten its origin". (See "Indøvelse i Kristendom" [Practice in Christianity], the newspaper articles, and "Øjeblikket" [The Moment]).

⁵⁶ Already *Hume* (Treatise II, 3, 1) and *Bichat* (La vie et la mort. p. 47–56) have insisted on the different effect of repetition on knowledge and activity on the one hand, and on feeling on the other. — Of later writers *Fries* (Neue Kritik der Vernunft. § 36) treats this matter in a very able way.

in the moment of awakening can also be released later, only distributed, so to speak, into several streams, no longer in a concentrated state. Feeling then gains in manifoldness and inwardness under repetition what it loses in freshness. In the course of adaptation the object of feeling discloses its nature and presents itself under different points of view, and on the other hand the different elements in the individual's own nature are brought into interaction with the object. Within the fundamental relation, then, a manifold of changing relations may be found. Feeling is thereby spread over an ever greater domain of life and can be fed from many more sources than at first. So it goes, e.g. in our relation to a person with whom we live. The inner growth that the feeling has undergone often first becomes clear when the relation is put to the test; there may then again take place a conversion from the distributed to the concentrated form, by which it appears that the capital invested has borne interest.

From Goethe's "Briefe aus der Schweiz" a fine example is given here to illustrate the influence of repetition on feeling. — The subject is the sublime impressions of a mountain journey in Switzerland. — "A young man whom we took with us from Basel said that he did not have the same feeling as the first time, and gave the credit to novelty. I would, however, say to this: when we see such an object for the first time, the unaccustomed soul first expands, and this brings about a painful joy, an overflowing fullness, which moves the soul and draws voluptuous tears from us. Through this operation the soul becomes greater in itself without knowing it, and is no longer able to have that first feeling. Man believes he has lost, but he has gained: what he loses in delight, he gains in inner growth."

Much, then, is inevitably lost that cannot be regained. It is with the budding feeling as with the first breath of the newborn, by which the lung is first distended in such a way that it is never afterwards quite emptied; no later breath, then, can be like the first. To that extent repetition is impossible. But pessimism *wrongly* regards this as a mere loss; seen from another side it is a great gain. It depends on each individual's view of life [334] from which side he chooses to see the matter.

5. This contrast between the influence of repetition and adaptation on the passive and on the active side of feeling leads us to bring up the contrast that the older psychology (especially after Kant's excellent presentation in "Anthropologie" § 71) set up between *affect* and passion. By affect or emotion is then understood a sudden surging up of feeling, which for a while overwhelms the mind and checks the free and natural connection of the elements of knowledge. Passion, on the other hand, is the movement of feeling that has become nature, that is ingrained by habit. In passion the element of impulse in

feeling comes especially to the fore, so that it stands at the transition between feeling and will (cf. IV, 7 d and VI B, 2 c). What affect is in the single moment, with violent, expansive movement, passion is in the depths of the mind as a saved-up sum of force that lies ready for use. That is why passion does not exclude sober deliberation; on the contrary, it finds its expression in a thought that governs the whole circle of ideas. "Affect", says *Kant*, "acts like water that breaks through a dam, passion like a stream that digs itself ever deeper into its bed. Affect is like an intoxication that can be slept off, passion like a madness that broods over a thought that strikes ever deeper root."

Feeling thus begins as affect and – if it finds constant nourishment – passes over into passion. Anger and grief are affects, vindictiveness and melancholy are passions. The deepest and most central current in a man's being is his ruling passion, which is conditioned and prepared by his innate nature, and fed, developed and clarified by all movements of feeling and experiences. It can grow both by satisfaction and by disappointment. Just as the affects pave the way for passion, so passion finds vent in violent outbursts and thus leads to affect; but it can also find its satisfaction in a well-considered and calm fashion.

[335] By the way in which a passion can be satisfied, it can awaken affects of another kind than itself. Thus when love of the fatherland in the moment of danger awakens warlike courage and the lust for battle. Or when, in carrying out a plan of murder that springs from cold self-interest, cannibal impulses are awakened, so that the murderer maltreats his victim in a way that is not necessary for his purpose.⁵⁷

The word passion is taken in this connection in a narrower sense than in ordinary usage, which does not make this definite distinction between affect and passion. At the same time we here keep every moral valuation away from the concept of affect and passion. There are good passions or affects as well as bad ones; but both kinds have the same general psychological characteristics, just as nourishing and poisonous plants can belong to the same family. Everything that is to be able to assert itself as a power in us must express itself as affect and passion. Pure reason is powerless over against the actual life of the soul, where it is always feelings that struggle with one another. It is therefore psychologically incorrect when people so often speak of the struggle of reason with the passions. Directly, such a struggle cannot take place at all. A thought can displace a

⁵⁷Cf. *Anselm von Feuerbach*: Aktenmässige Darstellung merkwürdiger Verbrechen. Giessen 1828. I. p. 93: "It is therefore not to be regarded as a lying invention and an empty evasion when murderers in their confessions tell how, in the execution, they were seized by rage or savagery, were deprived of all presence of mind and carried away by an irresistible power, so that they no longer knew what they willed or did.... Everything that happens in the execution of the crime cannot therefore be derived from the main motive."

feeling only by awakening another feeling that is able to push the former aside. Among modern psychologists *Spinoza* and Hume especially have insisted on this, whereas it is overlooked by Kant (in his systematic writings)⁵⁸ and still more by his followers. *Greek* psychology was even inclined to regard the matter from the entirely opposite side. When it is said that passion and reason struggle with each other, it ought, according to *Plato* [336] (Protagoras. Heise's transl. p. 196 ff.), properly to be said that thoughts struggle with thoughts: passion is at bottom a false knowledge, which is displaced by the true. This consistent elimination of the element of feeling forms an illuminating contrast to the conception that results from many-sided observation.

Repetition thus has a different influence on affect and on passion; the former is weakened, the latter nourished. And this is to be derived from the fact that passion is active in its nature and more closely bound up with definite and distinct ideas than affect is.

6. The law of relation gives each single feeling its fixed, definite form and makes it possible for the one feeling to pave the way for, and become of significance for, the other. But just as a superficial conception of the law of relation in the intellectual domain draws the conclusion that everything is equally true or untrue, so in the domain of feeling the conclusion has been drawn that all feeling of pleasure is an illusion, because it exists only through its contrast with the feeling of displeasure. On this assertion *Schopenhauer* has built his pessimistic philosophy. At the bottom of all consciousness there lies, according to his theory, a blind but unbridled will or impulse, which clings to life and brings conscious beings to preserve their existence and propagate it further. Everything that lives and feels wills first and foremost to live. This "will to live" is not motivated by life's being a good, by its causing pleasure and joy; the relation is the reverse, that we fancy that life brings happiness because we are constantly led by an obscure power to cling to it.⁵⁹ So-called pleasure arises through the satisfaction of a want; the want, bound up with displeasure, is the positive thing; pleasure is on the other [337] hand something negative, and signifies only the cessation of the want. Life's greatest

⁵⁸Kant's psychology, as is well known, does not quite agree with his ethics. Thus he says in his *Anthropology* (§ 70): "Die durch die Vernunft des Subjects *schwer oder gar nicht* bezwingliche Neigung ist Leidenschaft". But according to Kant's moral principle there is an absolute imperative and an *unconditional capacity* to obey reason.

⁵⁹*Schopenhauer's* forerunner, the Italian poet *Leopardi*, expresses the same thought: "All men want to live; therefore they must find life beautiful and pleasant; they want to find it beautiful and are therefore angry with the disturber of the peace who allows himself to think otherwise." (*Opusculi et Pensées*. Trad. de l'italien par A. Dapples. Paris 1880. p. 114). The theory that all pleasure rests on the relief of a want was held in antiquity in part by *Epicurus*, in modern times especially by Italian thinkers (*Cardanus*, *Verri*), from whom *Kant* took it over (*Anthropol.* § 57), without, however, drawing the consequences that *Schopenhauer* later drew.

goods, health, youth and freedom, we therefore do not prize as long as we possess them; only when they have vanished does their value dawn on us.

Schopenhauer's theory conflicts entirely with the biological significance of feeling. If there really existed such an unquenchable ardor, cooling only in single moments to flare up the more strongly afterwards, it would soon consume itself. Pain proves to be throughout bound up with what harms life or threatens its existence. Therefore pain and displeasure cannot be the first, the positive thing, which is checked and broken by experiences in the struggle of life. If we want to call the tendency to movement that is present already before the awakening of consciousness impulse or will, then in itself it cannot lead to displeasure or pain. These can arise only when this impulse meets a greater resistance than it can overcome. To the normal performance of the organic functions there is bound a happy fundamental mood, a feeling of lightness and freedom, to which we certainly only seldom direct our attention directly, and which as a rule we notice only when it succeeds or is succeeded by indisposition. But health, youth and freedom are goods precisely in that they make possible the full development and employment of our powers. A morbid feeling of life soon paralyzes all our faculties; mental illness as a rule begins with disturbance and morbidity in the feeling of life. Its significance and positive value rests not on its pushing itself into the foreground, but on what it carries and conditions.

The Schopenhauerian theory fits best the feelings that are immediately bound up with the preservation of the individual and the species. The theory might seem to fit best the very earliest period of life; at least an unbiased observer has also come to the result that in the first half-year of human life the feelings of displeasure are the preponderant ones, and that the feelings of pleasure in this period arise predominantly through the removal of states of displeasure, not through the production of positive states of pleasure.⁶⁰ But on the other hand Schopenhauer teaches precisely that the [338] feeling of displeasure at existing first really awakens when consciousness has developed. If that were so, we should expect the first period to be filled with the feeling of pleasure, and not predominantly with displeasure.

Schopenhauer denies the whole psychological process by which the development of knowledge also benefits feeling. He is, it is true, inconsistent enough to assume ideal and disinterested feelings (joy in art and science, sympathy, especially compassion); but he cannot explain the difference between them and the feelings bound up with purely physical impulses. In spite of all his thoughts and ideas, man is still just as brutal as the

⁶⁰Preyer: *Die Seele des Kindes*. p. 91. 93.

animal, strives after the same thing as it — only the animal reaches its goal far more easily and with less torment and pain.⁶¹ — In this respect Schopenhauer stands as a decided opponent of the hypothesis of evolution, not quite without justification over against optimistic exaggerations of “wie wir’s so herrlich weit gebracht”. That the brutal instincts of self-preservation (Schiller’s “Hunger und Liebe”) still in great part govern life cannot be denied; but the hypothesis of evolution can also very well admit this without giving up its main teaching. (Cf. C, 3). It is one thing that a development really takes place; another, how much has already been accomplished at the stage at which we now find ourselves. It would be unthinkable if the development of ideas and thoughts, which is after all admitted by all, should have had no influence whatever on the life of feeling. It is in any case a significant advance that those impulses are now compelled to prove their justification by being attached to more or less ideal and universal ends (in the family, the state, etc.).

It is a misunderstanding of the law of relation to assume that pleasure always needs pain as its background. It comes forward most strikingly when it succeeds pain; but it can quite well have a weaker feeling of pleasure as its background, and this holds especially wherever the impulse of self-preservation does not act immediately. The struggle for a higher and nobler human existence therefore aims at pushing back the immediate instinct of self-preservation and opening as wide an access as possible to the goods that are more than the relief of the bare want to live. [339]

7. It is in itself a meaningless usage to call a pleasure or displeasure negative. Every feeling as such is a real, and hence positive, state. Even “illusory” or “imagined” joy is real joy. The hypochondriac really feels displeasure and will not let it be argued away. What is meant by expressions like positive and negative, true and untrue, can in this connection only be the reality or non-reality of the object of the feeling. One cannot criticize the feelings without criticizing their causes and their objects.

Just as people have spoken of positive and negative feelings of pleasure, so it has also been held that there is a zero point that signifies indifference, unconcern, a point, that is, where neither pleasure nor displeasure is felt. It is a big question whether such neutral states really exist. A purely theoretical consideration might certainly lead to the assumption that on the line leading from the highest pleasure to the strongest pain there must be a midpoint lying equally far from both extremes. But this theoretical midpoint cannot be the expression of an actual state of consciousness. If we come to it from the

⁶¹“Am Ende und im Realen handelt es sich nur um dieselben Dinge, die auch das Thier erlangt, und zwar mit unvergleichlich geringerem Aufwande von Affekten und Quaalen.” *Parerga und Paralipomena*. Berlin 1851. II. p. 260.

side of pain, it will look like pleasure; from the side of pleasure, on the other hand, it will appear as displeasure — both so long as no adaptation has taken place. This is a simple consequence of the law of relation.

Immediate experience is difficult to interpret here, because the law of relation itself brings it about that we declare certain states indifferent when we look back on them in very agitated moments. Yet those psychologists seem to be right who assert that sharp and exact observation will always discover fine quiverings of pleasure or displeasure in the apparently neutral states. Even to the most abstract processes of thought there are bound faint moods of liking or disliking. Bain cites wonder as an example of neutral states or excitations of feeling (neutral excitements). As we have seen, it is a border phenomenon, where elements of knowledge and of feeling often seem to be in equilibrium. But as an [340] awakening and movement of spiritual powers it is surely to be regarded as a feeling of pleasure, like all free and fresh activity. When wonder stands as the introduction to fear, grief, scorn or indignation, on the other hand, it is a feeling of displeasure. Wonder does not belong to the violent feelings that let the object fall entirely into the shadow of the movement of the mind itself; but it is not therefore neutral.

We can very well conceive an unbroken series of transitions from pleasure to pain without there needing to be any neutral point in the series. One finds by experiment that when, e.g., one holds one's hand on a body whose degree of heat increases, faint feelings of displeasure arise at a certain point beside the pleasant sensation of warmth; they gradually grow and at last entirely displace the pleasant feeling.⁶²

The background of our life of feeling at any time, the general feeling of life, has under normal conditions the character of pleasure, although as a rule we do not become conscious of it, since, as just remarked, we use our health, freedom and youthful strength to give ourselves up to activities that awaken stronger and more definite feelings. Only in the period of convalescence, or when attention is for some reason or other drawn to our condition, does the feeling of pleasure bound up with health become really conscious to us. Of *Kant*, who rejoiced greatly in his health, which was so good a basis for his energetic mental work, one of his biographers says: "His health was not merely a complete absence of all pain; it was a real sensation and a true enjoyment of the highest well-being."⁶³ — It is surely a correct remark of *Horwicz* that real indifference would be sensed as unpleasant apathy; the normal mood of the feeling of life is a faint feeling of pleasure, which lies midway between emptiness and dissatisfaction on the one side, and particular good form

⁶²*Horwicz*: Psychologische Analysen. II, 2. Magdeburg 1878. p. 26.

⁶³*Wasianski*: Kant in seinen letzten Lebensjahren. Königsberg 1804. p. 32.

or elation on the other side.⁶⁴

At the bottom of the assumption of neutral states there lies not only an overlooking [341] of the weaker degrees of pleasure and displeasure, but also a confusion of the state of the mind as a whole with the effect that single ideas and experiences have. Many impressions and ideas come and go without awakening any noticeable feeling or having any distinct influence on our general state of feeling. But this general state itself is at every moment determined by the predominance either of the feeling of pleasure or of the feeling of displeasure.

8. As an example of feelings whose character is determined by the law of relation, we shall here go somewhat more closely into the feelings of the sublime and the ludicrous. The consideration of these feelings will at the same time be able to serve to show how a feeling is developed to higher forms by displeasure passing over into pleasure, egoism into sympathy, and what significance the development of knowledge has for feeling.

The feeling of the sublime is in its simplest forms akin partly to wonder, partly to fear. Sublime is that which comes before us with such a fullness of impressions that the ordinary measure of intuitability is far exceeded. We try to take it in with our intuition, but are unable to, or able only with difficulty. So act high, steep mountains, the desert, the surface of the sea, the starry sky. The idea of infinite time, too, gives the impression of the sublime, as when it is said that every thousand years a bird comes and whets its beak on an enormous mountain, and that when the mountain has in this way been worn away, only the first second of eternity has passed. But even before immense extension in space and time, overwhelming force acts. The feeling of the sublime surely first asserts itself as a consequence of the idea of extraordinary and superior human power. The savage has this feeling over against a mighty warrior, whose strength and skill in battle make his own vanish for his imagination, just as we feel our existence as a nothing in relation to infinite space and infinite time. The sublime in extension in time and space (what *Kant* called the mathematically sublime) no doubt acts not only by setting our faculty of intuition in more than ordinary activity, but also and especially by giving the impression of an immense power that acts in the enormous mass of parts of space and [342] time. As remarked in an earlier connection, interest in the human expresses itself before the feeling for nature (B, 13). The feeling of superior power as sublime arises within the human world⁶⁵ and spreads from there out over nature, the powers of nature being more or less distinctly conceived in analogy with human power. In the roaring of the winds

⁶⁴*Horwicz*: Zur Lehre von den körperlichen Gemeingefühlen. (Vierteljahrsschrift für wissenschaftl. Philosophie. 1880). p. 303.

⁶⁵Cf. *Grant Allen*: "The Origin of the Sublime." (Mind 1878).

and the sea, in the chasing clouds and the mightily towering masses of the mountains, the spirits of the dead and the gods of the tribe reveal themselves.

The feeling of superior power will not always be disinterested wonder or admiration. In the struggle of all against all the mighty warrior may just as well turn his weapons against his own tribesmen as against enemies. He protects his countrymen, it is true, but demands their submission in return. Of a similar kind, at the lowest stage, is the relation to the gods. Since evil beings are worshipped and honored before good ones, the feeling toward the divine must bear predominantly the stamp of fear. Only where the deity stands essentially as the protecting and gracious power does fear become reverence (see B, 12). There is sublimity not only in the Jehovah of the Old Testament, who brings forth a world by his word and gives his people the stern law on the mountain top amid lightning and thunder, but also in the teaching of Buddhism and Christianity about the infinity of the divine compassion and love, before which human sin and suffering vanish like dew before the sun.

The displeasure bound up with the feeling of the sublime in its primitive forms disappears with higher development, partly because the intellectual elements of the feeling become richer and nobler, partly because the object of the feeling is embraced with sympathy. There is, as was stressed earlier, an element of sympathy in reverence that separates it from fear. It is therefore hardly correct when some psychologists regard an element of displeasure as abiding in the feeling of the sublime, whether one then [343] holds, with *Edmund Burke*,⁶⁶ that fear or terror more or less definitely asserts itself before the sublime, or, more idealistically, with *Kant*,⁶⁷ finds a moment of displeasure in our lower, sensuous nature being pressed down in order that our supersensible being may assert itself the more clearly. The moment of displeasure falls away when the feeling of the sublime takes on purer and higher forms. At the same time the egoistic moment contained in fear also disappears.

The contrast that is the presupposition for the feeling of the sublime to arise need not, then, have any painful character. We can even recognize our smallness with joy, when our soul is at the same time enlarged and filled by the great. We sacrifice life in order to gain life.

Here, as everywhere (V A, 6; — B, 8 a; — C, 7 b. — VI B, 2 e), the successive precedes the simultaneous. The feeling of our smallness and the feeling of the greatness and loftiness of the object will for some time first alternate rhythmically with each other,

⁶⁶Philosophical Inquiry into the origin of our ideas of the sublime and beautiful. Part. II. Sect. 1–2.

⁶⁷Kritik der Urtheilskraft. § 27.

before they fuse into the feeling of the sublime proper.⁶⁸ It is especially at this more restless stage of the development of the feeling that a moment of displeasure will be able to assert itself. The painful feeling at the overwhelming belongs at this stage.

9. While the feeling of the sublime presupposes a certain spiritual development, the feeling of *the ludicrous* is possible at the lower stages of the development of the life of consciousness, as soon as definite ideas can assert themselves. Indeed, even before such a stage is reached, we meet the physiological expression of this feeling; laughter appears before anything ludicrous can yet assert itself for consciousness.

a. Laughter can arise from purely physical causes, so that it is then not at all the expression of an emotion. Severe cold can call forth not only shivering but also laughter. The ancients mention a convulsive laughter that was called forth by eating a herb growing in Sardinia (hence “sardonic laughter”). Laughter is often bound up with hysterical convulsive fits.⁶⁹ Gladiators who were wounded in the diaphragm died [344] laughing. *Ludvig Vives* (de anima lib. III) tells of himself that at the first mouthfuls he took after a rather long fast he could not hold back his laughter. Here belongs also laughter from tickling; already the child in its 8th week laughs when one tickles the soles of its feet.

These phenomena must most nearly be counted under reflex movement. Only as regards the last observation cited can there be *doubt*, since the child, according to *Preyer*, laughs only when it is in a *pleasant mood*. Such a mood is in itself enough to call forth smiles and laughter. Already in the very first days a faint raising of the corners of the mouth can be observed in the satisfied and sleeping child (the counterpart of the lowering of them that expresses displeasure and is often the prelude to crying). Real smiling, however, is first observed in the fourth week, in connection with beaming eyes and a general, indescribable expression of well-being, together with some bleating sounds, recurring in jerks, which evidently also express well-being. Laughter is only an intensification of the smile and of these sounds, and is originally, like them, only the expression of a well-being grounded immediately in the general sensation.⁷⁰

Like the infant, the idiot too, who, as *Eschricht* has said, is an infant in the body of an older child or of an adult, gives expression to his well-being by smiles and laughter. Many idiots go about with a constant smile on their faces and frequently break out in laughter, especially when they get food, are caressed or hear music. In most of them this

⁶⁸Cf. on such fusion VI B, 2 e.

⁶⁹*Laycock*: On the reflex functions of the brain (British and Foreign Medical Review. 1845. 19de Bd. p. 306) mentions a case of reflex laughter as the consequence of an epilepsy brought on by a brain tumor.

⁷⁰*Darwin*: Expression of Emotions, p. 211 f. — *Preyer*: Die Seele des Kindes. p. 141. 184–186.

laughter cannot be called forth by definite ideas.⁷¹

*Ewald Hecker*⁷² has attempted a physiological explanation of laughter from tickling. Experiments seem to prove that tickling has as its consequence a reflex influence on the sympathetic nerves, which shows itself in a rhythmical narrowing of the blood vessels [345] corresponding to the rhythmically interrupted touch. From this it then follows that the blood supply of the brain is likewise rhythmically checked; and laughter now stands as a purposive reflex movement, the chest being compressed by the strong expirations, so that the outflow of blood from the brain is checked. The movements of expiration are interrupted and rhythmical, like the touches of tickling.

This explanation, however, does not cover all the phenomena of laughter, especially not those where laughter appears as the expression of immediate joy and well-being. While the jerkiness of expiration in laughter finds a fine explanation in Hecker's theory, it is not clear why an immediate feeling of well-being, which contains no contrast or rhythm, should be expressed precisely in this convulsive way. — Besides, it must be remarked that already in laughter from tickling a central or psychical influence can assert itself. As a rule we do not laugh when we are prepared to be tickled, and therefore we cannot tickle ourselves either. There must, then, be something unexpected and sudden in the impression, and when the first touch has taken place, an obscure expectation must be awakened that it will continue (with the same strength), an expectation that is disappointed, since the touch ceases the next moment, only to begin again at once. It will appear that herein lies an analogy between laughter from tickling and the laughter called forth by the idea of something ludicrous.

Laughter has often been regarded as something peculiar to man, so that it could serve as a means of defining man (*animal risibile*). If one requires for laughter the idea of something ludicrous, this would also be correct. But apes too smile and laugh, not only when they are tickled or get especially good food, but also when they see again someone they are fond of, or are reconciled again with their keeper.⁷³ Besides, *Ludvig Vives* (de anima lib. III) already remarked that what we express by laughter can be expressed in other ways by the animals (e.g. by wagging the tail).

b. The feeling of pleasure, which is the simplest psychical cause of laughter, will by the nature of the case very often, and at a primitive stage almost exclusively, be [346] called forth by impressions that satisfy the impulse of self-preservation and appeal to

⁷¹Cf. *Eschricht*: Om Muligheden af at helbrede Idioter [On the Possibility of Curing Idiots]. Kbhvn. 1854. p. 76.

⁷²*Die Physiologie und Psychologie des Lachens und des Komischen*. Berlin 1873.

⁷³*Darwin*: Expression of Emotions. p. 132–135.

the feeling of self. Life is first and foremost a struggle to survive.

A strong and suddenly awakened feeling of self easily breaks out in laughter. In idiots at a somewhat higher stage than the one just mentioned, personal vanity is the most frequent cause of outbursts of laughter. Two conditions now come together here: the sudden and unexpected, and the strong feeling of self. This kind of laughter we shall meet especially in characteristic forms where a hard and doubtful struggle has been waged and is suddenly crowned with victory, so that what threatened life now all at once lies powerless and harmless. Then arises the gloating in which the Homeric heroes break out over their vanquished enemies. What finds vent here is not only a feeling of rescue and deliverance. The image of the adversary in his full strength and dangerousness is suddenly succeeded by the image of the same adversary crushed and clipped in his great plans. The urge to dwell on the nothingness and impotence of the vanquished enemy, in contrast to his former bearing, often expressed itself in *barbarous mockery* of prisoners or corpses. In more or less brutal forms this kind of feeling of the ludicrous constantly finds occasion to arise, as long as life, in the human world too, is a struggle for existence. *Thomas Hobbes* therefore seized an essential point of view when he explained laughter as a suddenly awakened feeling of superiority (sudden glory. Human Nature IX, 13). Hobbes, with his doctrine of the state of nature as a war of all against all, was naturally bound to be led to lay the main weight on the feeling of self *or of power*. The one-sidedness of his conception was that he did not inquire whether the feeling of power that expresses itself in laughter does not undergo essential changes and take on a different character according to the nature of that on which it is grounded. But wherever there is struggle, in the service of whatever cause, there may be in those struggling an *element of scorn* in the way in which they regard their adversaries' plans and efforts. Even in religious poetry and religious polemic we find this form of the feeling of the ludicrous again. Thus it says in the 2nd Psalm: "The kings of the earth rise up, and the rulers have taken counsel together against the Lord and his Anointed. — 'Let us break their bands asunder and cast their chains from us!' — He that dwelleth in heaven laugheth; the Lord mocketh them." And when *Pascal* defends himself against the Jesuits' complaints that he had used laughter as a weapon in his "Provincial Letters", he says that just as there are two things in religious truth, a divine beauty and a divine majesty, so there are also two things in error: the impiety that makes it terrible, and the impudence that makes it ludicrous; therefore the saints feel both hatred and contempt toward error and show their *zeal*, not only by forcefully repelling the wickedness of the impious, but also in refuting their error with laughter, following in this God's own

example. (Lettres écrites à un Provincial. XI.).

The very fact that laughter can be used as a weapon shows that it has power as its presupposition. One can laugh at another only because he is the weaker. It is naturally not always real superiority that finds vent in this way. Precisely *embarrassment* and *shyness* can express themselves in smiles and forced laughter. Indeed, the consciousness of complete impotence, when one nevertheless will not bow, breaks out in the same way. In Paludan-Müller's "Lucifers Fald" [The Fall of Lucifer], Lucifer laughs for the first time when "with trembling beyond compare — yet with *defiance* that bends not" he takes possession of his kingdom.

That no one likes to be the one laughed at is naturally explained by Hobbes's theory. Only few would have taken the matter as humorously as Socrates, who at the performance of the "Clouds" stood up so that the original and the caricature could be compared. The mere fact that a thing has a side from which it can be made ludicrous is enough to show that it does not represent absolute power. Precisely for that reason, on the other hand, it is so tempting to find something ludicrous wherever claim is made to authority and complete recognition. Everything sublime, all authority and dignity, has in laughter its dangerous enemy. Laughter here is not so much the expression of superior power as of liberation. Even what under ordinary conditions would not seem ludicrous to us becomes so when it has a situation of enforced seriousness as its background. Boys in a school lesson can find the most insignificant thing ludicrous. *Merely* the fact that [348] something can happen without the will of the ruling authority is enough to awaken a *consciousness of freedom*. Authorities that are on the point of losing their dominion naturally become the object of the feeling of the ludicrous. The appearance of comic poetry (Aristophanes, Molière, Holberg) therefore always marks a turning point, where the consciousness of freedom breaks forth.

c. Scorn, however, is not a necessary element in the feeling of the ludicrous. This feeling undergoes an essential change when it has sympathy as its basis. When an inward bond unites the one who laughs with the object of the laughter, a new and peculiar feeling arises. This is already the case when we laugh at what we ourselves have said or done at an earlier time; we give ourselves up, though we must necessarily have another understanding of our absurd conduct than one who merely sees it from outside or as an adversary. To laugh at someone for whom one feels sympathy is like laughing at oneself. Here there is a doubleness in the feeling: one sees the worth of the object in the midst of its smallness. In one and the same moment one applies a double standard. So when one laughs at the child's helplessness or groundless fear, or at its naivety. There

is here a subterranean connection between the one who laughs and the object of the laughter. The sting is taken out of the laughter. The feeling of the ludicrous on the basis of sympathy is called humor. This can be more than a passing mood. It can develop into being the fundamental mood of a view of life which indeed has an open eye for the finite, painful, meaningless and disharmonious in the world and sets it in sharp contrast to the great and significant, but which, in its heartfelt fellow-feeling with all that lives and in its firm faith in the powers that rule in nature and history, has got beyond all bitterness. The humorous conception of life has reconciled itself to the experience that even the great and lofty has its limitation, its finite side, and its laughter at the small and finite does not forget that this can be the form of a valuable content.

Comic poetry rests not only on the consciousness of freedom, but also on a more or less distinctly apparent humorous conception of life. Only thereby does it become different from the lampoon and satire, where the egoistic, or at any rate the antipathetic, moment rules, in that a single definite person or a single cause is struck at. Only on [349] the basis of humor does art become really free. The power on which the humorist and the comic poet rely is no longer their own ego in contrast to other egos; it is truth and reason, whose power they maintain through the exposure of the low and meaningless. The feeling of power is thus not egoistic here; what is seen in its nothingness is only that in the object of the laughter which conflicts with the true and right. After one has laughed at a person, one therefore often feels the urge to declare that he is at bottom a decent fellow after all.⁷⁴

d. Whether the basis is antipathetic or sympathetic, it is common to everything ludicrous that something powerless *suddenly* appears in its nothingness through its *contrast* with a superior power. The ludicrous presupposes that for a moment we have let ourselves be duped, bewildered, caught in an illusion or held in the tension of an expectation, and that the whole thing then all at once dissolves into nothing.⁷⁵ The psychical element in tickling is, as already remarked, the expectation that is awakened but at once disappointed. The child laughs at every sudden movement, especially when it is repeated in jerks, as when one quickly squats down in front of it, then gets up, and repeats this sinking and rising. Sudden deliverance from painful or dangerous situations acts in the same way. All wit rests on the expectation being strained by a knot being

⁷⁴Rousseau reproaches Molière (in his “Lettre à d’Alembert”) for having presented a respectable person as ludicrous in the “Misanthrope”. Lessing on that occasion drew attention to the difference between laughing and laughing at (lachen — verlachen). (Hamburgische Dramaturgie Nr. 28–29). — Already Aristotle distinguishes between the lampoon and comedy. (Poët. c. 4).

⁷⁵The conception to which we here adhere has been stated by Kant (Kritik der Urtheilskraft. § 53 Anm.), Zeising (Ästhetische Forschungen. Frankfurt 1855. p. 282–290) and Spencer: L’atténuation de la physiologie [The Physiology of Laughter]. (Mindre Afhandlinger. Danish transl. Kbhvn. 1877).

tied or the solution of a riddle being announced. He who prepares his witticism robs it of its effect. The Horatian verse: "The mountains are in labor: a ludicrous mouse will be born" can serve as a motto for everything ludicrous. — In a farce that was played some [350] years ago at the "Bouffes Parisiens" there was a character who right from the beginning of the piece sat quite still at one side of the foreground, wholly untouched by the action. Now and then one wondered to oneself what he was sitting there for. At last someone addresses him, but is interrupted by another: "Don't talk to him; he is deaf!" Now, then, one had got the explanation of his passivity. But in the same moment the character adds with a very melancholy expression: "and dumb!" This awakened a storm of laughter.⁷⁶ — To the Emperor Charles V is attributed the remark about his relation to Francis I of France (with whom he was constantly at war, especially over Milan): "My brother Francis and I are in complete agreement; we both want Milan!" That in which the agreement is to consist is precisely what causes the strife; an expectation is awakened of seeing the relation between the two contending princes from a new side; but this expectation is dissolved the next moment.

The ludicrous rests on an *effect of contrast*. Two thoughts or two impressions, each of which awakens its own feeling, and of which the latter tears down what the former builds up, suddenly meet each other. No closer logical grounding of the effect of this can be given, any more than one can ground logically why two complementary colors set each other off. When the ludicrous has often been explained (among us especially by S. Kierkegaard in "Uvidenskabelig Efterskrift" [Unscientific Postscript] p. 394 ff.) as resting on a contradiction, this is too abstract and at the same time too narrow a conception. Even if one can often formulate the clash by which the feeling of the ludicrous arises as a logical contradiction (e.g. being dumb and deaf — and yet answering oneself that one is), the real process proper is not thereby hit upon, which consists essentially in a contrast of feeling. A certain intellectual development is naturally necessary in order to apprehend the ludicrous, just as it is a presupposition for the contrast effect of colors that the eye can apprehend each single color by itself. The feeling of the ludicrous is in [351] this respect akin to surprise and wonder, which also rest on opposition and contrast (cf. 3). That the moment of feeling is the essential thing in these phenomena can be seen from the fact that they do not bear repetition and habit. While knowledge (e.g. insight into logical contradictions) is practiced and strengthened by repetition, feeling is dulled (cf. 4). The ludicrous does not bear too frequent repetition.

Even where we laugh at the logically contradictory, the feeling of power (the antipa-

⁷⁶As far as I remember, the farce was a parody of a diplomatic congress on the occasion of the Greek affairs, and Greece had access to the congress — without daring to take part in the negotiations.

thetic or the sympathetic) plays a part. For we have our own reason as the firm ground from which judgment is passed, and we become more or less clearly conscious of it as we laugh at the meaningless. Even if we do not celebrate our own victory in laughter, we celebrate the victory of reason.

e. The feeling of the ludicrous thus rests, like the feeling of the sublime, on a contrast. But the two feelings also stand, besides, in a relation of contrast to each other. They build on one and the same fundamental relation, the relation between greatness and smallness, seen from different sides. It was probably this that *Socrates* was thinking of when (at the end of the banquet *Plato* has depicted in the “Symposium”) he insisted that it was the business of the same man to write comedy and tragedy. This proposition is confirmed already by the old Greek tragedians, who wrote satyr plays alongside tragedies; but it first stands before us in its full truth through Shakspeare’s great example. It has been aptly said that Shakspeare’s humor was a part of his fidelity to the world. For man’s real position is this, that he must assert his power outwardly, abolish and annihilate resistance, while he himself must feel his smallness over against the great powers in nature and history. He has fought his way to full victory over himself and over the world who neither gloats nor fears. Much laughter, says *Hobbes* already, is a sign of a small mind, which feels its strength only in comparison with those who stand beneath it, while it is the sign of a great mind to help others and to compare oneself with the greatest (*Leviathan*. cap. 6). In humor we feel ourselves at once great and small, and it is sympathy that makes laughter humorous, just as it makes fear into reverence.

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E. The Influence of Feeling on Knowledge

In an earlier section (B) it has been shown how the connection of ideas becomes useful to the feelings and makes their development possible. But deeper than this influence of the life of ideas on feeling lies the influence of feeling on the life of ideas. The original connection of the element of feeling and the element of knowledge is always the beginning of the whole higher psychological development; but during this — that is, during the psychological process depicted in section B — the element of feeling by no means behaves wholly passively.

1. In considering the development of feeling by means of knowledge, we presupposed that there was nothing that hindered the connection of ideas. But feeling itself can act as a *hindrance*. When the feeling α has grown inwardly together with the idea a , it will

be able to hinder the connection, in itself natural, between a and $a_2 a_3 \dots$ and still more between a and b ; that is to say: we cannot complete the train of thought consistently, because the feeling will not extend beyond its original object. Here the inertia of feeling is at work, which thus becomes the source of many inconsistencies in history and in daily life. When the Greeks could not extend love of humanity to the barbarians, this did not come from purely intellectual narrowness (though the limited historical experience contributed here); but full ethical consistency was checked by national feeling. Christianity overthrew these barriers, not by intellectual superiority, but by the deep movement of feeling that it awakened. Within Christianity, intolerance has set up new barriers and checked the consistent unfolding of the religion of love. The result to which logical thinking seems able to lead in a single moment thus takes long periods in history: a revolution in the life of feeling is the result of the experiences of life of great spans of time. Therefore historical criticism may use logical contradiction as a criterion only with caution; historical development would be impossible if inconsistencies were not possible. Historical research has to seek out the real feelings that have acted to hold back and check in the particular cases. The measure of the courage and energy of thought can [353] be taken from the barriers of feeling in spite of which the new thoughts have worked their way forward.

But once the step has been taken, feeling is the faithful preserver of what has been won. Here its inertia benefits knowledge. By b fusing through a with α , it strikes deep roots in the soul. Knowledge gains in firmness and certainty, and it is in reality the property of the personality only when it has in this way its root in feeling, in the immediate state of the individual.

When a certain idea or a certain circle of ideas is carried by strong interest or passion, its position with regard to other ideas is changed. It becomes a stronger center of association than it otherwise would be. In all experiences regard is had only to what in one way or another concerns the idea that is carried and reinforced by the interest. All other elements in the world do not exist for consciousness. Feeling here brings about a *selection of quality*. All ideas that do not harmonize with the ruling feeling are displaced, just as those living beings perish that are unable to adapt themselves to the given conditions. Lotze has particularly stressed the fusion of ideas with the feeling of life given at the same time. If the feeling of life is altered, the way to the ideas bound up with it is also barred to us; even when new experiences again call forth single ones of them, the common bond is lacking. "In this way it could", Lotze continues, "be explained why things we experience during severe illnesses, or before their outbreak, while the

feeling of life was already altered, are wholly forgotten when we get well, and why the series of ideas in a paroxysm of fever are not remembered in the free interval, but are continued in the next attack, when the sick feeling of life has returned.”⁷⁷

2. Some psychologists have ascribed to the single ideas an impulse of self-preservation, by which they seek to assert themselves in consciousness with a certain strength and to displace one another from it. The psychology of knowledge has proved the incorrectness of this assumption (V B, 6). *The strength of the single idea* rests in the first place on its relation to the other ideas in consciousness. The idea that can rely on the greatest [354] number of experiences and memories will have the greatest possibility of becoming the ruling one. But besides the mutual relation of the ideas, their relation to the feelings is also of importance. In strong tension or deep and lasting interest, even ideas that hang together with very comprehensive and often repeated experiences can be pushed entirely aside. The fetish-worshipper lays greater weight on the few cases in which he can believe he has been helped by his holy stone than on the many where this belief is wholly impossible. He who loves a person inwardly does not see the unlovely features in him. Very beautifully and movingly such a relation is depicted by *Prevost* in his “*Manon Lescaut*”. Love makes blind — but only because it makes extraordinarily seeing on a single point.

In the biographies of criminals countless examples are found of the blinding power of feeling and passion. The violent covetousness for an object — be it a royal crown or a silver watch — overwhelms thought, or rather concentrates all thoughts on the object and on the means of obtaining it. In “*Macbeth*” *Shakspeare* has shown in masterly fashion how the idea of the criminal act can govern the mind to such a degree that it stands as the only reality:

This thought,
whose murder yet is but a dream, so shakes
my whole being that all my vital power
perishes in surmise, and *nothing* is
except that *which is not*.

(Act I. Sc. 3).

From this comes the often incredibly foolish way in which crimes are carried out. “In most criminals”, says the famous jurist *Anselm v. Feuerbach*,⁷⁸ “it can quite definitely be shown that and how the criminal’s understanding has been dazzled, darkened, taken captive by

⁷⁷Drei Bücher der Metaphysik. p. 600.

⁷⁸Aktenmässige Darstellung merkwürdiger Verbrechen. II. p. 342.

desire through the magic power of overmastering incitements, and restricted in the free
 [355] use of its activity, and how precisely this restriction has been a main contributing cause of his committing his deed." Such crimes come about through a drowning out not only of the demands of conscience but also of those of prudence.

In the particular cases it is impossible to know beforehand whether the impulse of feeling or the connection of ideas will be the stronger. It comes down to a trial of strength. But other conditions can make the matter still more complicated. Precisely a very strong impulse of feeling can, by an effect of contrast, call forth a vivid idea of the moral law and of the interest we are about to violate; on such an effect of contrast rest some of the most drastic utterances of conscience. In other cases the ordinary association of ideas can still call forth images that are able to check the stream of the ruling feeling. When Lady Macbeth would lay hands on the old king, she was held back by his resembling her father. ("I had done it, had he not resembled my father as he slept." Act II, Sc. 2). A little more passion — and the law of similarity could not have operated!

We stand here at the boundary between feeling and will. For blindness and the absolute dominion of passionate feeling seldom arise through the purely involuntary play of feelings and ideas; man can, strange as it sounds, *will* to deceive himself, can for the sake of his passion push sober deliberation back. When the satisfaction of passion meets resistance in a man's mind, he can set his understanding and his fancy in motion to find reasons to deaden the inner voice. The inner contradiction is not to be endured and must be removed in one way or another. Man therefore comes forward over against himself as a sophistical rhetorician. In all passion something of such sophistry can be shown.⁷⁹ In a milder form the same thing shows itself in the way in which favorite opinions are held fast, painted out and carried through, very often with desperate reasons.

3. It is originally a practical interest that sets ideas in motion. The task of primitive
 [356] knowledge is to discover the means of satisfying impulse. Only gradually does an impulse to knowledge for knowledge's own sake develop (see V D, 4. — VI B, 13). And even the attempts at thought called forth by contemplative enthusiasm emancipate themselves only slowly and incompletely from the dominion of practical feeling, and therefore as a rule bear a teleological stamp. The resignation by which feeling holds itself back in order to let thought go its own ways and follow its own laws is the work of hard struggles in the history of the individual and of the whole human race. Man wants to know nature as it is; — but he also wants his own ends to be the ends of nature. All causal laws are for him in the last instance means to the realization of the highest good. This tendency

⁷⁹Cf. *Anselm v. Feuerbach*. II. p. 331 f.

appears at the crudest stage of fetishism as well as in the highly developed world of thought of the idealist philosopher. The content and value of the feelings and ideas are extremely different at the different stages; but the psychological relation between feeling and knowledge is the same from the lowest to the highest stage.

The concept of necessity is originally practical: thought seeks the conditions without which the valuable end cannot be realized. Thought thereby acquires from the first a negative character over against feeling. Feeling is sanguine and impatient; only reluctantly does it give thought room. — One easily sees that this impatience is only another side of the slowness and inertia mentioned earlier; impatience proceeds from inertia. When *Tertullian* conceived original sin as impatience, *Fichte* as sloth, they were not so different in their thinking as the wording might suggest.

The definite, unavoidable connection between means and ends, defying even the highest passion — that if one wills a, one must also will b — is man's first collision with necessity. When such experiences are repeated, thought at last draws the conclusion that such a necessary connection is an essential part of all given reality. Now it can become an independent task, the object of an immediate interest, to know these necessary laws and, forgetting oneself, to become absorbed in the great connection of the phenomena. Then the relation between cause and effect takes the place of that between means and end. [357] The history of the sciences shows us a progressive transition from teleology to mechanism — under the constant protest and resistance of feeling. Even if science had explained the whole universe according to its laws, it would still not be able to forbid feeling to subordinate the whole system of causes and effects to a highest teleology, inconceivable to us. The last questions in the domain of views of life, the questions of the value and significance of reality and of life — these questions, to which thought has never been able to give a conclusive answer, are decided in the last instance at the dictation of the feelings. This comes forward with particular clarity in our day in the great significance that the contrast between the optimistic and the pessimistic view of life has acquired. Here in the end it is always our own innermost nature and our personal experience of life that turns the scale.

4. Feeling acts on ideas not only by holding back and narrowing, preserving and selecting, but also by *altering*. Just as one and the same landscape looks different according to the light that falls on it, so the same things and events appear to us quite different according to our different moods. Here the feeling of life with its fluctuations plays an especially important part. *Lichtenberg* says: "I have very distinctly noticed that I often have one opinion when I am lying down and another when I am standing,

especially when I have eaten little and am weary." Feeling does not change at once with ideas, but spreads out over the new ideas, even if these stand in no connection with those that have awakened the feeling. This "dominion of a feeling that surges on" (to use Lotze's apt expression) is sometimes striking and unintelligible to ourselves, especially when the feeling is due to inner, organic states; but for the most part it exerts its influence on the new content of ideas without our noticing it. We owe much to this influence; for by it a heightening of mental life at one point can also further other mental activities. Thus, e.g., music, wine and brisk bodily movement can help the activity of thought along. Here belong a great many of the examples in Feilberg's book "Om størst Udbytte af [358] Sjælsevner". Among other things he rightly stresses how new impressions and situations can produce a movement in the whole of consciousness, quite apart from the content of what is new. "New situations bring the soul into what the chemists call *status nascendi*, the state in which substances find themselves at the moment of their production. Just as chemistry teaches that substances in this state possess a particularly great tendency to enter into new combinations, so it is also in the soul. The as it were floating state, freed from something old, contains heightened chances for the beginning of new states (value of possibility)" (p. 30).

This phenomenon might be called *the expansion of feeling*. Every strong feeling strives for sole dominion in the soul and gives its color to mental activities.⁸⁰ It is the same law according to which feeling procures itself outward expression. There is, so to speak, both a mental and a bodily mimicry: the former is the influence of feeling on ideas, the latter its effect on the muscular movements. — We describe in what immediately follows some particular forms of this law.

a) *The anticipating effect of feeling*. In a strong tension of mind one is inclined to take impressions that are expected as given before they really occur. When, e.g., we are impatiently awaiting a carriage, we believe every moment that we hear a rumbling. — The experiments on physiological time give good examples of this. When, e.g., a certain impression is to be signaled, attention can be so strained that one signals another impression instead of the expected one, not from confusion, but because any impression whatever, under the strong tension, leads to the action one was on the point of performing. [359] When one follows with the eye the movement of a pointer on a circular dial, and at

⁸⁰This law was already pointed out by *Hume* (Treatise I, 3, 8) and plays an important part in his theory of knowledge. Later *Beneke* developed it in an interesting way (Psychologische Skizzen. I. p. 362 ff.) — Cf. also *Spencer*: Princ. of Psychol. § 260–261. — A fine example is found in Goethe's "Erster Epistel" (Ernst und wichtig scheint mir die Frage; doch trifft sie mich eben in vergnüglicher Stimmung. — — Und dem heitern erscheint die Welt auch heiter). — Cf. also Mme de Staël's remark (Corinne I, 1): "Quand on souffre, on se persuade aisément que l'on est coupable, et les violents chagrins portent le trouble jusque dans la conscience."

each revolution a stroke of a bell takes place without one's knowing beforehand at what moment it will sound, in most cases the stroke of the bell is combined with a position of the pointer that lies before the real sound, so that the latter is apparently heard before it really occurs. How much earlier it is heard differs in different individuals and depends on "the personal equation".⁸¹ — A large part of the so-called mesmeric and spiritualist phenomena can be explained by the strained expectation that the experimenter calls forth in the persons he operates on.⁸² — In the domain of thought, too, feeling can act anticipatorily, in that it decides the matter by its postulates instead of following the lengthy and sober course of the development of thought. Contemplative enthusiasm sets up its demands for unity and harmony in the view of the world and often believes it has won a result by the mere setting up. A great example is Plato's doctrine of ideas and its after-effects. It is the peculiarity of great geniuses to act anticipatorily, led by a deep feeling; but critical revision must not be omitted if admiration for the profundity of genius is not to lead to dogmatism. Besides, only experience can decide whether the great ideas really are anticipations.

b) *The realizing effect of feeling.* An idea can, under strong movement of feeling, be transformed into an apparently real observation. Images of memory and fancy differ only in degree from real observations, and a strong feeling can efface this difference of degree. When the memory is bound up with a feeling akin to the original feeling, the difference between ideality and reality is at once diminished. Illusions and hallucinations arise especially in very exalted or morbid states of feeling. The critical distinction between idea and reality becomes altogether impossible the more man lets himself be led by his immediate feeling.

c) *The idealizing effect of feeling.* It lies in the nature of feeling not to ask about [360] differences, conditions and limits. It has an absolute character and finds vent in superlatives (always, never, only, etc.). Closer determination, the fixing of conditions and limits, is the task of knowledge. But now, since feeling constantly presses on thought and is never quite satisfied with the given, it drives thought ever further, at last beyond the limits of knowledge. This pressure feeling exerts through the concentration on a single thought mentioned earlier, which is thereby loosened from its connection with and limitation by other thoughts. The practical and ethical interest acts especially in this way. Feeling thereby leads to the formation of an ideal world, in which the imperfections and sufferings of the given world are removed. The idealizing speculation that stems from

⁸¹ *Wundt*: *Physiol. Psychologie*, II. p. 226. 265.

⁸² A detailed investigation of the influence of "expectant attention" in such phenomena is found in *Carpenter's Mental physiology*. p. 279 ff. 618 ff.

this source has its great significance. The great philosophical systems stand as great thought experiments, as acute workings-out of a series of the fundamental thoughts of the human spirit. The struggle of views of life that has led to these bold attempts will itself in turn reap manifold benefit from them. In any case criticism will never be able to stop up the source of idealization. Feeling is bent back into itself by criticism, and resignation will perhaps temporarily check its flight. But the idealizing impulse can be satisfied without any dogmatizing. In the ethical and social relations of human life, in artistic and scientific work, there is employment and outlet enough for the ideal impulses. Feeling will then lead to increasing what is valuable in the world, and with it the very basis of an ideal faith (cf. VI B, 12).

d) The *spurring and awakening influence* of feeling. By its very obscurity and inexplicability feeling exerts a great influence on knowledge. The single stones in the building of knowledge can as a rule easily be pointed out. But feeling has its source in the original instincts, and we know only a small part of its course. The quiet power of the conditions of life, whose effect first shows itself after long periods of time, means more than the single, clearly apparent and conscious experiences. But in whatever way feeling [361] may have arisen, it will not only spread out and govern everything; it will also itself be explained and justified. This need of explanation hangs together with man's impulse of self-preservation. He experiences in pleasure and pain the action of the world on his life-process; they are signs that he must interpret and whose causes he must track down, if his life is to endure and advance. At higher stages of development too, the individual finds his innermost being expressed in the feelings and therefore seeks to prove their justification. But feeling cannot justify itself. In every man there is a mystical vein; there will always be something in his immediate moods that he can express to other men only very imperfectly. If he appeals directly to his feeling, all discussion stops. Feeling is in itself no source of knowledge. The struggle of views of life must be waged with clear thoughts. Feeling plays an important part here as releasing and helping forward; it asks, or leads to asking, but does not itself give the answer. — While feeling, as purely individual and inexpressible, isolates individuals, it brings them together through the need of explanation and justification. Only in union can they find the full explanation. Thus feeling acts to found communities, calls forth congregations, parties, schools and scientific societies.

Illuminating examples of the need of feeling for explanation and justification can be taken from the effect of music and from the general course of development of mental illnesses.

Feeling exerts an attractive force not only on ideas that are of the same kind as its original cause, but also on other ideas that call forth similar feelings. Feeling can thereby become an intermediate link between ideas of different kinds. A man born blind, to whom one tried to describe the color scarlet, cried out: “it must be something like the blast of a trumpet!” It is the kindred effect on feeling that lets the idea of a shrill sound arise when the impression of a dazzling and strong color is described. (Cf. VI A, 3 e). *Wundt*⁸³ has called this phenomenon *the analogy of sensations*. It is in virtue of it that music calls forth images and ideas from other domains of sense. *Hanslick*,⁸⁴ who argues [362] against the conception that music acts directly on feeling, maintains that the analogy concerns not feeling but sense-sensation itself. Without analogy between the sensations there would surely be no analogy between the corresponding feelings. But that feeling itself plays a decisive part cannot be denied when one compares the feeling one has in light and free breathing, at the breaking forth of light after darkness, and at the sound of pure and clear tones after dissonance and confused noise. The feeling awakened by music involuntarily procures itself — though perhaps most in those who are not specially and technically trained — a more or less distinct symbolic expression by means of the analogous sensations. Events and experiences from inner and outer nature serve for a more concrete painting-out of the general mood.⁸⁵ Musicians warn against giving oneself up to such reveries, lest one be lost to the specific effect of music; but it is no doubt impossible to avoid them entirely. Music owes its great power over men precisely to the circumstance that the moods it awakens can, by countless hidden threads, touch all our experiences of life and branch out toward all sides of our being.

According to a conception developed especially by *Guislain*, the beginning of all mental illness consists in a morbid disturbance of the life of feeling. The intellectual disturbances, as well as the abnormal, now violent, now bizarre, now convulsive determinations of will, are then to be only consequences of a primary disturbance in sensibility. “For me”, says *Guislain*, “mental derangement is in most cases a pain in the heart (*une douleur du sens affectif*).”⁸⁶ A *stadium melancholicum* forms in most cases the first period [363] in the history of the mental illness. The intelligence is for the time being untouched.⁸⁷

⁸³Physiol. Psychologie, I. p. 486. — Cf. *Nahlowsky*: Das Gefühlsleben. p. 142 ff.

⁸⁴Vom Musikalisch-Schönen. 6te Aufl. p. 48.

⁸⁵Cf. what *Goldschmidt* tells of himself (*Livs Erindringer og Resultater* [Memories and Results of a Life]. I. p. 47): “From the earliest I can remember, song has not acted on me with an invitation to sing along, but has transformed itself into images”.

⁸⁶Neue Lehre von den Geistesstörungen. Nach dem Französ. von Kanstatt. Nürnberg 1838. p. 4. — Cf. *Griesinger*: Die Pathologie und Therapie der psychischen Krankheiten. 2. Aufl. p. 65. 214. — *Prichard*: Om Sindssygdomme [On Mental Illnesses]. Translated by Selmer. Kbhvn. 1842. p. 14 ff. — See also above V B, 5 (the cases described by *Krishaber*).

⁸⁷Thus a patient said of his state before the actual outbreak of the illness: “Es war mir damals recht

But the sick man involuntarily searches for the cause of his painful feelings. Only with great self-control can he hold fast to the conviction that the only cause lies in his own morbid state. Soon he forms an explanation for himself. He accuses those around him or other innocent persons (especially when they have something mysterious in their conduct, like the secret police, the Freemasons, the Jesuits). He believes himself persecuted, misjudged and molested. These ideas fix themselves in the sick man's consciousness especially when they are favored by illusions and hallucinations.

These false ideas are an attempt to explain the new, altered and abnormal feelings. The individual has been torn by his morbid-passionate mood out of normal harmony with his surroundings; feeling is now determined wholly from within and is no longer the subjective announcement of man's position in the world. In a healthy state, too, objectively unmotivated feelings can arise; but they then easily find their correctives; what is morbid consists precisely in their not being able to be corrected and governed. The altered feeling now becomes the basis of a new conception of the world, in which the sick man absorbs himself more and more and so constantly feeds his despair.

But often, in the midst of the greatest despair, a sudden turn from darkness to light can occur (see D, 2). By a certain impulse of self-preservation the mind finds in an imagined world a substitute for what was lost. "Such a state of mind", says *Ideler*⁸⁸ of a woman who had gone mad from unhappy love, "can, when some composure again becomes possible, have only a twofold outcome: either the soul sinks into the darkest [364] melancholy, when the certainty of its loss presses it to the ground; or, if it does not lack the power of resistance, it forces upon itself a delusion that promises it the fulfillment of its warmest wishes... The whole striving of the madman then goes toward developing that imagining ever further in accordance with the need of the heart... and removing with sophistical dialectic all contradictions into which it has come *over against* the real world."

bang um's Herz, aber noch klar im Kopf." (*Krafft-Ebing*: Die Melancholie. Eine klinische Studie. Erlangen 1874. p. 57).

⁸⁸Biographien Geisteskranker. p. 18.

Chapter 7

The Psychology of the Will

[365]

A. The Primitiveness of the Will

1. Just as Eros in Greek mythology appears both as one of the oldest and as one of the youngest of the gods, so in psychology, according to the point of view, the will can be presented as the most primitive and as the most composite and derived of psychical manifestations. If we will admit will only where there is a conscious choice between possibilities, the will presupposes a higher development of knowledge and feeling and stands as something derived in relation to these, so that it cannot be given with the first awakening of consciousness. But all willing presupposes a content, which must be given in order to be chosen. This content must be developed and assert itself in and for consciousness. This holds not least of the movements that are the condition of all outward action, and it has therefore rightly been held that a germ of the will is found in the capacity for independent movement, which is present even before the awakening of consciousness. In order to understand the nature of the will it will then be necessary to go back to this original germ and follow the course of development from unconscious movement to conscious but involuntary movement, and from this again to movement chosen with consciousness.

2. Even the simplest organisms possess the capacity to set themselves in motion without awaiting influence from outside. The source of movement lies in the individual organism itself. Such an “*automatic*” or “spontaneous” movement is of course not without a cause. It is called forth by internal changes, through which accumulated potential energy is released. Of this kind are the movements that an amoeba or a hydra constantly performs, and which even seem to be peculiar to all organic cells, those too that are elements in higher organisms (the white blood corpuscles, for example). The amoeba has

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got its name (“change-animal”) from its constant movements, motivated from within, and from this, in turn, such constant change of form has been called amoeboid movement.¹ The internal changes that release the potential energy must in turn proceed from the activity of nutrition, which is the fundamental organic process. Spontaneous movement in living beings is possible only because life itself is a constant process, a taking up and working over of elements.

But this in turn implies that spontaneity, the capacity for self-movement, means only independence of the external influences of the *moment*, not total and constant independence of them. Life rests on a definite relation of interaction between the organism and its surroundings, and it soon ceases when this relation is entirely abolished. An absolute spontaneity would be a sucking of one’s paw, which can keep life going only for a short while.

Spontaneity is therefore bound up with irritability, the capacity to answer external influences in a peculiar way, i.e. by a movement of another strength, and perhaps also of another kind, than the impression. Only through this capacity does independence benefit the organism, since adaptation to circumstances becomes possible. The ultimate ground of irritability, too, must be sought in the organic process, in particular in the great decomposability of organic matter. “A truth,” says *Pflüger*,² “which meets all biologists at every step, is the altogether astonishing decomposability of all living matter. . . . This decomposability is the cause of irritability. Is it not in truth vanishingly small vis vivas [367] which, acting in a ray of light, call forth the mightiest effects in the retina and in the brain? Is not the gentle vibration produced by the point of a needle passing over an exposed muscle sufficient to occasion an instantaneous contraction, with simultaneous formation of carbonic acid and lactic acid? How altogether minimal are the vis vivas of the nerves, by whose help they are able to heighten the processes, and hence also the chemistry, in the organs in the mightiest way?”

3. In higher organisms too, endowed with a central nervous system, it is justified to distinguish between spontaneity and irritability. It does not seem possible to explain all the movements of such an organism as reflex movements. The potential energy stored up in the central ganglia can be discharged without any influence through centripetal nerves. This holds in the greatest degree of the centers of respiration and circulation situated in the medulla oblongata. An immediate influence of the altered state of the

¹ *Waldeyer*: Om Organismernes simpleste Livsytringer [On the Simplest Vital Manifestations of Organisms]. (Tidsskrift for popul. Fremstillinger af Naturvidenskaben. 1878). p. 245. 249. — *M. Forster*: Text Book of Physiology. p. 2. 74.

² Ueber die physiologische Verbrennung in den lebendigen Organismen. (*Pflüger’s Archiv für Physiol.* 10ter Bd. 1875). p. 311.

blood (through the accumulation of carbonic acid) takes the place of influence from outside. The blood acts as a stimulant at the same time as it acts as nourishment. Likewise the condition of the blood can immediately affect higher centers, whereby automatic ideas (dream-images, hallucinations) and movements arise.

*Alexander Bain*³ has maintained that the first movement is always of this kind, and that this kind of movement plays a part in every form of movement. The energy stored up by the process of nutrition itself seeks discharge, and finds it along the motor nerve-paths that are originally laid down. The organism sets itself in motion without waiting for impulses from outside. Bain, who in this carries further what Joh. Müller had indicated, names as supports for this view the first movements of the fetus, — awakening that is not due to impressions from outside, — the strong impulse to movement in young animals and children, — the greater liveliness and delight in movement in men and animals in the morning and after good nourishment, — and finally the peculiar energy that some individuals possess, what might be called the “temperament of will” (volitional constitution). Movement thus precedes sense-perception and originally sets [368] in independently of external influences. It is more intimately and inseparably bound up with our nature than sense-perception is. A man is capable of extraordinary activity independently of what he sees, hears and thinks; his sense-perception and thinking have importance as determining the direction of the movement, but do not call it forth in the first place. — *Fichte* already taught that what is original in us is an urge to action; this urge is given prior to the consciousness of the actual world and cannot be derived from it.

In most recent times *Preyer* has joined Bain’s view. — “How does the *first* embryonic movement come about? That it cannot be caused by passive contact I have learned from a close observation of the chick in the egg, which, as I found, moves from the fifth day of incubation on. Here the trunk moves first, then also the limbs and head, without the least change in the surroundings, and long before reflex irritability is present at all. The cause of these remarkable primitive movements of the body of unborn animals must therefore lie in themselves, and cannot be derived from a reaction of the superficial parts upon the central ones.... The cerebrum has no part whatever in this, since brainless monsters and headless animal embryos move their limbs in the same way. In the formation of the motor ganglia in the spinal cord and cervical cord a certain quantity of potential energy must be stored up, which already through the flow of blood or of lymph, or even through the rapidly advancing formation of tissue, is uncommonly easily converted into

³The Senses and the Intellect. Book I. Chap 1. — Emotions and Will. II. Chap. 1.

kinetic energy.”⁴

The independence of sense-impressions that these automatic movements express cannot, of course, be complete. Just as important as it is that the activities most significant for the maintenance of life can be initiated without waiting for external impulses, just so important is it, on the other hand, that an adaptation to external circumstances can take place. Just as important as it is that the organism from the first comes forward actively over against its surroundings, just so important is it that it also lets its activity be [369] determined by the nature of its environment. Even before the awakening of consciousness such adaptation and determination can take place by means of reflex actions. (Cf. II, 4 b). In these it is not the immediate inner state that arouses the movement, but an impression from the surroundings or from a part of the organism. By a purely mechanical route more or less purposive movements are thereby released in answer to impressions. There has been dispute as to whether such purposive reflex movements are connected with consciousness or not; but even if a dawning consciousness accompanies them (as probably already in the fetal state), they are not the fruit of conscious deliberation. Elementary feelings and sensations may perhaps be present; but they are subjected to no further working over. What is characteristic of reflex action is precisely the immediate transition from impression to movement. Reflex action takes place just as involuntarily as spontaneous movement.

The simplest reflex movement would be one released by a single impression. As soon as several impressions coincide, their effect will depend on whether the movements that each of them strives to release harmonize or not. If the impressions tend to release different and incompatible movements, it depends on which of them is the strongest; this is no doubt somewhat weakened by the resistance, but nevertheless determines the result. A frog deprived of its cerebrum croaks when the skin of its back is gently stroked; but when at the same time its hind foot is strongly irritated, the croak fails to come. The most important inhibitions of reflexes proceed from the cerebrum. In the first period of the child's life, when this organ does not yet intervene, no reflexes are therefore inhibited.

4. When the cerebrum takes part in determining the movement, a *spontaneity* of a *higher order* than that elementary one appears. Through its wealth of cells this organ can both subject the impressions propagated to it to a thorough working over and initiate active movements independently of the impressions of the moment. In the movements that arise here, consciousness undoubtedly asserts itself.

⁴Die Seele des Kindes. p. 127 f.

In so far as a boundary can be drawn between reflex movement and instinct, it must be marked by instinct's being more composite, more active and more conscious than reflex movement. Instinctive movements appear as a combined system of means for attaining an end that lies beyond the momentary horizon of the individual, and whose realization it perhaps does not even live to see itself. It is not an instantaneous discharge, as in elementary spontaneity and simple reflex movement, but an application of various forces toward a more or less distant end. Impressions are necessary to call forth the instinctive action, but this is nevertheless determined far more by the tendencies to movement laid down in the organization than by the impression. The impression acts only as the opening of a valve. Hence the animal is easily deceived, as when insects, deceived by the smell, lay their eggs in the carrion-flower. The impulse is so strong that the impression is subjected to no control. When the instinct is not immediately satisfied, it takes on the character of a drive and appears as an obscure feeling, which is attached to certain indistinct ideas and turns into displeasure when the movement is inhibited (cf. VI B, 2 c). [370]

It is not clear whether instinct is bound to the cerebrum. *Flourens* already showed that removal of the cerebrum abolished instincts such as the nutritive drive and the sexual drive. *Goltz* reports of several of his dogs that, after larger parts of the cerebral cortex had been removed, they no longer showed, as before, any aversion to eating dog's flesh. But on the other hand, instincts manifest themselves clearly even in beings in whom the cerebrum plays no part (the nutritive instinct of newborn children). It is therefore probable that instinctive movements can also proceed from the midbrain (the corpora striata and optic thalami).⁵

On the other hand, *will proper* is bound to the cerebrum. Will proper is psychologically characterized by ideas of the purpose of the action and of the ways that lead to its realization, and by a lively feeling of the value of the purpose. The phenomena of will are therefore so intimately bound up with knowledge and feeling that only by renewing the errors of phrenology could one ascribe to them an independent center. On this the most recent investigators, too, are agreed (see II, 4d). It is otherwise with the question of the initiation of movements. If those investigators are right who assume motor centers within the cerebrum, the transition from organs of consciousness to organs of movement takes place already here, since those centers are necessary intermediate links for the conduction of the impulses of will to the motor nerve-paths.⁶ If, on the other hand, those are right who with *Goltz* deny the right to assume centers of movement in the cerebrum, [371]

⁵Vulpian: *Physiol. du syst. nerv.* p. 692 f.

⁶*Wundt*. I. p. 156 f.

that transition first takes place through the connecting paths between the cerebrum and the more deeply lying parts of the brain.⁷

5. The prehistory of the will necessarily had to lead us beyond psychology, in accordance with the general law that the unconscious precedes the conscious. But on the other hand we understand this prehistory rightly only when we know the manifestations of will from our own consciousness. The task now is to show the psychological connection of the will with the other elements of consciousness.

Psychologically we speak of will wherever we become conscious of an activity and do not behave in a purely receptive way. But according to what the psychology of knowledge and of feeling has taught us, we never behave in a purely receptive way. In all sense-perception and thinking, and in all feeling, a self-activity stirs. Only if it were possible to conceive consciousness as a series of sensations with passive after-effects would we arrive at an absolute passivity. But an absolutely passive sensation stands as an abstraction, which is only approximately realized in certain abnormal states. Since it has turned out to be precisely the fundamental peculiarity of consciousness that all its individual elements and states are united in a synthetic activity, the very existence of consciousness may be said to be an activity of will.

The stronger the individual sensations and ideas are, each taken by itself, the more this activity of will recedes. Exclusive sensations tend to call forth hypnotic states. *Bonnet* rightly said that a being which throughout its whole life had only one single sensation, [372] and had it only in one single degree, would have no will at all. We may add that it would have no consciousness either (see II, 5). Only at the moment when the sensation itself set in would an elementary activity of will be aroused, as the attention turned to the impression. No impression occurs without laying claim to attention and calling forth a greater or lesser activity, which helps to apprehend it as clearly and distinctly as possible. Along with the sensation we also become more or less conscious of this *involuntary attention*; in any case it helps to give the momentary state its character. This appears with particular energy when the simple sensation already calls forth a strong feeling, such as astonishment or fright.

The alternation of sensations brings a somewhat higher form of activity of will to light. When a new sensation turns up, it will be more or less welcome according to the relation in which it stands to its predecessors. If it stands in glaring contrast to them, or if for other reasons it arouses displeasure, the mind will strive to push it back and turn away from it. It is otherwise if (as with complementary colors) it brings a beneficial

⁷ *Goltz* in *Pflügers Archiv*. 26de Bd. 1881. p. 36–37. — Munk: *Ueber die Functionen der Grosshirnrinde*. p. 52.

variety, or for other reasons carries with it an element of pleasure. These elements of pleasure and displeasure are of course most often so weak that we do not become clearly conscious of them, and yet in each individual case they determine the way in which things look to us, since they guide an involuntary selection among the impressions that present themselves to us. Just as the plant moves toward the light, so our observations move toward what arouses pleasure and interest and away from the opposite.

The transition from involuntary to *voluntary attention* and selection takes place as memory and free imagination develop (cf. V B). The choice among the impressions presenting themselves can now be determined by earlier experiences. From being a mere drive, attention becomes a process of selection guided by a conscious purpose. We set ourselves, e.g. to follow certain tones or a certain theme through a piece of music, or to observe the timbre of a single instrument. The investigations into physiological time have shown how great an influence strained attention can have on the speed of [373] observation, indeed that it can even anticipate observation (cf. VI E, 4 a). The difference between involuntary and voluntary attention lies here in this, that in voluntary attention the strain, the directing of the powers of the mind in a certain direction, is present prior to the impression, while in involuntary attention this strain is first aroused by the impression itself. We see to a great extent what we will to see, and on the whole we can see only when we will to see. The original sanguine tendency in human nature anticipates experience and only gradually, often reluctantly, lets itself be taught by it. Fortunately experience is able to get us to open our eyes, compels us to will to see. But activity of will is always the presupposition.

Voluntary attention can also be directed to mere ideas, images of memory or of fantasy. The endeavor to call up and hold fast such images is connected with a sensation of effort similar to that felt when we try to observe an obscure object. Yet this sensation seems to be differently localized. *Fechner* has pointed out — what everyone's own observation will confirm — that while in voluntary observation of external phenomena the strain is directed toward the sense organ that plays the chief part, in the activity of memory or fantasy the strain withdraws entirely from the external sense organs and appears as a strain and contraction of the scalp and a pressure, going from outside inward, on the whole skull.⁸ The muscular strain that asserts itself here is, however, not always present, and in any case does not constitute the whole physiological process

⁸Elemente der Psychophysik. II. p. 475. 491. — The expression of a person absorbed in his thoughts is especially characterized by the indefinite direction of the eyes; their lines of sight often even become divergent, because of the relaxation of certain eye-muscles, in consequence of the strain having moved to other organs. The head is commonly lowered, likewise because of a muscular relaxation. *Darwin*: Expr. of Emot. p. 229.

connected with attention. Possibly impulses go back from the centers to which the voluntary concentration of consciousness is bound to the centers of sense-perception [374] (just as they otherwise go to the centers of movement), whereby the activity of the latter can be strengthened.⁹ This would be the physiological form of the psychological fact that an idea stands out more clearly before us through our setting ourselves the task of imagining it. Some individuals can even call forth hallucinations in themselves at will. Finally, *Carpenter* has sought to ground the assumption that in attention, as in all activity of will, there takes place a flow of blood to the organ whose function is strengthened. In the endeavor to hold fast an idea or a train of thought there thus takes place an increased flow of blood to certain parts of the cerebrum (ideational hyperaemia). With this agrees the influence that every emotion has on the circulation of the blood. In every act of willing an emotion stirs.¹⁰

An activity of will asserts itself not least in the preservation of the coherence among our ideas and in all thinking. It is the presupposition of the arrangement of the elements of consciousness not coming to be governed by purely accidental associations of ideas. Just as the waking state is distinguished from the sleeping one by the stronger “latent innervation” that keeps the body from approaching a posture determined by mere gravity, so waking consciousness differs from dream-consciousness in that all thoughts are more or less consciously directed toward one end. In its most primitive form this end is orientation in the surroundings, in order to find means for preserving existence. But even at the highest stages of spiritual development a purpose, and a feeling aroused by it, governs the course of thought. The more such a spiritual center of gravity is lacking, the more incoherent the life of consciousness becomes, until at last it dissolves into momentarily shifting ideas.

In both involuntary and voluntary attention, feeling and will work immediately together. Several of the phenomena belonging here could therefore already be mentioned under the psychology of feeling (cf. VI B, 2 c. 7–9. 11. — D, 4–6. — E, 2). The unity of psychical life comes clearly to light here when one considers the importance of attention [375] for sense-perception and thinking, the intimate connection of feeling and will, and the connection of feeling and idea, which lies deeper than all association of ideas (V B, 8 c. — VI E).

6 a. Before we follow the development of the will further in its interaction with the life of knowledge and of feeling, we must see how the will gradually gains power over bodily movement. This is the first important course of exercises the will has to

⁹ *Wundt* I. p. 218 f.

¹⁰ *Mental physiology*. p. 382 f.

run through, so important that *ideas of movement* have even often been regarded as an essential element in all conscious will. Apparently all action is directed outward; but it nevertheless always presupposes an inner action, a determination of the ideas by the thought of the purpose. The thought of the purpose is thus the most important element, and it is only a special, if frequent, case when it is ideas of movement that are attracted and determined by the thought of the purpose.

The presupposition for the formation of ideas of movement is that movements have been performed. As we have seen, the organism does not wait for impulses from outside to set itself in motion; it is itself a little world, which can produce impulses in its own interior. These spontaneous movements, together with the reflex movements, are the material that underlies the ideas of movement. By an involuntary urge we are thus led to have the experiences that are necessary for the development of voluntary movement.

Just as the psychology of knowledge begins with sense-sensation, so the psychology of the will ends with the impulse to movement. What precedes sense-sensation and what follows the impulse to movement we come to know only by roundabout ways. In the simplest voluntary movements sense-sensation and impulse to movement immediately coincide. The state of consciousness present in involuntary movements consists, as already remarked, in a feeling of pleasure or displeasure with indefinite sense-sensations and a certain restlessness called forth by the awakening urge to movement, a restlessness that rises to pain when the movement is hindered. But through the sensations of movement (cf. V A, 7) and through observation of our own movements by the help of sight and touch, memory-images of the movements performed are formed. Between these memory-images and the motivating state of consciousness a natural connection is formed. [376] If the movement has had unfortunate, pain-causing consequences, its memory-image will be connected with displeasure, which can hinder the repetition of the movement when the same state of consciousness sets in again. On the other hand, there will be a tendency to repeat movements whose performance and consequences have been connected with pleasure.

What makes possible the will's mastery over bodily movements is thus, first, the connection between the feeling of pleasure or displeasure and a certain impulse to movement, and second, the more precise determination of this impulse to movement by memory-images of movement (*ideas of movement*). If these two conditions were not possible, all outward voluntary movement would be impossible. But on them too rests the whole development of outward voluntary action from its lowest stage to the highest.

When attention has once been directed to an idea of movement, as we have prepared

ourselves to perform the movement, the performance takes place easily and quickly. This preparation or inner readiness, by which the movement is as it were adopted or fixed in consciousness, as we become completely one with the idea of movement and the corresponding feeling, — cannot be described more closely. It is the fundamental element in the consciousness of a willed movement and can be known only by immediate self-observation, in the same way as the inner process by which we call up and hold fast an idea or a train of thought. Physiologically regarded, it no doubt rests on a similar process.

The investigations into physiological time have shown that when one has once prepared oneself to perform a certain movement as soon as a definite signal is given, the movement sets in involuntarily, without its release taking any time, and without any new impulse of will being needed. On the other hand it takes some time if one wants to cancel that state of preparation again, or as it were to annul the first impulse of will; and if the signal occurs before this has succeeded, the movement can be released without or against our will. It happens, e.g., that a signal awaited with strained attention fails to [377] come, and that the experimenter turns round impatiently to see whether the apparatus is not out of order; now the signal sounds, and with it follows the willed movement, although one had already given it up.¹¹ By preparing oneself to perform the movement on the supposition that a certain impression will come, one has thus brought oneself into a state analogous to that in which the animal guided by an instinct finds itself. The willed action is released just like the instinctive one; the difference between them lies in what goes before.

b. Nature paves the way for our will. But nature gives us both too much and too little. The original spontaneous movement is strong; but it must be guided in a definite direction and adapted as regards degree and form in order to serve our purposes. In involuntary movements several muscles are set in activity at once. Here the task is then sometimes to dissolve these accompanying movements and to form others in their place. Under the influence of the laws of voluntary movement mentioned, thus under the discipline of pleasure and pain, a process of selection takes place, which leads partly to *isolation*, partly to *combination* of movements.

Thus the vocal organs are moved at first involuntarily, as the child gives vent to its displeasure or its well-being. But of the sounds so produced, those are later held fast especially which prove to bring agreeable effects. This is the child's first language; only after it comes the period in which it tries to imitate what it hears. It goes similarly with

¹¹Sigm. Exner: Experimentelle Untersuchungen der einfachsten psychischen Prozesse. (Pflügers Archiv. 1873). p. 616 f. (Also in Hermann's Handbuch. II, 2. p. 286 f.).

many other bodily movements; they are first produced involuntarily and at random, and then they are held fast and repeated, or inhibited and displaced. At first no reflex is inhibited; but discipline pushes back more and more of them (as when the child is accustomed to cleanliness). The infant, in whom the cerebrum does not yet actively intervene, lacks a central apparatus of inhibition, which is the condition of self-control. The development of voluntary movement presupposes a certain development of ideas, and hence also of the brain.

The child performs many movements quite instinctively as soon as it has strength [378] enough for them. Sucking, biting, smacking, chewing and licking are, according to *Preyer*, just as instinctive as the chick's pecking at grains and insects. The same holds for the most part also of sitting, standing, crawling, walking and running. Imitation plays a subordinate part here too; it has most importance as encouragement. Even a child who had never seen anyone crawl or walk would perform these movements as soon as it had strength enough.¹²

Voluntary movement proper sets in when these movements are called forth with a definite intention and guided toward a definite end. The first and most important examples of this are the grasping movements, where the wish to appropriate an object makes the hand move toward it and close around it. The limit of how far the isolation and combination of movements can go lies in the original organization. There are accompanying movements that cannot be dissolved, and independent movements that cannot be combined. — How far practice and adaptation can go is seen from the fact that the grown-together Siamese twins had brought their movements into agreement to such a degree that they walked, ran and jumped as necessity required, without making any arrangement, just as if they formed only one single individual.

c. This process, by which the will becomes master of the body in such a way that the individual can come forward with energy and compactness over against its surroundings, takes place more slowly in man than in the animal. Kittens go through the course of education necessary for them in less than a month, while children need almost two years for theirs. This difference shows the importance of the innate foundation. This importance appears just as clearly when we compare normal and idiot children. The latter are not able, without special and laborious guidance, to learn the use of their organs of sense and movement.

“Even in the lesser degrees the idiot is easily recognized by the lack of a real gaze and of a firm bearing. The lack of gaze shows itself in some idiots as a dead staring, in others [379]

¹²*Preyer*: *Die Seele des Kindes*. p. 146–176.

as a wild rolling of the eyes; the lack of a firm bearing in some in the form of a persistent immobility and torpor, with a certain inclination to monotonous swinging or circular movements of single parts of the body, in others in the form of an everlasting restlessness and purposeless working of the arms and legs.”¹³ Education here too therefore consists in a selection, carried out partly by strengthening some movements, partly by inhibiting and neutralizing others. Sluggishness is to be counteracted, and the restless and disordered play of the muscles tamed. In education one then commonly starts from the given foundation. Thus an idiot girl who ceaselessly rocked her body and swung her arms or legs was got to wind yarn, whereby the restless movement was guided to a definite end, which, through the pleasure it gave, could afterward itself motivate the movement. The excessively strong movement was sought to be inhibited by having the patient led up and down in the garden until fatigue compelled her to keep entirely still.¹⁴

B. The Will and the Other Elements of Consciousness

1. The higher development of the will, *conditioned by the development of knowledge and feeling*.

a. We cannot, as has often been remarked in the foregoing, conceive of any consciousness that is entirely dissolved into single, momentary sensations. A certain degree of memory, and with it also a certain duality, a weaker or stronger contrast between two currents, must always be assumed to be present. But the relation between the two currents can vary without end. We have already seen that the contrast between sense-perception and thinking runs through a whole scale of degrees, and that the same was the case with the contrast between physical and ideal feelings. In the domain of the will we meet a related contrast: between drive and willing proper.

[380] *Drive* had already to be described within the psychology of feeling (VI B, 2 c), and we have likewise mentioned it in the inquiry into the primitiveness of the will; instinct, involuntary attention and the urge to movement are different manifestations of drive, when the word drive is taken in the widest sense.

Already in drive consciousness is freed from the impression of the moment. A striving beyond the momentary state announces itself; the equilibrium is abolished. Drive is psychologically conditioned by a more or less clear idea of what can heighten the pleasure of the moment or diminish its pain becoming connected with the momentary

¹³*Eschricht*: Om Muligheden af at helbrede og opdrage Idioter [On the Possibility of Curing and Educating Idiots]. p. 7.

¹⁴Op. cit. p. 66.

feeling and sensation. Drive presupposes a contrast between the actual and something possible or future. By this drive differs from reflex movement and instinct (A, 4), where the impression is perhaps sensed, but where an idea of what ought now to follow does not assert itself. In drive proper there always lies a more or less conscious demand. — In a wider sense of the word, drive comprises all tendencies to movement connected with feeling and sensation. It would be best if in psychology we could avoid using the word drive otherwise than in the narrowest sense, of something different from reflex movement and instinct. But usage does not allow us to; we cannot, e.g., avoid speaking in psychology too of a drive of self-preservation, a drive to movement, etc., as a short and comprehensive designation for tendencies to movement that are partly unconscious, partly half conscious. And to this is added, as will have become clear from the foregoing, that experience shows us manifold transitions between the different stages and kinds of movement. These can to that extent be presented as different forms of development of drive, which culminate in *desire*, where the purpose of the movement is an object of distinct consciousness. — Drive proper and desire are distinguished from mere instinct in that they contain an idea of the end, but not always of the means, while instinct leads to the application of means for furthering an unconscious end. Instinct is a manifestation of drive in so far as the actions and movements that lead to the end become objects of idea and interest.

It is a fateful turning point when a definite idea is connected with the feeling of pleasure and displeasure and thereby becomes the expression of the object of drive. (Cf. [381] VI B, 2a–b). The movement is thereby guided in a definite direction and will not be able to be altered without a definite exertion of force in another direction. As soon as drive is aroused, the equilibrium is abolished, and it then depends on whether one can master the movement. — It is not those who know, nor yet the ignorant, who strive after knowledge; and whether this striving is aroused depends on whether ignorance is felt with displeasure and with the idea of something that is better than ignorance. — It is at this stage that revolutions arise in the inner and the outer world. The greatest degree of suffering inhibits and strikes to the ground; only when so much relief and progress has been attained that the idea of a better state can assert itself — however it may stand with the means of attaining it — only then does the movement break out. It is neither the free nor the unfree who make revolution, but the half free. Hence, as *Tocqueville* has remarked, the most dangerous moment for a bad government is when it begins to reform itself. The slightest arbitrary acts on the part of Louis XVI seemed harder to bear than

the whole of Louis XIV's despotism.¹⁵ The most recent history of Russia offers entirely corresponding examples.

b. If the life of ideas is somewhat more developed, thoughts of purposes can arise whose attainment would be connected with pleasure, but which nevertheless do not immediately set drive in motion. Such purposes, and the feelings determined by them, correspond to the free memory-images in the domain of knowledge: just as these need not be called forth by the momentary sensations, so those need not lead at once to action. It is by this that wish differs from drive. To have wishes is, from a purely practical point of view, a luxury. Wish is contemplative in relation to drive. But on the other hand wish can be the first form of drive. What at first stands as a distant possibility, filling the mind with pleasure at the mere thought, can, when by being a constant thought it passes more into flesh and blood, arouse a drive.

[382] But the same higher intellectual development that makes wish possible will also call forth the consciousness of the importance of drive's not being immediately followed by action, but of there being an *interval* between thought and execution, so that the ideas that are naturally connected with the thought of the purpose can come forward and gain influence on the action. (Cf. IV, 4–6). Such an interval can arise simply through the action's being hindered and experience teaching that it was good that it was not carried out; its importance can also be impressed upon us by our growing wise through harm, namely by seeing what hasty action leads to; and it can finally be brought about by the idea of the purpose standing in such close connection with other ideas (e.g. the idea of the necessary means) that these at once turn up and thereby inhibit the impulse to movement. In each case — whether it is the consequences of the action or the means to its execution that assert themselves — it is the laws of the association of ideas that are at work. Jeppe would dearly like to drink for one more shilling; but his back reminds him of the consequences. "My belly says, you shall; my back, you shall not." It is the association of ideas and the feelings aroused by it that come forward as inhibiting over against the drive or wish of the moment.

Here there takes place, as regards the will, the same turn as when feeling passes over from physical to ideal feeling. The action is determined by more comprehensive considerations than drive and wish permit. Naturally these more remote considerations too must first assert themselves in the form of drive or wish; but the decisive thing is that a single incitement does not become immediately and solely determining. The process that is thereby begun, of which Jeppe's struggle with himself before the innkeeper's

¹⁵L'ancien régime et la révolution. Livre II. chap. 1. — Livre III. chap. 4.

door is a simple example, can develop into higher forms the more comprehensive the associations of ideas become. Here the strength of memory,¹⁶ the life of fantasy and the clarity of thought become of immediate importance for the development of the will. The more firmly and clearly the thought of more comprehensive purposes than those of the moment, or the thought of the difficulty or the calamitous character of the action demanded by drive and wish, stands out before consciousness, the sooner will the momentary incitement be inhibited and the will be determined by more remote or higher considerations. Even if the wish stirs, it now becomes a “pious” wish, as the consciousness of the impossibility or impropriety of realizing it asserts itself. There is then here a trial of strength between the logic of drive and the higher logic. Drive, according to its nature, goes straight at its object and is able to rake up sophisms against other considerations. (Cf. VI E, 2). Jeppe asks himself: “Is not my belly more than my back? I think so,” — and consoles himself at last that Jacob the Shoemaker will give him credit, which he knows very well he will not. The more the wish fixes itself, the more difficult it is for other thoughts and feelings to inhibit its unfolding. The wish then becomes a resolve. Psychologically it depends only on the strength, not on the value, of the forces that determine the action. The momentary incitement can be pernicious; but it can also be ethically justified and yet succumb, as when enthusiasm is displaced by egoistic and prosaic considerations. It then depends on what thoughts and memories the idea connected with the drive arouses, and what strength of feeling these command in relation to the immediate incitement. [383]

c. When the momentary incitements rule, there is really no inner center, no ego in consciousness. But the more memory, and the ruling feelings determined by it, assert themselves, the more the man’s whole nature, and not merely a single side of it that momentarily predominates, gains influence on the action. A man’s real ego has its expression in the thoughts and feelings that have taken the deepest root in him in the course of life. (Cf. V B, 5). And only when the action has been determined by this firm kernel can the man be said to have willed his action, to have determined himself. The will is in so far determined by the ruling passion, as the expression of the result that the man’s development has deposited up to the moment of action. Now it is the elements of knowledge, now the elements of feeling, that predominate, so that one can distinguish between a will of thought and a will of feeling. We must remember [384]

¹⁶“Our purpose is bound to our memory” — or, as it says more energetically in the English text (Hamlet Act III. Sc. 2): »Purpose is but *the slave* to memory«. — Cf. *Spinoza*: Eth. III. Schol: “We can do nothing by the decision of the mind unless we hold it fast in memory.” (Nihil ex mentis decreto agere possumus, nisi ejus recordemur).

here that it is not always the feeling most violent at the moment that is in reality the strongest, i.e. has the deepest root in the individual's nature. All the more important is it that the moment should not become solely determining; hence, again, the formation of the interval mentioned is so important. It is filled by deliberation, the inner debate in consciousness, in which all the ideas and feelings psychologically possible for the individual can have a say. Through this debate the mere resolve becomes *decision*, and a choice is made among the possibilities at hand. The die is cast.

This is the highest form of willing. It is predominantly determined from within, not by the single sensation. Drive knows only one possibility, one motive; will proper unfolds itself through the interaction or struggle of several motives and possibilities. It is often determined by something that lies far beyond the moment, indeed beyond the whole actuality of the individual, but which nevertheless asserts itself in the individual's consciousness, is represented within it. The psychological understanding of the voluntary action depends on whether it is possible to trace the individual's course of development and follow it step by step to the moment of action, each single transition being explained by the general psychological laws. This is of course an ideal; but it states the principle and the presupposition of all psychological inquiry and of all practical dealings among men.

2. *The reaction of the will upon knowledge and feeling.*

Such a reaction takes place at all stages of the will's development. We are thereby led into a web of psychological processes that it is impossible to unravel and survey. We must be glad to find certain guiding points of view, and, referring to earlier sections as regards the elementary phenomena, we shall here chiefly bring out how the more highly developed will can act in a determining and guiding way upon knowledge and feeling.

[385] a. *The reaction of the will upon knowledge.* Just as with outward movement, nature also helps us to get going in the activity of ideas and thought before we ourselves can consciously intervene. The will is nowhere creative, but altering and selecting. And this altering and choosing is in turn determined by the will's own nature, so that Shakspeare's word is confirmed, that "over that art which you say adds to nature, is an art that nature makes." — We cite a few single examples of this.

1) It is, as shown earlier, only through manifold disappointments that we work our way forward to the apprehension of what is real. Prejudices, passions and unclear observations easily lead us astray. And besides, involuntary hallucinations and illusions can sometimes arise, with which, when the kernel of the mind is otherwise sound, a hard struggle is fought. One such sufferer struggled for 27 years with hallucinations

that tempted him to lay violent hands on other people; even his best friends suspected nothing before the day came when he declared himself overcome and sought out a lunatic asylum. His physician says: "When he told us the torments that had assailed his soul in this long series of years, we could not but wonder *that* he had not succumbed much earlier. Seven years have now passed since his apparent cure; he has been able to attain a high position, but the inner struggle has not ceased."¹⁷ — Conversely, some persons can call forth hallucinations in themselves at will, whereby it then often goes with them as with Goethe's "Zauberlehrling," in that the phantoms gain power over them and cannot be driven away.¹⁸

2) The course of memories and ideas is subject to definite laws. When we call forth certain ideas at will, this happens only by virtue of these same laws, just as it is only by virtue of the laws of external nature that we can alter it and subject it to our purposes. The presupposition of the will's intervention in the course of ideas is that a seeking, an interest, asserts itself. The idea that has aroused this interest now becomes a specially accented and energetic center of association (cf. VI E, 1), which is able to loosen given [386] associations of ideas and to form new ones. Isolation and combination are the chief forms of the will's influence on the life of ideas, as on outward movement. — In the spiritual domain it is certain that "he that seeketh shall find"; for the very fact that there is seeking marks a great spiritual advance, a concentration that will always exert its influence, however great or small the result may turn out to be. — This seeking arises when, of two connected ideas a and b, one has vanished or is unknown to us, but we notice an empty place; attention is then concentrated on a, whereby all connections of similarity and of contiguity with a are favored. Hereby an impulse is given which — often, however, only through unconscious after-effects — can lead to the goal. — The will acts here, in Goldschmidt's apt expression, *boringly*.¹⁹

Such boring and voluntary intervention may be necessary, but does not always lead to as valuable a goal as the involuntary promptings. *Willed* thoughts and fantasies have as a rule a more abstract and pale character than those that turn up "of themselves." Thinking goes best when "it thinks" in us. It is of the greatest importance for spiritual health and the right conduct of life that there be a harmonious relation between the involuntary unfolding of the life of thought and voluntary intervention. The harmonious relation is no doubt different for each single individual.

3) In moral respects it is of great importance that the thoughts and ideas that underlie

¹⁷ *Brierre de Boismont*: Des hallucinations. 3 éd. p. 525.

¹⁸ *Op. cit.* p. 27. 39. 462 f.

¹⁹ *Goldschmidt*: *Erindringer* [Memoirs] p. 183 f.

the conduct of life be brought forward as objects of *repeated* contemplation and deliberation.²⁰ They thereby gain firmer ground in consciousness, so that they are recalled more easily under the changing circumstances of life and are displaced with more difficulty by the impressions and affects of the moment. They gain at the same time in clarity and coherence and thereby become better fitted to guide action. Finally, only thereby can an anticipatory adaptation to more difficult and more intricate circumstances of life take place.

[387] 4) Besides the will's being able thus in single cases to gain direct or indirect influence on the course of ideas, *the development of the life of will as a whole* acts back upon the life of thought, strengthening and clarifying it. A firm decision, carried through with determination and without wavering, cleanses the whole spiritual atmosphere and drives away the mists that hinder the clarity of thought. It takes courage to think; one must be not only free of prejudice, but also free of fear. Consistency in thought and firmness in character are related. And only through firm willing does real self-consciousness become possible. What manifests itself in the unity and coherence of memory and in the immediate feeling of self is completed by the act of will, the strongest psychical experience proceeding from within that we can have. Only in our self-activity do we become rightly conscious of ourselves.

b. *The reaction of the will upon feeling.* It might seem to be so peculiar to feeling to arise and develop involuntarily that no intervention of the will could be thought of here. The will's activity is indeed even more indirect here than in the course of ideas, and what it can accomplish here is surely also more limited and conditioned. But it is nevertheless of great theoretical and practical interest to see what ways stand open here.

1) Even if we cannot prevent a feeling from arising in us, we can perhaps prevent it from spreading by *inhibiting* the *organic movement* connected with the feeling, through which the feeling strengthens itself. The art of self-control consists chiefly in this, since it cannot immediately strike the feeling at its first stage. Help in this is given by the presence of other people, which brings us "to pull ourselves together." Adam Smith²¹ already stressed the importance of the way in which the involuntary outburst and expression of feeling is inhibited by the thought of other persons, really or ideally present. Often this inhibiting perhaps goes too far and leads to a false shame or to blasé indifference. But in any case here is a way by which the very germ of the feeling can be struck, for it is weakened by being frequently hindered in its organic spreading. On the

²⁰Cf. on this *James Sully*: On some elements of moral self-culture. (In his book: Sensation and Intuition. London 1874).

²¹Theory of Moral Sentiments. Part. III. Sect. 3.

other hand, the fact that the feeling is dammed up can also make it bore itself so much [388] the deeper into the man's nature. Which consequence follows in the single cases will depend on the man's character; in the long run, however, the inhibition of outburst and expression will surely always have a reacting influence.

Conversely, a feeling can be aroused by one's beginning by putting oneself in the posture corresponding to the feeling, by producing the corresponding looks and movements. Savages excite themselves for battle by violent dances. Participation in the external ceremonies can, in Pascal's opinion, form the introduction to real conversion. One is certainly in a different mood when one clenches one's hands than when one folds them, — when one stretches out one's arms than when one sets them against one's breast. A characteristic contrast appears especially between the mood under muscular tension and under muscular relaxation. — It is by this route that persons in a hypnotic state can be brought by the experimenter into various moods.²² *Campanella* maintained that by imitating others' looks and movements one could also clearly enter into their feelings. A stranger who came in to him found him writing a letter with the same expression on his face as the man to whom he was writing used to have.²³

2) By *alteration of external circumstances of life* the arising of many feelings can be prevented, or they can at any rate be cut off from nourishment. The power of customs and institutions rests on the influence of the circumstances of life, and political reforms are indirectly reforms of the life of feeling. Even our daily habits and surroundings are often of great importance here. By bringing ourselves under certain definite conditions we can further or inhibit the arising of certain feelings. Much that our will cannot directly bring about is attained when we bind ourselves in such a way that it is afterward not in our power to release ourselves. There is a spiritual as well as a bodily hygiene.

3) If the feeling cannot be altered in these ways, it may be possible to succeed by [389] directing attention to another *end*. But when this is to happen through one's own inner endeavor, it presupposes that the momentary feeling does not lay claim to the whole of consciousness. The will cannot "bore" without definite starting points. The first condition is thus that there be seeking, that a "hunger and thirst" be present. When the man is wholly absorbed in the present state, "laughs and is full" (Luke VI, 25), no ground can be laid for any new feeling. Yet the individual can sometimes feel the need of a change in his life of feeling, but be unable without help to work toward it. Thus *Lichtenberg* wished for himself "the first differential of a push" in order to get the better

²²*Carpenter*: Mental physiology. p. 602–605. — *Preyer*: Die Entdeckung des Hypnotismus. p. 36–41. 85.

²³*Vita Campanellæ*. Autore E. S. Cypriano. Amstelod. 1722. p. 48. — Cf. on this also *Burke*: On the sublime and beautiful. Part. IV. Sect. 4. — *Fechner*: Vorschule der Ästhetik. I. p. 156 f.

of his hypochondria.

The feelings for which a ground can be laid by exertion of the will all belong, of course, to the ideal feelings. The endeavor of will develops the life of feeling by the help of the laws of the association of ideas, which make it possible that, once one new idea can be held fast although it stands in conflict with the mighty feeling, several can soon follow after. The feelings that are most easily remembered, and that suffer least by repetition, are also those that can most easily be called forth by way of inner endeavor. Ideas, by their greater mobility, become means in the service of the will. What I make my constant thought will gradually also determine my feeling. Naturally it holds of such feelings, as of ideas produced at will, that they hardly get the life of those that arise involuntarily.

It is the intervals between the strong emotions that must be used to gain mastery over the life of feeling. Education must here necessarily precede self-education, and even when we take the matter into our own hands, we often need a helping hand ("differentials of push") so as not to sink back.

4) A clear insight into the causes of the feeling acts back upon it, clarifying and purifying it. *The endeavor to understand* the feeling that masters me will therefore be able to set me freer over against it. Feeling as a rule has an indefiniteness that forms part [390] of its power, and that can vanish before clear knowledge, as ghosts before the light of day. It is a common experience that the conviction of the inevitability of grief softens it. Knowledge of causes acts most of all on feelings such as hypochondria, which nourish illusions and suspicion. *Kant* became master of his hypochondria, "which in earlier years bordered on weariness of life," through the knowledge that it was due to his flat and narrow chest. "The oppression," he says,²⁴ "has remained; for its cause lies in my bodily build; but I have become master of its influence on my thoughts and actions." *Lichtenberg*, who has been called "the Columbus of hypochondria,"²⁵ relates that when, during his nervous illness, he stopped his ears with his fingers, he felt better, since he now regarded the morbid buzzing as something artificial.

When knowledge leads to the conviction of a fixed order of things, into which we with all our wishes and sorrows are woven in such a way that its laws are the laws of our life, the way is paved for resignation. There are those who, instead of feeling day by day "the slings and arrows of outrageous fortune," choose with *Spinoza* and *Goethe* to resign once for all. "These convince themselves of the eternal, the necessary and the

²⁴Von der Macht des Gemüths durch den blossen Vorsatz seiner krankhaften Gefühle Meister zu sein. (Kehrbach's edition p. 26).

²⁵*Feuchtersleben*: Zur Diätetik der Seele. 10te Aufl. p. 169.

lawful, and seek to form for themselves such concepts as are irrefutable, indeed as are not abolished but confirmed by the contemplation of the transitory.”²⁶

c. Through its influence on knowledge and feeling *the will acts back upon itself*. Ideas and feelings are the motives of the will, and it is thus possible, according to what has now been developed, that our motives themselves can become objects for the will. In this sense one can will one’s own will. One can do so also in the sense that one can set oneself the task of training in oneself the capacity to make decisions, to put an end to the inner debate and deliberation. Finally, *to will* can also mean that one wills to maintain and carry through one’s decision and not let it be overturned by moods turning up later. [391] This finds application especially when the chosen end requires the application of a whole series of means, a multiplicity of single actions. One wills a, and therefore wills also b, c, d...

The will is never either something concluded or something absolutely beginning. It is impossible to point out any point where receptivity, passivity, gives way entirely to activity, or conversely. No psychological dissecting knife, however sharp and surely handled, will be able to hit a line that separates the drawing power of fantasy and feeling from voluntary surrender. There can be an infinitely varying relation between the two sides, but at no moment does either of them fall away entirely. When it says in Goethe’s “Fischer”: »Halb zog sie ihn, halb sank er hin«, the doubling repeats itself within the second member; for to *sink* is to *let oneself sink*. Here, then, there is no first and no last, but an infinite relation of interaction between acting and suffering.²⁷

3. Relations of opposition between the will and the other elements of consciousness.

The higher development of the life of will is possible only through the influence of knowledge and feeling. But during the transition from lower to higher forms of will, stoppages and paralyses can set in as a consequence of a disharmonious relation between the different elements of the life of consciousness. In the main the general law of inhibition is at work here, in that sensations and ideas that cannot fuse with the given motive of action tend to cower and displace it.

The very “will to live” can be cowed in this way. With increasing civilization, emancipation and enlightenment the frequency of suicide increases; it is greater in Protestant than in Catholic countries, in the towns than in the country.²⁸ Reflection weighs the value of life, which instinct does not engage in. The more possibilities, wishes and needs assert themselves, the more difficult it can become to get a favorable result from the

²⁶Aus meinem Leben. 16tes Buch. — Cf. “Spinoza’s Liv og Lære” [Spinoza’s Life and Teaching]. p. 139 f.

²⁷Cf. V A, 8; B, 11. — VI B, 12; E, 2.

²⁸*Legoyt*: Le suicide ancien et moderne. Paris 1881. p. 293–315.

calculation. Over against possibility actuality pales, and the reflective, contemplative life
 [392] can therefore lead to weariness of life, as was already the case in the monasteries of the Middle Ages, where spiritual slackness and enervation (*acedia*, *animi remissio*, *mentis enervatio*) were counted among the cardinal sins. In more recent times this feeling seems to be even more frequent. Naturally suicide cannot always be derived from lack of will; but it is certain that it is most often a capitulation before a resistance that one regards as insuperable. The energy of the will is measured by the resistance.

When children are to learn to walk, self-confidence is an essential support. An action proceeds more easily when one has a strong assurance that it will succeed. All reflection and *doubt* paralyzes, and in any case begins a period in which energy is dissolved and divided. The child can perhaps walk briskly when no one speaks to it, but totters and falls when various impressions divide its attention. The will must according to its nature always be limited. Its object is one definite thing, and thoughts and feelings attached to other things must always act in a more or less paralyzing way. Hence a higher moral faith, which opens wider spheres and horizons, often brings at first a decline in energy, so that there comes to be an inverse relation between extent and strength. If only strength is taken into account, instinct and authority evidently stand as the foremost forces determining the will. With immediate certainty the way is here traced out; the future almost loses the character of possibility. There are, as Turginjew says somewhere, no stronger powers in us than precisely those feelings that we ourselves know only as dimly surmised secrets. Enlightenment and reflection always contain, to that extent, a danger. The question is whether the will of instinct can become a will of reason with the same energy, just as within the life of feeling the task is to develop the ideal feelings to the same degree of strength as the physical ones possess. In the opinion of many the Negroes have gone backward since their emancipation, and much distrust of the consequences of political freedom is explained by the same cause. All spiritual development takes time.

Shakspeare's Hamlet stands as the famous type of such a stage of transition. The poet has presented in him his own and his age's life of thought and feeling, and yet has set him in an age in which purgatory is believed in, and in which blood-revenge is a
 [393] duty. Even if this should be only the consequence of the frame and main features of the old legend having been preserved, although the character of the chief person is altered, Hamlet, as he appears to us in the tragedy, gives the impression of a life of thought and feeling that does not fit the task set him. Saxo's Amleth has no scruples, although he behaves circumspectly enough. Each of the steps in his intricate plan he takes with certainty, guided by the drive of self-preservation and by the instinct of revenge. Hamlet

has Amleth's cleverness and wit, but not his ruthlessness, although he is placed far more favorably for being able to act, since, as he says himself, he has both "cause, and will, and strength, and means" to do it. What is only hinted at in Orestes (in *Aeschylus*, and, characteristically enough, somewhat more strongly in *Euripides*), a *doubt* and wavering in the decision, fills out the whole of Hamlet's character. Lack of energy and of the capacity for decision can hardly be ascribed to him, since on several occasions he shows presence of mind enough; but there is a dividedness in his being, an inclination to sink away into reflections and feelings, which are no doubt aroused by his circumstances and his task, but which lead far away from these and consume a part of the energy that in Amleth and Orestes goes immediately to the benefit of the action. It is what he himself expresses, that "conscience [i.e. reflection] does make cowards of us all," and "the native hue of resolution is sicklied o'er with the pale cast of thought." Hamlet is rightly regarded as the representative of the man of the modern age, in so far as it is peculiar to this age that reflection has awakened and turns, searching and testing, against instinct and authority. The problem is whether a new certainty can be won in place of the one lost.

It is originally in the very interest of will and action that an interval be established between the emergence of the motive and the decision. Consciousness in general arises, of course, only when the impression does not immediately release movement (see IV). In the interval the motives are to contend with one another, so that the innermost being of the soul may become determining for the action. But this play of possibilities can exert an alluring or alarming power over the mind, so that it loses itself in it without coming to decision and action. The way to madness lies here. "As long as passion preserves its original energetic character, it only rarely leads to insanity, because it summons [394] understanding and will precisely to the highest exertion of force, whereby both mutually keep each other in the path of sobriety. Only passive passion, which dissolves into empty longing, vain craving, foolish hoping or cowardly despondency, only that is the root of insanity."²⁹

The task is therefore to return from the world of possibilities to what is necessary according to circumstances. This limitation and narrowing demands a resignation. S. *Kierkegaard* has aptly said that what is lacking in such cases is really "power to obey, to bow under what is necessary in one's self, what must be called one's limit."³⁰ To will is to bind oneself to something entirely definite. Deliberation must therefore, however many winding ways it strikes into, lead back at last to a simple starting point immediately given in our own nature. A concentration must set in after the expansion. *Aristotle*

²⁹*Ideler*: Biographien Geisteskranker. p. 156.

³⁰*Sygdommen til Døden* [The Sickness unto Death]. Kbhvn. 1849. p. 32.

already says: "If one is always to deliberate, one is led into infinity. Everyone stops inquiring how he is to act *when he has led the principle back to himself*."³¹ If the will is not sick, such a principle or beginning (ἀρχή) must be able to be found. The misfortune of the doubter and the dreamer is that he cannot find his way back to himself. If he cannot himself remedy this, he must call on the help of others, so that he may undergo a new education that can give the motives their natural power over his mind.³²

[395] The art of life consists on this point in bringing about the right relation between expansion and concentration, freedom and boundness.

4. *The consciousness of the will.*

a. It turned out to be the distinguishing mark of will proper, in contrast to instinct, that we know *what we will*, become conscious of the purpose and content of the will. On the other hand, we have not yet raised the question of how we know that we *will*, or what it really is that stirs in us when we will something.

In the case of the phenomena of knowledge and feeling there was no reason to raise a corresponding question. Sensations, ideas and feelings are clearly prominent elements of consciousness. But it is not so easy to point out the elements of will immediately.³³ What we experience in our consciousness when we will can, on closer investigation, be traced back to elements of knowledge and feeling. In drive there is given a feeling of pleasure or displeasure with a certain restlessness, brought about by dawning sensations of movement. By these latter drive differs from mere feeling. In decision, the typical expression of will proper, there is given the thought of a purpose and of the means that are to be applied to reach it, the feeling of pleasure at the thought of the realization of the purpose, and more or less lively sensations of strain and collectedness. Neither in drive nor in decision, then, do elements appear to our inner experience that do not occur elsewhere too. What is peculiar to these phenomena is the definite concentration or pointedness that arises when we regard the possible action as ours. We adopt or anticipate the action, regard as an accomplished act what, outwardly regarded, still stands only as a possibility. In contrast to the inner action that the decision expresses,

³¹Eth. Nicom. III, 5. (ed Bekker p. 1113 a). — This concentration, or return to an active starting point, can take place with a jerk. Thus *Plutarch* has Caesar tear himself out of deliberation in *passion*, to give himself up to the future and cross the Rubicon (μετὰ θυμοῦ τινος ὥσπερ ἀφείς ἑαυτὸν ἐκ τοῦ λογισμοῦ πρὸς τὸ μέλλον. *Cæsar*. cap. 22).

³²The use of opium produces lack of initiative, paralyzes the capacity to begin and to attempt. De *Quincey*: *En Opiumsæders Bekendelser* [Confessions of an Opium-Eater]. p. 157: "The opium-eater by no means loses his moral consciousness, his higher longing and desire; he wishes and longs as vehemently as ever before to be able to accomplish what he regards as possible or feels to be the demand of duty; but his apprehension of what is possible infinitely exceeds his capacity not only to carry out, but even to make an attempt."

³³As regards the physiological side of the matter, see V A, 7 and VII A, 4.

the many changing wishes and fantasies stand only as possibilities.

b. But here, then, the problem of reality turns up in the domain of inner experience. ^[396] What criterion can be given that there has really been *willing*, that an inner action has taken place? How do we here distinguish possibility (wish and fantasy) from actuality?

Concentration differs only in degree from expansion. In every lively wish a certain concentration also takes place. Wishes are related to decision as the nebula to the articulated star system. But here it is not so easy to make sure how far the formation has advanced. If the outer action does not follow upon the inner, how can I then be certain that I have really willed? Here it appears how much we lack a psychical dynamometer. — The decision stands as the conclusion of an inner debate. But what certainty is there that the debate will not be taken up anew, so that the decision perhaps “is sicklied o’er with the pale cast of thought”? And if that happens, was that decision then more than a wish?

In practice we rely here on our immediate feeling. We believe that we can immediately notice in ourselves that a determination has taken place. We feel a peculiar active expectation, feel ourselves set for a certain movement. People whose calling compels them to make firm and unalterable determinations (such as military men and judges) acquire more certainty in this respect than others, though often, to be sure, only within the circle of their professional activity. — Full certainty can never be had. We can here arrive only at a practical faith in ourselves, supported by our self-experience, our acquaintance with our character. In any case it is an inference that we draw, not an immediate fact, as a sensation, a thought or a feeling is.

Although in practice sharp boundaries are drawn between wishes and decisions, it is on the other hand also admitted that it is exceedingly difficult to know oneself. Socrates confessed of himself that he did not know whether it was a flaming and composite being that dwelt in him, or whether it was a milder and nobler nature. Nowhere do confusion and illusion lie so near as in the question of possibility or actuality in the domain of inner nature. The anticipating influence of feeling (VI E, 4 a) manifests itself nowhere ^[397] more easily and more dangerously than here, where external control is lacking. Many regard themselves as great heroes of will because they have wallowed in “great decisions,” although these have never received the tangible and prosaic form of outer action. Here, therefore, a constant criticism must be exercised, of the same kind as that exercised over against facts in the domain of external nature (see V D). The consciousness of the will, our inner actuality, is just as little immediate as the consciousness of external actuality. And just as little there as here can our knowledge reach a complete conclusion. New depths and new riddles always show themselves.

c. The ethical view agrees entirely with the psychological one here, in that it pronounces its judgment on thoughts and wishes as well as on decisions and actions. But on the other hand the ethical view likewise rightly starts from the premise that our innermost being lies in the will. The difficulty of finding the will clearly and distinctly in a single phenomenon arises precisely from the fact that the will does not begin at any single point, but from the beginning of the life of consciousness stirs in every association of ideas and every emotion (cf. A, 5). What is concentrated in the decision is an energy that is applied in less intensive form in all knowledge and feeling. The psychology of the will therefore really embraces the whole domain of the life of consciousness (cf. IV, 7 e). The phenomena specially so called phenomena of will mark only the high points of a process that extends through the whole life of consciousness.

5. *The will and unconscious psychical life.*

a. It holds of the will even more than of the other forms of psychical life that it is not fully understood as long as we keep to the clear daylight of consciousness. Even if our decisions and actions are determined by motives that spring from our own innermost nature, it is not said that these motives always stand out clearly before our consciousness. In such cases we know well enough that we will, and *what* we will, but not distinctly *why* we will it.

Here as everywhere (see III) there is a whole scale of transitions from the conscious to [398] the unconscious. At every moment there is no doubt one thought, one mood, at the *center of consciousness*, from which the other thoughts and feelings asserting themselves at the moment shade off into the unconscious. What is determined by the constant thought and springs from the ruling feeling is what both we ourselves and others understand best. But what at the moment, or at most moments, occupies the center of consciousness need not be what is in reality most deeply grounded in our nature. *The center of individuality* does not always coincide with the center of consciousness. Especially in weak and little-developed individuals the center of consciousness shifts, even if certain thoughts and feelings are those that most frequently rule. When it then comes to action, it is no wonder if something happens that surprises both the onlookers and the agent himself. Something can turn up that has never before been forward in the foreground of consciousness, and which the individual therefore does not rightly own to.

Nature gives us from the first a push forward, in that we already find ourselves in activity when consciousness awakens. Only little by little can consciousness gain influence on activity (inner and outward), and this influence never becomes complete. The spontaneous drive to movement, the reflex movements, and the half-conscious

movements accompanied by obscure feeling always preserve a certain independence, even after conscious thought has officially taken the chair. They play a part, varying in the single cases, in all conscious willing, and sometimes they break out in open revolt. Here belong, e.g., the obscure incitements, which everyone no doubt knows, to smash objects, to interrupt a serious speech, or to do similar unreasonable and unmotivated things. This phenomenon has been called the “antilogy of the will.”³⁴ Against the rational will there rises up a drive inexplicable to ourselves, which is nevertheless as a rule overcome, but in many cases with great effort. Such phenomena establish the correctness of the proposition that we first come to know our character from our actions. Since our nature or our character is more comprehensive than the small part clearly lit up by consciousness, and since our actions too can never fully and many-sidedly reveal our nature to ourselves and others, there always remains a possibility of new experiences. [399]

As long as the unconscious tendencies to activity work in the same direction as the conscious thoughts and feelings, they are not noticed. Their strength fuses with that of the conscious motives, to which we give the honor or the shame for the whole action. We feel free and unhindered in our activity. Only when they work against the end of conscious endeavor do we become aware that there is something in us of which we are not, or at any rate not yet, fully masters. The inner dividedness, the feeling of a resistance, is at the same time a feeling of unfreedom.

Such a feeling of unfreedom often marks the transition to a higher stage of the life of will. It is the condition for a judgment being passed on our earlier willing. For as long as our will works with undivided energy, there is no room for any valuation or judging, but we work our way forward in a straight line. The interval spoken of earlier here shows its great importance. It can bring stoppage, also wavering and disharmony in the mind, but it is necessary for a higher development. It can lead to a complete condemnation and rejection of the earlier direction of will. And when in memory we go back to such times, when conflicting forces moved in us, we often take the part of the vanquished. In regret and repentance we acknowledge that it was the better, the justified part of us that succumbed. Just as deliberation can lead to losing oneself in infinite reflection, so repentance can lead to morbid brooding over what cannot be undone. But in repentance there stirs, on the other hand, a drive to higher development, a drive that, like all drive, is aroused by the feeling of the contrast between ideal and actuality. The self-rejection and self-contempt to which repentance can lead would be unbearable if repentance itself were not precisely a sign that nobler forces dwell in us. What wounds, then, also heals.

³⁴*Spitta*: Die Willensbestimmungen und ihr Verhältniss zu den impulsiven Handlungen. Tübingen 1881.

That self-rejection has its great psychological and ethical importance in being a sign that we are working our way out of the formerly ruling direction of will, just as it is a sign of recovery from mental illness when one acknowledges that one has been insane.

[400] **b.** If one keeps purely empirically to what asserts itself in and for consciousness before and during the action, one cannot establish the validity of the law of causality in the domain of the will, or of psychical life in general. The law of causality has here, as everywhere, only the significance of a presupposition, a postulate, with which we approach every domain of our inquiry. Where we understand anything at all, it is by finding a causal connection, and when we want to understand the life of will, we must therefore also assume that the law of causality can find application here as in other domains. This assumption is in itself justified and cannot be refuted. For even where we cannot, not even hypothetically, find any explanation, it is nevertheless most natural to assume that the causes lie so deep, or that the circumstances under which they act are so intricate, that we are unable to see through them. That is the conclusion we draw in every other domain of inquiry when explanation fails us. *Determinism*, which assumes the validity of the law of causality for the life of the soul and of the will as well, therefore need not come forward in dogmatic form. *Indeterminism*, on the other hand, proceeds dogmatically, in that, in good scholastic fashion, it disguises its ignorance by deriving inexplicable actions from a cause-free will presupposed for this purpose. This concept is a psychological counterpart to the “vital force,” now pretty well abandoned, in physiology. That we cannot find the cause does not prove that there is no cause, and proves still less that there exists a cause-free force. Observation certainly cannot settle the dispute, since from the indeterminist side it can always be maintained that it rests on an illusion when one believes one experiences that the victorious motive was also the strongest beforehand. But determinism is nevertheless more favorably placed with respect to experience and observation than indeterminism; for there is a possibility that it can be experienced and known that there are causes; but it is meaningless to speak of an experience that there is *no* cause.

Precisely the more highly developed the will is, the clearer an understanding of it can be won. The whole-cast, energetic characters we can understand, because every utterance
[401] and action is determined by the thoughts and feelings that give the character its stamp.³⁵ Where psychological understanding fails, we stand as a rule before what is restlessly

³⁵Conversely, it is also precisely especially pronounced and energetic characters that have embraced determinism. Thus the Stoics were determinists, while the Epicureans held to freedom from causes; energetic and practical Calvinism is deterministic, in contrast to soft, more passive Lutheranism; the English philosophical school, with its practical tendencies, is deterministic, in contrast to the speculative German school.

fermenting, wild and uncontrolled — thus precisely the opposite of self-determination and true will. In the incoherent ideas of the insane and of idiots (“Ideenjagd”) the cause-free would sooner have to be sought, if one wanted to find it within the psychological domain.

What we call motives in the psychology of the will is, by the nature of the case, not anything different from ourselves, from our ego or our nature. Usage, to be sure, lets a motive “act upon us,” as if it were an external power that pushed against us. Or one uses the image of the scale-pan, where the weight is put on from outside. The motives act in us and are themselves determined by our nature, partly by our earlier willing and acting. The voluntary and the involuntary, the conscious and the unconscious stand in an infinite interaction. Even the actions that are not the fruit of an express choice spring from our being, and, if we will not entangle ourselves in illusions (illusions which the theory of “the cause-free will” can easily encourage), we must ascribe them to ourselves. And on the other hand deterministic psychology impresses upon us that every change in our character is possible only through the influence of definite causes. The task, then, is not to will in general, but to will something entirely definite. How much we can accomplish we can never know beforehand. Our self-knowledge is always limited, and we cannot know how much energy still rests unused in our nature. The experimental way is the only one that can be applied here. Psychological determinism is experimental and thereby differs from fatalism.³⁶ The conviction that in a single case we have used all [402] our powers can only be a practical faith, a postulate.

According to indeterminism all “free” actions must arise without connection with the rest of the individual’s nature. Consistent indeterminism cannot help itself with phrases such as “relative independence” or such (in this connection) self-contradictory expressions. When it asserts that certain phenomena have no cause, it must absolutely and ruthlessly declare the psychological connection broken. But that is a denial of the principle of psychology. Such actions can be summed up, but they cannot be links in an inner development. We would then get the individual’s psychical life dissolved into a series or sum of members, a conception that we have constantly combated in the foregoing. The life of consciousness would be dissolved into atoms, into moments that had nothing to do with one another.³⁷ Our investigations have, on the contrary, at every

³⁶The fatalistic proposition that what is to happen will happen, whatever we do, is a self-contradictory superstition, which ought not constantly to be set up as an argument against determinism. Indeterminism can in any case be hit by an entirely similar sophism. “For,” one might say, “if I have a cause-free will, then I need only sit quite still and wait for it to stir; and if it will not stir, what can I do about it?”

³⁷Cf., e.g., J. G. *Fichte*: *Sittenlehre* § 10. “In a doctrine of nature every member can be explained. In a series of determinations of freedom no member can be explained, but each is a first and absolute one. Every free decision is itself substantial; it is what it is absolutely through itself.” The consequence of this

point established it as the peculiarity of the life of consciousness that past, present and future do not stand strange and isolated over against one another. Everything belonging to psychical life stands in the most intimate interaction. And, as we have seen, purpose is “the slave to memory.” If memory has its psychological laws, purpose and decision must also have theirs. One cannot take the single act of will out of the whole course of development of the life of will, or the will as such out of the intimate connection in which it stands with the other sides of psychical life.

[403] Further, indeterminism tears the individual loose from its natural foundation. Every willing being has certain presuppositions, with which it begins and on which it always builds. When the individual comes into possession of itself, it finds itself at an entirely definite point; this point it must take as something given. The task here for the individual is then to reconcile himself with his nature, so that he sets up his tasks with constant regard to the starting point given him. He must put up with the fact that there is something that has taken the initiative in him before he himself could; at any rate he could not take the initiative to his own existence. The resignation demanded here by the nature of things can take on a religious character, in that it appears as trust and confidence in the world-law that has governed the coming into being of the individual with this definite natural foundation. (Cf. VI B, 12).

The general reasons that lead to the assumption of the identity hypothesis concerning the relation between soul and body (II, 8) also speak decidedly against the assumption of a cause-free willing: such a willing would come into irreconcilable conflict with the law of the conservation of energy.

Finally, in so far as indeterminism relies on ethical grounds, the debate about the “freedom” of the will does not concern psychology. It becomes ethics’ own affair whether it will erect its building on dogmatic ideas instead of on the guiding thoughts of the science of experience. It must, however, be remarked that the dogma of a cause-free will will prove on closer investigation to be a presupposition that ethics itself can do without, indeed even a questionable one.³⁸

Psychology must, according to its principle, start from the premise that every spiritual phenomenon, and hence also every decision and action, has its cause. It maintains that we ourselves are the cause of our actions, and seeks only to find in the single cases what

must be a psychological atomism. The decision is then no longer the conclusion of a preceding process, but stands in an entirely indifferent relation to what goes before: so it turns up in a blind and accidental way! — On closer study of Fichte’s profound work one will, however, see that he is too much of a psychologist to be in earnest with this doctrine.

³⁸See the last chapter of my book “Om Grundlaget for den humane Etik” [On the Foundation of Humane Ethics].

this “we ourselves” is like, and what elements within our individuality they are that can make the action understandable. It seeks to describe the different stages of development that the active side of our nature can run through, and the transition from the one to the other. For the highest stage within this course of development there has been used, from the time of Socrates on, the expression inner or spiritual freedom, in a sense that [404] must surely be kept apart from the cause-freedom of indeterminism. Freedom in the Socratic sense of the word stands in contrast not to necessity, but to chance, blindness and spiritual bondage. It is that concentration and independence in the will which brings it about that a man in his whole conduct is in harmony with his innermost conviction, with his deepest feeling and his clearest knowledge. It is in this sense of the word that it is said: “Ye shall know the truth, and the truth shall make you free.” Freedom is thus for psychology a result, which is reached, and can only be reached, by definite ways.

Just as it is the business of psychology to explain spiritual phenomena, so it is the business of ethics to *evaluate* them. Evaluation and explanation need by no means meet each other as hindrances (cf. I, 9). When my ethical feeling is sufficiently alive, I condemn my evil thought or my evil action, however well I understand how they came about under the given circumstances. I condemn them even the more strongly the better I understand them: for the more closely must they hang together with my nature, and the more concerned about myself must I become.

C. The Individual Character

1. All life of consciousness is individual. Memory and thinking, pleasure and pain, drive and decision all presuppose a common inner center. It is the task of psychology to present the elements, forms and laws common to all life of consciousness. But these appear in reality in infinitely many different combinations and nuances. The general, abstract individuality of which psychology speaks is only a schema, which in every given case is filled out in a different way. This multiplicity general psychology cannot exhaust; that is the business of immediate experience of life, of poetry and of history. Psychology has only to point to certain typical differences, which are conditioned by the relation among the various psychical elements and forms of activity.

In the first place it will make a characteristic difference whether it is the elements of knowledge, of feeling or of will that predominate in the individual. Next, within each [405] single kind of ruling elements, there can again be a single direction that predominates. Thus within the domain of knowledge sense-perception and thinking stand as opposites;

within sense-perception the different senses (cf., e.g., the difference between the painterly and the musical gift); and within thinking the associations by similarity and by contiguity. Within the domain of feeling the contrast between physical and ideal feelings is of great importance; next, the contrast between pleasure and displeasure, egoism and sympathy. As regards, finally, the will, there are individuals who are predominantly guided by drive and instinct, while others work their way laboriously forward through a series of decisions. There are some in whom the will appears especially as inhibiting, others in whom the positively selecting and potentiating activity of will is most prominent. — And to all these differences are added, in each single domain, differences in strength, speed and extent.

From of old, psychology has laid most weight on the original dispositions of feeling that determine the fundamental tone of the spiritual life, whatever direction it may otherwise take. Both *talent* and character are determined by *temperament*, just as feeling occupies a central position over against knowledge and will. Temperament is conditioned by the organic constitution and shows itself immediately in the feeling of life, the fundamental mood that, apart from definite external experiences, rules the mind. In temperament the vegetative life of man, so to speak, becomes determining for his spiritual life. As a background given from the first, it conditions the way in which all experiences are taken up by him, and therefore also the way in which he acts back upon his surroundings.

Already in antiquity *Galen* set up the doctrine of the four temperaments, corresponding to the four elements of Empedocles and the four organic humors of Hippocrates. The physiology on which the doctrine of temperaments relied has long since been abandoned. For the most part, irritability is now taken as the basis, following *Haller*.³⁹ The differences of temperament thus rest on the different strength, speed and liveliness [406] with which influences from outside are received and answered. *Kant* thus distinguished between temperaments of feeling and of activity: the temperaments of feeling are the light-blooded (sanguine) and the heavy-blooded (melancholic), the temperaments of will the warm-blooded (choleric) and the cold-blooded (phlegmatic). In most recent times *Wundt* has taken the relation between the strength of emotion and its speed as the basis, and has thereby come back by another way to the old fourfold division: choleric (strong and quick), melancholic (strong and slow), sanguine (weak and quick) and phlegmatic (weak and slow).⁴⁰

³⁹Cf. H. *Hertz*: Temperamentslærens Historie [History of the Doctrine of Temperaments]. Kbhvn. 1856.

⁴⁰*Kant*: Anthropologie. 2te Aufl. p. 255 ff. — *Wundt*: Physiol. Psychol. II. p. 347 f. — *Sibbern* comes closest to *Kant*. Psykologisk Patologi [Psychological Pathology] § 55.

However essential a point of view has been grasped here, irritability or the capacity for reaction being the fundamental property of everything living, there is nevertheless something that the old fourfold division does not properly bring out, namely the disposition toward the one or the other of the great contrasts of the life of feeling; and yet the whole color and direction of the life of moods is conditioned by this. To the four old temperaments one might therefore add the bright and the dark temperament. And this contrast is more fundamental than the one underlying the first four temperaments, since it has its root in the very fundamental conditions of the existence of the individual organism. Physiologically regarded, the contrast between the bright and the dark temperament points to the influence of the vegetative functions on the brain, while the first four temperaments must be traced back to the greater or lesser ease with which the nervous central organs are set in activity by external influences.⁴¹

2. The origin of the individual character points back to the origin of the individual organism. In another connection (VI B, 7) we have had occasion to mention how early the germ arises from which a new organism develops. When fertilization has taken place and growth begins, the result is determined at every point by the relation between growth and differentiation determined from within, on the one side, and the *mechanical conditions* under which the development takes place, on the other. In the single cases it [407] is extraordinarily difficult to say whether the formation of a form is conditioned by inner processes or by the influence of the “mechanical factors.” Malformations often arise from an originally healthy germinal rudiment, which becomes diseased during development.⁴² — After birth too, physical conditions (nourishment, climate, etc.) intervene determiningly. Bodily stunting, e.g., leads also to spiritual stunting, and statistics establishes the influence of external circumstances on men’s actions. But for each single individual the power of the physical conditions is nevertheless always conditioned by the inner conditions with which the individual meets them. Individuality is thus here always presupposed.

The same holds as regards *social* causes. Imitation, education, relations of authority play an extraordinarily great part in the spiritual development of every individual. Even *Fichte*, who so one-sidedly and energetically maintained the inner originality of personality, its capacity to determine itself, cannot explain the development from lower to higher stages without presupposing influence from outside, even if this is to consist only in the release of the spring. In spiritual growth it is still more difficult to keep

⁴¹Cf. *James Sully*: Pessimism. A history and a criticism. London 1877. p. 405–414.

⁴²*Kölliker*: Entwicklungsgeschichte des Menschen. 2te Ausg. p. 385 ff. — Panum: Bidrag til Kundskab om Misfostrenes fysiologiske Betydning [Contribution to the Knowledge of the Physiological Significance of Monstrosities]. Kbhvn. 1877. (Universitetsprogram). p. 70 ff.

the inner and the outer influences apart than it was in bodily growth. While some regard individuality as given from the first, so that what is lived through, experienced and learned gets only an entirely subordinate importance, others have sometimes (thus *Helvetius*, and in our days *Stuart Mill*) derived all difference in mental powers from different education. Against this speaks the experience that education acts most on intermediate natures. The stubbornness of racial characters in spite of uniform education likewise shows that an original foundation constantly asserts itself.

[408] A more deeply lying point of view is given us by *heredity*. Organic individuals arise through propagation. The germs of the new organisms develop from earlier organisms, and since it now turns out that they inherit the constitution of these in a greater or lesser degree, it seems to be a natural assumption that the origin of individuality and its properties are due to the stock from which it springs. The properties that cannot be explained by physical and social causes will perhaps find their explanation when one goes back to the earlier generation. What stands as inexplicable in the individual can become explicable in the race. There is no individual trait for which one will not be able to get an explanation by investigating the history of the race. Such an investigation becomes difficult and intricate because heredity ramifies without end, and because several generations can be skipped (atavism).⁴³ What has once been deposited in the human organization is not easily rooted out again. Through this interaction between a fixed type (which, even where it is broken, can turn up again by atavism), — properties implanted by crossing, — and properties acquired by adaptation to new circumstances of life and experiences, by practice and suffering, — a prospect opens of an infinity of different combinations or syntheses.

Heredity in the race has often been compared with memory in the single man. But just as memory does not preserve everything, and therefore does not explain everything in a man's life either, so heredity too is no more than a tendency in nature to preserve what has been acquired. How far this tendency goes, and how far it is able to overcome new circumstances and experiences, is a question that must be taken up anew in each single case. Empiricism, which derives the content and peculiarity of consciousness from individual experiences of life, therefore always keeps a justified place open. There are in the race, as in individual consciousness, two currents or tendencies (cf. V B, 2), which can stand in the most various relations to each other. A special class of individual [409] differences rests precisely on whether the inherited, originally given constitution or what

⁴³The expression "atavism" was first used by *Duchesne* of plants. *Proper Lucas*: *Traité philos. et physiol. de l'hérédité naturelle*. Paris 1847–50. II. p. 43, — *Aristotle* already knew this phenomenon. *Hist. anim.* VII, 6. (ed. Bekker p. 585 b). *De gener. anim.* I, 18 (p. 722 a).

is experienced in the course of life plays the greater part in the formation of character. The sanguine and choleric temperaments thus let external influences exert a greater influence than the melancholic and phlegmatic.

Without going here more closely into the theories of heredity, we shall merely stress a couple of general results.⁴⁴ — 1) The more deeply something is taken up into the organization, the sooner it is inherited. What has just been acquired is in unstable equilibrium and is easily cancelled by crossing influences. — 2) Physical properties are inherited more easily than spiritual ones, simple talents more easily than those that rest on the cooperation of several mental powers. Instincts are inherited soonest; next come the powers of sense; dispositions of feeling are inherited sooner than intellectual capacities. — 3) Only elementary forms and dispositions are inherited. What is innate thus has a greater or lesser indefiniteness, and it depends on individual experiences in what degree and in what direction it is developed.

3. In the physical, social and inherited circumstances we have the elements from which the building of individuality is from the first erected. Our experience shows us no other domains where we could seek the stones for this building. Only by a purely mystical expedient would one escape the interminable investigations that are thus necessary in the case of every individuality. Individuality has sometimes been ascribed an absolute originality, regarded as an “Urposition” or an eternal monad. If one takes such a presupposition as basis, the problem of the origin of individuality naturally falls away entirely. But such a presupposition is arbitrary and unscientific. It can certainly be said that there is in nature a law of individuality, in so far as development in all domains leads to the formation of individual wholes at different stages (see above p. 102). But it is precisely the task of inquiry to find the elements of which these wholes are formed, and [410] the laws according to which they arise. Inquiry can and must admit that over against the individual it does not succeed in dotting the *i*, that here, as in all synthesis (cf. V C, 8), there is an *x* that escapes it — that individuality therefore stands as an irrational number, which can only be approximately determined. But it is of decisive importance here, as in all domains, to keep empirical inquiry free of mystical and speculative interventions.

From a purely psychological standpoint we must go a step further. Even if the individual organism, which in spite of its wholeness and relative closedness is yet a republic of cells, were explained as a synthesis of elements, and its origin could be understood according to the law of the conservation of energy, the individual consciousness, the

⁴⁴Cf. Decandolle: *Sur la part d'influence de l'hérédité, de la variabilité et de la selection dans le développement de l'espèce humaine*. (In the book: *Histoire des sciences et des savants depuis deux siècles*. Genève, Bâle, Lyon. 1873. p. 308–402). — Ribot: *L'hérédité psychologique*. 2e éd. Paris 1882.

formation of a peculiar center for memory, acting and suffering, would not thereby be explained. That such an inner center can come into being is the greatest problem, the fundamental problem in all our knowledge. Each single trait, each single property, we can perhaps explain to ourselves by the help of the power of heredity and the influence of experiences; but the inner unity to which all single elements are referred, and by which individuality is a *psychical* individuality, stands as the constant riddle. It is impossible for us, as we have remarked in an earlier connection, to carry through an analogy to the conservation of energy in the spiritual domain. Psychical individuality is one of the actual boundary points of science.

4. Not only individual properties and peculiarities of family or race, but also the forms and characteristic marks valid for the whole human race, for all human consciousness, have in most recent times been sought to be explained by way of heredity. Even before Darwin's hypothesis of the origin of species came forward, *Herbert Spencer* (in the first edition of "Principles of Psychology," 1855) set up the theory that the fundamental forms and powers of consciousness have developed through the adaptation of the primeval generations to the conditions under which their life was led. The forms of thought and feeling that are typical of the human race are therefore a priori for the single individual: i.e. [411] they cannot be fully explained by its experiences, but these, conversely, are conditioned by an original foundation. On the other hand, those forms find their explanation when one goes back to the whole race and the infinite series of experiences it must have had in the course of its development. What is à priori for the individual is thus a posteriori for the race.

This hypothesis is an attempt to lead the debate between idealism and realism into a new path and to settle it by giving each its due. Idealism is right as regards the single individual, while realism holds as regards the whole race. The earlier treatment of the problem (cf. the dispute between *Locke* and *Leibnitz*, *Hume* and *Kant*) really took account only of individual consciousness. Spencer has given up one of realism's most advanced positions, in order the more energetically to hold a more withdrawn position, a position that the narrower and purely empirical standpoint of the older realism prevented it from taking up.

This attempt at a solution nevertheless arouses misgivings of a twofold kind.

In the first place, race is a collective concept. At any given time the race consists of a certain number of individuals. It is these individuals who struggle for existence, exercise their powers, and in adaptation to the conditions acquire an organization that can be inherited by the next generation. But the individuals — however far back we go —

always come with an organization, thus with certain forms and powers that they have not themselves acquired, thus with something *a priori* for them. At every stage of the great process of development there is a given foundation that determines the effect of all experiences. It must therefore hold for the race as for the individual that the outer always presupposes the inner, that what is acquired is conditioned by what is originally laid down. This is a fundamental relation that always repeats itself. We here enter upon an infinite series; thought describes no limit, but has its footing in the ruling law.

Spencer's theory recalls Plato's mystical doctrine of knowledge as a recollection from pre-existence.⁴⁵ There is indeed the great difference that, while Plato explains what in [412] our consciousness cannot be derived from our experiences as acquired in a spiritual pre-existence, where the souls in the company of the gods beheld the eternal ideas, according to Spencer the foundation on which our spiritual life is built has been brought about by the labors, sufferings and struggles of millions of generations. But Plato and Spencer both overlook that what is at issue is a question of principle, and that if existence presupposes an *a priori*, such an *a priori* is also presupposed for every pre-existence, whether we think of it in a realistic or in a mystical way. An absolutely first or last we do not reach, according to the nature of our knowledge. The fundamental relation between an inner and an outer, which is the general law of life, and therefore also of psychical life, is a special example of the law of relation, like the relation between subject and object.

In the second place, a definite distinction must be made between the psychological and the epistemological point of view. Psychologically regarded, the hypothesis of development is a great advance. It opens for us a wide horizon, a prospect of explanation which was hitherto closed. It is, psychologically as physiologically, an entirely justified principle, and will surely also prove more and more to be a fruitful one, that what is inexplicable in the individual can become explicable in the race. But epistemologically the matter stands otherwise. The ultimate principles to which the analysis of our knowledge leads us are also the ultimate presuppositions that we can reach. All explanations, proofs and hypotheses — thus also the hypothesis of development itself — rest on them. How comprehensive this logical foundation of all our knowledge is, it is the business of the theory of knowledge, not of psychology, to investigate. Psychology is a special discipline, which itself presupposes the general principles of our knowledge and cannot explain their validity. The unassailable standpoint of idealism is given in the necessity of thought that underlies every realistic hypothesis.

⁴⁵Cf. my treatise on Plato's psychology in *Tidsskrift for Filologi* 1876, p. 209 ff.